Novel Acceleration Methods and Improved Transition State Finding Approaches for the Automatic Exploration of Reaction Networks

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Entwicklung von Beschleunigungsmethoden und Übergangszustandssuchen für die automatische Explorierung von Reaktionsnetzwerken

Von der Fakultät für Maschinenwesen der
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Contents

List of Abbreviations

Acronyms

AMS Amsterdam Modeling Suite

BDE bond dissociation energy

CDD chlorinated dibenzodioxin

CDF chlorinated dibenzofuran

COC chlorinated organic compound

CREST Conformer–Rotamer Ensemble Sampling Tool

CTY ChemTraYzer

CTY-TAD ChemTraYzer-Temperature-Accelerated Dynamics

CV collective variable

CVHD Collective Variable-Driven Hyperdynamics

DFT density functional theory

DOE design of experiment

DS dividing surface

eq equation

FSM Freezing String Method

GSM Growing String Method

IRC intrinsic reaction coordinate

NEB Nudged Elastic Band

NTC negative-temperature coefficient

pAD pressure-accelerated Dynamics

pbc periodic boundary conditions

PES potential energy surface

QM quantum mechanics

rMD reactive Molecular Dynamics

RMSD root mean squared deviation

RSEM reaction space exploration method

SC-AFIR Single-Component Artificial Force Induced Reaction Method

TAD temperature-accelerated Dynamics

TS transition state

TSSMM two-state spin-mixing model

TST transition state theory

Symbols

BO_{cut}	bond-order cutoff
BR_i	branching ratio of reaction i
c_i	concentration of species i
c_i^{ref}	concentration of species i in the real system
c_i^{sim}	concentration of species i in the simulation
ΔX	difference between two states in property X
E	energy
E_{elec}	electronic energy
$E_{\mathrm{elec+ZPE}}$	zero-point corrected electronic energy
E_{pot}	potential energy
G	Gibbs free energy
IRC	intrinsic reaction coordinate
k_{B}	Boltzmann constant
k_i	reaction rate constant of reaction i
k_{low}	lower bound of the reaction rate constant uncertainty
K_p	reaction equilibrium constant
k_{up}	upper bound of the reaction rate constant uncertainty
N	number of species
N_{backward}	number of reaction events for a backward reaction
N_{forward}	number of reaction events for a forward reaction
$N^{\mathrm{par.\ reactions}}$	number of parallel reactions

N_{REs}	number of reaction events
$N_{\text{reactants}}$	number of reactants
N_{replica}	number of replica
P	probability to observe a reaction at least once
ϕ	equivalence ratio
$p{*}_{\text{x}}$	partial pressure of component x in chemical equilibrium
R	ideal gas constant
r_i	reaction rate of reaction i
τ_{delay}	ignition delay time
T	temperature
t	time
T^{low}	low target temperature
T^{high}	elevated simulation temperature
Δt_{min}	required simulation time
t_{wait}	waiting time for a reaction event to occur
$\frac{t_{\text{wait}}(T^{\text{low}},c_i^{\text{ref}})}{t_{\text{wait}}(T^{\text{high}},c_i^{\text{sim}})}$	boost factor
V	volume
V_{barrier}	reaction barrier
$V_{\text{barrier}}^{\text{threshold}}$	reaction barrier threshold
V_{DS}	potential energy of a molecular geometry in the dividing surface
$V_{\text{DS}}^{\text{min}}$	reaction barrier estimate
$V_{\text{DS}}^{\text{RE}}$	potential energy of a trajectory snapshot at the time of a reaction event
$\mathbf{v}_i$	velocity vector of atom i
V_{reaction}	reaction energy
$\mathbf{x}_i$	position vector of atom i
χ	confidence level
ZPE	zero-point energy

Abstract. Reaction models are important for many engineering tasks, such as the optimization of internal combustion engines or production lines in the chemical industry. The automated reaction space exploration method ChemTraYzer attempts to alleviate the often time-consuming reaction model development process. ChemTraYzer uses reactive Molecular Dynamics (rMD) simulations to find the underlying reactions of the reaction process. Coupling the rMD output to quantum-mechanical reoptimizations of reactant, product and transition state (TS) geometries and to transition state theory, allows ChemTraYzer to produce kinetic and thermochemical data with the low uncertainties,which are required for many reaction model applications. In this thesis, I address two major issues of the ChemTraYzer method. First, I introduce two acceleration techniques for the rMD simulations, the pressure-accelerated Dynamics (pAD) and ChemTraYzer-Temperature-Accelerated Dynamics (ChemTraYzer-TAD) methods, to extend the applicability of rMD to reaction processes that occur on longer time scales than the nanosecond. The acceleration techniques are designed to work with minimal *a-priori* knowledge of the reaction systems, as they are meant for the exploration of unknown reaction space. With the pAD method, I describe a methodology to choose elevated simulation pressure and temperature to speed up the occurrence of reaction events without simulating a biased reaction network. ChemTraYzer-TAD is a progression of the pAD method. ChemTraYzer-TAD allows for higher simulation temperatures than pAD and allows for high boost factors of 10^8 as demonstrated in our ChemTraYzer-TAD case study. Both methods are applied to case studies of low-temperature ignition reaction processes. Second, I improve the performance of the automatic TS searches, which are a bottleneck in the ChemTraYzer methodology, by introducing a recovery method for failed TS searches. Our new recovery method improves the overall TS search success rate by up to 10 percentage points to a total of 48%. Finally, I demonstrate the practical usability of ChemTraYzer for automated reaction space exploration using case studies of chlorinated dibenzofurane formation and decomposition processes.

Kurzfassung. Reaktionsmodelle sind für viele Ingenieursaufgaben wichtig, wie z. B. für die Optimierung von Verbrennungsmotoren oder Produktionslinien in der chemischen Industrie. ChemTraYzer wurde entwickelt, um den oft zeitaufwändigen Prozess der Reaktionsmodellentwicklung zu automatisieren. ChemTraYzer verwendet reaktive Molekulardynamiksimulationen (rMDs), um die dem Reaktionsprozess zugrunde liegenden Reaktionen zu ermitteln. Durch Kopplung der rMDs mit quantenmechanischen Geometrieoptimierungen von Edukten, Produkten und Übergangszuständen (ÜZ) und der Übergangszustandstheorie kann ChemTraYzer kinetische und thermochemische Daten mit den häufig benötigten geringen Unsicherheiten erzeugen. Diese Arbeit beschäftigt sich mit zwei Problemen der ChemTraYzer-Methode. Es werden zwei Beschleunigungstechniken für die rMDs entwickelt, pressure-accelerated Dynamics (pAD) und ChemTraYzer-Temperature-Accelerated Dynamics (CTY-TAD), um die Anwendbarkeit der rMDs auf langsame Reaktionsprozesse zu erweitern. Die Beschleunigungstechniken sind so konzipiert, dass sie mit minimalem Wissen über die Reaktionssysteme funktionieren, da sie für die Erkundung eines prinzipiell unbekannten Reaktionsraums gedacht sind. pAD ist eine Methodik, bei der erhöhte Drücke und Temperaturen für die Beschleunigung der Reaktionsprozesse in den rMDs so gewählt werden, dass das resultierende Reaktionsnetwerk möglichst wenig verzerrt wird. CTY-TAD baut auf pAD auf. CTY-TAD erlaubt höhere Simulationstemperaturen als pAD und ermöglicht hohe Boostfaktoren von bis zu 10^8, wie anhand einer Fallstudie gezeigt wird. Beide Methoden werden auf Fallstudien aus dem Bereich der Niedertemperaturzündung angewendet. Daneben wird eine Wiederherstellungsmethode für fehlgeschlagenen ÜZ-Suchen des ChemTraYzers beschrieben. Die Wiederherstellungsmethode verbessert die Erfolgsquote der ÜZ-Suchen um 10 Prozentpunkte auf insgesamt 48%. Die Arbeit schließt mit einer Demonstration der Praxistauglichkeit von ChemTraYzer für die Erkundung von Reaktionsräumen anhand von Bildungs- und Zersetzungstudien von chlorierten Dibenzofuranen.

1. Introduction

Complex reaction networks consisting of reactions that occur on microscopic scales of time and space, make up many macroscopic reaction processes, such as ignition processes [1–4], or production processes in chemical industry [5, 6]. Detailed knowledge about the microscopic reaction network is vital for many engineering tasks, such as the optimization of internal combustion engines, the development of new engine concepts [7, 8], or the optimization of the product quality of chemical industry products. Such knowledge may come from experimental data or reaction models. Reaction models can be used to complement experimental results. They allow, for example, for the extrapolation of the experimental data to different reaction conditions, or can contribute hard-to-obtain data, such as undesirable side products at concentrations below the detection limit. Reaction models in combination with design of experiment (DOE) approaches [9] can be used to reduce the number of required experiments, which is of high interest, *e.g.* if reaction processes involving hazardous compounds, like phosgene or other polychlorinated organic compounds are involved.

Often, a large number of reactions have to be included in a reaction model (*e.g.* 10.000 reactions in [4]), which all need to be identified and quantified. The need for automatization of the reaction modeling process is well-known and the development of reaction space exploration methods (RSEMs) that automate the exploration and quantification of relevant reaction space is an active field of research. AFIR, AutoTST, CHEMOTON, EXGAS, the Heuristics-Guided Exploration of Reaction Mechanisms, the Heuristics-Aided Quantum Chemistry method, KinBot, Rule Input Network Generator, and Reaction Mechanism Generator are prominent examples [10–20]. Several RSEMs based on molecular graphs [21–27] have been proposed as well. Automated RSEMs based on reactive Molecular Dynamics (rMD) are also already known: The (*ab initio*) nanoreactor [28, 29], the nebterpolator [30], AutoMeKin [31–33], and ChemTraYzer (CTY) [34, 35], which I continue to develop in this thesis. Some of the abovementioned RSEMs have already been discussed and compared in a series of reviews [36–41].

A natural strength of rMD-based RSEMs is that rMD can, in theory, provide all relevant reactions within the limits of the employed force field models. Therefore, rMD-based RSEMs have a natural potential to discover reactions that have never been described before. In contrast, only reactions that can be derived from the implemented heuristics can be detected with rule-based approaches. To alleviate the restriction, first-principles chemical descriptors have been proposed for a more general applicability of rule-based RSEMs [12, 42]. Grimmel and Reiher [43] discuss the limitations of the first-principles chemical descriptors.

rMD-based RSEMs may be, in theory, more generally applicable than many of their rule-based competitors. In practice, the high computational effort for the computation of interatomic forces limits the applicability of rMD simulations to chemical processes that occur on time scales below the microsecond. Acceleration techniques allow for an extension of the rMD applicability to slower processes that occur on the time scale of seconds [44]. Many of the most widely used acceleration techniques, such as the metadynamics [45], hyperdynamics [46], and the Collective Variable-Driven Hyperdynamics (CVHD) [44, 47] methods, require the construction of a collective variable (CV), *i. e.* an approximate reaction coordinate, or even an entire bias potential surface to drive the rMD simulation towards a pre-defined state, *e.g.* a reactive complex. In other words, the user must have *a priori* knowledge about the system to set up the simulation. The search for a CV is a difficult task for many applications [48]. However, it has been demonstrated by several authors that metadynamics is principally applicable to reaction space exploration [49–53].

As the main purpose of the rMD simulations in the CTY workflow is the exploration of unknown reaction space, I develop acceleration techniques that work with minimal *a priori* user knowledge about the system of interest in the second and third chapter. Instead of using bias potential energies to lower reaction barriers, my methods rely on the elevation of pressure, or in the gas phase equivalently, the reactant concentration, and simulation temperature. The pressure-accelerated Dynamics (pAD) acceleration method that I present in the second chapter is based on a suitable choice of the simulation parameters pressure and temperature for simulation systems that do not allow for an independent choice of the two parameters. I discuss the choice of simulation temperature and pressure for the simulation of low-temperature ignition reaction processes as an example in which the reaction chemistry is dependent on the two parameters. In other words, a wrong choice of elevated temperature and pressure may lead to the observation of reactions in the rMD simulations that are irrelevant at the real target conditions. This risk of discovering an irrelevant reaction network will be called "kinetics bias issue" from now on. Chapter two starts with a more illustrative discussion of the kinetis bias issue using the example of *n*-pentane low-temperature ignition before I continue to derive the pAD methodology. In the third chapter, I present the ChemTraYzer-Temperature-Accelerated Dynamics (CTY-TAD) method for the example of 1,3-dioxolane low-temperature ignition. Like pAD, CTY-TAD uses elevated temperatures and reactant concentrations to accelerate reactions. The method is designed to explore new reactions by blocking already known ones from reoccurring in the rMD simulations. This way, I reduce the risk of obtaining biased reaction networks. CTY-TAD builds on pAD but allows for higher simulation temperatures than the pAD method and allows therefore for larger boosts of the simulated reactions than pAD. Both parameters, concentration and temperature, also have to be specified by the user in every conventional, *i. e.* non-accelerated rMD simulation in the canonical ensemble. In other words, no additional parameters compared to a conventional canonical rMD simulation have to be specified to use my acceleration techniques.

Principally, kinetic properties like reaction rate constants can be derived directly from rMD simulations [54]. Uncertainties in the potential energies and vibrational frequencies of typical rMD force fields are not small enough, though, to satisfy the high standards of modern reaction models, in most cases. As a consequence, all of the above-mentioned rMD-based RSEMs employ rMD-quantum mechanics (QM) coupling schemes that requantify the kinetic properties of the observed reactions using higher level-of-theory potential energy surfaces (PES) and transition state theory (TST). I describe conceptual updates to the existing CTY rMD-QM coupling scheme in the fourth chapter, among them a self-learning transition state (TS) search recovery method for failed TS searches. One of the conceptual updates addresses the spin multiplicity guesses for reactant, product, and TS geometries, which are a known shortcoming of the CTY rMD-QM coupling scheme [35, 55]. Other conceptual updates enhance the computational resource efficiency and the TS search success rate. Furthermore, I integrate the CTY-TAD acceleration method into the CTY rMD-QM coupling scheme and demonstrate the practical usability of the new CTY coupling scheme in case studies of chlorinated dibenzofurane reaction processes. The chlorinated dibenzofurane reaction systems are selected as case studies, because their toxicity leaves us almost no alternative than to study them with predictive computational methods [56].

2. Exploring the Chemistry of Low-Temperature Ignition by Pressure-Accelerated Dynamics

Chapter 2 is based on the publication "Exploring the Chemistry of Low-Temperature Ignition by Pressure-Accelerated Dynamics" by Krep et al. [57]. In this chapter, parts of the publication [57] are reused directly or adapted with permission by the authors and ChemSystemsChem, Wiley-VCH Verlag GmbH & Co. KGaA. My contribution to the publication [57] include the generation of the results and, under consultancy by Dr. Wassja Kopp and Dr. Leif Kröger, the compilation of the text. Dr. Malte Döntgen contributed the idea for the work. The work was supervised by Prof. Kai Leonhard.

Early on in chemistry classes at school, we learn that reaction rates can almost always be increased either by an increase of the temperature or the reactant concentration, or equivalently by an increase of the reactant partial pressure in gaseous system. In other words, increasing the simulation temperature and the reactant concentration is almost always a simple rMD simulation strategy that achieves a boost of the simulated reaction process. I have chosen the example of *n*-pentane low-temperature ignition as case study in chapter two, for two reasons. First, the simulation of the ignition process would be too slow under technically relevant conditions to be observed within nanoseconds. Multiple nanoseconds are currently the limiting time span that can be simulated with the computationally most efficient reactive force fields, such as

ReaxFF [58, 59], within days of computer time [60]. Therefore, the simulation of low-temperature ignition processes requires the use of acceleration techniques. The fact that the low-temperature ignition process is too slow for a conventional rMD study makes it easy to measure the success of the choice of simulation temperature and pressure. The parameter choice is successful if the known important reactions become visible in the simulations. The second and main reason has already been mentioned in the introduction. It is the fact that in rMD simulations of low-temperature ignition, simulation temperature and pressure cannot be chosen arbitrarily and independently from each other without biasing the underlying reaction chemistry. I discuss the choice of the two simulation parameters temperature and pressure (concentration) for the simulation of *n*-pentane low-temperature ignition using only *a priori* information about low-temperature ignition reaction processes that is known to be valid for many fuels [8, 61].

In low-temperature ignition processes, overall reactivity is determined by meta-stable reaction equilibria (O_2-additions), and slow subsequent reactions, such as the internal H-atom abstractions in Fig. 2.1 [61]. The simple use of elevated temperatures will push the meta-stable O_2-addition reaction equilibrium to the reactant side. The shift of the O_2-addition reaction equilibrium to the reactant side diminishes the number of peroxy radicals that can undergo further low-temperature reactions. The decreasing number of peroxy radicals counteracts the accelerating effect of increasing temperature on subsequent low-temperature reactions. The shift of the O_2-addition reaction equilibrium, but also an increasing flux through inhibiting side reactions lead to a lower overall reactivity and consequently, to an increase in the ignition delay time at certain temperatures [61]. The increase in ignition delay time can be seen, for example, in Fig. 2.2 at 10 atm between approximately 750 and 900 K. A further increase of temperature eventually paves the way for pyrolysis and other reactions with high-energy reaction barriers. That means, at sufficiently high temperatures, the ignition delay decreases again. This complicated interplay of shifts in O_2-addition reaction equilibria and acceleration of high-barrier reactions leads to s-shaped ignition delay time graphs (c. f. Fig. 2.2) for many hydrocarbons. A pronounced increase of ignition delay is called negative-temperature coefficient (NTC) behavior. Fig. 2.2 shows ignition delay times for different initial temperatures and pressures for lean pentane-oxygen mixtures with an equivalence ratio of $\phi = 0.09$, which is the equivalence ratio I chose for all simulations in this chapter. At the lowest temperatures, *e.g.* below 800 K at 10 atm in Fig. 2.2, ignition is dominated by O_2-additions and internal H-atom abstractions [8]. At the lowest pressures this fuel-oxygen mixture shows NTC behavior. From the pronounced NTC behavior, two more regimes besides the low-temperature regime can be identified that come along with the described changes in O_2-addition reaction equilibria and a change in predominant reactions from low-temperature reactions to high-temperature reactions (*e.g.* β-scissions) - the medium-temperature and the high-temperature regime. According to mechanistic simulations the NTC behavior becomes less pronounced with increasing pressure, until it eventually vanishes at pressures between 250 atm and 1250 atm (c. f. Fig. 2.2). After the NTC behavior has vanished at the highest pressures, both low-temperature and high-temperature reactions remain present. These findings are qualitatively validated by high-pressure experiments up to 530 atm [62]. An overview of important reactions in all three regimes can be found in Curran *et al.* [61].

Besides a vanishing NTC behavior, we observe that the low-temperature regime extends to higher temperatures and shorter ignition delay times with increasing pressure. The observation that elevated pressure accelerates the low-temperature ignition and that the low-temperature regime extends to higher temperatures is another motivation to use elevated pressures in rMD simulations of low-temperature ignition. I have mentioned already at the beginning of this chapter that low-temperature ignition processes occur on time scales that are too large to be feasible with modern rMD simulation tools at technically relevant conditions. For that reason, I modify the standard rMD approach from searching for the reactions that are present at the techi-

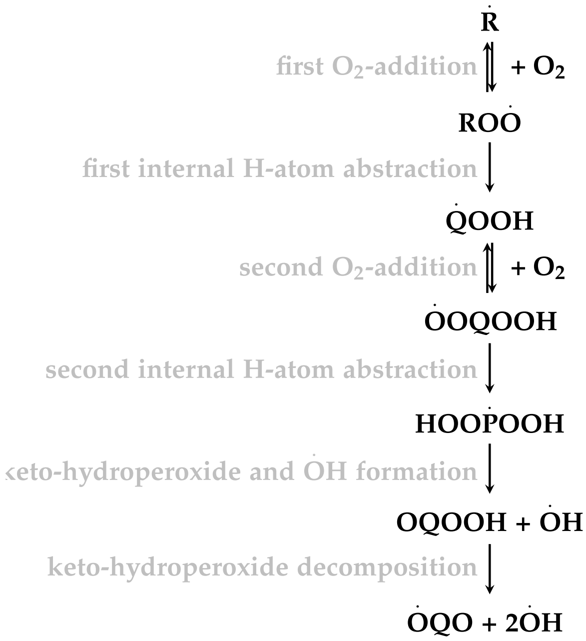


Figure 2.1.: The key low-temperature chain branching path starting from the alkyl radical $\dot{\mathrm{R}}$. A detailed review on low-temperature ignition chemical processes can be found in Zádor *et al.* [8].

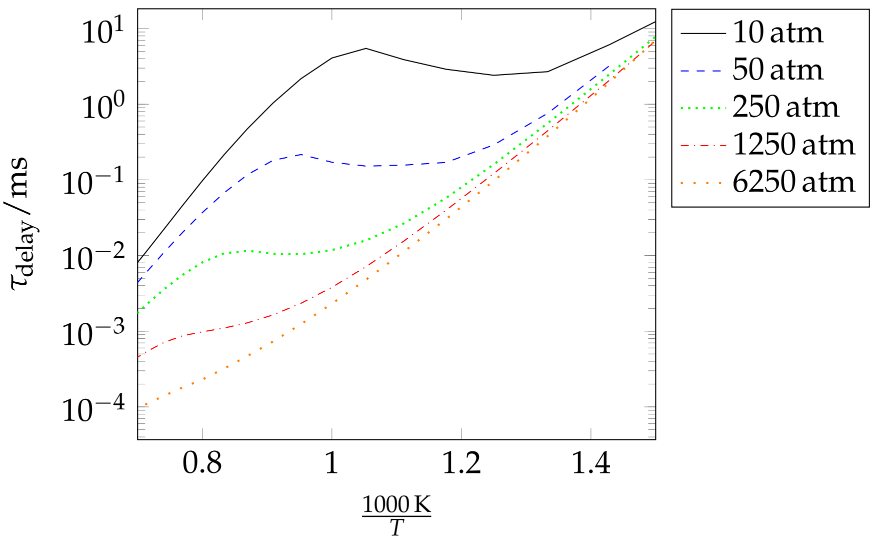


Figure 2.2.: Ignition delay time of *n*-pentane for different initial pressures modeled with the reaction mechanism from Bugler *et al.* [1–3] in adiabatic isochoric simulations. The simulations were conducted with Cantera [63]. Ignition delay times of lean *n*-pentane-oxygen mixtures with an equivalence ratio of $\phi = 0.09$ as in the rMD simulations are shown.

cally relevant target conditions to finding conditions that allow for the observation of the reactions that are relevant at the target conditions. Certainly, the new approach requires more *a priori* knowledge of the system than the set-up of a standard rMD simulation but compared to many of the other existing acceleration methods that have been mentioned in the introduction, much less detailed knowledge of the underlying processes on the atomic scale is required. In this chapter, I use knowledge of the reactions that are generally known to be important for low-temperature ignition [8, 61], to find a simulation set-up that increases the probability to observe the expected reactions. The key idea of the pAD methodology is to increase the partial pressures of the reactants. That way, the reaction equilibria of the O_2-addition reactions are pushed to the product side. Hence, the product, the peroxy radical, is available for a larger part of the simulation. As a consequence, the probability to observe subsequent reactions, such as the first internal H-atom abstraction, is increased. The extension to higher temperatures is another important ingredient of the pAD methodology, as it allows to accelerate reactions subsequent to the O_2-addition not only by the increase of reactant concentration, but also by temperature. A similar approach has already been employed in the (*ab initio*) nanoreactor method [28]. They use a virtual piston to increase the pressure periodically [28]. As the authors note, their goal is to find reactions that are generally possible on the given potential energy surface instead of generating kinetic models that are consistent with any given temperature and density directly. Meaningful kinetics have to be generated through reoptimization of barrier heights and thermochemistry using density functional theory (DFT) or higher-level methods and subsequent reaction model building. Only then, the kinetics are extrapolatable to other temperatures. With pAD, consistent high-pressure limit kinetics are directly generated for the simulated temperatures. With a second set of pAD simulations at different temperatures, extrapolation of the kinetics is possible without the need of reoptimizations. Apart from the periodically high concentrations in the nanoreactor, equilibrium simulations with high concentrations have already been successfully used to accelerate high-temperature oxidation and pyrolysis reactions [64], or have been used to study the oxidation of cyclohexane in combination with the metadynamics acceleration method [49]. Bal and Neyts [47] use the CVHD acceleration method and ReaxFF to investigate the oxidation processes of *n*-dodecane. From their simulations, they note that the average reaction time decreases with pressure. In this chapter, I exploit this observation for acceleration purposes and show that an increase in pressure is already sufficient to observe low-temperature ignition reaction pathways. I present pAD as a method to investigate low-temperature ignition processes. Nevertheless, the examples from the literature show that the applicability of pAD is not limited to low-temperature ignition processes.

In the case study, I present the reaction pathways observed during pAD simulations of *n*-pentane low-temperature ignition processes using the ReaxFF reactive force field [59]. It will be shown that inaccuracies of the ReaxFF force field can have a considerable impact on the branching ratios. Still, with ReaxFF, it is possible to find the expected, important low-temperature reactions, and other side reactions, whose importance for reaction models might have been overlooked so far.

2.1. Methods

The alkyl radical and especially the oxygen concentration c_{O_2} are the key parameters here. A higher concentration of these two species pushes the reaction equilibrium of the first O_2-addition to the product side, *i. e.* to the peroxy radical. As the peroxy concentration is thereby increased, the subsequent reactions, most notably the first internal H-atom abstraction (c. f. Fig. 2.1), are accelerated. For the optimal choice of the simulation box composition and the simulation temperature in *NVT* simulations, the existence of reactions parallel to the reactions I intend to observe, especially the O_2-additions and the internal H-atom abstractions, must be taken into account. The β-scission is the most important parallel reaction that becomes competing with the first

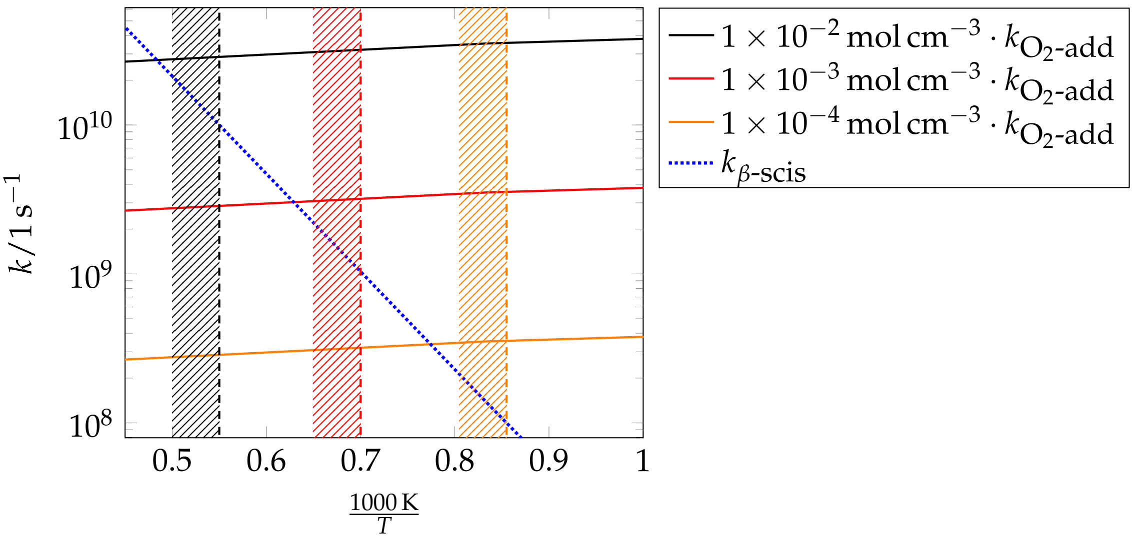


Figure 2.3.: The gap between the high-pressure limit pseudo-first-order reaction rate constants of the O_2-addition reaction $k_{O_2\text{-add}} \cdot c_{O_2}$ and the reaction rate constant of the β-scission $k_{\beta\text{-scis}}$ of the 2-pentyl radical is shown for different oxygen concentrations c_{O_2}. O_2-addition rate constants are taken from Asatryan and Bozzelli [65], the β-scission rate constants from Comandini *et al.* [66]. At temperatures left of the vertical lines β-scissions are competing to the pseudo-first-order reaction rate constants with a c_{O_2} of the same color or even become dominating.

O_2-addition. That means, the β-scission reaction rate constant $k_{\beta\text{-scis}}$ is of the same order of magnitude as the pseudo-first-order reaction rate constant $k_{O_2\text{-add}} \cdot c_{O_2}$. Fig. 2.3 illustrates that an increase of c_{O_2} by two orders of magnitude (0.0001 mol cm^{-3} - 0.01 mol cm^{-3}) shifts the temperatures at which the β-scission and the O_2-addition are competing from less than 1200 K to more than 1800 K (c. f. the areas left of the vertical lines in Fig. 2.3). A c_{O_2} of 0.0001 mol cm^{-3} corresponds to the oxygen concentration of a stoichiometric ideal pentane-oxygen mixture at approx. 10 atm.

To increase the reaction rate of the first O_2-addition, I could also increase the alkyl radical concentration. An increase of the alkyl radical concentration would also increase the reaction rate of the β-scission, though. Therefore, I decided to choose a highly diluted system. In order to focus on the key low-temperature reactions, I decided to put only one fuel radical into the simulation box. A side effect of the high dilution is that self-interactions, which might appear in small boxes with periodic boundary conditions, are shielded by the large number of O_2 molecules. The box size was adjusted to fit the respective c_{O_2}. The appropriate order of magnitude of O_2 concentration depends on the simulation temperature. For the choice of temperature, I take the most important parallel $RO\dot{O}$ reactions in the low-temperature chain-branching process (c. f. Fig. 2.1) into account (c. f. Fig. 2.4). In the high-pressure limit, the reaction rate constants of the first internal H-atom abstraction $k_{\text{int H-atom abstr}}$, and the reverse reaction of the first O_2-addition, the $R-O_2$ bond dissociation, $k_{R-O_2 \text{ bond diss}}$ only depend on the temperature. $k_{\text{int H-atom abstr}}$ and $k_{R-O_2 \text{ bond diss}}$ intersect at approx. 1200 K. Below, $k_{\text{int H-atom abstr}}$ is higher than $k_{R-O_2 \text{ bond diss}}$. In this regard, temperatures lower than 1200 K are preferable for the simulation of low-temperature ignition processes. On the other hand, for the chosen box composition with one fuel radical, which means, there will also be only one C5 peroxy radical in the box, I have to simulate at higher temperatures than 1200 K to achieve expected waiting times of less than 10 ns for the first internal H-atom abstraction to occur at least once with a 90% observation probability. A derivation for the expected waiting time can be found in Kröger *et al.* [54]. These considerations show that for the choice of temperature a trade-off is necessary between a low reactivity of reactions subsequent to O_2-additions at low temperatures and increasing importance of $R-O_2$ bond dissociations, and β-scissions at high temperatures. With a c_{O_2} of $0.01\,\text{mol}\,\text{cm}^{-3}$, $k_{O_2\text{-add}} \cdot c_{O_2}$ is more than two orders-of-magnitude larger than $k_{\beta\text{-scis}}$ at 1200 K which gives safety margin in case the β-scission is faster and the O_2-addition is slower than expected in the simulations. Also, at 1200 K $R-O_2$ bond dissociations are at least not faster than the first internal H-atom abstraction, making 1200 K a possible trade-off. Choosing pentane ignition as case study allowed me to determine favorable *NVT* simulation conditions from literature reaction rate constants [65, 66]. The chosen simulation conditions should be transferable to other fuels, as the change in regimes in the ignition delay time often appears at similar temperatures (c. f. [67–69], a comparative plot is given in the appendix in section A.1.5).

I have conducted 200 simulations with 90 oxygen molecules and one pentane molecule

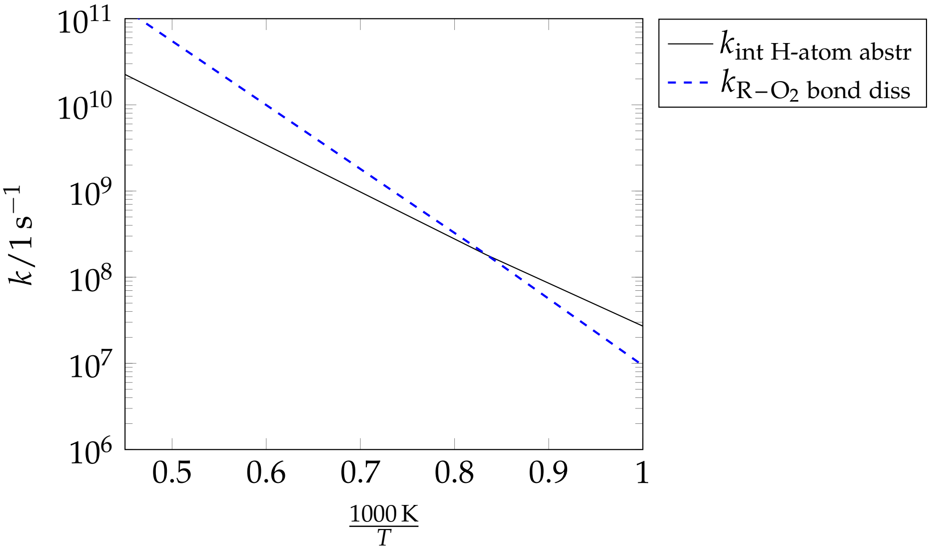


Figure 2.4.: The reaction rate constants of the first internal H-atom abstraction and the reaction rate constant of the $R-O_2$ bond dissociation reaction of the 2-peroxide-pentane, obtained from Asatryan and Bozzelli [65], intersect at approx. 1200 K.

($\phi = 0.09$) in the simulation box at 1200 K. The simulation boxes were created with packmol [70]. After a minimization and a thermalization of 500 ps, I removed an H-atom from the pentane molecule and from the system. That means, the analyzed part of the trajectory with a length of approx. 2 µs in total starts with a 2-pentyl radical. Simulations were run using the USER-REAXC package and fix qeq/reax [71] implemented in LAMMPS version 17Nov16 [72] and were analyzed with the ChemTraYzer [34]. The simulations were thermostated with the Nose-Hoover thermostat implemented in LAMMPS with a damping constant of 10 fs. A timestep of 0.1 fs was used. Back and forth reactions occuring within 1 ps were considered to be recrossing events and were therefore not counted.

I chose the well-established C/H/O-2008 parametrization by Chenoweth *et al.* [59], which has been designed to study combustion processes. The C/H/O-2008 parametrization has been applied to various systems [73–81]. For all these cases, "generally a good agreement with experimental results in terms of the initiation mechanism and barriers were observed" [64]. A root mean squared deviation (RMSD) of 18 kcal mol^{-1} of the ReaxFF reaction energies from B3LYP energies can be derived from the reactions provided by Chenoweth *et al.* in their Supporting Information [59]. I use the RMSD of 18 $\mathrm{kcal\,mol^{-1}}$ as an estimate for the deviations of both ReaxFF reaction energies and barriers from B3LYP energies in the following. Goerigk *et al.* report an RMSD of 4.5 $\mathrm{kcal\,mol^{-1}}$ of B3LYP reaction barriers from W1-F12 and F2-F12 reference barriers in their DFT methods benchmark study [82]. Combining the RMSDs of ReaxFF reaction barriers from B3LYP barriers and B3LYP barriers from the benchmark reference barriers results in a standard error of 19 $\mathrm{kcal\,mol^{-1}}$ for the ReaxFF reaction barriers using the C/H/O parametrization.

In order to verify observed reaction paths, TST reaction rate constants were calculated using the RRHO model implemented in tamkin [83]. Transition states were optimized with the Berny algorithm [84] implemented in Gaussian09 at the CBS-QB3 [85, 86] level of theory. The CBS-QB3 model chemistry has been evaluated on test sets, *e.g.* in the DBH24/08 database of barrier heights. Therein, Zheng *et al.* [87] report an error of 1.62 $\mathrm{kcal\,mol^{-1}}$ in CBS-QB3 barrier heights. Additional uncertainty arises from the other RRHO ingredients of the TST rate constant formulation. Kopp *et al.* [88] evaluated these uncertainties for B3LYP geometries and frequencies, which are also used in CBS-QB3. Combining these uncertainties with the uncertainty in barrier height leads to an uncertainty in the rate constants of a factor of 10 at 1000 K. By rotating C–C and C–O single bonds, conformers were searched with B3LYP and the CBSB7 basis set. B3LYP and the CBSB7 basis set are both also used for geometry optimizations and frequency calculations in the CBS-QB3 compound method [85]. Transition states on ReaxFF level were optimized with an in-house python implementation of the Berny algorithm [84]. Additionally, I conducted rMD simulations in a temperature range from 1200 K up to 1450 K in steps of 50 K (10 simulations per temperature) for comparison of reaction rate constants.

All simulations with the model by Bugler *et al.* [1–3] (c. f. Figs. 2.2, 2.5) were conducted with Cantera [63] and the IdealGasReactor model. Simulation conditions are given in the Figure captions.

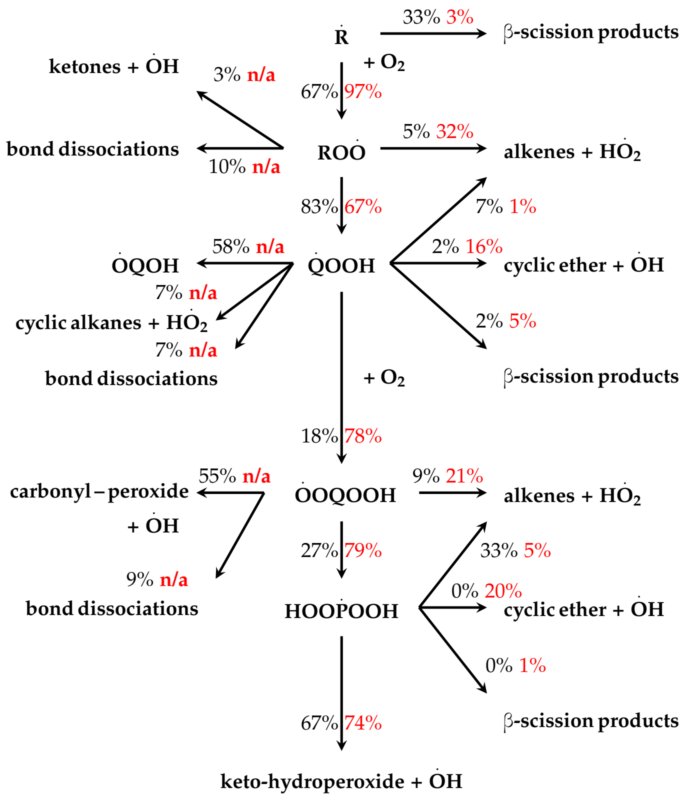


Figure 2.5.: Lumped C5 reaction scheme with branching ratios from the simulations (black) and from the model by Bugler *et al.* [1–3] (red) at 1200 K, 1300 atm and up to 1% fuel consumption. Fluxes are derived from lumping all isomers to classes of species. Thus, isomerizations, except internal H-atom abstractions and the ȮH-migration (ROȮ → Q̇OOH, Q̇OOH → ȮQOH,ȮOQOOH → HOOṖOOH), are not taken into account. Fluxes obtained from simulations with the reference model are time-averages over the simulation time up to 1% fuel consumption.

2.2. Results and Discussion

Fig. 2.5 shows the lumped reaction scheme of the low-temperature chain branching mechanism with branching ratios of C5 species from the simulations and from the reaction model by Bugler *et al.* [1–3] applied at 1300 atm. A detailed list of observed reactions is provided in the appendix in section A.1.1. In order to obtain the depicted branching ratios in Fig. 2.5, isomers were first lumped into species classes. As the internal H-atom abstractions are important reactions in the low-temperature ignition process, the peroxy radicals and the hydroperoxy radicals were lumped into two different species classes. Unlike Q̇OOH, ȮQOH species do not follow the key reaction path anymore. Therefore, also $\dot{\mathrm{Q}}$OOH and $\dot{\mathrm{O}}$QOH are seperate species classes. Branching ratios were then obtained by dividing the net number of escaping reaction events from one species class into another $N_{\text{forward}} - N_{\text{backward}}$ by the sum of all net number of reaction events that escape from one species class. For consistency, the fluxes generated with the reference model, from which the reference branching ratios have been obtained are averaged over the simulation time up to 1% fuel consumption. 1% fuel consumption is a trade-off between 20% fuel consumption, which usually serves as the reference time for the rate-of-progress analysis (c. f. [2, 89]) and the initial, pre-ignition reactor composition, which is represented by the rMD set-up (low fuel radical concentration). The propagation of the error in ReaxFF reaction barriers of 19 kcal mol^{-1} yields substantial uncertainties in the branching ratios. Hence, the branching ratios shown here can only serve as an orientation, whether the simulations produced mostly reasonable results, *i. e.* the majority of observed reactions are members of well-known reaction classes. An error propagation scheme is given in the appendix in section A.1.4. It should be mentioned that also the reference branching ratios must be expected to have uncertainties given the high pressure. Rigorously, real-gas effects would have to be considered. Here, I neglect them for two reasons. First, Zhukov [62] showed for the *n*-heptane model by Curran *et al.* [61] that only marginal reoptimizations of reaction rate constants are required to adjust the reaction model to high pressures. Second, as already argued, the branching ratios are only for qualitative comparison considering the large uncertainties in the rMD branching ratios. To resolve the largest deviations between reaction pathways from the simulations and from the model by Bugler *et al.* [1–3] I recalculated the 2-pentyl radical $\rightarrow$ 2-methyl-but-1-yl radical isomerization and the $\dot{\mathrm{O}}$H-migration of the 2-hydroperoxide-pent-4-yl radical with QM methods as described above (c. f. Figs. 2.6, 2.7). The optimized geometries of reactants and TS are given in the appendix in section A.1.2.

In the lumped reaction scheme, Fig 2.5, a majority of 67% of the members of the alkyl radical species class follow the O_2-addition pathway. The rest proceed through β-scissions. The high O_2-addition branching ratio comes at no surprise, because it was one the main targets of the rMD simulation set-up to obtain high fluxes through the major low-temperature ignition channels. The finding that the majority of alkyl radicals follows the O_2-addition pathway is also in qualitative agreement with the simulations using the reference model. However, before proceeding through these reaction channels, the 2-pentyl radical isomerizes first, mainly to the 2-methyl-but-1-yl radical in many rMD simulations. Intuitively, the branching ratio of the isomerization to the 2-methyl-but-1-yl radical of 41% seems to be very high compared to O_2-additions and β-scission. The isomerization branching ratio is not shown in Fig 2.5 as both the 2-pentyl radical and the 2-methyl-but-1-yl radical are both lumped into the alkyl radical species class. Comparison of the isomerization reaction rate constants derived from the rMD trajectories $k_{\mathrm{Iso,ReaxFF}}$ and CBS-QB3 TST reaction rate constants $k_{\mathrm{Iso,TST,CBS\text{-}QB3}}$ in Fig. 2.6 shows that the rMD reaction rate constants of the isomerization are approximately seven orders-of-magnitude higher than the ones obtained from TST. The statistical errors are too small to be the cause of the deviation. The lower slope of the Arrhenius fit to the data points from the trajectory $k_{\mathrm{Iso,fit}}$ compared to the CBS-QB3 TST reaction rate constant shows that the deviation is caused by a too low reaction barrier. The too low ReaxFF reaction barrier of approx. 15.3 kcal mol^{-1} instead of 60.71 kcal mol^{-1} on the CBS-QB3 level of theory makes the isomerization competing to the β-scission. The O_2-addition is two orders of magnitude faster than both the isomerization and the β-scission at 1200 K. Both the O_2-addition reaction rate constants $k_{\mathrm{O_2\text{-}add,ReaxFF}}$ and the β-scission reaction rate constants $k_{\beta\text{-scis,ReaxFF}}$ have met the reaction rate constants from the literature ($k_{\mathrm{O_2\text{-}add,CBS\text{-}QB3}}$, $k_{\beta\text{-scis,lit}}$) almost within statistical uncertainty (c. f. Fig. 2.6).

Further down the low-temperature reaction path, most $\mathrm{RO\dot{O}}$ species isomerize via internal H-atom abstractions. A minority reacts to alkenes and $\mathrm{H\dot{O}_2}$ radicals, decomposes into ketones and $\mathrm{\dot{O}H}$ radicals, or decomposes into radicals and diradicals

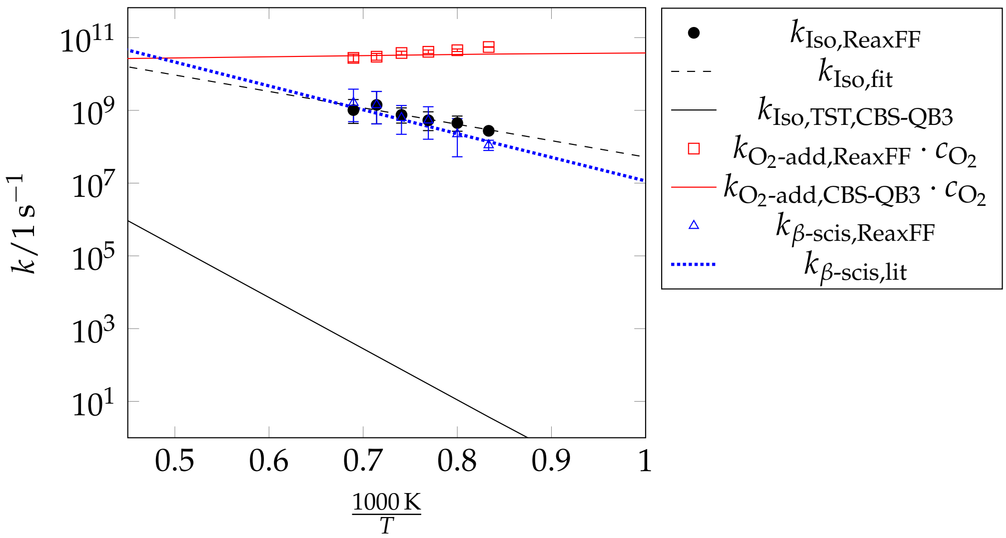


Figure 2.6.: (Pseudo-) first-order reaction rate constants of the isomerization of the 2-pentyl radical to the iso-pentyl radical and the parallel reactions O_2-addition, and β-scission. k_{fit} is an Arrhenius fit to the data points from the trajectory. $k_{O_2\text{-add,CBS-QB3}}$ was obtained from Asatryan and Bozzelli [65], $k_{\beta\text{-scis,lit}}$ from Comandini *et al.* [66].

("bond dissociations" in Fig. 2.5). The bond dissociations appear at multiple steps in the low-temperature path, *e.g.* reactions R54, R143, R157 in the reaction list A.1 in the appendix in section A.1.1. Neither the bond dissociations nor the formation of ketones or aldehydes and $\dot{O}H$ radicals are included in the reference model (c. f. "n/a" in Fig. 2.5), but the latter have been described in the literature [90, 91].

After the first internal H-atom abstraction, 18% of $\dot{Q}OOH$ species undergo the second O_2-addition. In the reference model, though, this is the main reaction channel with a branching ratio of 78%. The main difference comes from the $\dot{O}H$-migration reaction $\dot{Q}OOH \rightarrow \dot{O}QOH$. $\dot{O}H$-migration is not included in the reference model. Fig. 2.7 illustrates that in the simulations, the $\dot{O}H$-migration and the O_2-addition are competing. While the pseudo-first-order reaction rate constant of the O_2-addition of the 2-hydroperoxide-pent-4-yl radical to 2-hydroperoxide-4-peroxide-pentane has been met almost within statistical uncertainty (c. f. $k_{O_2\text{-add,ReaxFF}} \cdot c_{O_2}$, $k_{O_2\text{-add,CBS-QB3}} \cdot c_{O_2}$ in Fig. 2.7), the MD reaction rate constant of the $\dot{O}H$-migration $k_{OH\text{-migration,ReaxFF}}$ of the same molecule is too high by four orders of magnitude compared to the CBS-QB3 TST reaction rate constant $k_{OH\text{-migration,TST,CBS-QB3}}$. Principally, both too-low and too-high

ReaxFF energy barriers may cause deviations between the reference model and my simulations considering the uncertainty of approximately 20 kcal mol^{-1} in ReaxFF energy barriers. Furthermore, in extreme cases, too-high barriers may result in reactions not being observed in the simulations at given simulation temperatures and pressures. Here, however, the deviation can again be explained by the too low ReaxFF energy barrier of approx. 5.33 kcal mol^{-1} compared to approx. 28.44 kcal mol^{-1} on the CBS-QB3 level of theory. Green *et al.* [92] and Villano *et al.* [93] discuss the

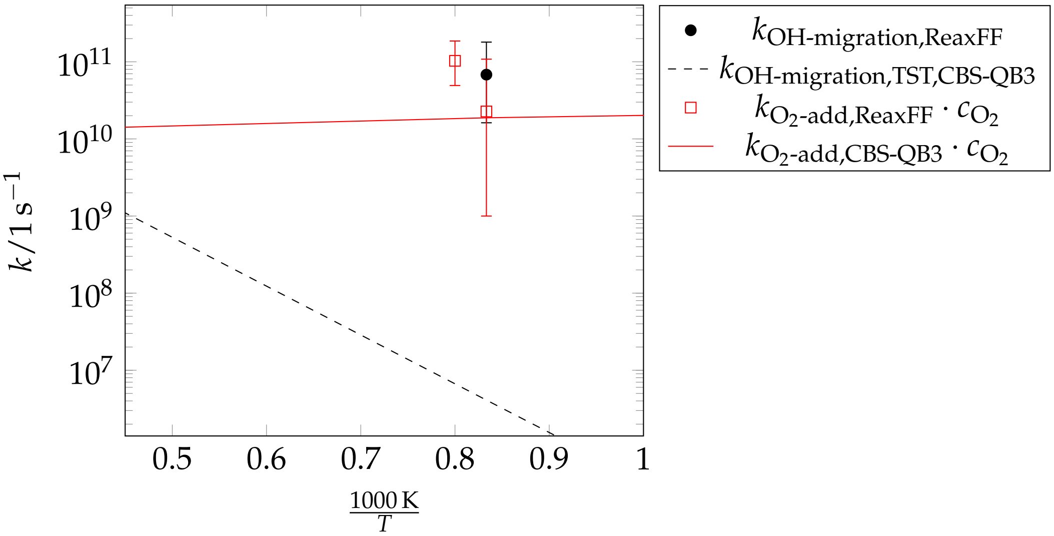


Figure 2.7.: (Pseudo-) first-order reaction rate constants of the O_2-addition and the competing $\dot{O}H$-migration of the 2-hydroperoxy-pent-4-yl radical. The CBS-QB3 O_2-addition high-pressure reaction rate constant is taken from Asatryan and Bozzelli [65].

$\dot{O}H$-migration reaction channel. Green *et al.* [92] found that "this channel has a significant rate", but the competing cyclic-ether + $\dot{O}H$ formation is about an order of magnitude faster. The authors state that a final assessment of the importance of this $\dot{O}H$-migration channel is only possible by taking into account subsequent reactions of $\dot{O}QOH$ [92]. Similar to the well-known R$\dot{O}$ species [61], I have found that $\dot{O}QOH$ decomposes into aldehydes and alkyl radicals via β-scission but also isomerizes into ethers (c. f. Fig. 2.8). Other $\dot{Q}OOH$ side reactions are the well-known cyclic ether formation, β-scissions, and formations of alkenes + $\dot{O}H$ (c. f. Fig. 2.5). Again, I observed bond dissociations, and formations of di-methyl-cyclopropane + $\dot{H}O_2$ (cyclic alkanes + $\dot{H}O_2$ in Fig. 2.5). I found the latter reaction neither in the reference model nor in the literature. From the literature model, the products of the second O_2-addition are

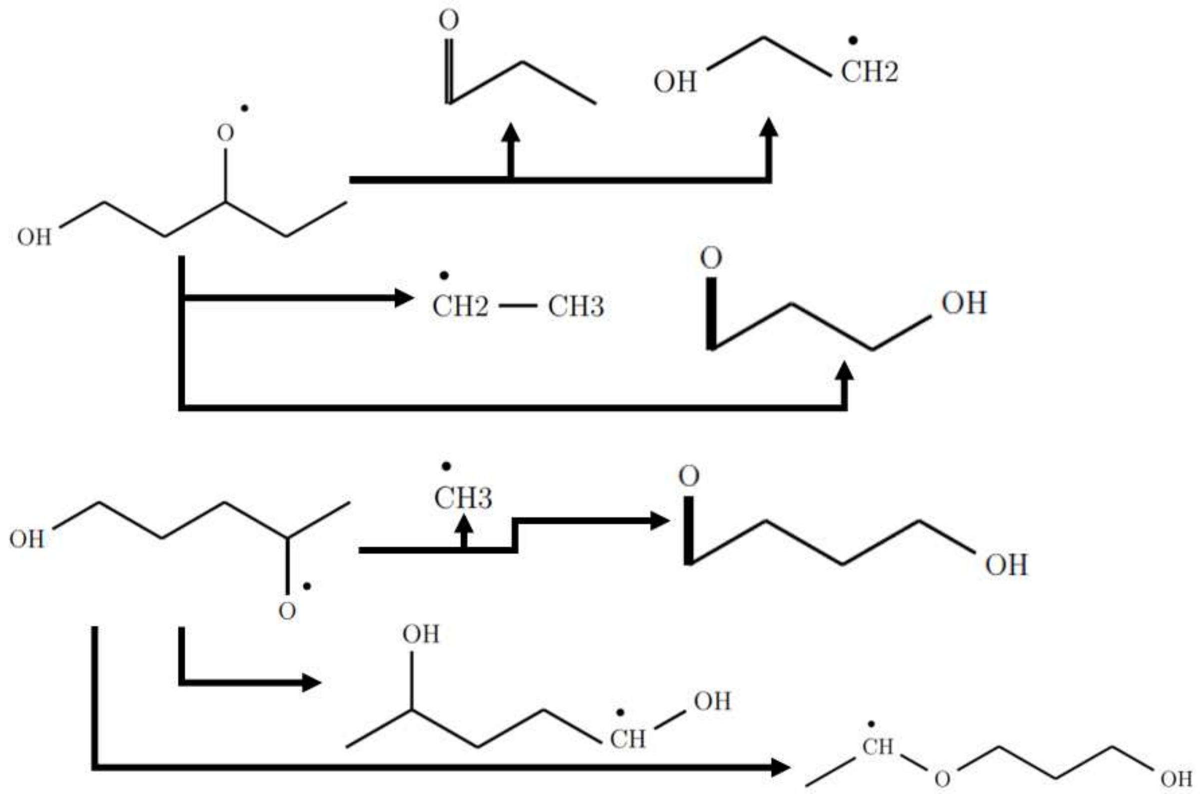


Figure 2.8.: Examples of observed reactions of $\dot{O}QOH$.

expected to undergo a second internal H-atom abstraction in 79% of all cases (c. f. Fig. 2.5). Instead, I find a branching ratio of 27% in the simulations. Again, the main difference comes from a reaction that has not been included in the reference model. Asatryan and Bozzelli [65] describe the barrierless carbonyl-peroxide + $\dot{O}H$ formation that can further decompose in a number of ways. They describe the decomposition of carbonyl-peroxides into oxygen, propene (A propyl radical was observed in my simulations, c. f. Fig. 2.9) and acetyldehyde, the elimination of CH_3, and the formation of oxetanes with the elimination of O_2, a reaction that I did not observe. Instead, I found a ring closure connecting the carbonyl radical site and the peroxy radical site (c. f. Fig. 2.9). A comparison of reaction rate constants of the carbonyl-peroxide + $\dot{O}H$ formation from the simulations and from the literature can be found in the appendix in section A.1.3. After the second internal H-atom abstraction, 67% of $HOO\dot{P}OOH$

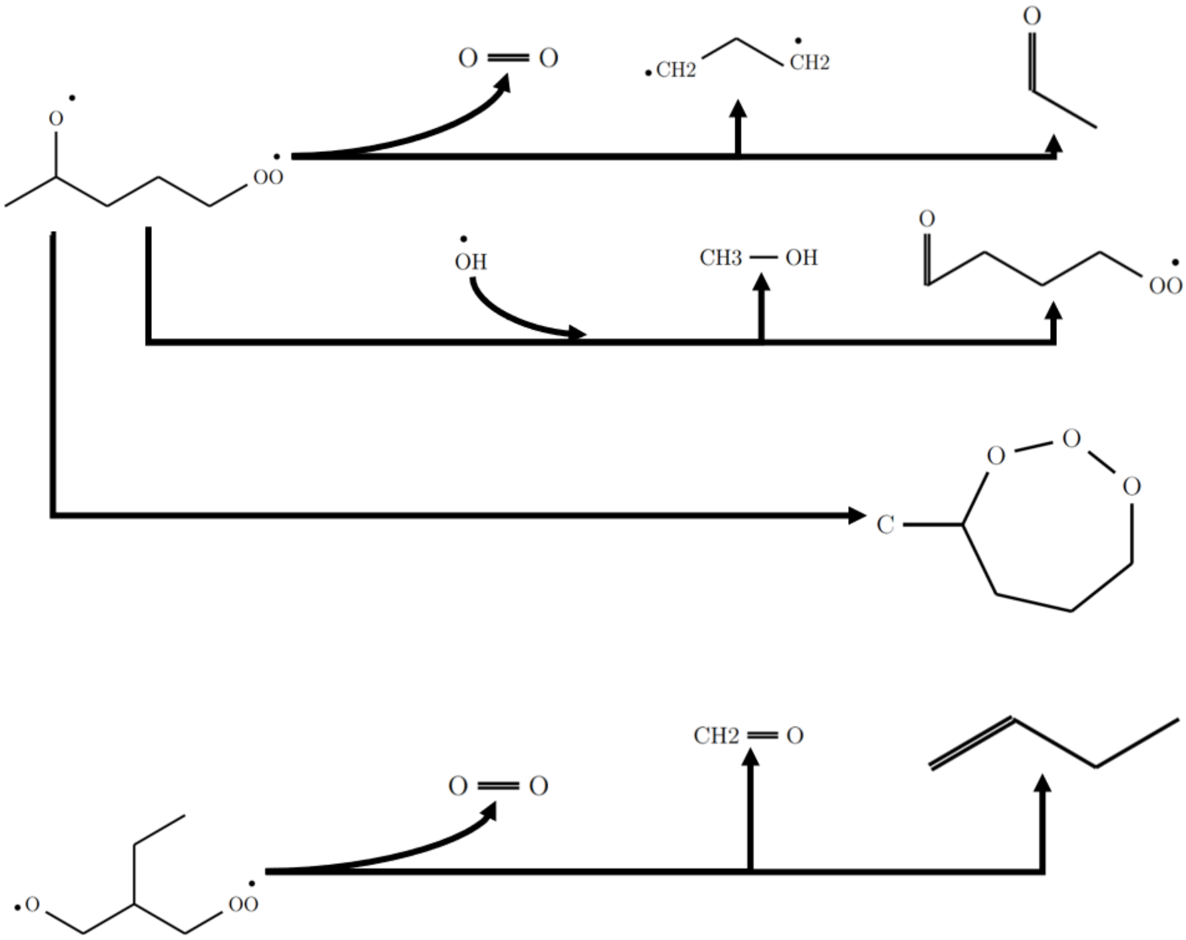


Figure 2.9.: Examples of observed reactions of carbonyl-peroxides.

species in the simulations or 74% in the literature model, respectively, decompose into keto-hydroperoxides and $\dot{O}H$ radicals. Other side reactions than the formation of alkenes and $H\dot{O}_2$ radicals side have not been observed.

I have shown that the major differences between the ReaxFF simulations and the reference model start with the $\dot{Q}OOH$ species in Fig. 2.5 as the second O_2-addition channel is less pronounced in the simulations than suggested by the reference model. The second O_2-addition channel is diminished by the large number of $\dot{Q}OOH$ side reactions, which reduce the number of $\dot{Q}OOH$ species that can undergo the second

O_2-addition. Also, the computation of the reaction equilibria of the first and second O_2-additions is computationally expensive, as the system spends much time hopping back and forth between the reactant and the product state. With high concentration, the representation of the reaction equilibria remains expensive, but, as the meta-stable O_2-addition reaction equilibria are pushed to the product side, and subsequent reactions are accelerated by high temperature, the remaining simulation time is not only sufficient to observe the remaining key low-temperature reactions but also to observe a variety of verified side reactions.

2.3. Conclusions

With pAD, an acceleration method has been presented that allows for the simulation of low-temperature ignition processes, which are dominated by meta-stable O_2-addition reaction equilibria, and internal H-atom abstractions. The method offers a physically substantiated way to increase the probability of escaping meta-stable reaction equilibria. While working without a bias, a suitable choice of parameters as ϕ, c_{O_2}, and temperature can optimize pAD performance. With only little information on three reactions, which are commonly known to be important in ignition processes, the three parameters could be tuned to increase visibility of the key low-temperature reactions in the pentane low-temperature ignition test case. The observed pathways were verified against an established pentane ignition reaction model from the literature [1–3]. In addition to the key low-temperature reaction pathways, other reaction classes not included in that literature model [1–3], such as $\dot{O}H$-migration or the formation of carbonyl-peroxides, showed up in the simulations. The applicability of pAD is not restricted to the low-temperature ignition process, but pAD can be employed more generally in cases in which association reactions in the gas phase play an important role.

3. Efficient Reaction Space Exploration with ChemTraYzer-TAD

Chapter 3 is based on the publication "Efficient Reaction Space Exploration with ChemTraYzer-TAD" by Krep et al. [94]. It is reprinted (adapted) with permission from Journal of Chemical Information and Modeling, American Chemical Society. Copyright 2022 American Chemical Society. My contribution to the publication [94] include the development of the concept and its implementation. I have further contributed the results, with the exception described in the following sentence, and, under consultancy by Indu Sekhar Roy, Dr. Wassja Kopp, Felix Schmalz, and Dr. Can Huang, the compilation of the text. Indu Sekhar Roy also contributed DFT energies of the known reactions. The work was supervised by Prof. Kai Leonhard.

I have developed the pAD methodology in the previous chapter, which is essentially a strategy to choose elevated simulation pressure and temperature in a way that the underlying reaction network is not biased towards a high-temperature reaction mechanism. In this chapter, I search for a more general solution for the kinetics bias issue. Before I start with the discussion of the new CTY-TAD method, I look again briefly at the kinetics bias issue from a more general perspective than in the previous chapter and give an overview of solutions from the literature.

An inherent issue of acceleration through higher temperature is the biasing of the system's kinetics. Not all reactions are accelerated by the same factor. With increasing temperature, the entropic reaction barrier becomes rate-determining, and high-energy barrier reactions, such as thermal dissociations, might become faster than reactions with lower reaction barriers. In contrast, low-energy barrier reactions are more relevant at lower, technically relevant temperatures. The authors of the temperature-accelerated Dynamics (TAD) method [95] solve the kinetics bias issue by reflecting the system back to the reactant PES basin after the detection of a transition to a product basin. A transition to a product basin is detected using periodic geometry optimizations of the system and comparisons of the potential energies of the reactant state and the new state. Using TST to derive waiting times for the observed reactions, the waiting times are extrapolated to a lower target temperature. The product state of the reaction with the fastest target-temperature waiting time is the starting point for a new simulation. Selecting the products of the fastest target-temperature reactions to relaunch simulations, allows for the approximate reconstruction of the target-temperature kinetics [95]. Besides the TAD method, many other methods can be found in the literature that block access to specific regions of the PES, thus exploring phase space more efficiently with MD simulations. Chekmarev and Krivov use a PES confinement strategy very similar to the confinement strategy in TAD for efficient determination of thermodynamic and kinetic data from MD, such as self-diffusion coefficients, densities of states [96, 97], and isomerization rates [98]. Pracht and Grimme [99] use bias potentials to block known conformers from being revisited by their CREST conformer search. Martìnez-Núñez and Shalashilin [100] use phase space constraints for faster quantification of non-RRKM effects on the reaction rate constant with MD simulations. Based on the work of Martìnez-Núñez and Shalashilin [100], the BXD method [101–104], which uses phase space constraints for efficient free energy mapping, has been proposed as an alternative to umbrella sampling [105]. BXD has since been repurposed to reaction searching [106, 107] in the now-called BXDE method. While the original BXD method [101–104] relies on user-specified CVs to determine the location of the confining walls, BXDE [106] uses potential energies for the automatic localization of the walls. The authors [106] observe, though, that their potential-energy-based procedure favors reactions with low entropy barriers. To overcome the entropy barrier issue, they again resort to user-specified BXD CVs to further reduce accessible reaction space.

In this study, I adopt the ideas to use elevated temperatures and basin constraints for the acceleration of reactions from the TAD method to design a method that finds as many reactions as possible that are relevant for a range of technically relevant temperatures for a given set of reactants. As opposed to the TAD method, I use bond-order processing steps from CTY [34, 35] instead of periodic minimizations to detect reactions. The use of CTY bond-order processing steps allows for the detection of both reactions with and without a reaction barrier. I do not adopt the TAD mechanism to launch simulations from the fastest target-temperature reactions at the rMD level of theory, as reaction rate constants on higher levels of theory are required for a trustable extrapolation of waiting times to target temperatures. As a consequence, the user must specify follow-up simulation boxes starting from reaction products manually. CTY-TAD simulations are easier to set up than most of the existing acceleration methods, as no additional parameters compared to a standard *NVT* simulation, such as the CVs used in the metadynamics or BXDE methods, are required.

I use CTY-TAD to study the 1,3-dioxolane-4-hydroperoxide-2-yl radical oxidation reaction network in the Results section in order to demonstrate how the new method can help reaction model developers to systematically search for new potentially relevant reactions. For this reason, I compare the observed reactions with the reactions included in the 1,3-dioxolane ignition model by Wildenberg *et al.* [108], which they currently continue to develop [109]. 1,3-dioxolane has been identified as a promising bio hybrid fuel candidate [108] and has been shown to reduce soot formation if added to diesel fuels [110]. Hydroperoxide radical ($\dot{Q}OOH$) oxidation processes are often of particular interest in the field of combustion modeling as they are known to be at the center of the kinetic bottleneck of the low-temperature oxidation mechanisms of many fuels as discussed in chapter 2.

3.1. Methods

ChemTraYzer Summary

Originally, CTY was developed as a post-processing tool for ReaxFF simulations in lammps [34, 71, 72]. In its first step, CTY processes the recorded bond orders [59]. Bond orders are a continuous representation of the bond strength. For the determination of molecules, the recorded ReaxFF bond orders [59] are sorted in descending order. Bonds with a bond order below a bond order cut-off value, in this work of 0.5, are not regarded as a bond. The remaining bond orders are rounded in steps of 0.5, *i. e.* to values of 0.5, 1, 1.5, etc. If the sum of rounded bond orders of accepted bonds exceeds an atom's valency, no further bonds are accepted. In the next step, CTY passes the accepted bonds to Open Babel [111], which converts the bonds into human readable SMILES code. In a final step, CTY calculates the reaction rate constants with equation (eq) (3.1) [34]. For a unimolecular reaction i of reactant A and reaction rate r_i in a reactor volume V, the reaction rate constant can be calculated as:

$$k_i(T) = \frac{\int_{t_0}^{t} r_i \mathrm{d}t' V}{\int_{t_0}^{t} c_\mathrm{A} \mathrm{d}t' V} \tag{3.1}$$

The numerator $\int_{t_0}^{t} r_i \mathrm{d}t' V$ is equivalent to the number of REs of reaction i in a trajectory of length $t - t_0$. c_A is the concentration of A. In a second publication [35], CTY was extended from a post-processing tool to a Python framework to run ReaxFF simulations with lammps [71, 72], automatically perform the described analysis of the bond orders, and reoptimize (meta-) stable species and TS at higher levels of theory on-the-fly. CTY derives input geometries for the reoptimization on higher levels of theories from geometries that are cut out from the rMD trajectory. The CTY TS reoptimization process can be summarized as follows: First, the geometry from the trajectory at the time of a RE is cut out and serves as a TS guess for the TS at the higher level of theory. In a second step, the geometry is relaxed at the higher level of theory with a frozen active core to reduce the number of imaginary frequencies, thereby producing a better TS guess. The active core is the geometry of the active atoms, *i. e.* the atoms involved in the bond changes of the RE. In the third step, the relaxed geometry is used as TS guess for the eigenvector-following algorithm that searches the TS. In the current implementation, molecules are assumed to be neutral. Spin multiplicities are guessed based on the number of electrons. The lowest allowed spin multiplicity is then used in the computations at the higher levels of theory. In some cases, *e.g.* O_2 , the lowest spin multiplicity does not correspond to the most stable configuration. Therefore, a predefined list of commonly known exceptions from the minimum spin multiplicity rule is provided in CTY and can easily be extended by the user. In the new CTY framework, total simulation time is split into batches of a user-specified number of time steps. A simulation batch is followed by the bond-order processing described above. The simulation time split into batches is an important feature for CTY-TAD, as will be explained in the following subsections. In a final step after the higher-level calculations have finished, the high-level data is collected, added to a database, reaction rate constants are computed and written to a CHEMKIN-formatted reaction model.

ChemTraYzer-TAD

The inner workings of CTY-TAD are shown in a in Fig. 3.1. CTY-TAD decides on-the-fly in each timestep if a new reaction has been found and consequently needs further CTY processing, or if the reaction is already known and a virtual, reflective wall, similar to the walls applied in other basin confinement implementations [95–98, 101, 112] needs to be applied. For the decision of whether a reaction occurred, the CTY bond-order processing and the valency check described above has been implemented into the lammps timestep integration as a new lammps fix. The lammps fix determines all the bonds, but not the molecules, present in the system based on bond orders that

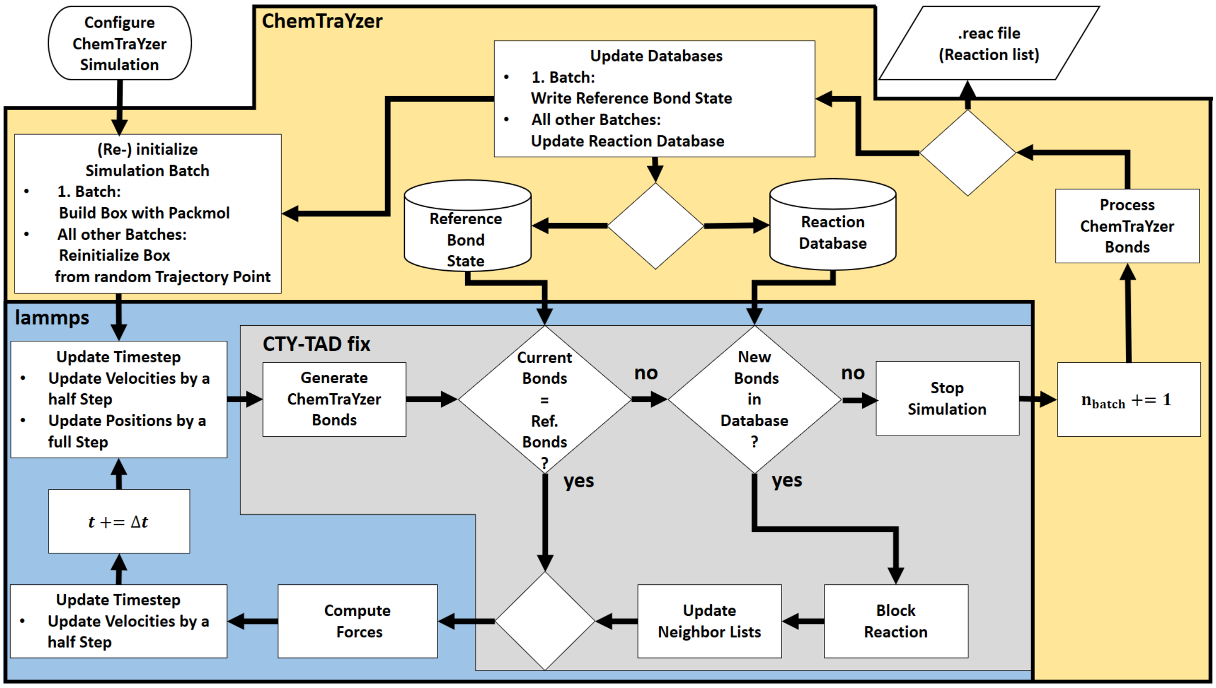


Figure 3.1.: The implementation concept of CTY-TAD.

are in computer memory. The decision of whether a reaction occurred is based on the comparison of the bonds currently present in the system and a reference list of bonds, the ”Reference Bond State” in Fig. 3.1. The Reference Bond State is read-in by the lammps fix from a text file. The text file is written automatically at the beginning of a new simulation and contains for all atom ids the atom type, the valency, and the atom ids of the bond partners.

CTY-TAD automatically builds a reaction database (“Reaction Database” in Fig. 3.1) to decide if a reaction has been observed before and the system must be reflected back to the reactant state. The reaction database is shown in more detail in Fig. 3.2. The `blocktransitions.dat` file is the database file that is read-in by the lammps fix. The file contains the atom ids of the active atoms and the active atoms’ bond partner atom ids in the product state. If the lammps fix detects a change in the system’s set of bonds that matches the bonds in the database, the placement of a virtual wall is triggered to reflect the system back to the reactant state. The bond partner atom ids of every atom are automatically available from the CTY bond-order processing

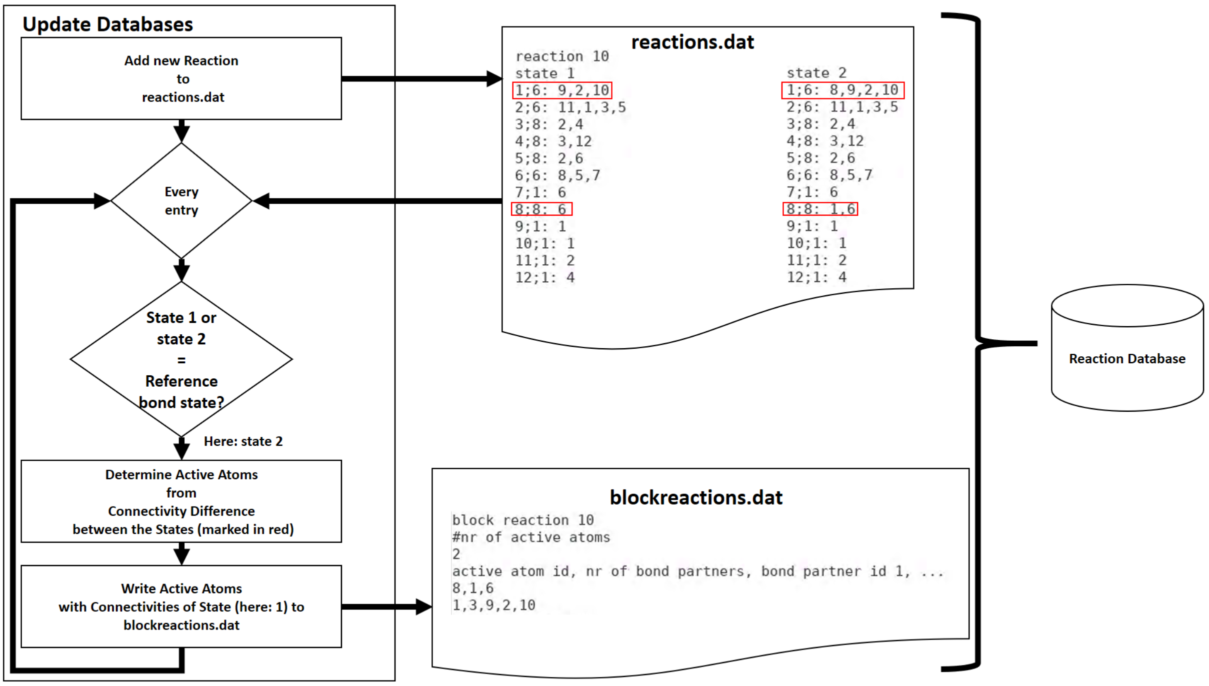


Figure 3.2.: Update scheme for the reaction database. Example entries for the text-file-formatted database files are shown, too. Active atoms, determined via the connectivity differences between the reactant (here: state 2) and the product state (here: state 1) in the `reactions.dat` file are marked with a red box.

in the "Process ChemTraYzer Bonds" operation in Fig. 3.1. Note that the bonds are defined for individually labeled atoms and not for atom types. In other words, my code cannot handle indistinguishability of atoms at present. As shown in the Results section, the atom labelling has only a small impact on the method's performance for the example system.

The reaction database can be initialized with generic bond-breaking reactions. In this work, I initialized the reaction database with all reactions that only break a C−H or an O−H bond. The choice of generic bond-breaking reactions that are to be blocked should be made with care. Including C−C and C−O bond breaks in the example system would have led to a blocking of important β-scissions as well, although they include not only a bond break, but also the formation of a double bond. The reason for CTY not being able to distinguish a bond break from a β-scission is that CTY, like other rMD post-processing tools [28, 31–33], too, detects REs based on connectivities but not based on changes of bond strength, i. e. changes between single, double, etc. bonds. My choice of generic bond-breaking reactions (C−H, O−H) bonds, of course, also results in β-scissions involving a C−H or O−H bond break not being observed. As the C−H and O−H bonds have a higher bond dissociation energy (BDE) than the C−C, C−O and O−O bonds that are present in the 1,3-dioxolane-4-hydroperoxide-2-yl radical (c. f. the BDEs in [108, 113]), β-scissions involving a C−H or O−H bond break are not likely to be competitive to the β-scissions involving a C−C, C−O or O−O bond break.

In total, the reaction detection and discrimination process can be summarized as follows: a reaction is detected if the set of bonds that are present in the current timestep differs from the reactants' set of bonds. If the new set of bonds is already known to the reaction database, the system is reflected back to the reactant state. On the other hand, if the new set of bonds is not yet in the database, the simulation is stopped, the reaction is processed by CTY and the reaction database is updated with the new reaction.

The reflection of the system back to the reactant state is implemented with virtual walls. The virtual walls, which are placed in the "Block Reaction" operation in Fig. 3.1, are implemented as two-step processes. First, the positions $\mathbf{x}_i$ of all atoms in time step t in the system are integrated backwards using $\mathbf{x}_i(t+1) = \mathbf{x}_i(t) - 2\mathbf{v}_i(t) \cdot \Delta t$, which corresponds to an elastic collision with a rigid wall perpendicular to the trajectory. Index i is the atom id. In the second step, the sign of the velocities of only the active atoms is flipped, which results in a distortion of the reacting molecules. $\mathbf{v}_i(t)$ is the velocity vector of atom i in timestep t. I have implemented these two steps to speed up the sampling of phase space and to reduce the risk of oscillations of the system around the placed wall as described for other phase space constraining methods [96, 106]. Oscillations can occur, if the reflection is unsuccessful in forcing the system back into the reactant state. In case of an oscillation, another wall is placed shortly after the first, changing the velocity signs back and forth. Besides the two-step reflection process, I have implemented further countermeasures to break an oscillation. First, walls cannot be placed in subsequent time steps. Second, if the system has not moved back into the reactant state after two subsequent timesteps, the system is reset into the initial configuration but with velocities from the current timestep, followed by a sign flip of the active atoms velocities. If a wall is still placed two timesteps after the reset to the initial configuration, the system is given one more chance to recover by repeating the reset to the initial configuration, again with the velocities from the now current timestep, *i. e.* two timesteps after the first reset to the initial configuration, and the active-atoms sign flip. In case the system has still not recovered then, the simulation finishes.

If the lammps fix finds a new reaction, the simulation batch is stopped and a bond order file containing bond orders of the reactant state and bond orders of the new product state is written in the "Stop Simulation" operation in Fig. 3.1. The bond order file is then processed by CTY as described in the ChemTraYzer Summary section. Ideally, the lammps fix and CTY always find the exact same bonds and bond changes. Slight differences between the bond-order processing implementations in the lammps fix and CTY resulting from different languages (lammps fix in C++ *vs.* CTY in python) impair the basin confinement to a negligible extent only, as I show in the Results section. After bond-order processing with CTY, the reaction database is updated and a new simulation batch is initialized unless a user-specified stopping criterion is fulfilled. A new simulation batch is launched from a random configuration sampled during the previous simulation batch. The velocities are reinitialized with the lammps velocity command. If a new batch run is accidentally initialized from a product state, the batch run is stopped and reinitialized with the simulation box that was built with Packmol [70] and was used to set up the simulation. The simulation finishes if a user-specified batch length has passed without any new reaction. Here, I use a batch length of 100 ps as the stopping criterion. Using a batch length as the stopping criterion instead of a total simulation time means that the total simulation time is also variable. The more reactions occur in a simulation, the longer the simulation will run. The user needs to specify a minimum waiting time for a reaction to occur, i. e. the time after which a reaction can be considered irrelevant, to find a suitable batch length as stopping criterion.

Simulation Set-Up

I ran *NVT* simulations at temperatures of 1000, 2000, 3000, 4000, and 5000 K with the ReaxFF force field and C/H/O parameters developed by Ashraf and van Duin [64], which has been used in numerous studies, for example [114–116], with a timestep of 0.1 fs. The simulations were thermostatted with the Langevin thermostat with a damping constant of 10 fs. 10 replica were run per temperature. The system was composed of one O_2 and one 1,3-dioxolane-4-hydroperoxide-2-yl radical. I have not used periodic boundary conditions (pbc), as I am interested in gas phase interactions only and not in bulk phase properties. Not using pbc allows for smaller systems as self-interactions can be avoided. The atoms were confined to the simulation box with a spherical repulsive harmonic potential with a force constant of $100\,\mathrm{kcal\,mol^{-1}\,\AA^{-2}}$. Spherical repulsive potentials have been used before by the *ab-initio* nanoreactor [28] and the nanoreactor implementation by Grimme [49]. The choice of the confining potential parameters ensures a bias-free volume consistent with the chosen system density of $10^4\,\mathrm{mol\,m^{-3}}$, which corresponds to an ideal gas pressure of 800 bar at 1000 K. The high density and pressure make bimolecular reactions more likely to occur [57]. The cubic box size was chosen to fit in the bias-free volume plus the confining potential. I obtain a diameter of 8.6 Å for the bias-free volume and a box length of 18.6 Å. The penetratable repulsive potential effectively reduces the density, though. For the determination of the density reduction factor, I first estimate the penetration depth. The potential energy from the repulsive potential exceeds $8\mathrm{R}T$, which is the expected energy stored in the motions relative to the center-of-mass obtained from the equipartion theorem, after less then 1 Å at 5000 K. From the penetration depth, I obtain a density reduction factor of 0.72. For comparison, the same number of rMD simulations have been run without the new lammps fix but with the standard CTY simulation set-up [35]. A simulation time of 300 ps was chosen. The simulation time was not split into batches. Otherwise, I used the same simulation set-up as described above for the "conventional simulations", as the standard CTY simulations are called from now on.

Post-Processing

I use a reaction identification test to assess the quality of the CTY bond-order-, or alternatively, connectivities-based dividing surface (DS). The ideal DS allows for the discrimination all possible parallel elementary reactions, *i. e.* transitions from one (meta-) stable molecular geometry to another (meta-) stable molecular geometry with different connectivities, and has a minimal number of recrossing events. The requirement of the ideal DS to have only a minimum number of recrossing events has already been addressed in CTY by the introduction of a recrossing filter [34]. The recrossing filter corrects the rMD-trajectory-based reaction rate constants for recrossing. The first requirement of the ideal DS, *i. e.* the ability to discriminate parallel reactions, is particularly important for CTY-TAD, as the method should not unintendedly block yet-unobserved reactions. CTY-TAD allows for the testing of the DS's ability to discriminate parallel reactions due to the high trajectory resolution of 0.1 fs between consecutive trajectory snapshots and due to CTY-TAD's design that finds a large amount of different parallel reactions. In the reaction identification test, I use geometry optimizations of the reaction products and test if the reaction product molecular identifiers, such as SMILES codes, obtained from the CTY trajectory analysis correspond to stable ReaxFF geometries. Starting geometries for the reaction product optimizations are the reactive trajectory snapshots, *i. e.* the snapshots after a change in connectivities was detected in the bond-order processing. Van-der-Waals complexes of reaction products are split into the bonded structures that form the complex before the geometry optimization. After the geometry optimizations of the products, I use Open Babel [111] to regenerate SMILES and InChi codes. For comparison of the connectivities only, the InChi codes are truncated after the main layer. If I retrieve the same molecular identifier (SMILES code, truncated InChi code) from the set of ReaxFF-optimized product geometries as CTY obtained from the trajectory analysis, the DS correctly identifies the reaction and discriminates the (meta-) stable reactants and products of the reaction. On the other hand, if I obtain a set of optimized product geometries with different molecular identifiers from REs of the same reaction type, I have identified cases in which the CTY DS definition is insufficient for the discrimination of the reactions. The set of ReaxFF-optimized product geometries includes the product geometries from all simulations combined. I believe that my findings will not be only be relevant for CTY, but also for other rMD-based reactions mechanism generators, like nebterpolator [28, 30] and AutoMeKin [31–33], as both define the DS based solely on connectivities as well.

For the set of reactions that are discussed in the Results section, I have computed the reaction rate constants from eq (3.1) to determine the boost factor obtained with my method. The reaction rate constant calculations are based only on the first occurrences of a reaction in a simulation. This method is used to avoid biasing the reaction rate constants due to an artificial increase of the entropic reaction barrier. I artificially increase the entropic reaction barrier, if I block only some but not all of the multiple ways in which some reactions can occur with the virtual, reflective walls. An example is the O_2 addition, where both O atoms can react with the radical site. After the first occurrence of the reaction, the reflective wall that blocks further occurrences of the O_2 addition is incomplete. As the accessible part of the O_2 addition DS is shrunk by the incomplete wall, the entropic reaction barrier is increased. The wall will automatically complete itself after the second occurrence of the O_2 addition. First occurrence observation times from the replica at the same temperature were combined as described in the publication by Kröger *et al.* [54] to compute the reaction rate constant. Reaction rate constants of reactions that have been observed at three or more temperatures have been extrapolated to lower temperatures using an Arrhenius fit.

I have computed reaction barriers with ReaxFF as a first assessment of which reactions can be expected to be fast and potentially relevant. The potential energies of the reactive trajectory snapshots and of the optimized reactants are both standard output from CTY. From the set of potential energies of the reactive trajectory snapshots of every RE, $V_{\mathrm{DS}}^{\mathrm{RE}}$, belonging to the same type of reaction, I have picked the minimum value $V_{\mathrm{DS}}^{\mathrm{min}}$ to serve as a reaction barrier estimate. I have based the $V_{\mathrm{DS}}^{\mathrm{RE}}$ on the arithmetic averages of the potential energies of the ReaxFF-optimized reactant geometries as a simple, temperature-independent means to consider the contributions from conformers and to reduce the impact of numerical noise in ReaxFF forces and energies [117]. Other options for referencing the $V_{\mathrm{DS}}^{\mathrm{RE}}$ are choosing the minimum potential energy from the set of optimized reactants or using temperature-dependent Boltzmann averages of the potential energies from the set of optimized reactants. The root-mean-square deviations of the reaction barrier estimates $V_{\mathrm{DS}}^{\mathrm{min}}$ from the energy parameters in the Arrhenius fits differ by less than $1\,\mathrm{kcal\,mol^{-1}}$ in a comparison of the three potential energy referencing options. That means, all three options can be used, but I prefer either the arithmetic average or the reactants' minima due their temperature-independence for simplicity. I use a standard error of $19\,\mathrm{kcal\,mol^{-1}}$ for the ReaxFF reaction barriers as explained in the methods section of the previous chapter. For a small number of hand-picked reactions, which are discussed in the Results section, I have recomputed the reaction barrier manually for a better assessment of their potential relevance in a reaction model. For the recomputations I used the B3LYP method with Grimme's D3 empirical dispersion correction [118] and the 6-311+G(d,p) basis set implemented in the Gaussian software package [119]. CTY-TAD can principally be coupled to the automatic CTY TS reoptimization scheme [35] that I have briefly summarized above, so TS can be searched for automatically on higher levels of theory. As I have needed higher-level TS for only very few reactions and many reactions

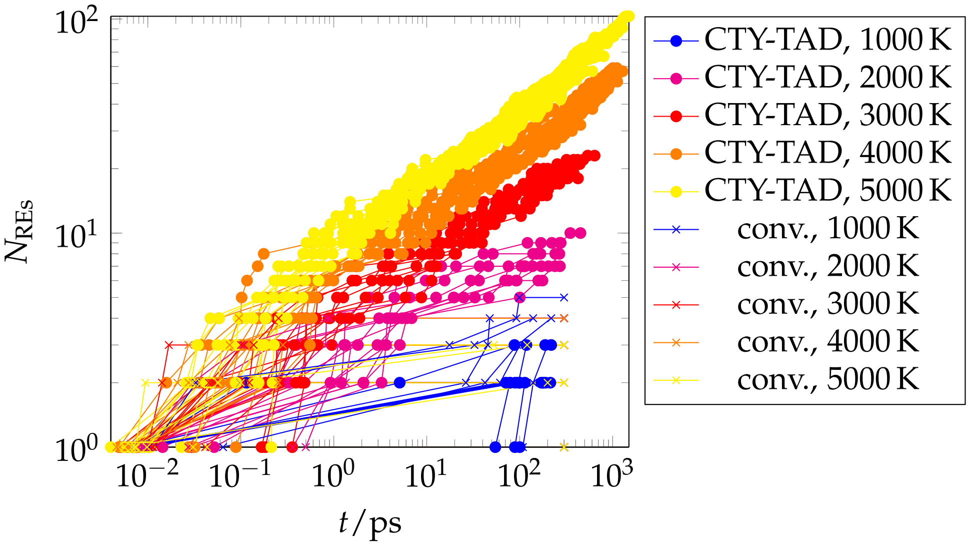


Figure 3.3.: Number of observed uni- and bimolecular REs of the 1,3-dioxolane-4-hydroperoxide-2-yl radical and O_2 over simulation time. The Figure shows results for 10 replica at each temperature. Connected data points are derived from the same simulation.

are already known in the literature [108], I decided that a manual TS search would be computationally more efficient. Also, I would like to emphasize the modularity of CTY. The search for reactions using rMD, for example using CTY-TAD, and the TS reoptimization scheme can be used coupled as one program or seperately. The value of using CTY-TAD as a seperate module is demonstrated in the 1,3-dioxolane-4-hydroperoxide-2-yl radical oxidation case study discussed in the results and discussion section.

3.2. Results and Discussion

Method Validation

The main objective of CTY-TAD is to find as many different parallel reactions as possible in short time. In comparison to conventional simulations, I find 10 times more uni- and bimolecular reactions of the 1,3-dioxolane-4-hydroperoxide-2-yl radical ($\dot{\mathrm{Q}}$OOH) and O_2 per simulation with CTY-TAD at 5000 K within a simulation time of

300 ps, which is the simulation length of the conventional simulations. With CTY-TAD, the number of REs per simulation time increases with temperature. At 1000 K only a handful of reactions are fast enough to be observed within the batch length of 100 ps, which I chose as the stopping criterion. This number of REs is similar to the number of $\dot{Q}OOH$ (+ O_2) REs observed in the conventional simulations. At 5000 K, a maximum of 103 REs has been found within a total simulation time of 1.5 ns in a single simulation. In the conventional simulations, the number of $\dot{Q}OOH$ (+ O_2) REs is low for all temperatures, as the reactants dissociate after a short time, *i. e.* within less than 1 ps at temperatures above 3000 K. The large difference between the number of $\dot{Q}OOH$ (+ O_2) REs observed in the CTY-TAD and the conventional simulations underlines the importance of the reflective walls. They improve the sampling of the local configurational space of the reactants, or in other words, prevent the simulations from only finding the same, at high temperatures mostly dissociative reactions in every simulation.

The REs observed in the individual CTY-TAD simulations on the other hand, *i. e.* the connected data points in Fig. 3.3, should be unique reactions, as the reflective walls prevent reactions from occurring twice in a simulation. In the case study, duplicates occur mostly because of the neglect of indistinguishability (185/1312 REs) and, in much smaller quantity (23/1312 REs), because of differences between the bond-order processing implementations in the lammps fix and in CTY. I count REs as duplicates if they have been observed before in the same simulation. In total, duplicates make up less than 20% of the set of REs from all CTY-TAD simulations combined. To sum up my findings up to this point, I have established that the implementation of the concept meets the method's target of finding a large number of different parallel reactions in short simulation time. It was shown that the use of increased temperature allows for finding much more REs in the same simulation time than with conventional simulations and that my implementation of the virtual walls is capable of preventing the observation of duplicate reactions.

CTY-TAD does not only have to prevent the occurrence of duplicate reactions, it also needs to filter the new reactions from the already known. For the determination of the filtering capabilities of the CTY bond-order-based DS, I have implemented the reaction identification test described in the methods section. In the first step, I select the molecular identifier (SMILES code or truncated InChi code) that is better suited for the reaction identification test. The observed REs belong to 144 different reactions. In the reaction identification test, 35 from the 144 reactions pass the reaction identification test if SMILES codes are used, and 79 from the 144 reactions pass if truncated InChi codes are used. While truncated InChi codes only contain connectivity information, which CTY obtains from the bond-order processing, SMILES codes also contain formal charges and bond types, which are inferred by Open Babel. Therefore, the information encoded in truncated InChi code is more consistent with the CTY DS, which is solely based on connectivity changes, than the information encoded in SMILES code. Therefore, truncated InChi code is the better suited molecular identifier for the reaction identification test and forms the basis for the test. The difference between the number of correctly identified products using SMILES codes and truncated InChi codes shows also that for many molecules representated by a given truncated InChi code, multiple SMILES codes can be found, as multiple electronic states, and very related to the electronic states, multiple spin quantum numbers can be assigned. Fig. 3.4 gives an example of the representations of two reaction products with identical connectivities but different electronic states.

Besides the lack of information about the most stable electronic state, more reasons can be identified that cause reactions to fail the reaction identification test. In 11% of the 144 reactions, the unoptimized product geometries of all REs of the same reaction type, which are equivalent to the reactive trajectory snapshots, optimize back to the reactant geometry. This means that for reactions with a reaction barrier, the reaction is detected before the energy barrier is crossed. In 26%, the truncated InChi codes obtained from the bond-order processing of the trajectory cannot be retrieved after the

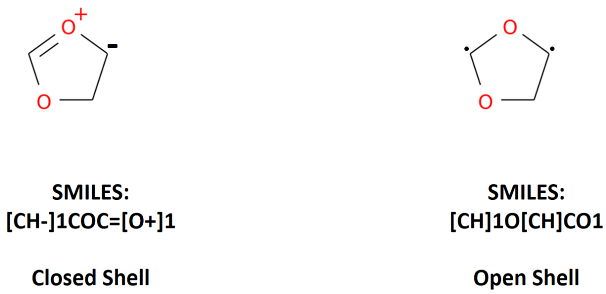


Figure 3.4.: Example of two representations of two reaction products with identical connectivities but different electronic states.

product geometry optimization, but for all REs of the same reaction type, the same molecular structures are obtained after the optimization. In all the cases described up to this point, I have not observed that the virtual walls have accidentally blocked yet-unobserved reactions in the attempt to block an already-known reaction. In 8% of the 144 different reactions, however, product geometries of the same reaction type do not all optimize to the same stable structure. Note that also in the set of 79 reactions that passed the reaction identification test, 26 reactions can be found that have single REs with products that optimize to a different species than the reactant or the product obtained by the bond-order processing. So, in total, I obtain 38 reactions with one or multiple product geometries, or in other words, reactive trajectory snapshots that optimize to other molecules than the reactant or the product, which CTY obtains from its bond-order processing. These 38 reactions are the cases in which virtual, reflective walls might have blocked unobserved reactions. The unintended blocking of unobserved reactions can be caused by bifurcations in the reaction coordinate as described in literature [120]. Another possible case in which the bond-order based DS definition might fail occurs if the RE is not detected in the vicinity of a first-order PES saddle point but in the vicinity of a higher-order PES saddle point. I illustrate one additional case, in which unobserved reactions may be blocked as a consequence of the bond-order processing failing to detect all the bond changes that are involved in the reaction in the appendix, section A.4. As compensation for the imperfect reaction differentiation, multiple independent, parallel simulations that do

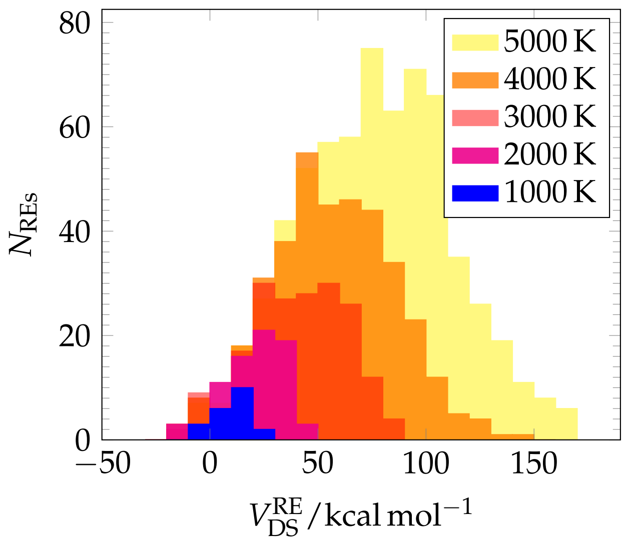


Figure 3.5.: Histogram of reaction barrier estimates $V_{\mathrm{DS}}^{\mathrm{RE}}$ of the $\dot{\mathrm{Q}}$OOH + O_2 system. All observed REs are included.

not share information about the observed reactions should always be run. I have added the reaction identification test results in the reaction list in the appendix in section A.2.1.

I cannot expect all of the 144 different reactions to be equally important for the low-temperature ignition process of 1,3-dioxolane. The comparison of reaction barriers provides the means for a first assessment of which reactions are the fastest at lower temperatures and, as a consequence, can be expected to be important for the low-temperature ignition process. The distribution of reaction barrier estimates $V_{\mathrm{DS}}^{\mathrm{RE}}$ also allows for the assessment if even higher simulation temperatures can be expected to deliver more potentially relevant reactions. With increasing temperature, especially the number of reactions with higher reaction barriers grows compared to the lower temperatures (see Fig. 3.5). In the area below 50 kcal mol^{-1} in which all reactions found in the simulations at the technically relevant temperatures below 2000 K lie, I have not observed significantly more reactions at 5000 K than at 4000 K. I would therefore not expect to find more potentially relevant reactions going to even higher temperatures.

The largest amount of reactions with a small $V_{\mathrm{DS}}^{\mathrm{RE}}$ below 50 kcal mol^{-1}, *i. e.* the more relevant reactions for the test system, are found at 4000 and 5000 K. The larger amount of potentially relevant reactions comes at the cost of finding a larger amount of high-barrier reactions as well. As the total simulation time increases with every reaction found, a lower simulation temperature than 5000 K can be found that delivers a better ratio between the observed number of potentially relevant reactions and total simulation time. The ratio of the number of observed RE with a $V_{\mathrm{DS}}^{\mathrm{RE}}$ below 50 kcal mol^{-1} divided by the sum of total simulation times of all simulations with the same temperature increases with temperature from 0.012 ps^{-1} at 1000 K to 0.04 ps^{-1} at 3000 K and decreases again at even higher temperatures, which makes 3000 K the simulation temperature with the most efficient use of simulation time for finding potentially relevant reactions for the test system. While it is trivial to pick the best simulation temperature *a posteriori*, the question arises if any conclusions for a good choice of simulation temperatures can be drawn already *a priori*. At the lower temperatures 1000 and 2000 K only 10% and 37%, respectively, of the number of REs with a $V_{\mathrm{DS}}^{\mathrm{RE}}$ below 50 kcal mol^{-1} that are observed at 5000 K are found. This shows that the simulation temperature cannot be chosen independently from the stopping criterion. The reaction rate constant can be approximated using TST to estimate if the chosen stopping criterion is sufficiently long to observe reactions with a given reaction barrier threshold $V_{\mathrm{barrier}}^{\mathrm{threshold}}$ at a given simulation temperature. I can then write $k(T, V_{\mathrm{barrier}}^{\mathrm{threshold}}) \approx \frac{k_{\mathrm{B}}T}{h} \exp\left(-\frac{V_{\mathrm{barrier}}^{\mathrm{threshold}}}{\mathrm{R}T}\right)$. Note that I have neglected the recrossing correction factor in the TST reaction rate constant and have replaced the free energy in the exponential term by $V_{\mathrm{barrier}}^{\mathrm{threshold}}$. The replacement of the free energy with the potential energy reaction barrier $V_{\mathrm{barrier}}^{\mathrm{threshold}}$ means that I neglect the entropic reaction barrier. The entropy contribution to the reaction barrier often further reduces the reaction rate constant. With the approximate reaction rate constant as input, I can then resort to eq 5 in the publication by Kröger *et al.* [54] to estimate the minimum simulation time required to observe a reaction with reaction rate constant $k(T, V_{\mathrm{barrier}}^{\mathrm{threshold}})$ at least once with a given probability P. For reactions with the same potential energy

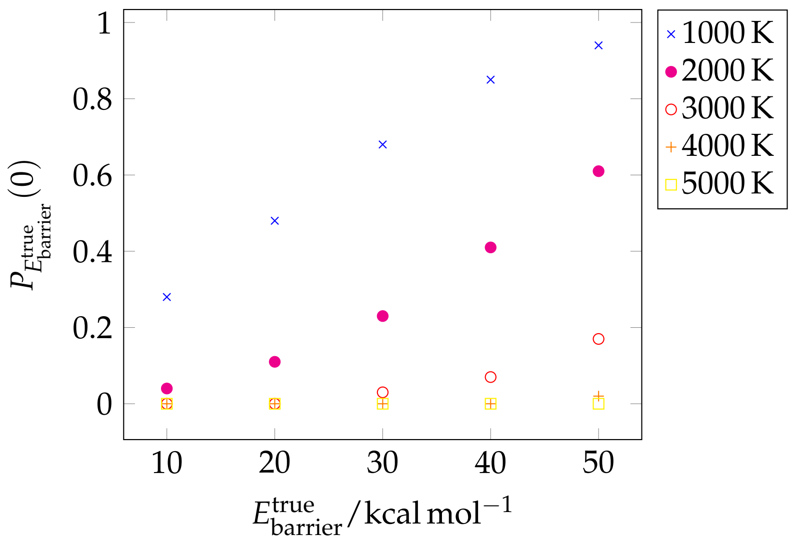


Figure 3.6.: Probabilities for missing a reaction with a given true physical energy barrier $E_{\mathrm{barrier}}^{\mathrm{true}}$ in a set of 10 replicas with simulation lengths of 100 ps.

reaction barrier and a high entropic reaction barrier, longer simulation times might be required, though. If I use $P = \frac{1}{N_{\mathrm{replica}}}$, i. e. here 10%, and search for a reaction with the reaction rate constant $k(1000\,\mathrm{K}, 50\,\mathrm{kcal\,mol^{-1}})$, I obtain a required simulation time of $\Delta t_{\mathrm{min}} = -\frac{\ln\left(1-\frac{1}{N_{\mathrm{replica}}}\right)}{k\Pi_i^{N_{\mathrm{reactants}}} N_i} = 4 \times 10^8\,\mathrm{ps}$ with $\Pi_i^{N_{\mathrm{reactants}}} N_i = 1$. For 1000 K and a $V_{\mathrm{barrier}}^{\mathrm{threshold}}$ of 20 kcal mol^{-1}, which is among the reaction barriers that I observed at 1000 K, I obtain a required simulation time of 117 ps. For 2000 K, required simulation times of 60 ps are obtained for reactions with a reaction barrier of 40 kcal mol^{-1} and 736 ps for reactions with a reaction barrier of 50 kcal mol^{-1}. This means that the stopping criterion of 100 ps was approximately long enough to observe reactions with a reaction barrier of 20 kcal mol^{-1} at 1000 K and 40 kcal mol^{-1} at 2000 K with a probability of 10%. On the other hand, the simulation time was not long enough to expect the observation of reactions with even higher reaction barriers than 20 kcal mol^{-1} at 1000 K and 40 kcal mol^{-1} at 2000 K.

After the simulation temperature, simulation time, and number of replica are chosen, the risk of missing potentially relevant reactions in the simulations can be estimated, thereby taking into account that the reaction barriers may not be represented by their

true physical value by the model potential. The probability of missing a relevant reaction, *i. e.* observing it $N = 0$ times, with a given true physical energy barrier $E^{\text{true}}_{\text{barrier}}$, $P_{E^{\text{true}}_{\text{barrier}}}(0)$, can be computed from the probability density that the reaction barrier is represented by the model potential energy $V^{\text{ReaxFF}}_{\text{barrier}}$, $f_{E^{\text{true}}_{\text{barrier}}}(V^{\text{ReaxFF}}_{\text{barrier}})$, and the probability of missing a reaction with the model barrier of $V^{\text{ReaxFF}}_{i,\text{barrier}}$, $P(0|V^{\text{ReaxFF}}_{\text{barrier}})$:
$P_{E^{\text{true}}_{\text{barrier}}}(0) = \int_{V^{\text{ReaxFF}}_{\text{barrier}}=-\infty}^{V^{\text{ReaxFF}}_{\text{barrier}}=\infty} P(0|V^{\text{ReaxFF}}_{\text{barrier}}) f_{E^{\text{true}}_{\text{barrier}}}(V^{\text{ReaxFF}}_{\text{barrier}}) \mathrm{d}V^{\text{ReaxFF}}_{\text{barrier}}$.
I approximate the integral using 10000 samples from an ReaxFF reaction barrier range from -50 to 1000 kcal mol^{-1}. That means, $\mathrm{d}V^{\text{ReaxFF}}_{\text{barrier}}$ is approximated with the finite integration step of $\Delta V^{\text{ReaxFF}}_{\text{barrier}} = 0.105\,\text{kcal}\,\text{mol}^{-1}$. $f_{E^{\text{true}}_{\text{barrier}}}(V^{\text{ReaxFF}}_{\text{barrier}})$ is computed from a normal distribution centered around $E^{\text{true}}_{\text{barrier}}$. The ReaxFF reaction barrier standard error of 19 kcal mol^{-1} is used as the standard deviation.

$P(0|V^{\text{ReaxFF}}_{\text{barrier}})$ can be computed from the Poisson distribution of rare events [54]:
$P(0|V^{\text{ReaxFF}}_{\text{barrier}}) = \exp(-\lambda)$, with $\lambda = k \times \Pi_i^{N_{\text{reactants}}} N_i \times \Delta t_{\text{simulation}} \times N_{\text{replica}}$,
$k \approx \frac{k_{\text{B}}T}{h} \exp\left(-\frac{V^{\text{ReaxFF}}_{\text{barrier}}}{\text{R}T}\right)$.
$P_{E^{\text{true}}_{\text{barrier}}}(0)$ is shown for unimolecular reactions and simulation temperatures from 1000 to 5000 K for a set of 10 replica with simulation times of 100 ps in Fig. 3.6. k_i of reactions with negative reaction barriers were approximated with $k = 1 \times 10^{12}$. The probabilities of missing a reaction with a given true reaction barrier decreases with temperature regardless of the reaction barrier height. At 5000 K the probabilities of missing reactions with true reaction barriers of 50 kcal mol^{-1} (or less) is smaller than 1%. In other words, less than 1 from 100 reactions with true reaction barrier of 50 kcal mol^{-1} (or less) must be expected to be missed in a set of 10 replica with a simulation time of 10 ps at 5000 K.

To conclude the discussion of simulation parameters, I would like to emphasize that the choice of temperature is a compromise between thoroughness of the search for potentially relevant reactions and efficient use of simulation time. While the TST-based estimate does not allow for the optimization of the simulation parameters *a priori*, it is still possible to use it to estimate *a priori* if the combination of simulation temperature

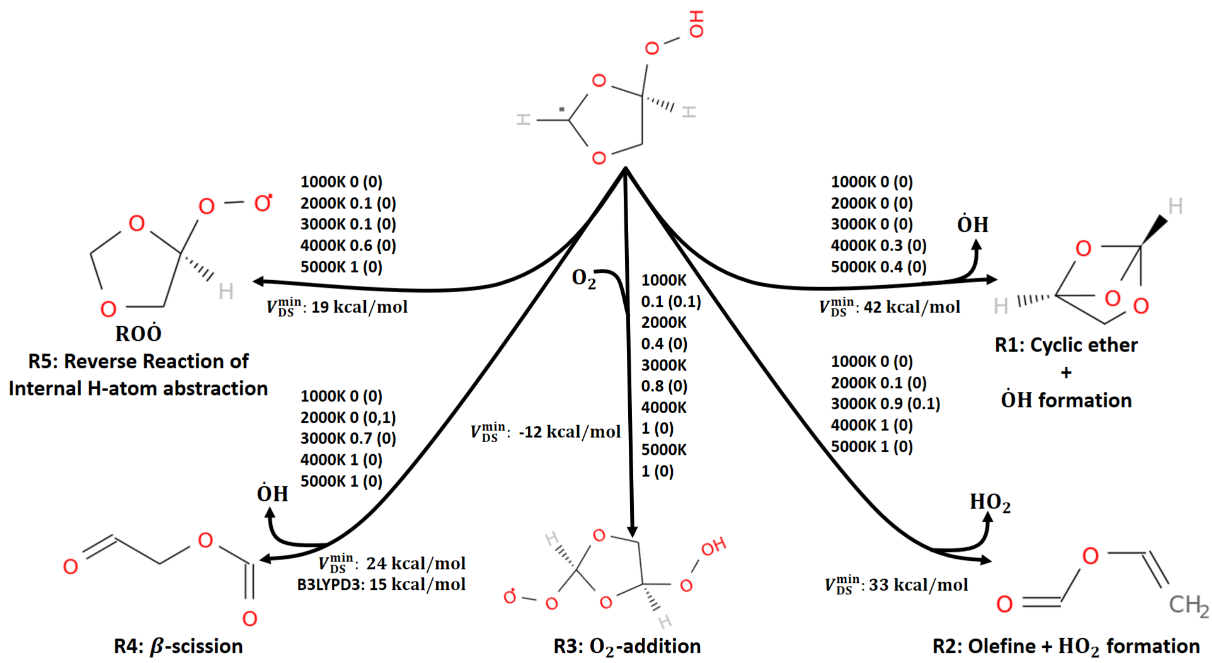


Figure 3.7.: Main low-temperature ignition Q̇OOH reactions, which are also known from other fuels [8, 121]. The arrows are labeled with the observation probabilities for each temperature and the reaction barrier estimates $V_{\mathrm{DS}}^{\mathrm{min}}$ from my CTY-TAD simulations. The observation probabilities from the conventional simulations are shown in parentheses.

and stop time is appropriate for observing reactions with a reaction barrier below a given threshold with a given observation probability.

Q̇OOH Oxidation

Among the 144 different reactions, I have observed the major Q̇OOH reactions that form the core of the low-temperature ignition reaction mechanisms of many fuels [8, 121]: β-scissions, cyclic ether plus ȮH, olefines plus HO_2 formations and the reverse reaction of the Q̇OOH production channel, the internal H-atom abstraction. A comparison of the shown reactions with the results from the reaction identification test described above shows that most of the reactions are no artifacts of the CTY bond-order processing and that the reaction products correspond to minima on the ReaxFF PES. The molecule depictions in Fig 3.7 are derived from the CTY bond-order processing of the rMD trajectories. For all of the depicted reaction products, except for the

R2 product, the same truncated InChi codes are obtained from optimized geometries and from the bond-order processing of the trajectory. The shown reactions, except for reaction R2, pass the reaction identification test as a consequence (see appendix, section A.2.1).

The conventional simulations have only found some of the low-temperature reactions (R2, R3, R4 in Fig 3.7) but always with little probability ($\leq$ 10%, see Fig 3.7). In the CTY-TAD simulations, increasing temperature delivered higher observation probabilities for all reactions in Fig 3.7. At 5000 K, all the main low-temperature ignition reactions except the cyclic ether plus $\dot{\text{O}}$H formation (R1) were observed in every single simulation. Despite the use of the same pressures and temperatures in the conventional and in the CTY-TAD simulations, the expected trend to larger observation probabilities of the major $\dot{\text{Q}}$OOH reactions can only be observed for the CTY-TAD simulations. The higher observation probabilities with CTY-TAD compared to the observation probabilities in the conventional high-temperature simulations underlines that not only the elevated temperatures that are used to increase the frequencies of reaction events but also the basin confinement strategy is a key ingredient of my new method for reaction space exploration.

Not all of the common $\dot{\text{Q}}$OOH reactions from Fig 3.7 are included in the 1,3-dioxolane reaction model [108]. Reactions R1 and R2 are not included because the dominating β-scission R4 suppresses the reactions. In the simulations, though, the β-scission does not have a significantly lower reaction barrier than the competing reactions in Fig 3.7, considering that I have estimated the reaction barrier estimate $V_{\text{DS}}^{\text{min}}$ to have an uncertainty of 20 kcal mol^{-1}. In summary, the reactions expected from literature models of other fuels and, more importantly, the reactions included in the 1,3-dioxolane reaction model [108] have been found. However, a discrimination between the relevant common $\dot{\text{Q}}$OOH reactions (see Fig 3.7) reactions and the negligible ones is not possible from the reaction barrier estimates due to the large reaction barrier uncertainty.

CTY-TAD helps reaction modelers find new potentially relevant reactions. Besides

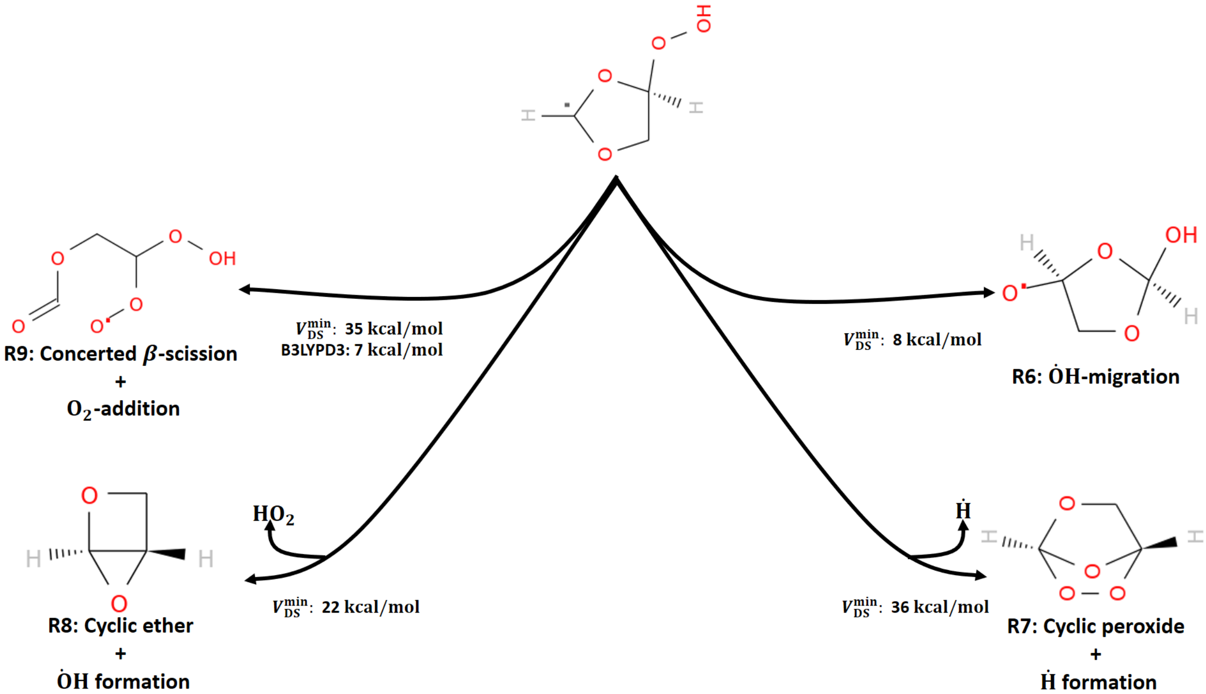


Figure 3.8.: Side reactions, observed in the CTY-TAD simulations, with a reaction barrier estimate $V_{\mathrm{DS}}^{\mathrm{min}}$ below the 38 kcal mol^{-1} reaction barrier cut-off that have not been included in the 1,3-dioxolane reaction model [108].

the reactions that are already included in the 1,3-dioxolane reaction model [108], I found 20 reactions that are not yet included in the reaction model and whose reaction barriers are comparable or lower than the reaction barrier of the known β-scission R4. This subset of the observed reactions is the set of new reactions that have a reaction barrier below the reaction barrier threshold of 38 kcal mol^{-1}. The reaction barrier threshold of 38 kcal mol^{-1} is the result of the addition of the R4 reaction barrier of 15 kcal mol^{-1} at the B3LYP level of theory and the reaction barrier uncertainties of reaction R4 and of the new reactions. For the reaction barrier uncertainty of R4, the 4.5 kcal mol^{-1} reaction barrier uncertainty reported in the literature for B3LYP [82] is used. For the reaction barrier uncertainty of the new reactions, I use the uncertainty of 19 kcal mol^{-1} as explained in the methods section.

Within the set of new potentially relevant reactions, there are four reactions that have passed the reaction identification test with both molecular identifiers. The four reactions are shown in Fig. 3.8. From the reactions shown in Fig. 3.8, the concerted

β-scission plus O_2 addition R9 is of particular interest as it has a lower B3LYP reaction barrier than the β-scission R4. XYZ coordinates of the R9 TS are provided in the appendix in section A.2.2. The other reactions R6-R8 can probably be neglected in the reaction model as they are similar to other reactions that have been discussed in the 1,3-dioxolane kinetic modeling study by Wildenberg *et al.* [108] and have been found to be negligible. The reactions R6 and R7 have similar reaction coordinates to the already-known and neglected cyclic ether plus $\dot{O}H$ formation R1. Wildenberg *et al.* [108] argue that none of the possible cyclic ether formations of the 1,3-dioxolane-4-hydroperoxide-2-yl radical, i. e. reactions R1 and R8, are relevant for the ignition processes of 1,3-dioxolane. Also, $\dot{O}H$-migrations like R6 have been found in literature to be slower than cyclic ether plus $\dot{O}H$ formations [92]. The concerted β-scission plus O_2 addition R9, on the other hand, has a B3PLYPD3 reaction barrier of only $7\,\mathrm{kcal\,mol^{-1}}$, which is approximately half the size of the R4 reaction barrier of $15\,\mathrm{kcal\,mol^{-1}}$. R9 should therefore be studied in more detail to determine its relevance for the low-temperature ignition process, which is beyond the scope of this chapter, though. As the R9 reaction rate is not only dependent on the $\dot{Q}OOH$ concentration like R4 but also on the O_2 concentration, R9 might be particularly interesting at high pressures and/or lean conditions.

Boost Factor

Although the ReaxFF reaction rate constants shown in Fig. 3.9 have an uncertainty too large for further assessment of the relevance of the reactions at technically relevant temperatures, we can still use them to investigate the acquired boost for the system. In literature, the boost factor is defined as the ratio of the fictional time required to simulate the reaction process at the low target temperature and the actual high-temperature simulation time [95]. The authors of the original TAD method [95] approximate the boost factor as the ratio of the fastest low-temperature waiting time and the simulation time spent in the reactant states at the high simulation temper-

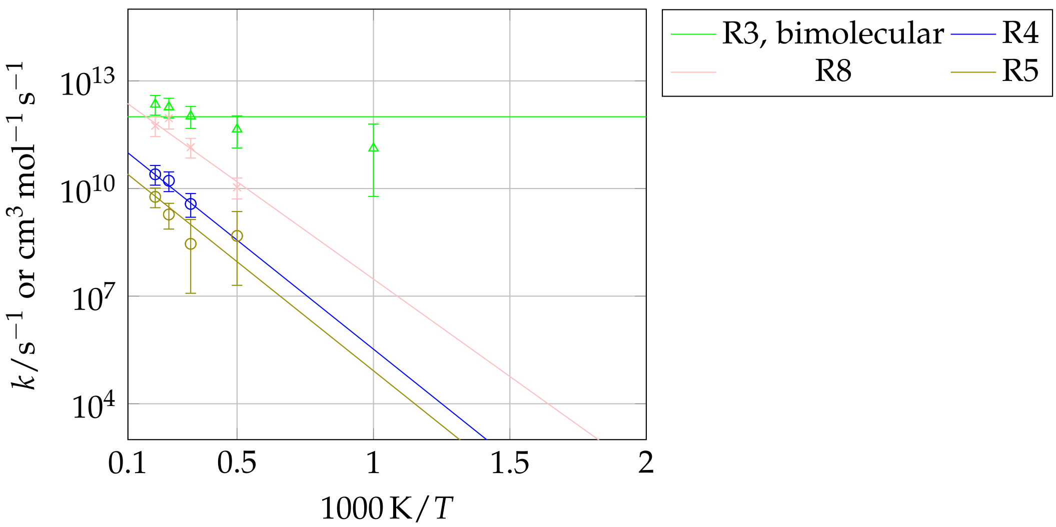


Figure 3.9.: Reaction rate constants for the reactions discussed in this work. The reaction labels are the same as in Figs. 3.7 and 3.8. The error bars represent the statistical uncertainties that depend on the number of observed reaction events for the given reaction at the given temperature [54]. The reaction rate constants were extrapolated to lower temperatures using an Arrhenius fit. The Arrhenius energy parameter was set to zero for the barrierless reaction R3. We were not able to find a temperature trend for all the discussed reactions and were consequently not able to provide an extrapolation scheme for them. Therefore, these reactions are not shown here.

ature. The TAD boost factor approximation is a good approximation if the goal of the simulation is to find a minimal reaction network consisting of the fastest low-temperature reactions only.

Considering that the purpose of CTY-TAD is not to find a minimum low-temperature reaction network but to find as many relevant reactions as possible, I do not specify a single boost factor for the system. Instead, I specify the range of individual boost factors for the reactions discussed above. The boost factor for a reaction can be written as the ratio of waiting time at the low target temperature $t_{\mathrm{wait}}(T^{\mathrm{low}}, c_i^{\mathrm{ref}})$ and the waiting time at the high temperature $t_{\mathrm{wait}}(T^{\mathrm{high}}, c_i^{\mathrm{sim}})$ or equivalently $\frac{t_{\mathrm{wait}}(T^{\mathrm{low}}, c_i^{\mathrm{ref}})}{t_{\mathrm{wait}}(T^{\mathrm{high}}, c_i^{\mathrm{sim}})} = \frac{k^{\mathrm{high\ T}} \prod_i^{N_{\mathrm{reactants}}} c_i^{\mathrm{sim}}}{k^{\mathrm{low\ T}} \prod_i^{N_{\mathrm{reactants}}} c_i^{\mathrm{ref}}}$. c_i^{ref} are the reactant concentrations at the reference pressure and c_i^{sim} are the reactant concentrations in the simulation. I use 720 K and 20 bar as reference temperature and pressure, which are the conditions used in the 1,3-dioxolane oxidation kinetic modeling study by Wildenberg *et al.* [108] to determine the low-temperature reaction mechanism. With the simulation set-up, the reactant concentration ratio contributes a factor of 21.6 per reactant assuming the same reactant concentration ratio of 1:1 in both the accelerated simulation and a simulation in the reference state. Due to the wide range that the reaction rate constants extrapolated to low temperatures (see Fig. 3.9) span, I obtain a wide range of boost factors from 4.7×10^2 for the O_2 addition (R3) to 3.7×10^8 (R4) at 5000 K compared to the reference conditions.

3.3. Conclusion

I have presented the rMD acceleration method CTY-TAD that is designed for the efficient exploration of local reaction space of a given set of reactants. CTY-TAD uses elevated temperatures to accelerate reactions. The efficiency of the reaction space exploration is increased by blocking already-observed reactions with virtual walls. Within the same simulation time, the use of virtual walls results in the observation of about 10 times more parallel reactions in a single simulation at the same temperature than in the same simulation without them. The positions of the virtual walls in configuration space are determined from an automatically constructed database of observed reactions. The system's current position in configuration space for comparison with the reaction database is determined in every simulation time step by integrating CTY bond-order processing steps into the rMD simulation. High boost factors up to the order of 10^8 were obtained in a case study of oxidation processes of 1,3-dioxolane-4-hydroperoxide-2-yl radical. I have demonstrated that the method finds the most important low-temperature ignition reactions known in literature [8, 121], which was not possible with conventional simulations that do not use the virtual walls at the same temperatures. Furthermore, I have compared the observed reactions with a reaction model for the ignition of 1,3-dioxolane [108], which the authors of the model currently continue to refine [109], to give an example of how CTY-TAD can help reaction modelers systematically search for new potentially relevant reactions. My CTY-TAD method has found about 100 new reactions not yet present in the 1,3-dioxolane reaction model [108]. Using reaction barriers as an indicator, I have found that many of the new reactions are not relevant for low-temperature ignition processes of 1,3-dioxolane, though. However, I have found the new concerted β-scission plus O_2 addition with a small reaction barrier of $7\,\mathrm{kcal\,mol^{-1}}$, which makes it potentially competitive to the dominant β-scission included in the 1,3-dioxolane reaction model. A more detailed study of the new reaction in the context of low-temperature oxidation of 1,3-dioxolane is therefore highly desirable to determine its impact on the reaction model performance.

4. A ChemTraYzer Study of Chlorinated Dibenzofuran Formation and Decomposition Processes

Chapter 4 is based on the publication "A Reactive Molecular Dynamics Study of Chlorinated Organic Compounds. Part II: A ChemTraYzer Study of Chlorinated Dibenzofuran Formation and Decomposition Processes" by Krep et al. [122]. In this chapter, parts of the publication [122] are reused directly or adapted with permission by the authors and ChemSystemsChem, Wiley-VCH Verlag GmbH & Co. KGaA. My contribution to this chapter include the conception and the implementation of the described methodological improvements. I have further contributed the results, with exceptions described in the following sentences, and, under consultancy by Leonid Komissarov, Dr. Wassja Kopp, and Felix Schmalz, the compilation of the text. Florian Solbach has implemented the TS search recovery method described in this chapter under my supervision. Thomas Nevolianis has contributed information about conformations of the discussed species. The work was supervised by Prof. Toon Verstraelen and Prof. Kai Leonhard.

In the following chapter, I integrate the CTY-TAD method into the CTY rMD-QM coupling scheme. I describe four performance updates for the coupling scheme, but focus on the application of the coupling scheme to dibenzofuran reaction chemistry and discuss practical aspects of the use of CTY for automated reaction space exploration. The choice of the dibenzofuran case studies has already been motivated briefly in the introduction. In the following, I pick up the thread and begin with a more detailed motivation to apply automatized RSEMs, like CTY, for the study of hazardous species like dibenzofurans.

The use of chlorinated organic and inorganic compounds in the chemical industry is a double-edged sword. On the one hand, they have large economic value, with phosgene, for example, being a valuable intermediate for the production of insecticides, plastics, dyes, and industrial resins [123–126]. On the other hand, working with chlorinated compounds often requires substantial safety precautions to protect the work force and close-by residents from exposure to highly hazardous compounds. After industrial accidents and food scandals involving toxic members of the chlorinated organic compounds (COCs) family, such as chlorinated dibenzodioxins (CDDs) and chlorinated dibenzofurans (CDFs) [127, 128], the use of COCs has also been subjected to greater public scrutiny and stricter regulations [129, 130].

The toxicity of many COCs does not only affect their employment on an industrial scale but also in the laboratory. As an established and safe alternative to practical experiments, computer methods have been proposed for the study CDDs and CDFs reaction processes [56]. A variety of computational studies of formation and decomposition processes of CDDs and CDFs, for example, is readily available in the literature [131–145]. Two major formation mechanisms of CDDs and CDFs have been proposed in the literature [56, 143]: Homogenous gas-phase formation from structurally similar precursors and the heterogenous *de novo* synthesis route from carbon matrices in fly ash. The presence of metals, such as copper and copper oxides, has been found to catalyze both routes. Others have investigated chlorination inhibition strategies for waste incinerators [145]. However, despite the large number of studies of CDDs and CDFs reaction processes, many questions have remained unanswered. The contributions of the homogeneous and heterogenous formation routes to final

CDDs and CDFs yields are not finally clarified, yet, due to high complexity of the reaction mechanisms [56]. The impact of chlorination sites in CDDs and CDFs on their thermochemical and reactive properties, for example, is not yet fully understood. While group additivity methods for thermochemical properties of CDDs and CDFs are already available in the literature [146, 147], the impact of chlorination on CDF reaction barriers has, to our knowledge, not yet been studied. Better understanding of the chlorination processes will aid to understand the catalytic CDDs and CDFs formation processes [148] and will aid the development of chlorination inhibition strategies. Additionally, the main channels of dibenzofuran low-temperature oxidation are still up for debate [56, 131, 149]. The review by Altarawneh *et al.* [56] is an excellent introduction to the topic. While the reactions involving CDDs and CDFs investigated with computer methods so far have been compiled manually, rMD simulations allow exploring reactions automatically with principally little to no *a priori* knowledge about the system required by the user. rMD simulations provide the means to discover reactions, but the reactions must be quantified using higher level-of-theory PES, to obtain the low uncertainties that are required for many applications. In this chapter, I demonstrate the use of CTY for the study of reaction processes of hazardous species in a systematic and highly automated fashion in three case studies of formation, decomposition and oxidation processes of CDFs. I first introduce four conceptual updates to the CTY rMD-QM coupling scheme in the methods section, among them a self-learning method for the recovery of failed TS searches. The conceptual updates address known shortcomings of our coupling scheme [35], enhancing its computational resource efficiency and TS search success rate. We further demonstrate that CTY can be used to systematically study reaction processes of hazardous species in a highly automated fashion; we do so using three case studies of formation and decomposition processes of chlorinated and unchlorinated dibenzofurans. The reactions derived in this study by the updated CTY scheme are used as validation data for the force field parameters described in part I of our series [150]. The chemical

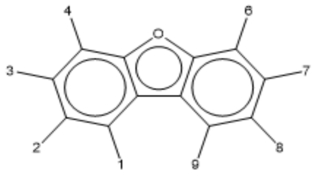


Figure 4.1.: Dibenzofuran chemical structure and atom numbering convention in our work. We use the same atom numbering convention as in [56, 127].

structure of dibenzofuran is given in Fig. 4.1. I present reaction pathways for (i) the formation of CDFs from the precursors phenoxy radical plus 1,3,5-trichlorobenzene, (ii) the pyrolysis of singly-chlorinated and unchlorinated CDFs, and (iii) the oxidation processes of singly-chlorinated and unchlorinated 1-peroxy-dibenzofuranyl radicals. The phenoxy radical plus 1,3,5-trichlorobenzene system has been chosen as an example for dibenzofuran precursor reaction processes and as an extension to the 2-chlorophenoxy radical plus chlorobenzene study by Altarawneh *et al.* [151]. The impact of chlorination on CDF reaction barriers is addressed by the pyrolysis and oxidation case studies. Furthermore, we add further insights to the very core of the debate on the main dibenzofuran low-temperature oxidation channels [131, 149] with our 1-peroxy-dibenzofuranyl radical oxidation case study, as the proposed reaction mechanisms differ in their 1-peroxy-dibenzofuranyl radical reactions [56].

4.1. Methods

The ChemTraYzer Concept

CTY has a modular design. The first CTY module, which is shown in the left swimlane of Fig. 4.2, is an automated rMD simulation and trajectory processing procedure. Trajectory processing can be used separately as a post-processing script for rMD trajectories that have been generated with the implementations of ReaxFF [58, 59] in LAMMPS [34, 71, 72] or AMS [153]. CTY detects reactions based on changes in ReaxFF bond orders, which are continuous representations of chemical bond strengths, *e.g.* to map the differences between single or double bondsreplaced, such as a single

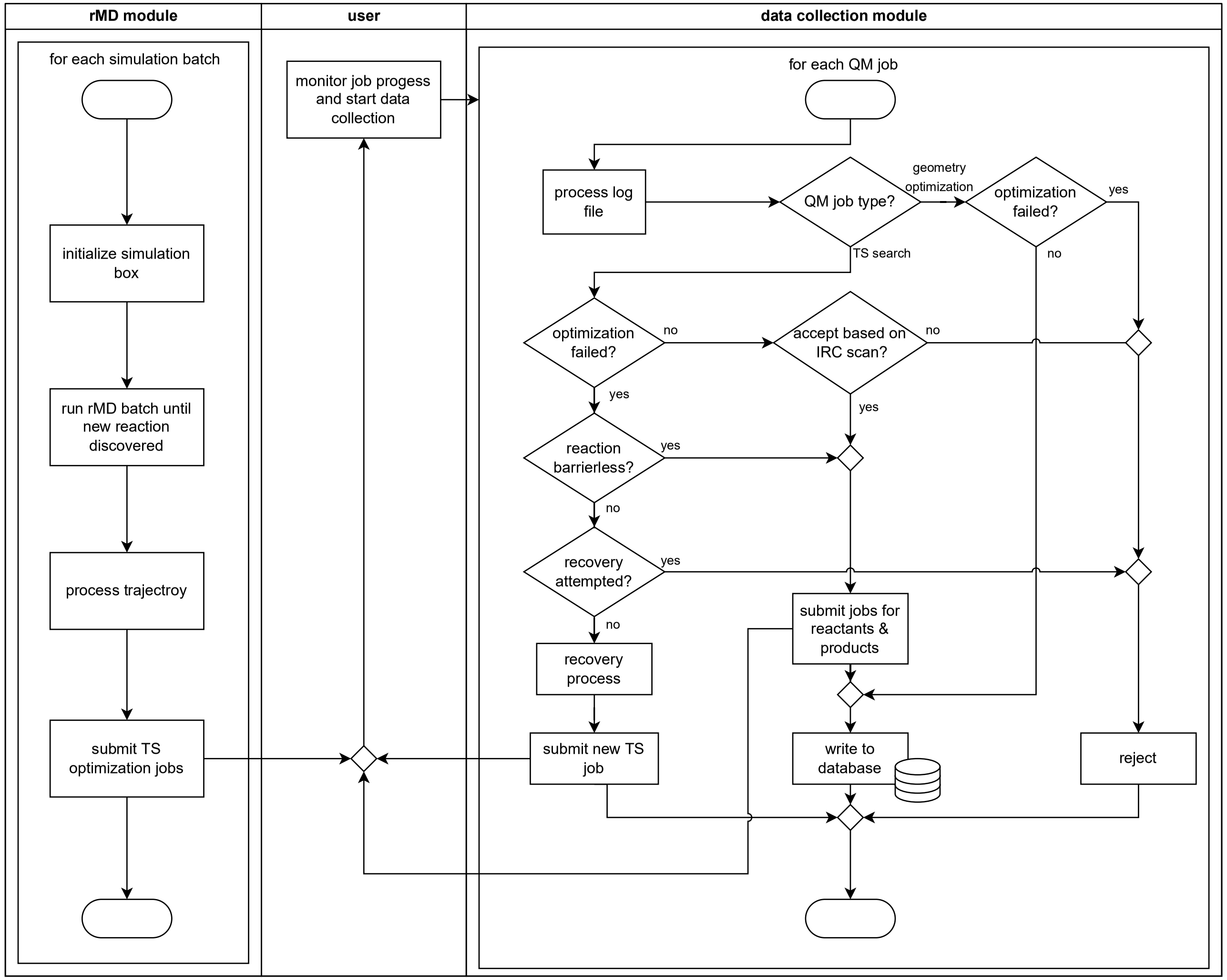


Figure 4.2.: Simplified scheme of the new CTY simulation process. After every iteration of the outer data collection iteration loop, a CHEMKIN [152]-formatted reaction model is written using the reactions in the database. For simplicity, I have not shown the reaction model output step.

or double bond. The first CTY module allows for the estimation of reaction rate constants from the points in time at which reactive events are detected [34, 54].

The rMD simulation step of the first CTY module can also be coupled with CTY-TAD acceleration technique for more efficient reaction searching, as I do here. The CTY-TAD implementation of the basin-constraint strategy [94–104, 106, 107, 112] has been described in detail in the previous chapter.

The PES data presented in the results and discussion section is obtained using the second CTY module. The second module is an optional rMD-QM coupling scheme [35] for the reoptimization of stationary points on the PES, followed by an automatic production of thermochemical and kinetic data using TST. This second module is shown in the right swimlane of Fig. 4.2 in the "data collection module" panel. The depicted coupling scheme shows four conceptual improvements compared to the original from Döntgen *et al.* [35], all four of which are described in the next subsection. As a common basis for both the original and the new rMD-QM coupling scheme, whenever a reaction is detected in the rMD trajectory, molecular geometries are are extracted and used as initial guesses for geometry optimizations and TS searches. After the geometry optimizations and TS searches are completed, the user initiates a final data collection processing step, in which molecular properties, such as electronic energies, symmetry numbers, frequencies, and optimized geometries, are written to a database. Molecular properties are first used to compute thermochemical properties and reaction rate constants with TST, are then subjected to Arrhenius and NASA polynomial fits, and are finally written to a CHEMKIN [152]-formatted reaction model.

Conceptual Updates to the ChemTraYzer RMD-QM Coupling Scheme

In the following, I discuss four conceptual updates to the CTY rMD-QM coupling scheme [35]. The conceptual updates are introduced to improve TS-search success rates, to increase the computational efficiency of the CTY concept, and to address the known issue of spin multiplicity guessing for the molecular geometries extracted from the rMD trajectory [35].

The first conceptual update is the simplification of the CTY acceptance criterion for optimized TS.. In the original coupling scheme, a TS is only accepted, if its intrinsic reaction coordinate (IRC) connects the same reactants and products as observed by CTY's trajectory processing. By contrast, in the new implementation, I accept every reactive TS, *i. e.* every first-order saddle point on the PES, whose IRC connects species with different connectivities. The new TS acceptance criterion accounts for the fact that reaction detection procedures that are used for the analysis of the rMD trajectories do not always identify the type of the observed reactions correctly [94, 154]. More importantly, the new implementation also accounts for the fact that reactions that haven't been observed in the rMD trajectory can also be of value, as will be demonstrated in the case studies.

ReaxFF is a classical force field that is oblivious to the concept of electron spin. Hence, when post-processing ReaxFF geometries with *ab-initio* calculations, charges and spin multiplicities must be guessed. In the original implementation, the smallest allowed multiplicities were always assigned, despite being well aware that such a procedure does not always yield ground-state configurations [35], *e.g.* for triplet oxygen *vs.* singlet oxygen. As a solution for this issue, I decided to follow the example of Schmitz *et al.* [55] and probe multiple multiplicities for reactants, products and TS. This is the second conceptual change compared to the original implementation of the coupling scheme. Similar to Schmitz *et al.* [55], my implementation only accepts combinations of multiplicities for reactants, TS, and products that result in a spin-conserving reaction. However, while Schmitz *et al.* only accept reactions that connect ground-state reactants and products, I also accept reactions connecting reactants and products with higher electronic energy states. Although for most applications, only the electronic ground states of molecules are relevant, and these are usually connected to the minimum multiplicity, it is still reasonable to consider reactions connecting higher electronic states as well. There are some exceptions to the minimum multiplicity rule-of-thumb, such as for metal-oxo species, whose minimum multiplicity and higher multiplicity configurations often have similar energies [155]. The user has two options to adjust the CTY multiplicity probing. One is to add pairs of species and multiplicities to the `spin` list. Species that are in the `spin` list are not subject to the multiplicity probing. Instead the assigned multiplicities from the `spin` list are always used for the species. The list currently contains only the elements C, O, H, and Cl, along with their ground-state multiplicities. The other option is to adjust the list of probing multiplicities, named `testmultiplicities`. In this study, I probe either singlet and triplet states or doublet and quadruplet states for each species and each TS. In the case studies, I demonstrate that the multiplicity probing yields potentially relevant reaction paths, even for systems that have no known exceptional electron spin or multi-reference characteristics. All charges are currently still assumed to always be zero.

As the third conceptual update, I switch to a TS-centered procedure. In the original implementation, TS, reactant, and product initial guess geometries are taken from the rMD trajectory and geometry optimizations are automatically started after the trajectory processing. As TS searches do not always converge, some reactants and products become isolated nodes in the reaction network. In order to conserve computational resources that would otherwise be wasted on the computation of molecular properties of these isolated nodes, I only start TS searches automatically after the rMD trajectory processing. Reactant and product geometry optimizations are obtained only after a given TS search and the subsequent IRC scan converged based on the fully-optimized IRC endpoints. With this procedure, conformers of products and TS are sampled if the same reaction is observed multiple times. To improve the sampling of conformational space, we employed the Conformer–Rotamer Ensemble Sampling Tool (CREST) [99] to search for conformers of reactants, products, and TS for the reactions discussed in this work.

The last conceptual update is the introduction of a recovery process for TS searches.

It is well-known that TS searches are difficult to converge, and multiple attempts with different TS guess geometries or optimization parameters are often needed. Many approaches in the literature that attempt to facilitate or automate the TS searching process focus on improving the TS guess geometry. Chain-of-state TS search methods, such as Nudged Elastic Band (NEB) [156] and Growing String Method (GSM) [22, 157], apply interpolation techniques to generate TS guess geometries from given (ideally fully optimized) reactant and product geometries. Requiring the atoms of the product geometries to be mapped to the atoms of the reactant poses a combinatorial challenge for the automated use of chain-of-state methods. However, a rMD-QM coupling scheme using the chain-of-state method NEB has already been proposed by Wang *et al.* with their nebterpolator method [30]. They report a success rate, *i. e.* the number of converged TS searches per observed reaction event, of 49%. A second group of rMD-QM coupling schemes directly uses trajectory snapshots from the rMD simulation as TS guesses. Two representatives of this second group are CTY [35] and AutoMeKin [31–33]. The authors of AutoMeKin report being able to converge TS for approximately 70% [31] of the observed reactions in their case studies. A major difference between the original CTY rMD-QM coupling scheme and theirs is that AutoMeKin initiates between three and eight TS searches per reaction event, starting from snapshots close to a detected reaction event [31]. In contrast, CTY uses only one TS search per reaction event. A more data-driven TS search method has been recently introduced by Pattanaik *et al.* [158]. They report a deep-learning TS search method that yields inspiring success rates between 71% and 89%. However, the approach is currently limited to isomerization reactions containing the elements $\mathrm{C, H, O, N}$. Like chain-of-state methods, the approach still requires the mapping of atoms between reactant and product states [158]. Grambow *et al.* [36] compare success rates of multiple automated TS search approaches. Their reported success rates range from 23%-66%.

As no single available automated TS search algorithm yields satisfactory success rates yet, I decided to add a second TS search method to the rMD-QM coupling scheme as a way to recover failed TS searches. The recovery method is only spawned after the primary method fails to converge a TS. For the recovery method, I take an intermediate stance between a data-driven approach, such as the deep-learning TS search method by Pattanaik *et al.* [158] with its high success rates and the application flexibility of the more "classical" methods mentioned above, which can function without extensive training data. The new recovery method is inspired by a strategy that is often successful and efficient for manual TS searches. From my experience, it is often helpful to first search for the TS of a simpler, *e.g.* smaller, but similar reaction system and then to use the active atoms, *i. e.* the atoms that are involved in bond-breaking and -formation processes, as a template for the system of interest. In the automated implementation of this approach, I make use of the CTY database as a source for templates. As the CTY database grows with every accepted TS, the recovery process is self-learning.

After a CTY TS search has failed, the failed TS search is passed through a 2-step examination process to determine if a recovery should be attempted. The examination process is shown in the data collection module panel in Fig. 4.2. First, the TS search is subjected to the CTY virtual TS test heuristics [35]. A set of reactants - products spin multiplicity combinations that are often connected to barrierless reactions is implemented in these test heuristics [35]. For example, radical recombinations, in which two doublet state reactants recombine to a singlet state product, are flagged as barrierless. If the heuristics find a reaction to be barrierless, the recovery process is skipped, and reactant and product geometry optimizations based on the reaction SMILES, which are obtained during trajectory processing, are spawned. I have moved the virtual TS test to behind the initial TS search to account for the fact that for some reactions, the existence of a reaction barrier depends on the employed model chemistry. With the new position of the virtual TS test in the CTY data collection process, a reaction needs to fulfill two criteria to be found to be barrierless: (i) no TS must be found and (ii) the chemical intuition implemented in the CTY virtual TS test heuristics [35] must suggest a barrierless reaction. In a final examination step, I determine if the recovery process has already been attempted before. In case that the failed TS belongs to a reaction that is assumed to possess a reaction barrier and no recovery has previously been attempted, the recovery process is invoked.

In the first step of the recovery process, a set of TS templates is chosen from the database based on their active atom types and their bond-changing pattern. The bond-changing pattern defines changes of the bonds between the active atoms, for example, the simultaneous breaking of a C−O bond and formation of a C−C bond. To further distinguish between reactions, I also store bonds between active atoms that do not change in the bond-changing pattern. The active atoms, the active atom labels, and the bond-changing pattern are obtained from the trajectory processing of the to-be-recovered TS guess and from the optimized IRC endpoints of the templates. The IRC-derived information is stored in the database. The selection of templates is subjected to the active atom mapping and replacement process that is shown in Fig. 4.3. This starts with the aligning of the centers-of-mass of the active atoms of each template and the current TS guess. The Kabsch algorithm [159] then rotates the template geometries to minimize the RMSD between the active atoms of the TS guess and of the templates. I use the SciPy [160] implementation of the Kabsch algorithm[159]. The templates are ranked by their RMSDs between the active atoms of the TS guess and of the templates after the rotations. Next, the Cartesian coordinates of the active atoms of the TS guess geometry are replaced with the Cartesian coordinates of the active atoms of the optimal template. Using this new TS guess, a new TS search is then spawned. Like in the original CTY TS searching process, the TS guess is first subjected to a constraint optimization with frozen active atoms before the TS is converged with an eigenvector-following algorithm. As a result of the recovery process, multiple TS searches per reaction event are spawned, which is similar to the AutoMeKin approach. The novelty is twofold: CTY not only runs additional TS searches only when needed, but it also learns from similar successful searches in order to conserve computational resources.

Overview of the Conceptually Improved CTY Process

Following the introduction of the individual conceptual improvements, it is instructive to summarize the entire CTY process. The user initiates rMD simulations that are run within the CTY framework and opts for the use of the rMD-QM coupling scheme. The rMD simulations are run in batches, interleaved with on-the-fly trajectory processing [35, 94].

After trajectory processing, TS searches are initialized based on the trajectory snapshot in which a reaction event was detected. When the TS search is converged, an IRC scan is spawned and its endpoints are fully optimized.

In the first round of an iterative data collection process, the user initiates a Python data collection script that processes the TS searches and IRC scans and determines if the TS searches have converged and if the TS connects reactants and products with different connectivities, *i. e.* if the TS are reactive. An example of a non-reactive TS would be a conformational transition. If a TS is converged and reactive, the script retrieves molecular properties of the TS and writes them to a database. If the TS search failed and the CTY virtual TS test ("Reaction Is Barrierless?" in Fig. 4.2) confirms that a TS exists and no recovery for this TS search was attempted before, the recovery process is invoked, and a new TS search is started. Based on the optimized IRC endpoints, initial guess geometries for reactants and products are generated, and geometry optimizations are started. Reactant and product geometry optimizations are run using all multiplicities from the user-defined range of probing multiplicities that allow for a spin-conserving reaction for the given TS. If the CTY virtual TS test determines that no TS exists, reactant and product geometries are generated using Open Babel [111] based on the SMILES of the reactants and products that have been generated during the rMD trajectory processing. In subsequent iterations of the data collection process, TS searches, reactants, and products are read into the database.

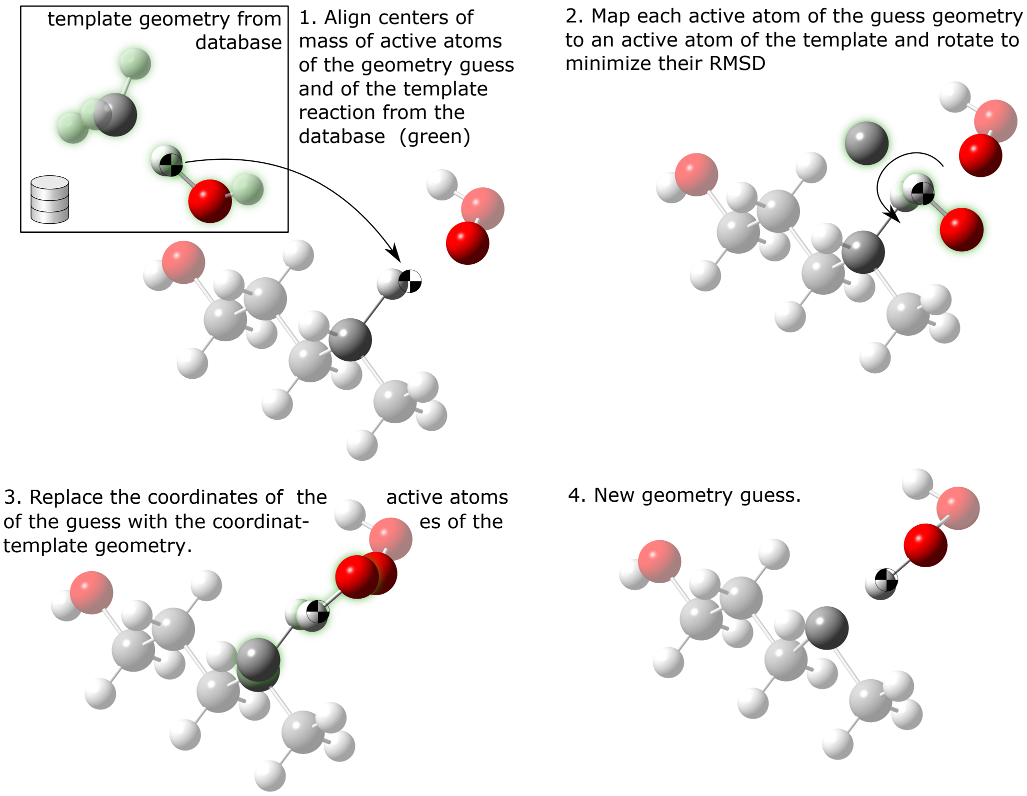


Figure 4.3.: Active atoms mapping and replacement process of the recovery method, shown using the example of a H-atom abstraction reaction. The four steps are executed for each reaction template that is selected from the database. Here, the template TS geometry from the database is highlighted in green. If there are ambiguous mappings of active atoms in step 2 (*e.g.* when there are two active C atoms), all of them are tested and only the one with the lowest RMSD is kept. The reference TS geometry from the database belongs to an H-atom abstraction from methane by an OH radical. The TS guess geometry that is being recovered belongs to an H-atom abstraction from butanol by an HO_2 radical. All non-active atoms are semi-transparent

All data collection iterations are initiated by the user re-invoking the Python data collection script. Effectively, the user is part of the data collection process and takes on the role of a job manager. If at least some *ab-initio* calculations that were spawned during the last data collection iteration have finished, the user initializes a new data collection iteration. New iterations are initiated until no new *ab-initio* calculations are spawned anymore. If no TS search needs a recovery, only two data collection iterations are necessary to collect the data of TS from the first iteration and of reactants and products from the second iteration. At the end of each data collection iteration, the CHEMKIN [152]-formatted reaction model is written using the reactions whose TS, reactants, and products are all already known to the database. The reaction model generation is not shown in Fig. 4.2 for the sake of brevity.

Simulation Setup

The rMD simulations were performed using the ReaxFF parameters developed by Komissarov *et al.* [150]. Some reactions discussed below were obtained using a preliminary version of the force field parameters. The preliminary version differs from the published parameters only by the parameters for Cl−Cl interactions, meaning that only the systems containing multiple chlorine atoms are affected. The preliminary version of the force field parameters is provided in the appendix in section A.3.1. I have run the rMD simulations with the acceleration technique CTY-TAD. The CTY-TAD simulations are *NVT* simulations at 2000 K using the Langevin thermostat with a damping constant of 10 fs and a timestep of 0.1 fs. The CTY-TAD simulations had a maximal simulation batch length of 100 ps. This combination of maximum batch length and simulation temperature is expected to result in the majority of observed reactions having a reaction barrier below 50 $\mathrm{kcal\,mol^{-1}}$, as I have discussed in the previous chapter. Due to the uncertainty in ReaxFF energies of $\approx 30\,\mathrm{kcal\,mol^{-1}}$ with the C/H/O/CL force field parameters by Komissarov *et al.* [150] higher reaction barriers can be found after recomputation of the reaction barrier with a higher-level PES. Note that the ReaxFF energy uncertainties are dependent on the parameterization [150].

For each of the three case studies, the first batch of rMD simulations started from a box containing the two reactants of interest. The initial boxes were created with Packmol [70]. The molecules were confined by a repulsive harmonic potential that is used in CTY-TAD to keep the molecules inside the simulation box [94]. The parameters of the repulsive harmonic potential were determined automatically by CTY-TAD [94] and depend on the box volume. The box lengths of the simulation boxes were chosen to be small to obtain a high enough density for pAD acceleration effects [57] to be used, but large enough that the molecules in the box could pass each other freely in all dimensions without being distorted by the repulsive wall potential. I used a box length of 20 Å for the phenoxy radical plus 1,3,5-trichlorobenzene simulations. For the (chlorinated) 1-peroxy-dibenzofuranyl radical plus O_2, as well as for the Cl_2 plus (chlorinated) CDFs and their radicals systems, box lengths of 24 Å were used. The simulation box sizes result in densities of $415\,\mathrm{mol\,m^{-3}}$ and an ideal gas pressure of 70 bar for the smaller boxes. The larger boxes have densities of $240\,\mathrm{mol\,m^{-3}}$ and an ideal gas pressure of 40 bar. I ran one replica for each system with both the old and the final force field parameters, the latter of which are reported in [150]. Only for the phenoxy radical plus 1,3,5-trichlorobenzene system and the final force field parameters, I increased the number of replica to five, to expand the number of observed reactions. As I have discussed in the previous chapter, it is generally advisable to run multiple replica of CTY-TAD simulations to obtain more complete reaction networks. Nevertheless, I decided to run only the minimal number of replicas per system. This was done because benchmarks for the computational resource requirements of the CTY rMD-QM coupling scheme were not available beforehand, and my computational resources were limited. While the CTY-TAD simulations themselves consumed only a few core-hours per simulation, the costs incurred by the accompanying rMD-QM coupling were hard to predict.

I chose the DFT method B3LYP with the TZVP basis set and the Gaussian 16 software package [161] for QM reoptimizations.

Goerigk *et al.* report an RMSD of 4.5 kcal mol^{-1} of B3LYP reaction barriers from W1-F12 and F2-F12 reference barriers in their DFT methods benchmark study[82]. The energy uncertainty is too large to predict reaction rate constants, but it is low enough to draw conclusions on the relevance of many of the observed reactions based on the relative differences of the reaction barriers. Furthermore, Saeys *et al.* [162] have shown that B3LYP predicts TS geometries similar to more expensive *ab-initio* methods, in their case QCISD. In conclusion, I opt for the B3LYP functional for the QM reoptimizations as a compromise between computational efficiency and reliably optimized molecular geometries, but also for easier comparison of reaction energies and barriers with literature values, as they have mainly been obtained from B3LYP studies. The ability of B3LYP to produce good molecular geometries also makes it a good basis for future calculations of reliable kinetic data involving higher-level single-point energies.

4.2. Results and Discussion

I start the discussion of the results with an overview of the performance of the new CTY rMD-QM coupling scheme. All simulations together, approximately 200 spin-conserving reactions with minimal reactant and product multiplicities were found and quantified automatically. Including higher reactant and product multiplicities, the reaction list totals about 500 reactions, from which more than 100 reactions include (chlorinated) dibenzofurans, -furanyl radicals and 1-peroxy-dibenzofuranyl radicals as reactants and products.

The total TS search success rate for the combination of original CTY and recovery process TS searches is 48% with reference to the number of observed reaction events. Failed TS searches that have been found to be barrierless by the CTY virtual TS test heuristics [35] are not included. The original CTY TS search has a success rate of 38%. The success rate of the recovery process is 27% with reference to the number of submitted recovery TS searches. Within this study, two possibilities to further enhance the success rate showed up: First, since only the coordinates of the active atoms are replaced, the distance between them and the non-active atoms may change. In extreme cases, this artificial change of inter-atomic distances can lead to breaking or forming of bonds between active and non-active atoms. In other words, the quality of the initial TS guess still matters. Second, if there are only two active atoms or the active atoms lie on a straight line, the Kabsch algorithm [159] cannot determine a unique optimal rotation in 3D-space as information about the third rotational degree-of-freedom is not available. In that case, the Kabsch algorithm returns one random rotation from an infinite number of possibilities. The issue is illustrated in the appendix in section A.3.2. At this point, I would also like to point the reader to the discussion of the connection between inaccuracies in the CTY reaction detection procedure and TS convergence issues and of strategies to improve the CTY reaction detection procedure in the appendix, section A.4. To conclude, the new CTY TS detection performs comparable to other automatic TS-search methods [30, 36] and delivers TS for about half the reactions detected. As no single TS search method available in the literature features success rates of 100% yet, combinations of methods, for example as implemented in a recovery process such as ours, are promising solutions. The entire process of rMD simulations and requantification at higher levels of theory consumed about 200.000 core hours on Intel Xeon Platinum 8160 Processors, but required only very little effort from the user. The simulation set-up and the data collection iterations consume only a few hours or even minutes, respectively, of the user's time.

A detailed discussion of all reactions I have found with CTY, is beyond the scope of this publication. I have, therefore, decided to focus on the subset of reactions that adds most to the state of research on dibenzofurans in my opinion. The reactions discussed here are parallel or consecutive reactions to the already known ones near the debated nodes in the reaction networks. A larger subset of the reactions of 150 reactions found with CTY in this study have been re-used as validation data for the

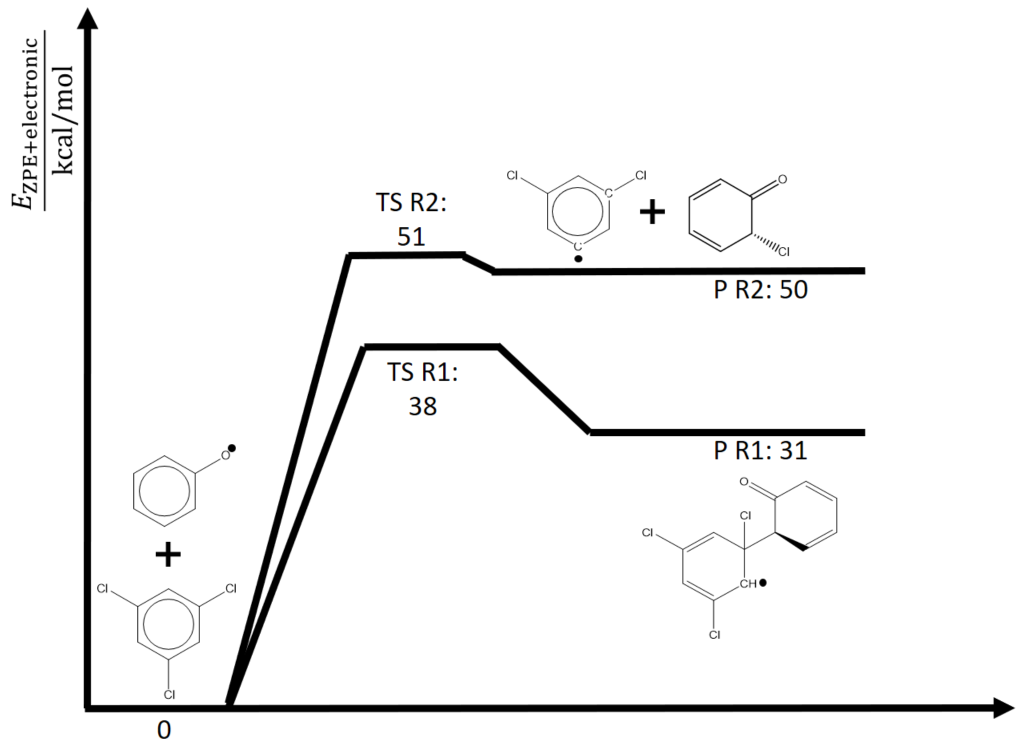


Figure 4.4.: ZPE-corrected electronic energy surface of the phenoxy radical and 1,3,5-trichlorobenzene reaction paths at the B3LYP level of theory.

employed ReaxFF C/H/O/Cl parameters [150]. An overview of all reactions including reaction barriers and reaction energies can be found in the appendix in section A.3.5. Optimized geometries of reactants, products and TS will be made available in the Supporting Information of [122] after publication.

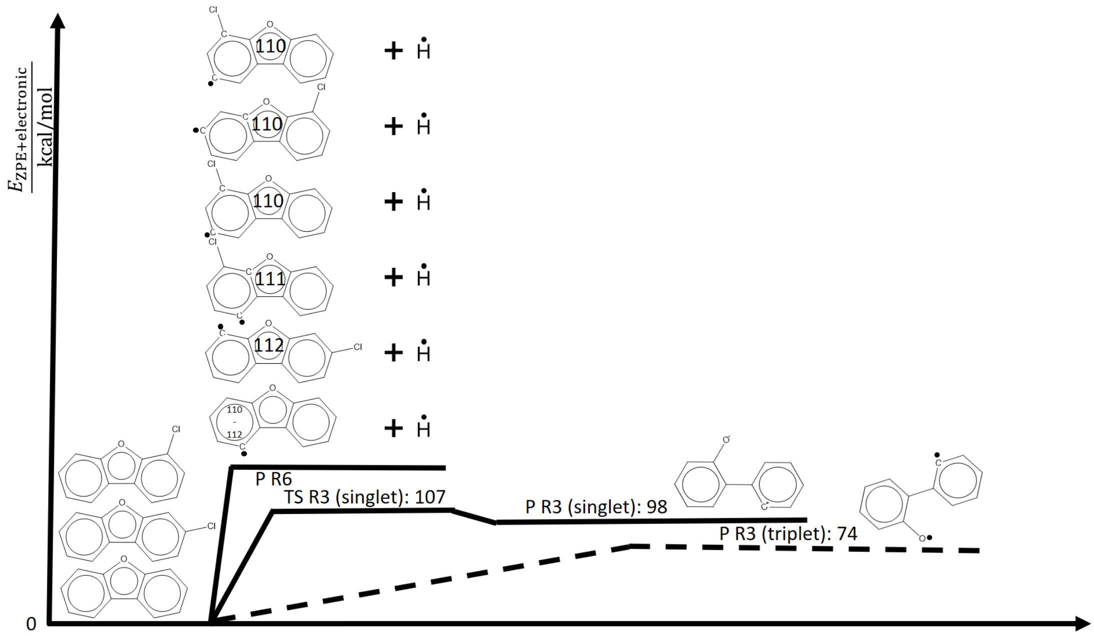
CDF Precursors Phenoxy Radical and 1,3,5-Trichlorobenzene

Phenoxy radicals are among the most important dibenzofuran and -dioxin precursors [56]. They can react, for example, with other benzene-like compounds to form cyclohexadienons, which can then continue to form cyclohexadieonon radicals and CDFs given that the C−C bridge between the two rings is in the α-position relative to the carbonyl group [56, 151] (see Fig. 4.4, reaction R1). Altarawneh *et al.* [151] studied the formation of cyclohexadienons and their radicals from the 2-chlorophenoxy radical and chlorobenzene. Dependent on the attack position (ortho, meta, or para) of the

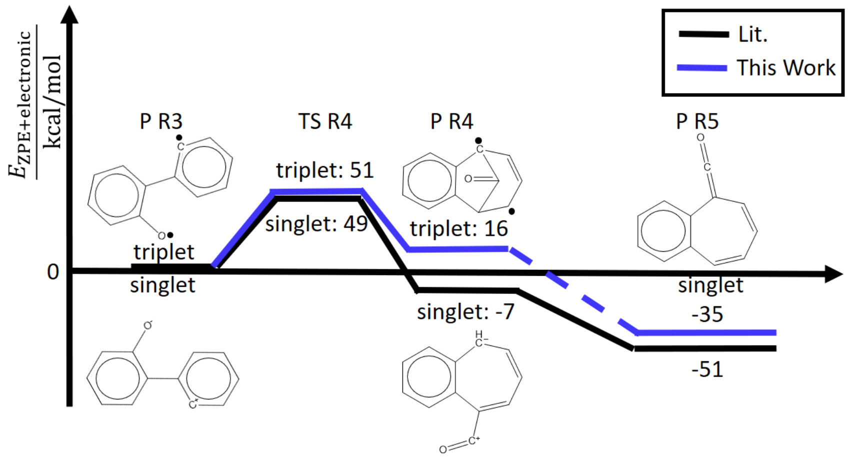
2-chlorophenoxy radical at chlorobenzene, the cyclohexadienon formation reaction barriers range from 30.7 $\mathrm{kcal\,mol^{-1}}$ up to 41.2 $\mathrm{kcal\,mol^{-1}}$ [151]. Despite one additional chlorine atom, the cyclohexadienon formation from 1,3,5-trichlorobenzene plus phenoxy radical (Reaction R1 in Fig. 4.4) has a reaction barrier of 38 $\mathrm{kcal\,mol^{-1}}$, which is within the same reaction barrier range. This observation suggests that the attack positions of the (chlorinated) benzene and phenoxy radicals may be more important for the reaction barrier height of the cyclohexadienon formation than the number of chlorination sites. In the simulations, I also find the parallel reaction path to the cyclohexadienon formation R1, which is reaction R2 in Fig. 4.4. In the chlorine-transfer reaction R2, the phenoxy radical abstracts a chlorine atom from the trichlorobenzene. The reaction barrier of R2 is only 12 $\mathrm{kcal\,mol^{-1}}$ larger than the competing reaction R1. Using CTY, I was able to find two bimolecular reactions for the phenoxy radical plus 1,3,5-trichlorobenzene system, which has not yet been subject to studies, to my knowledge. One of the reactions, reaction R1 in Fig. 4.4 is similar to the initial step of one of the major known dibenzofuran formation mechanisms. The second reaction has not been described in the literature, yet. The reaction barrier is not much larger than the known dibenzofuran formation channel and may therefore be a relevant side reaction that needs to be considered in models of dibenzofuran formation. I would like to stress that these two bimolecular reactions were obtained fully automatically, as the automatization of TS-searches for bimolecular reactions poses a special challenge for many automated TS search concepts.

Dibenzofuran Pyrolysis and Chlorination

Cieplik *et al.* [163] have studied the thermal hydrogenolysis of dibenzofurans at pyrolytic temperatures between 1100 and 1200 K. They find that the overall conversion rates of dibenzofurans do not depend on the partial pressure of hydrogen at conditions under study. They also pointed out that hydrogenolysis and pyrolysis of dibenzofurans give similar main products, *i. e.* napthalene and carbon monoxide [163].



(a) ZPE-corrected electronic energy surfaces of dibenzofuran unimolecular decomposition entrance channels. The P R6 energy is specified for every P R6 species with reference to the respective (chlorinated) dibenzofuran. For the unchlorinated P R6 species, the range of energies depending on the position of the radical site is given. The displayed radical site of the unchlorinated dibenzofuranyl radical is therefore only an example, *i. e.* the 1-dibenzofuranyl radical is shown. Energies for reaction R3 on the singlet spin surface and for the unchlorinated dibenzofuran radicals are taken from [132].



(b) ZPE-corrected electronic energy surface for the P R3 reaction pathway as proposed by Cieplik *et al.* [163]. Energies of the black pathway are taken from [132]. The stationary points on the blue path were compiled without the help of CTY using B3LYP and the TZVP basis set.

Figure 4.5.: Two competing unimolecular dibenzofuran decomposition mechanisms discussed in the literature [132, 163]. Dashed lines denote spin-changing reactions.

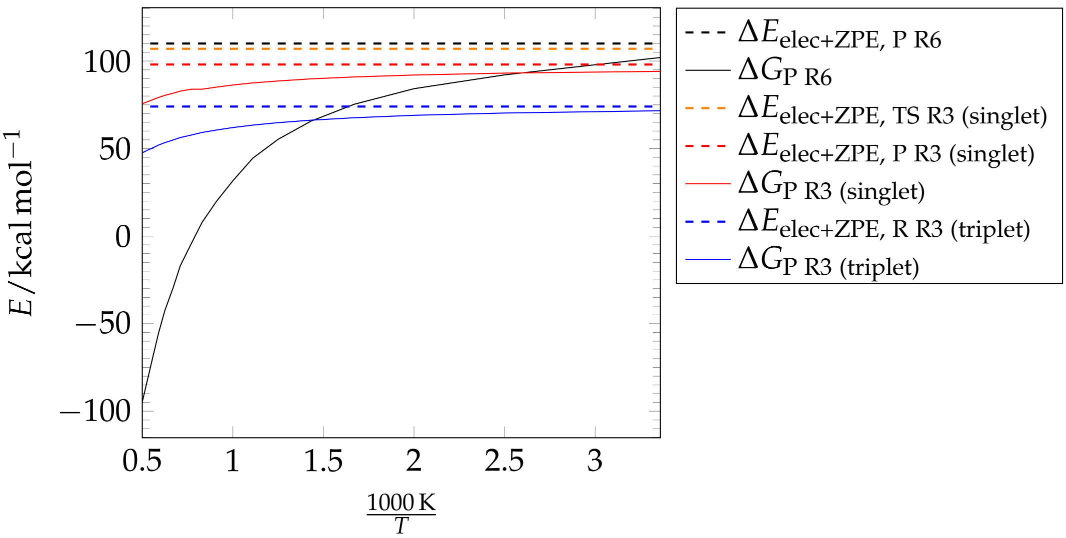


Figure 4.6.: P R6 Gibbs free energy and the zero-point corrected electronic energy (of the unchlorinated 3-dibenzofuranyl radical) in comparison to TS R3 (singlet) and singlet and triplet P R3. The molecular structures of P R3 and P R6 are given in Fig. 4.5a. The energies are referenced to dibenzofuran.

Therefore, they argue that the major decomposition mechanism must be unimolecular. The initial three steps of the decomposition mechanism Cieplik *et al.* propose are shown in Fig. 4.5. The final steps that lead to the formation of napthalene and carbon monoxide are not shown in the figure. Altarawneh *et al.* [132] have provided reaction barriers and reaction energies at the B3LYP level of theory for the proposed reaction mechanism on the singlet spin surface. Fig. 4.5a, however, shows that the reaction product P R3 in the triplet state is lower in energy by 24 $\mathrm{kcal\,mol^{-1}}$ than P R3 in the singlet state. The P R3 follow-up reaction R4, shown in Fig. 4.5b, has similar barriers on both spin surfaces. The kinetics, in particular of the mixed singlet-triplet spin surface channels, must be studied in more detail, *e.g.* using the two-state spin-mixing model (TSSMM) method by Yang *et al.* [164], to determine the dominant one from the two R3-R5 mechanisms. The reaction products of the barrierless C–H bond scission R6 have the highest energies from the three competing initial unimolecular decomposition channels, shown in Fig. 4.5a. The chlorination site has only a small impact on the R6 reaction energies as the R6 reaction energies of the two chlorinated dibenzofurans shown in Fig. 4.5a vary by only up to 2 $\mathrm{kcal\,mol^{-1}}$ with the chlorination and radical sites. Altarawneh *et al.* [132] argue that R6 in Fig. 4.5a is the main unimolecular decomposition channel, as they are entropically more favorable than reaction R3. The P R6 Gibbs free energy that is shown for the 3-dibenzofuranyl radical with reference to dibenzofuran in Fig. 4.6 decreases faster with temperature than the Gibbs free energies of both the singlet and triplet R3 products P R3. At temperatures above 700 K, P R6 is the most stable from the P R3 and P R6 product states on the Gibbs free energy surface. Suggesting reaction R6 as the major decomposition channel is only seemingly a contradiction to the finding by Cieplik *et al.* that the conversion of dibenzofuran is independent on hydrogen pressure. In reaction equilibrium of reaction R6, for example, the dibenzofuranyl radical partial pressure is reciprocal to the partial pressure of the $\dot{H}$ radical and therefore dependent on the partial pressure of hydrogen. The experiments by Cieplik *et al.* were conducted at partial pressures of H_2 of 1.1 to 35 bar and dibenzofuran/H_2 mixing ratios of 10^{-4}. As a result the $\dot{H}$ radical partial pressures can be assumed to be low even under the studied hydrogenolytic conditions, as the dissociation of H_2 is slow [163]. Estimates of the $\dot{H}$ radical partial pressures in $H_2 \rightleftharpoons 2\,H$ reaction equilibrium, which are given in the appendix in section A.3.3, show that $\dot{H}/H_2$ mixing ratios are more than 2 orders of magnitude smaller than the dibenzofuran/H_2 mixing ratio. As the $\dot{H}$ radical partial pressures are small, the reaction R6-initiated reaction mechanism may be relevant also under the hydrogenolytic conditions studied by Cieplik *et al.* [163].

Using CTY simulations of (chlorinated) dibenzofuranyl radicals (and Cl_2), I have searched for unimolecular (chlorinated) dibenzofuranyl radical reactions in order to extent the comparison of the two unimolecular decomposition mechanisms beyond the discussion of their initial channels, reactions R3 and R6. Fig. 4.7 shows reaction barriers and energies of unimolecular (chlorinated) dibenzofuranyl radical pathways that I have observed in the CTY simulations. I have observed most of the P R6 follow-up reactions in Fig. 4.7 for more chlorinated dibenzofuranyl radical isomers than just the displayed ones. Fig. 4.7 shows only those reactions of the (chlorinated) dibenzo-

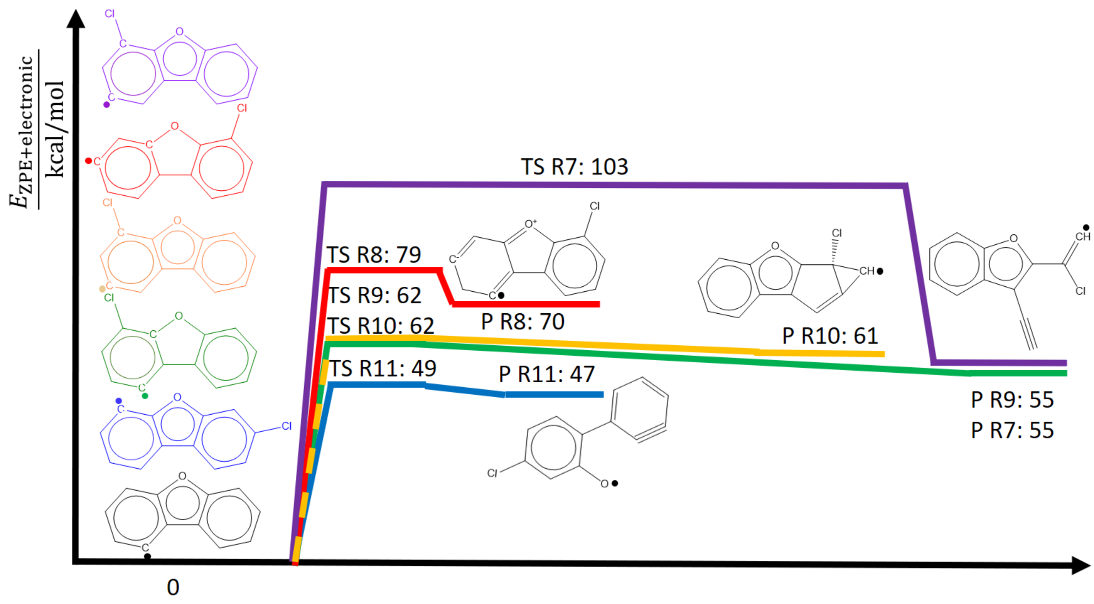


Figure 4.7.: ZPE-corrected electronic energy surfaces of chlorinated and unchlorinated dibenzofuranyl radical reactions. The (chlorinated) dibenzofuranyl radicals have the same colors as the reaction pathways that originate from them.

furanyl isomers with the lowest reaction barrier for simplicity. The α-scission R7 has the largest reaction barrier of more than 100 kcal mol^{-1}. α-scissions were observed for 4/36 (chlorinated) dibenzofuranyl radical isomers (36 includes unchlorinated radicals). The observed α-scission reaction barriers differ by up to 10 kcal mol^{-1} depending on the dibenzofuranyl isomer. The H-shift reaction R8 was observed for two dibenzofuranyl radical isomers, both with a reaction barrier of $\approx$ 80 kcal mol^{-1}. Both have in common that an H-atom shifted to a neighboring, already hydrogenated, C-atom. In the reaction R8 that is displayed in Fig. 4.7 an H-atom shifted from C-atom 1 to C-atom 2. As a result, the C-atom 2 becomes doubly hydrogenated. Other observed H-shifts that are not shown in the figure result in the change of the radical site. They have substantially larger reaction barriers of more than 100 kcal mol^{-1}. The β-scission R9 and the ring contraction R10 have similar reaction barriers of more than 60 kcal mol^{-1}. The ring contraction R10 was observed for three dibenzofuranyl isomers. The R10 reaction barriers vary by up to 30 kcal mol^{-1} with the isomer. β-scission R9 was observed for five isomers. Their reaction barriers vary by only up to

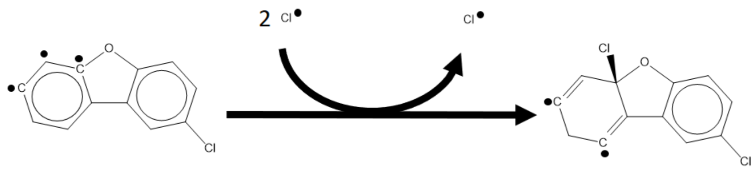


Figure 4.8.: Chlorination reaction observed with CTY. One of the reactants, the chlorinated dibenzofuranyl triradical, and TS are in a quadruplet state. The chlorine radicals are in doublet states. The dichlorinated dibenzofuranyl diradical is in a triplet state.

$6\,\mathrm{kcal\,mol^{-1}}$. Interestingly, β-scission R9 can lead to the same products as α-scission R7. Reaction R9 is not the only β-scission shown in Fig. 4.6. Reaction R11 also belongs to the β-scission reaction class and has the lowest reaction barrier of the observed dibenzofuranyl radical follow-up reactions with a barrier of $\approx 50\,\mathrm{kcal\,mol^{-1}}$. The difference between reactions R9 and R11 are the different bond types that are broken in the reaction process. While in reaction R9 a C−C bond is broken, it is a C−O bond that is broken in reaction R11. The dibenzofuranyl radical reaction with the lowest reaction barrier, β-scission R11, has a similar reaction barrier as the P R3 follow-up reaction R4 on both spin surfaces (c. f. Figs. 4.5b and 4.7). However, the R11 reverse reaction must be expected to be faster than the forward reaction, as the reaction barrier of the reverse reaction is low. The P R4 follow-up reaction R5, on the other hand, is more favorable than the R4 reverse reaction. For further comparison of the two competing unimolecular decomposition mechanisms, more reaction steps, especially those following the β-scission product P R9, should be included into the discussion. The differences of up to $30\,\mathrm{kcal\,mol^{-1}}$ in reaction barriers of (chlorinated) dibenzofuranyl radical reactions dependent on the chlorination site suggest that the chlorination site of the chlorinated dibenzofurans may be another factor that decides over the relevance of the unimolecular decomposition reactions.

I also observe reactions involving Cl_2 in the rMD simulations. During the QM reoptimizations, the TS in doublet states converged to complexes, in which the Cl_2 molecule is inactive. The H-shift reaction R8, for example, has been found including an inert Cl_2 as well. The reaction barrier of the H-shift reaction with inert Cl_2 does not differ significantly from the value reported above for reaction R8. A snapshot of the TS, optimized with B3LYP, is given in the appendix in section A.3.4. Another chlorination reaction was obtained on the quadruplet spin surface. A snapshot of the B3LYP-optimized quadruplet TS is again available in the appendix in section A.3.4. The automatic analysis of interatomic distances of the reactant and product van-der-Waals complexes with CTY suggests that a chlorinated dibenzofuranyl triradical reacts with two Cl radicals to a dichlorinated dibenzofuranyl diradical (see Fig. 4.8). Visual inspection of the occupied orbitals of the reactant van-der-Waals complex, however, shows that an electron-donor-acceptor complex is present, which transfers an unpaired electron from the dibenzofuranyl triradical Π-orbital system to Cl_2. That means, it is Cl_2 instead of the two Cl radicals suggested by CTY that reacts with the dibenzofuranyl triradical. The determination of reactants based on interatomic distances implemented using the state-of-the-art cheminformatics tool Open Babel [111] is in this case insufficient to determine the reactants correctly. While our new CTY concept allows for the construction of reactant and product van-der-Waals complexes via full optimization of the IRC endpoints, the determination of reactants and products requires in some cases, such as electron-donor-acceptor complexes, verification by the user.

Using CTY simulations of chlorinated dibenzofuranyl radicals and Cl_2, allowed me to report multiple unimolecular decomposition paths for CDFs and to add a new perspective to the discussion of the most favorable unimolecular dibenzofuran decomposition path [132, 163]. Including the new reactions into the discussion, the dominance of the unimolecular reaction mechanism initiated by C – H bond scission appears less clear than previously suggested as the fate of follow-up reaction products has to be taken into account as well. Consequently, I find that the inclusion of further reaction steps is necessary to clarify the picture. Furthermore, I find that the reaction barriers of the chlorinated dibenzofuranyl radical decomposition reactions vary by up to several tens of $\mathrm{kcal\,mol^{-1}}$ with the chlorination sites. The set of reactions needed to

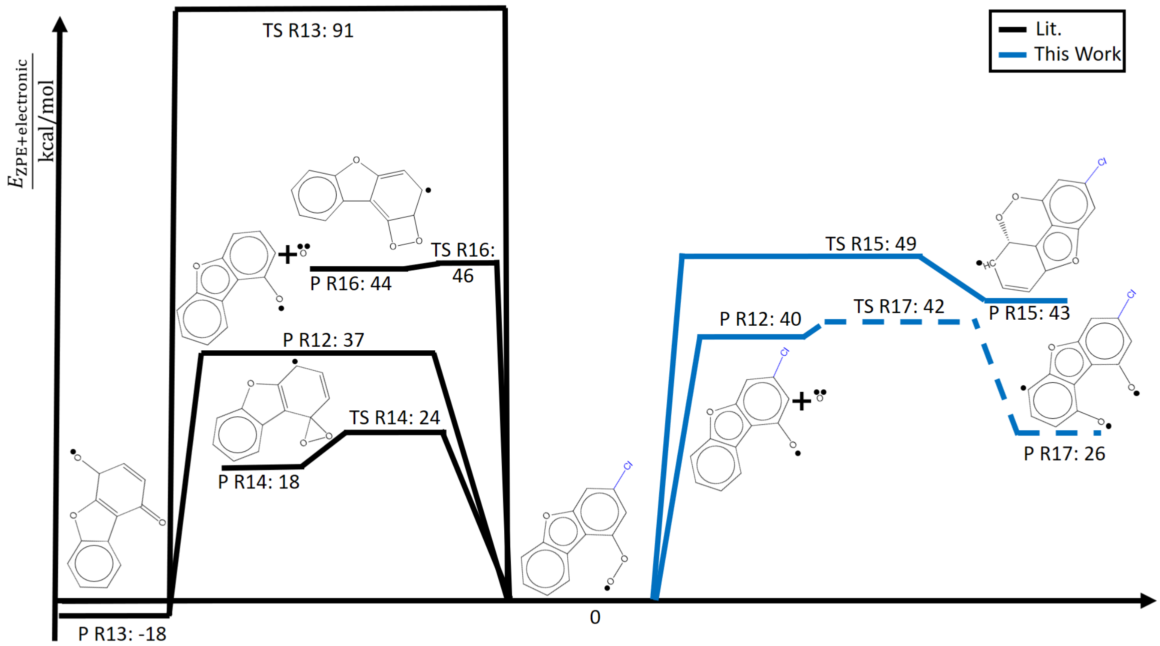


Figure 4.9.: ZPE-corrected electronic energy surfaces of 1-peroxy-dibenzofuranyl radical reaction pathways at the B3LYP level of theory. The black paths are taken from Altarawneh *et al.* [131]. Note that the pathways from this work (blue paths) are shown for the singly-chlorinated 3-chloro-1-peroxy-dibenzofuranyl radical. The dashed path shows a recombination channel of the 1-oxyl-dibenzofuranyl radical (P R12) with another $\dot{O}$ on the quadruplet spin surface.

describe the decomposition mechanism of the chlorinated dibenzofurans may therefore depend on the chlorination site as well. Relevant chlorination reactions were not observed.

CDF Oxidation

In the literature, low- and high-temperature oxidation pathways of unchlorinated dibenzofurans have been proposed from analogies to benzene and phenol oxidation [56, 131, 149]. Both low- and high-temperature mechanisms are initiated by an H-atom abstraction by O_2 or another radical [56, 149]. The H-atom abstraction is then followed by an O_2-addition to form a 1-peroxy-dibenzofuranyl radical. The isomer 1-peroxy-dibenzofuranyl radical is depicted in Fig. 4.9 as the reference structure of the PES and is the O_2-addition product of the most stable dibenzofuranyl radical.

The 1-peroxy-dibenzofuranyl radical is the point in the reaction network, at which the high- and low-temperature mechanisms start to differ [56]. At high temperatures, the 1-peroxy-dibenzofuranyl radical is suggested to mainly undergo the barrierless O–O bond scission R12 in Fig. 4.9 and form the 1-oxyl-dibenzofuranyl radical and an O atom. In the following reaction step, the bond scission R12 product 1-oxyl-dibenzofuranyl radical abstracts an H-atom from a second dibenzofuran and forms dibenzofuranol plus another dibenzofuranyl radical.

For the low-temperature mechanism, there is dissent about the major reaction channels of the 1-peroxy-dibenzofuranyl radical. Marquaire *et al.* [149] propose that the 1-peroxy-dibenzofuranyl radical Fig. 4.9 continues to form a dibenzofuran-quinone as the major intermediate [56, 131, 149]. Altarawneh *et al.* [131] suggest that the dibenzofuran-quinone is formed via a 4-hydroquinone-dibenzofuranyl radical, P R13 in Fig. 4.9. Altarawneh *et al.* [56, 131] also claim to have found an alternative, at least energetically more favorable route, *i. e.* R14 in Fig. 4.9. To conclude the discussion of the literature on dibenzofuran oxidation, I would like to emphasize that the 1-peroxy-dibenzofuranyl radicals, which I have studied with CTY, do not only play a central role for both low- and high-temperature oxidation mechanisms but also that the major reaction channel of the low-temperature mechanism is still up for debate.

With the help of CTY, I have found two new reactions that have not been discussed in literature so far. The first reaction R15 in Fig. 4.9 is a third alternative to the two cyclic peroxide formations R14 and R16 discussed by Altarawneh *et al.* [131]. The reaction barrier of the new reaction R15 is, with a difference of $3\,\mathrm{kcal\,mol^{-1}}$, not significantly higher than reaction R16 and is $25\,\mathrm{kcal\,mol^{-1}}$ higher than R14. R15 is therefore unlikely to be competitive to R14. It should be noted that I present energies for the two new paths R15 and R17 for the singly-chlorinated 3-chloro-1-peroxy radical, as I have not observed the reactions for all the singly-chlorinated isomers or for the unchlorinated 1-peroxy-dibenzofuranyl radical. rMD simulations stochastically sample reactions, which means that the degree of completeness of the discovered reaction network depends on the length and the number of replica. I have observed reaction R15 for four from the seven simulated chlorinated isomers and R17 for five chlorinated isomers. The reaction barrier differences and energy differences of the products between the isomers is less than 3 $\mathrm{kcal\,mol^{-1}}$ for reaction R15 and less than 2 $\mathrm{kcal\,mol^{-1}}$ for reaction R17. The differences are below the uncertainty of the DFT method. I therefore expect that the energies of the singly-chlorinated species are comparable to their unchlorinated counterparts. This expectation is strengthened by the fact that I observe the O−O bond scission R12 for both the unchlorinated 1-peroxy-dibenzofuranyl radical and the 3-chloro-1-peroxy-dibenzofuranyl radical. The energy difference between the two product states is 2 $\mathrm{kcal\,mol^{-1}}$.

Reaction R17 is an alternative route parallel to the barrierless recombination of one of the O−O bond scission products P R12, the 1-oxyl-dibenzofuranyl radical with another O atom, to the 4-hydroquinone-dibenzofuranyl radical P R13. The fact that CTY (plus CTY-TAD) finds this alternative route reaction R17 is remarkable for two reasons. First, neither the reactants nor the products have been present in the simulation box, although CTY-TAD searches for parallel reactions of the initially specified reactants only. Still, I regard this reaction R17 worth reporting, which supports the new CTY concept that accepts not only the QM-optimized reactions if they have been observed in the rMD simulations but accepts also reactions without a rMD counterpart. Second, both the TS and the product of reaction R17 have a quadruplet spin multiplicity. One of the P R12 products, the doublet 1-oxyl-dibenzofuranyl radical, may couple with another triplet O atom to both a product in the doublet state (P R13) or in the quadruplet state (P R17).

In conclusion, I discovered two new reactions in the low-temperature oxidation mechanism of 1-peroxy-dibenzofuranyl radicals. One of the discoveries was only possible with the new CTY reaction acception criteria and the multiplicity probing for both TS and stable species. The fact that I have not observed a lower reaction channel than the one proposed in the literature supports the idea that reaction R14 in Fig. 4.9 is the main unimolecular CDF-peroxy radical reaction.

The automated reaction space exploration and quantification tool CTY has allowed for the addition of many new reaction paths, even for already well-studied reaction systems. Due to the large number of reactions that are found and that have to be quantified, the user will likely opt for inexpensive PES, such as DFT methods for geometry optimizations and quantification of reaction energies, in order to keep the computational costs at bay. While DFT methods are known to be a good choice for geometry optimizations in many cases, they often have too large uncertainties to determine the reaction kinetics with low uncertainty.

4.3. Conclusion

I have demonstrated that the hybrid rMD-QM reaction space exploration tool CTY can be used to discover reaction paths of COCs and to quantify them automatically. I was able to find and quantify about 500 reactions using the CTY-TAD [94] acceleration technique in combination with the updated CTY rMD-QM coupling scheme. The new self-learning TS searching recovery method enhances the overall CTY TS search success rate by 10 percentage points to a total of 48%. While the whole CTY process took two weeks for the simulations to finish and only a few hours for me to set up the simulations, a manual approach would have probably taken multiple months if not years.

I have demonstrated ChemTraYzer's capability of finding new and potentially relevant reactions in three case studies of CDF precursors phenoxy radical and 1,3,5-trichlorobenzene, CDF pyrolysis and chlorination, and oxidation processes. In the CDF precursors study, I have not only observed known reaction steps [151] in the CDF formation mechanism from chlorobenzenes and phenoxy radicals, but also a possibly competitive intermolecular chlorine-transfer reaction that has not been described in the literature yet. In the CDF radical pyrolysis case study, I have found the following four unimolecular CDF radical decomposition channels, which are given in descending order of their reaction barriers: α-scissions of C−C bonds, H-shifts, ring contractions, and β-scissions of C−C bonds or C−O bonds. The new reactions add a new perspective to the discussion of the most favorable unimolecular CDF decomposition pathways that are either initiated by (i) C−O or by (ii) C−H bond scission [132, 163]. The comparison of the energetics of the two competing mechanisms including my new reactions shows that the dominance of the mechanism initiated by path (ii) is less certain than suggested in the literature. Further reaction steps have to be included into the discussion to clarify the picture. My findings also suggest that the chlorination site may be a factor that decides over the relevance of the unimolecular decomposition reactions. In the oxidation case study, I have described a third cyclic peroxide formation pathway for the 1-peroxy-dibenzofuranyl radical with comparable reaction barrier and reaction energy as the ones already known in the literature [131]. The high degree of automatization of the entire CTY reaction discovery and quantification process made it worthwhile to search for and find some of the rare possibly relevant reactions on higher spin surfaces, from which one was presented in the oxidation case study. The high degree of automatization also allowed me to study the impact of the chlorination sites in the case studies of CDF radical pyrolysis and CDF oxidation simply by repeating the CTY simulations for all possible chlorination sites. The reaction barriers of the majority of the reactions differ by less than $10\,\mathrm{kcal\,mol^{-1}}$ with changing chlorination sites. The impact of the chlorination site is largest for ring contraction reactions of the CDF radicals. The three observed representatives of the ring contractions differ by as much as $30\,\mathrm{kcal\,mol^{-1}}$, depending on their chlorination site. All in all, I have demonstrated that hybrid rMD-QM reaction space exploration tools like CTY can make a valuable contribution to the field of reaction model development, especially for difficult-to-study-experimentally compounds, with their capability to study specific portions in reaction space in detail with little human intervention.

5. Conclusions and Outlook

In this work, I have addressed two major issues of the CTY rMD-QM coupling scheme to extent the range of feasible investigation systems and to allow for more efficient reaction space exploration. I have introduced two acceleration techniques for the rMD simulations, pAD and CTY-TAD, to be able to extent the applicability of rMD simulations to gaseous reaction processes that occur on longer time scales than the nanosecond. I have also improved the performance of the automatic TS searches, which were and continue to be a bottleneck of rMD-QM coupling schemes such as CTY by introducing a recovery method for failed TS searches.

As opposed to many prominent acceleration techniques from the literature, such as the hyperdynamics, metadynamics, or the CVHD methods [44–47], my acceleration techniques require only a minimum of *a priori* knowledge about the reaction system of interest. As my acceleration methods are developed for the automatic reaction space exploration of unknown reaction systems, their ability to work with minimal a priori knowledge is an important feature. In order to keep the number of user parameters small, my acceleration methods simply use elevated simulation temperatures and pressures to boost the reactions. The use of elevated temperatures and pressures comes with the challenge to avoid the kinetics bias issue, *i. e.* the challenge to boost the reactions without changing the reaction chemistry. The reaction chemistry of low-temperature ignition, for example, is often sensitive towards temperature and pressure as different reaction mechanisms become relevant when the two parameters change. The sensitivity of their reaction chemistry to changes in temperature and pressure made low-temperature ignition systems ideal case studies for the testing and demonstration of the new acceleration methods. Furthermore, low-temperature ignition processes are challenging case study objects as they mostly take place in the order of microseconds at technically relevant conditions and are therefore beyond the feasibility of rMD simulations without the use of acceleration techniques. Using the pAD acceleration technique, I have shown that it is sufficient to know the kinetics of only three commonly known core low-temperature ignition reactions *a priori* to find suitable simulation temperatures and pressures for the simulation of the core low-temperature ignition reaction mechanism. The CTY-TAD method is a progression of pAD and is a less system-specific solution for the kinetics bias issue. CTY-TAD integrates the CTY reaction detection scheme into the rMD simulation to place virtual walls that block already observed reactions. This way, the simulation time is spent more efficiently on the exploration of new reactions. CTY-TAD requires even less *a priori* knowledge about the reaction system than the pAD method. The design choice of an upper threshold value for reaction barriers is all that is needed to estimate suitable simulation parameters. CTY-TAD allows for higher simulation temperatures than pAD and therefore for larger boost factors. I have shown that the CTY-TAD method achieves boost factors up to 10^8 for the example of 1,3-dioxolane low-temperature ignition, which is comparable to the boost factors obtained with known acceleration methods from the literature.

The high trajectory resolution that is inherent of the CTY-TAD method allowed for an analysis of the CTY reaction detection scheme. I found that in only 55% of the observed reactions, the detected change in connectivities is consistent with the optimized reactant and product geometries. In other words, some detected reactions do not correspond to transitions between basins on the underlying PES. As a consequence, there is a risk that CTY-TAD reports qualitatively wrong reactions. The risk that the implemented basin confinement method may block new reactions before they have been observed in the rMD simulation is another consequence as the method relies on the reaction detection scheme to distinguish new from the to-be-blocked known reactions. While simple compensation strategies are available for both issues, improvement of the reaction detection scheme is necessary to further reduce the risk of blocking new reactions. The alternative reaction detection scheme described by Hutchings *et al.* [154] may be an ansatz for improving the consistency of the detected reactions with transitions between PES basins. Hutchings *et al.* suggest to analyze first derivatives of bond orders [154] instead of comparing connectivities to user-defined thresholds as applied in CTY. They demonstrate that their approach accurately locates REs in rMD trajectories. However, their reaction detection scheme is still to be demonstrated for trajectories containing more than one molecule [154], which is essential for practical applications. As another alternative, information about the local curvature of the PES at the time of a RE could be integrated into the reaction detection procedure by analyzing frequencies. Reactions with a reaction barrier could only then be be returned by the method if not only the connectivities have changed but also if only one dominant imaginary frequencies is present. That way, the returned geometry from the trajectory snapshot would make for an excellent TS search start geometry for a TS search on the force field level of theory. The molecular identifiers of the reactants and products could then be derived directly from the optimized geometries that are connected by an IRC scan. This way, the detected reactions would be inherently consistent with transitions between PES basins. I would like to conclude the discussion of the acceleration methods that there is room for improvement, despite the success of the CTY-TAD method in extending feasible simulation times and in reproducing the relevant reaction space of its low-temperature ignition chemistry case study. The key for improvement is the enhancement of the CTY reaction detection scheme. Until a better solution for the CTY reaction detection scheme has been found, the simple reaction identification test described in chapter three can be used as a post-processing step to double-check the observed reactions.

I have shown in chapter four that the CTY rMD-QM coupling scheme has a TS search success rate of 38% for the dibenzofurane example systems without the new recovery method. By introducing a self-learning recovery method for failed TS searches, I was able to increase the CTY success rate by 10 percentage points. The new TS search procedure has a success rate well within the success rate range of 23%-66% that was reported for other automated TS search procedures in the literature [36]. The resulting success rate of 48% is still unsatisfactory, though, considering the large amounts of reactions that I have found in most of the case studies. Further improvement of the TS search success rate could be achieved with multiple approaches. The integration of the deep-learning TS search by Pattanaik *et al.* [158] for reactions within its application range into CTY appears promising due to the high reported success rates of more than 71%. The application range of the method includes currently only isomerization reactions of C, H, O, N systems, though [158]. The idea of integrating other automated TS searches within their optimal range of applications could be extended to other existing TS search methods from the literature, such as the GSM, the Freezing String Method (FSM), or the Single-Component Artificial Force Induced Reaction Method (SC-AFIR) [22, 24, 157, 165–167]. However, the definition of their optimal range of applications requires an extensive comparison of the methods on a wide range of reaction systems. A first comparison of the GSM, the FSM, and the SC-AFIR for unimolecular reactions of γ-ketohydroperoxide has already been done by Grambow *et al.* [36]. The second ansatz for improvement of the CTY TS searches is the improvement of the CTY reaction detection procedure discussed above. Both the original CTY TS search and the new recovery method use pre-optimizations with frozen active atoms to reduce the number of imaginary frequencies, in order to generate a better starting geometry for the eigenvector-following TS search algorithm. The CTY reaction detection method and the TS searches are directly coupled, as the active atoms are determined by the reaction detection method. In summary, the TS search success rate of 48% may still need to be improved, but CTY was already cabable of quantifying about 500 reactions in the chlorinated dibenzofurane case studies with only few hours of work by the user to set up the simulations and monitor their progress. It is therefore not far-fetched to say that CTY already has the potential to save months of work for the user, considering the large amount of automatically quantified reactions.

In all of the case studies that I discussed in the previous chapters, I have found reactions that were either not included in the reference reaction mechanisms from the literature or have not been discussed in the literature at all. Many of these new reactions had large reaction rate constants or low reaction barriers so that I deemed them potentially competitive to the known competing reactions and therefore potentially relevant. The fact that I observed new and potentially relevant reactions shows that CTY is already of practical use for reaction model developers, for example if relevant reactions may be missing in the reaction model. The dream of the full automatization of the exploration of all relevant portions of the reaction network has not yet been reached, though. Ironically, the CTY-TAD method is currently the part of the CTY concept that is prohibitive of fulfilling the dream of full automatization, although the method makes the exploration process generally more efficient. CTY-TAD is designed to explore the reaction space for a user-specified set of reactants only. A mechanism, as introduced in the TAD method [95] that progresses the reaction space exploration to the reaction space of the most relevant products requires reactive force fields with low reaction kinetics and equilibrium uncertainties. Force fields with low uncertainties for kinetic and equilibrium properties are necessary for the TAD method to guarantee that the discovered minimal reaction network actually contains the most relevant reactions at the target temperatures and pressures. CTY-TAD leaves the selection of species in the reaction network for further investigation to the user to keep the force field requirements at bay. Developing accurate force fields that are also efficient enough for the simulation of larger time scales is itself a challenging field of research, so a different approach is needed that selects the portions of reactive space for exploration to further automate the CTY methodology. Nicolaou *et al.* [168] make the connection between missing reactions in a reaction network or reaction model and apparent errors in the reaction rate constants. In their natural gas combustion case study they have to adjust reaction rate constants included in a reduced reaction model by up to a factor of ten to fit the properties predicted by the complete model. These variations are a result of missing reactions in the reduced reaction model [168]. The exploitation of their connection between apparent errors in reaction rate constants and missing reactions in the reaction model as well as the utilization of network completion methods from other fields of research, such as those referenced in [168], may provide an ansatz to develop a methodology that selects species for further investigation with CTY. To conclude the discussion and this work, I would like to emphasize that I have demonstrated that CTY provides an automatization ansatz that is already of use for reaction model developers in its current form. CTY can help reaction modelers find missing relevant reactions in their reaction model. Until the methodology is capable of fully automatic reaction model generation, issues on both sides of the rMD-QM coupling scheme, some of which I have discussed in this work, remain to be solved.

A. Appendix

A.1. Supplementary Information to Chapter 2

A.1.1. Reaction List

Table A.1.: The list of reactions discussed in Chapter 2 is given in table A.1. The table contains the reaction SMILES as well as the corresponding reaction rate constants from the rMD simulations, k, upper and lower bounds of a confidence interval with confidence level $\chi = 0.9$, k_{low}, k_{up} (c. f. [54]), and the number of reaction events N_{REs}. The reaction SMILES are colon-seperated lists of the reactant and product SMILES. If multiple reactants or products are present, they are seperated by comma. The reaction rate constants are computed at 1200 K using CTY [34]. Reaction rate constants of uni-, bi-, and of termolecular reactions are given in s^{-1}, $\mathrm{cm}^3\,\mathrm{mol}^{-1}\,\mathrm{s}^{-1}$, and in $\mathrm{cm}^6\,\mathrm{mol}^{-2}\,\mathrm{s}^{-1}$, respectively.

R<ID>	Formulas	SMILES	k	k_{low}	k_{up}	N_{REs}
R0	C3H6O2 -> C3H6O2	C1COOC1:[O]CCC[O]	2.78e+08	1.18e+07	1.33e+09	1
R0*	C3H6O2 -> C3H6O2	[O]CCC[O]:C1COOC1	0.00e+00	0.00e+00	2.18e+10	0
R1	CH2O + C4H9O + O2 -> C5H11O4	C=O,C[CH][C@H](O)C,O=O:[O]COO[C@H]([C@H](O)C)C	2.27e+14	9.63e+12	1.08e+15	1
R1*	C5H11O4 -> CH2O + C4H9O + O2	[O]COO[C@H]([C@H](O)C)C:C=O,C[CH][C@H](O)C,O=O	0.00e+00	0.00e+00	4.99e+12	0
R2	CH2O + HO2 -> CH3O3	C=O,O[O]:OOC[O]	1.78e+12	4.23e+11	4.72e+12	3
R2*	CH3O3 -> CH2O + HO2	OOC[O]:C=O,O[O]	2.63e+11	3.99e+10	8.42e+11	2
R3	CH2O + HO -> H2O + CHO	C=O,[OH]:O,[CH]=O	2.51e+12	1.06e+11	1.20e+13	1
R3*	H2O + CHO -> CH2O + HO	O,[CH]=O:C=O,[OH]	0.00e+00	0.00e+00	3.12e+14	0
R4	CH2O + HO -> CH3O2	C=O,[OH]:OC[O]	2.26e+13	1.08e+13	4.09e+13	9
R4*	CH3O2 -> CH2O + HO	OC[O]:C=O,[OH]	7.21e+08	3.06e+08	1.40e+09	7
R5	CH2O + O -> CH2O2	C=O,[O]:[O]C[O]	3.76e+15	1.59e+14	1.79e+16	1
R5*	CH2O2 -> CH2O + O	[O]C[O]:C=O,[O]	0.00e+00	0.00e+00	7.49e+12	0
R6	C4H8 + CH3 -> C5H11	CC(=C)C,[CH3]:CC([CH2])(C)C	2.10e+14	3.19e+13	6.72e+14	2
R6*	C5H11 -> C4H8 + CH3	CC([CH2])(C)C:CC(=C)C,[CH3]	4.65e+09	1.97e+08	2.22e+10	1
R7	C4H8O + O2 + H -> C4H9O3	CC(=C)CO,O=O,[H]:[O]OC[C@H](CO)C	1.04e+19	1.58e+18	3.33e+19	2
R7*	C4H9O3 -> C4H8O + O2 + H	[O]OC[C@H](CO)C:CC(=C)CO,O=O,[H]	0.00e+00	0.00e+00	2.35e+08	0
R8	C3H5O + O2 -> C3H5O3	CC(=C)[O],O=O:CC(=O)CO[O]	4.41e+12	1.05e+12	1.17e+13	3
R8*	C3H5O3 -> C3H5O + O2	CC(=O)CO[O]:CC(=C)[O],O=O	7.50e+10	1.78e+10	1.99e+11	3
R9	C3H5O -> C3H5O	CC(=C)[O]:CO[C]=C	1.36e+10	5.76e+08	6.47e+10	1
R9*	C3H5O -> C3H5O	CO[C]=C:CC(=C)[O]	0.00e+00	0.00e+00	2.34e+11	0
R10	C5H9O + O2 -> C5H9O3	CC(=O)C([CH2])C,O=O:C[C@H]1COO[C@]1(C)[O]	6.64e+11	2.81e+10	3.16e+12	1
R10*	C5H9O3 -> C5H9O + O2	C[C@H]1COO[C@]1(C)[O]:CC(=O)C([CH2])C,O=O	0.00e+00	0.00e+00	6.66e+11	0
R11	C5H9O -> C5H9O	CC(=O)C([CH2])C:C[CH]CC(=O)C	1.26e+10	1.91e+09	4.02e+10	2
R11*	C5H9O -> C5H9O	C[CH]CC(=O)C:CC(=O)C([CH2])C	0.00e+00	0.00e+00	5.57e+09	0
R12	C5H9O -> C5H9O	CC(=O)C([CH2])C:C[C]1OC[C@H]1C	6.28e+09	2.66e+08	2.99e+10	1
R12*	C5H9O -> C5H9O	C[C]1OC[C@H]1C:CC(=O)C([CH2])C	4.76e+11	2.02e+10	2.27e+12	1
R13	C4H6O + CH3 -> C5H9O	CC(=O)C=C,[CH3]:C[C@H]1C[C@]1(C)[O]	9.41e+16	3.99e+15	4.48e+17	1
R13*	C5H9O -> C4H6O + CH3	C[C@H]1C[C@]1(C)[O]:CC(=O)C=C,[CH3]	0.00e+00	0.00e+00	9.11e+10	0
R14	C4H8O2 + HO2 -> C4H7O2 + H2O2	CC(=O)CCO,O[O]:CC(=O)CC[O],OO	9.18e+13	3.89e+12	4.37e+14	1
R14*	C4H7O2 + H2O2 -> C4H8O2 + HO2	CC(=O)CC[O],OO:CC(=O)CCO,O[O]	0.00e+00	0.00e+00	2.20e+15	0

R15	C4H7O2 -> CH2O + C3H5O	CC(=O)CC[O]:C=O,CC(=C)[O]	7.81e+10	3.31e+09	3.72e+11	1
R15*	CH2O + C3H5O -> C4H7O2	C=O,CC(=C)[O]:CC(=O)CC[O]	0.00e+00	0.00e+00	3.83e+14	0
R16	C2H4O2 + C3H7O2 -> C2H3O2 + C3H8O2	CC(=O)O,CCCO[O]:CC(=O)[O],CCCOO	7.97e+12	3.37e+11	3.80e+13	1
R16*	C2H3O2 + C3H8O2 -> C2H4O2 + C3H7O2	CC(=O)[O],CCCOO:CC(=O)O,CCCO[O]	0.00e+00	0.00e+00	2.12e+14	0
R17	C2H3O2 -> CO2 + CH3	CC(=O)[O]:O=C=O,[CH3]	7.53e+09	3.19e+08	3.59e+10	1
R17*	CO2 + CH3 -> C2H3O2	O=C=O,[CH3]:CC(=O)[O]	0.00e+00	0.00e+00	6.60e+14	0
R18	C5H11 + O2 -> C5H11O2	CC(C)([CH2])C,O=O:[O]OCC(C)(C)C	6.08e+12	2.75e+12	1.14e+13	8
R18*	C5H11O2 -> C5H11 + O2	[O]OCC(C)(C)C:CC(C)([CH2])C,O=O	2.44e+09	1.03e+09	4.74e+09	7
R19	C5H11 -> C4H8 + CH3	CC(C)([CH2])C:CC(=C)C,[CH3]	1.47e+10	2.23e+09	4.70e+10	2
R19*	C4H8 + CH3 -> C5H11	CC(=C)C,[CH3]:CC(C)([CH2])C	2.10e+14	3.19e+13	6.72e+14	2
R20	C3H7O + O2 -> C3H7O3	CC(O)[CH2],O=O:[O]OC[C@H](O)C	1.21e+13	1.05e+13	1.39e+13	202
R20*	C3H7O3 -> C3H7O + O2	[O]OC[C@H](O)C:CC(O)[CH2],O=O	1.64e+10	1.42e+10	1.88e+10	198
R21	C3H7O -> C3H7O	CC(O)[CH2]:C[CH]CO	5.71e+08	2.42e+07	2.72e+09	1
R21*	C3H7O -> C3H7O	C[CH]CO:CC(O)[CH2]	0.00e+00	0.00e+00	1.38e+09	0
R22	C2H5O2 -> C2H5O2	CC(O)[O]:CO[CH]O	5.36e+08	2.27e+07	2.55e+09	1
R22*	C2H5O2 -> C2H5O2	CO[CH]O:CC(O)[O]	0.00e+00	0.00e+00	2.67e+11	0
R23	C2H5O2 -> CH2O2 + CH3	CC(O)[O]:OC=O,[CH3]	1.61e+09	3.82e+08	4.26e+09	3
R23*	CH2O2 + CH3 -> C2H5O2	OC=O,[CH3]:CC(O)[O]	0.00e+00	0.00e+00	3.79e+14	0
R24	C5H11 + O2 -> C5H11O2	CC([CH2])(C)C,O=O:[O]OCC(C)(C)C	5.35e+12	2.76e+12	9.20e+12	11
R24*	C5H11O2 -> C5H11 + O2	[O]OCC(C)(C)C:CC([CH2])(C)C,O=O	3.48e+09	1.73e+09	6.13e+09	10
R25	C2H5O2 -> CH2O2 + CH3	CC([O])O:OC=O,[CH3]	5.43e+10	2.30e+09	2.59e+11	1
R25*	CH2O2 + CH3 -> C2H5O2	OC=O,[CH3]:CC([O])O	0.00e+00	0.00e+00	3.79e+14	0
R26	C4H8 + O2 -> C4H8O2	CC1CC1,O=O:CC1COOC1	1.39e+10	5.88e+08	6.62e+10	1
R26*	C4H8O2 -> C4H8 + O2	CC1COOC1:CC1CC1,O=O	0.00e+00	0.00e+00	2.42e+09	0
R27	C4H8 + CH3 -> C5H11	CC1CC1,[CH3]:CCC(C)[CH2]	3.14e+16	1.33e+15	1.49e+17	1
R27*	C5H11 -> C4H8 + CH3	CCC(C)[CH2]:CC1CC1,[CH3]	0.00e+00	0.00e+00	9.08e+07	0
R28	C4H8 + CH3 -> C5H11	CC1CC1,[CH3]:C[CH]CCC	3.14e+16	1.33e+15	1.49e+17	1
R28*	C5H11 -> C4H8 + CH3	C[CH]CCC:CC1CC1,[CH3]	3.01e+06	1.27e+05	1.43e+07	1
R29	C3H6 + C2H5 + O2 -> C5H11O2	CC=C,C[CH2],O=O:C[C@H](O[O])CCC	5.34e+13	2.26e+12	2.55e+14	1
R29*	C5H11O2 -> C3H6 + C2H5 + O2	C[C@H](O[O])CCC:CC=C,C[CH2],O=O	0.00e+00	0.00e+00	9.18e+06	0
R30	C3H6 + C2H5 -> C4H8 + CH3	CC=C,C[CH2]:CC1CC1,[CH3]	5.07e+11	2.15e+10	2.42e+12	1
R30*	C4H8 + CH3 -> C3H6 + C2H5	CC1CC1,[CH3]:CC=C,C[CH2]	0.00e+00	0.00e+00	9.40e+16	0
R31	C3H6 + C2H5 -> C5H11	CC=C,C[CH2]:CCC([CH2])C	7.60e+12	4.36e+12	1.22e+13	15
R31*	C5H11 -> C3H6 + C2H5	CCC([CH2])C:CC=C,C[CH2]	3.97e+08	2.28e+08	6.35e+08	15
R32	C3H6 + C2H5 -> C4H8 + CH3	CC=C,C[CH2]:CCC=C,[CH3]	6.59e+12	3.61e+12	1.09e+13	13
R32*	C4H8 + CH3 -> C3H6 + C2H5	CCC=C,[CH3]:CC=C,C[CH2]	1.92e+14	9.13e+13	3.47e+14	9
R33	C3H6 + C2H3O -> C4H6O + CH3	CC=C,C[C]=O:CC(=O)C=C,[CH3]	8.55e+15	3.62e+14	4.08e+16	1
R33*	C4H6O + CH3 -> C3H6 + C2H3O	CC(=O)C=C,[CH3]:CC=C,C[C]=O	0.00e+00	0.00e+00	2.82e+17	0
R34	C3H6 + O2 + CH3O -> C4H9O3	CC=C,O=O,[CH2]O:[O]OC[C@H](CO)C	5.52e+16	2.34e+15	2.63e+17	1
R34*	C4H9O3 -> C3H6 + O2 + CH3O	[O]OC[C@H](CO)C:CC=C,O=O,[CH2]O	0.00e+00	0.00e+00	2.35e+08	0

R35	C3H6 + HO2 -> C3H7O2	CC=C,O[O]:OOC([CH2])C	2.07e+12	3.14e+11	6.62e+12	2
R35*	C3H7O2 -> C3H6 + HO2	OOC([CH2])C:CC=C,O[O]	1.23e+11	1.87e+10	3.95e+11	2
R36	C3H6 + CH3O -> C3H6O + CH3	CC=C,[CH2]O:OCC=C,[CH3]	1.01e+15	1.54e+14	3.24e+15	2
R36*	C3H6O + CH3 -> C3H6 + CH3O	OCC=C,[CH3]:CC=C,[CH2]O	0.00e+00	0.00e+00	7.05e+16	0
R37	C3H6 + HO -> C3H7O	CC=C,[OH]:C[CH]CO	6.75e+13	2.86e+12	3.21e+14	1
R37*	C3H7O -> C3H6 + HO	C[CH]CO:CC=C,[OH]	0.00e+00	0.00e+00	1.38e+09	0
R38	C2H4O + HO2 -> C2H5O3	CC=O,O[O]:[O]O[C@H](O)C	3.64e+12	8.65e+11	9.65e+12	3
R38*	C2H5O3 -> C2H4O + HO2	[O]O[C@H](O)C:CC=O,O[O]	5.26e+10	7.99e+09	1.68e+11	2
R39	C2H4O + HO -> C2H5O2	CC=O,[OH]:CC(O)[O]	5.24e+12	1.25e+12	1.39e+13	3
R39*	C2H5O2 -> C2H4O + HO	CC(O)[O]:CC=O,[OH]	0.00e+00	0.00e+00	1.60e+09	0
R40	C2H4O + HO -> C2H5O2	CC=O,[OH]:CC([O])O	3.50e+12	5.30e+11	1.12e+13	2
R40*	C2H5O2 -> C2H4O + HO	CC([O])O:CC=O,[OH]	5.43e+10	2.30e+09	2.59e+11	1
R41	C2H4O + HO -> CH2O2 + CH3	CC=O,[OH]:OC=O,[CH3]	1.75e+12	7.40e+10	8.33e+12	1
R41*	CH2O2 + CH3 -> C2H4O + HO	OC=O,[CH3]:CC=O,[OH]	0.00e+00	0.00e+00	3.79e+14	0
R42	C2H4O + O -> C2H4O2	CC=O,[O]:COC=O	8.33e+14	3.53e+13	3.97e+15	1
R42*	C2H4O2 -> C2H4O + O	COC=O:CC=O,[O]	0.00e+00	0.00e+00	1.75e+11	0
R43	C5H10 + O2 + H -> C5H11O2	CCC(=C)C,O=O,[H]:C[C@H](CC)CO[O]	6.26e+18	2.20e+18	1.36e+19	5
R43*	C5H11O2 -> C5H10 + O2 + H	C[C@H](CC)CO[O]:CCC(=C)C,O=O,[H]	7.52e+06	1.79e+06	1.99e+07	3
R44	C5H11 + O2 -> C4H8O2 + CH3	CCC(C)[CH2],O=O:CC[CH]CO[O],[CH3]	9.74e+09	2.31e+09	2.58e+10	3
R44*	C4H8O2 + CH3 -> C5H11 + O2	CC[CH]CO[O],[CH3]:CCC(C)[CH2],O=O	1.34e+16	5.69e+14	6.41e+16	1
R45	C5H11 + O2 -> C5H11O2	CCC(C)[CH2],O=O:C[C@H](CC)CO[O]	1.25e+13	1.22e+13	1.29e+13	3866
R45*	C5H11O2 -> C5H11 + O2	C[C@H](CC)CO[O]:CCC(C)[CH2],O=O	9.64e+09	9.34e+09	9.95e+09	3844
R46	C5H11 + O2 -> C5H11O2	CCC(C)[CH2],O=O:C[C@H]([CH]C)COO	3.25e+09	1.37e+08	1.55e+10	1
R46*	C5H11O2 -> C5H11 + O2	C[C@H]([CH]C)COO:CCC(C)[CH2],O=O	0.00e+00	0.00e+00	2.02e+10	0
R47	C5H11 -> C3H6 + C2H5	CCC(C)[CH2]:CC=C,C[CH2]	2.43e+08	1.10e+08	4.54e+08	8
R47*	C3H6 + C2H5 -> C5H11	CC=C,C[CH2]:CCC(C)[CH2]	3.55e+12	1.51e+12	6.91e+12	7
R48	C5H11 -> C5H10 + H	CCC(C)[CH2]:CCC(=C)C,[H]	9.10e+07	2.16e+07	2.41e+08	3
R48*	C5H10 + H -> C5H11	CCC(=C)C,[H]:CCC(C)[CH2]	1.18e+16	4.98e+14	5.60e+16	1
R49	C5H11 -> C5H11	CCC(C)[CH2]:CC[CH]CC	3.64e+08	1.94e+08	6.12e+08	12
R49*	C5H11 -> C5H11	CC[CH]CC:CCC(C)[CH2]	2.95e+07	7.00e+06	7.80e+07	3
R50	C5H11 -> C5H11	CCC(C)[CH2]:CC[C](C)C	3.03e+07	1.28e+06	1.44e+08	1
R50*	C5H11 -> C5H11	CC[C](C)C:CCC(C)[CH2]	0.00e+00	0.00e+00	1.05e+08	0
R51	C5H10O -> C2H4 + C3H6O	CCC(C[CH2])[O]:C=C,CCC=O	5.00e+12	2.12e+11	2.38e+13	1
R51*	C2H4 + C3H6O -> C5H10O	C=C,CCC=O:CCC(C[CH2])[O]	0.00e+00	0.00e+00	3.20e+14	0
R52	C5H11O2 -> CH2O + C4H9O	CCC(C[O])CO:C=O,CC[CH]CO	1.00e+13	4.24e+11	4.77e+13	1
R52*	CH2O + C4H9O -> C5H11O2	C=O,CC[CH]CO:CCC(C[O])CO	0.00e+00	0.00e+00	1.66e+13	0
R53	C3H7O2 -> C2H5 + CH2O2	CCC(O)[O]:C[CH2],OC=O	4.17e+11	1.77e+10	1.99e+12	1
R53*	C2H5 + CH2O2 -> C3H7O2	C[CH2],OC=O:CCC(O)[O]	0.00e+00	0.00e+00	1.69e+13	0
R54	C5H11O2 -> C5H10O + HO	CCC(OO)C[CH2]:CCC(C[CH2])[O],[OH]	2.59e+10	1.10e+09	1.23e+11	1
R54*	C5H10O + HO -> C5H11O2	CCC(C[CH2])[O],[OH]:CCC(OO)C[CH2]	0.00e+00	0.00e+00	1.41e+17	0

R55	C5H11O2 -> C5H11O2	CCC(OO)C[CH2]:OCC[C@H](CC)[O]	5.18e+10	7.86e+09	1.66e+11	2
R55*	C5H11O2 -> C5H11O2	OCC[C@H](CC)[O]:CCC(OO)C[CH2]	0.00e+00	0.00e+00	2.23e+10	0
R56	C5H11O2 -> C5H11O2	CCC(O[O])CC:CCC(OO)C[CH2]	4.25e+07	1.80e+06	2.02e+08	1
R56*	C5H11O2 -> C5H11O2	CCC(OO)C[CH2]:CCC(O[O])CC	0.00e+00	0.00e+00	7.76e+10	0
R57	C5H11 + O2 -> C4H8O2 + CH3	CCC([CH2])C,O=O:CC[CH]CO[O],[CH3]	5.66e+09	8.59e+08	1.81e+10	2
R57*	C4H8O2 + CH3 -> C5H11 + O2	CC[CH]CO[O],[CH3]:CCC([CH2])C,O=O	1.34e+16	5.69e+14	6.41e+16	1
R58	C5H11 + O2 -> C5H11O2	CCC([CH2])C,O=O:C[C@H](CC)CO[O]	1.25e+13	1.21e+13	1.29e+13	4419
R58*	C5H11O2 -> C5H11 + O2	C[C@H](CC)CO[O]:CCC([CH2])C,O=O	1.10e+10	1.07e+10	1.13e+10	4390
R59	C5H11 + O2 -> C5H11O2	CCC([CH2])C,O=O:C[C@H]([CH]C)COO	2.83e+09	1.20e+08	1.35e+10	1
R59*	C5H11O2 -> C5H11 + O2	C[C@H]([CH]C)COO:CCC([CH2])C,O=O	0.00e+00	0.00e+00	2.02e+10	0
R60	C5H11 + O2 -> C5H10O2 + H	CCC([CH2])C,O=O:C[C](CC)CO[O],[H]	5.66e+09	8.59e+08	1.81e+10	2
R60*	C5H10O2 + H -> C5H11 + O2	C[C](CC)CO[O],[H]:CCC([CH2])C,O=O	0.00e+00	0.00e+00	1.41e+17	0
R61	C5H11 -> C4H8 + CH3	CCC([CH2])C:CC1CC1,[CH3]	2.65e+07	1.12e+06	1.26e+08	1
R61*	C4H8 + CH3 -> C5H11	CC1CC1,[CH3]:CCC([CH2])C	3.14e+16	1.33e+15	1.49e+17	1
R62	C5H11 -> C5H10 + H	CCC([CH2])C:CCC(=C)C,[H]	5.29e+07	8.03e+06	1.69e+08	2
R62*	C5H10 + H -> C5H11	CCC(=C)C,[H]:CCC([CH2])C	2.35e+16	3.57e+15	7.53e+16	2
R63	C5H11 -> C4H8 + CH3	CCC([CH2])C:CCC=C,[CH3]	2.91e+08	1.50e+08	5.00e+08	11
R63*	C4H8 + CH3 -> C5H11	CCC=C,[CH3]:CCC([CH2])C	2.13e+14	1.06e+14	3.75e+14	10
R64	C5H11 -> C5H11	CCC([CH2])C:CC[CH]CC	3.44e+08	1.88e+08	5.68e+08	13
R64*	C5H11 -> C5H11	CC[CH]CC:CCC([CH2])C	6.87e+07	2.92e+07	1.34e+08	7
R65	C5H11 -> C5H11	CCC([CH2])C:CC[C](C)C	7.94e+07	1.89e+07	2.10e+08	3
R65*	C5H11 -> C5H11	CC[C](C)C:CCC([CH2])C	0.00e+00	0.00e+00	1.05e+08	0
R66	C3H7O2 -> C2H5 + CH2O2	CCC([O])O:C[CH2],OC=O	1.01e+10	4.27e+08	4.80e+10	1
R66*	C2H5 + CH2O2 -> C3H7O2	C[CH2],OC=O:CCC([O])O	0.00e+00	0.00e+00	1.69e+13	0
R67	C5H10 + HO2 -> C5H11O2	CCC1CC1,O[O]:CC[CH]CCOO	9.41e+15	3.99e+14	4.48e+16	1
R67*	C5H11O2 -> C5H10 + HO2	CC[CH]CCOO:CCC1CC1,O[O]	0.00e+00	0.00e+00	4.43e+09	0
R68	C4H8 + CH3O2 -> C5H11O2	CCC=C,CO[O]:CC[CH]COOC	3.64e+11	5.52e+10	1.16e+12	2
R68*	C5H11O2 -> C4H8 + CH3O2	CC[CH]COOC:CCC=C,CO[O]	3.51e+11	5.32e+10	1.12e+12	2
R69	C4H8 + O2 + CH3 -> C5H11O2	CCC=C,O=O,[CH3]:C[C@H](CC)CO[O]	1.37e+16	5.36e+15	2.80e+16	6
R69*	C5H11O2 -> C4H8 + O2 + CH3	C[C@H](CC)CO[O]:CCC=C,O=O,[CH3]	1.25e+07	4.41e+06	2.72e+07	5
R70	C4H8 + CH3 -> C5H11	CCC=C,[CH3]:CCC(C)[CH2]	1.49e+14	6.33e+13	2.90e+14	7
R70*	C5H11 -> C4H8 + CH3	CCC(C)[CH2]:CCC=C,[CH3]	1.82e+08	7.13e+07	3.72e+08	6
R71	C5H10 + O2 + H -> C5H11O2	CCC=CC,O=O,[H]:C[C@H](O[O])CCC	1.01e+19	1.53e+18	3.22e+19	2
R71*	C5H11O2 -> C5H10 + O2 + H	C[C@H](O[O])CCC:CCC=CC,O=O,[H]	0.00e+00	0.00e+00	9.18e+06	0
R72	C3H6O + HO -> C3H7O2	CCC=O,[OH]:CCC(O)[O]	9.11e+13	3.86e+12	4.34e+14	1
R72*	C3H7O2 -> C3H6O + HO	CCC(O)[O]:CCC=O,[OH]	0.00e+00	0.00e+00	1.25e+12	0
R73	C3H6O + HO -> C3H7O2	CCC=O,[OH]:CCC([O])O	9.11e+13	3.86e+12	4.34e+14	1
R73*	C3H7O2 -> C3H6O + HO	CCC([O])O:CCC=O,[OH]	0.00e+00	0.00e+00	3.02e+10	0
R74	C5H10O + HO -> C5H11O2	CCCC(=O)C,[OH]:CCCC(O)([O])C	1.06e+12	4.50e+10	5.06e+12	1
R74*	C5H11O2 -> C5H10O + HO	CCCC(O)([O])C:CCCC(=O)C,[OH]	0.00e+00	0.00e+00	2.90e+10	0

R75	C5H10O + HO -> C5H9O + H2O	CCCC(=O)C,[OH]:C[CH]CC(=O)C,O	1.06e+12	4.50e+10	5.06e+12	1
R75*	C5H9O + H2O -> C5H10O + HO	C[CH]CC(=O)C,O:CCCC(=O)C,[OH]	0.00e+00	0.00e+00	5.24e+13	0
R76	C5H11O2 -> C2H4O2 + C3H7	CCCC(O)([O])C:CC(=O)O,CC[CH2]	9.69e+09	4.10e+08	4.62e+10	1
R76*	C2H4O2 + C3H7 -> C5H11O2	CC(=O)O,CC[CH2]:CCCC(O)([O])C	0.00e+00	0.00e+00	3.42e+14	0
R77	C5H11O2 -> C5H11O2	CCCC(O)([O])C:CCCC([O])(O)C	9.69e+09	4.10e+08	4.62e+10	1
R77*	C5H11O2 -> C5H11O2	CCCC([O])(O)C:CCCC(O)([O])C	1.15e+10	4.87e+08	5.48e+10	1
R78	C5H10 + HO2 -> C5H11O2	CCCC=C,O[O]:CCC[CH]COO	1.24e+12	5.28e+11	2.42e+12	7
R78*	C5H11O2 -> C5H10 + HO2	CCC[CH]COO:CCCC=C,O[O]	2.26e+11	9.59e+10	4.40e+11	7
R79	C4H8O + HO2 -> C4H9O3	CCCC=O,O[O]:CCC[C@H](OO)[O]	6.68e+12	2.83e+11	3.18e+13	1
R79*	C4H9O3 -> C4H8O + HO2	CCC[C@H](OO)[O]:CCCC=O,O[O]	0.00e+00	0.00e+00	1.04e+11	0
R80	C5H11O2 -> C5H11O2	CCCCCO[O]:CC[CH]CCOO	1.97e+07	2.99e+06	6.31e+07	2
R80*	C5H11O2 -> C5H11O2	CC[CH]CCOO:CCCCCO[O]	0.00e+00	0.00e+00	4.43e+09	0
R81	C5H11O2 -> C5H11O2	CCCCCO[O]:C[CH]CCCOO	9.85e+06	4.17e+05	4.70e+07	1
R81*	C5H11O2 -> C5H11O2	C[CH]CCCOO:CCCCCO[O]	0.00e+00	0.00e+00	2.21e+09	0
R82	C5H11 + O2 -> C5H11O2	CCCC[CH2],O=O:CCCCCO[O]	1.64e+13	1.56e+13	1.72e+13	1452
R82*	C5H11O2 -> C5H11 + O2	CCCCCO[O]:CCCC[CH2],O=O	1.42e+10	1.35e+10	1.49e+10	1440
R83	C5H11 + O2 -> C5H11O2	CCCC[CH2],O=O:CC[CH]CCOO	1.13e+10	4.78e+08	5.38e+10	1
R83*	C5H11O2 -> C5H11 + O2	CC[CH]CCOO:CCCC[CH2],O=O	0.00e+00	0.00e+00	4.43e+09	0
R84	C5H11 + O2 -> C5H11O2	CCCC[CH2],O=O:C[CH]CCCOO	1.13e+10	4.78e+08	5.38e+10	1
R84*	C5H11O2 -> C5H11 + O2	C[CH]CCCOO:CCCC[CH2],O=O	0.00e+00	0.00e+00	2.21e+09	0
R85	C5H11 -> C2H4 + C3H7	CCCC[CH2]:C=C,CC[CH2]	2.11e+08	3.21e+07	6.76e+08	2
R85*	C2H4 + C3H7 -> C5H11	C=C,CC[CH2]:CCCC[CH2]	0.00e+00	0.00e+00	3.49e+13	0
R86	C3H8O2 + CH3O2 -> C3H7O2 + CH4O2	CCCOO,CO[O]:CCCO[O],COO	2.25e+13	3.41e+12	7.20e+13	2
R86*	C3H7O2 + CH4O2 -> C3H8O2 + CH3O2	CCCO[O],COO:CCCOO,CO[O]	3.66e+13	5.56e+12	1.17e+14	2
R87	C3H8O2 -> C3H7 + HO2	CCCOO:CC[CH2],O[O]	1.02e+09	4.30e+07	4.84e+09	1
R87*	C3H7 + HO2 -> C3H8O2	CC[CH2],O[O]:CCCOO	0.00e+00	0.00e+00	2.82e+17	0
R88	C4H9O3 + O2 -> C4H8O + O2 + HO2	CCC[C@H](OO)[O],O=O:CCCC=O,O=O,O[O]	3.73e+12	1.58e+11	1.78e+13	1
R88*	C4H8O + O2 + HO2 -> C4H9O3 + O2	CCCC=O,O=O,O[O]:CCC[C@H](OO)[O],O=O	0.00e+00	0.00e+00	2.14e+15	0
R89	C4H9O3 -> C4H9O3	CCC[C@H](OO)[O]:CCC[C@H](O[O])O	1.05e+11	2.48e+10	2.77e+11	3
R89*	C4H9O3 -> C4H9O3	CCC[C@H](O[O])O:CCC[C@H](OO)[O]	1.30e+11	5.50e+09	6.19e+11	1
R90	C4H9O3 -> C4H8O + HO2	CCC[C@H](O[O])O:CCCC=O,O[O]	2.60e+11	3.94e+10	8.31e+11	2
R90*	C4H8O + HO2 -> C4H9O3	CCCC=O,O[O]:CCC[C@H](O[O])O	6.68e+12	2.83e+11	3.18e+13	1
R91	C4H9O3 -> C4H9O + O2	CCC[C@H](O[O])O:CCC[CH]O,O=O	1.30e+11	5.50e+09	6.19e+11	1
R91*	C4H9O + O2 -> C4H9O3	CCC[CH]O,O=O:CCC[C@H](O[O])O	0.00e+00	0.00e+00	2.07e+13	0
R92	C5H11O2 + O2 -> C5H11O4	CCC[CH]COO,O=O:CCC[C@H](COO)O[O]	7.08e+12	1.07e+12	2.27e+13	2
R92*	C5H11O4 -> C5H11O2 + O2	CCC[C@H](COO)O[O]:CCC[CH]COO,O=O	3.92e+11	5.95e+10	1.26e+12	2
R93	C4H9O + O2 -> C4H9O3	CCC[CH]O,O=O:CCC[C@H](OO)[O]	1.38e+13	2.10e+12	4.42e+13	2
R93*	C4H9O3 -> C4H9O + O2	CCC[C@H](OO)[O]:CCC[CH]O,O=O	0.00e+00	0.00e+00	1.04e+11	0
R94	C3H7O -> CH2O + C2H5	CCC[O]:C=O,C[CH2]	2.11e+10	8.94e+08	1.01e+11	1
R94*	CH2O + C2H5 -> C3H7O	C=O,C[CH2]:CCC[O]	0.00e+00	0.00e+00	1.63e+12	0

R95	C2H5O2 + CH2O2 -> C2H6O2 + CHO2	CCO[O],OC=O:CCOO,[O]C=O	1.21e+14	5.11e+12	5.75e+14	1
R95*	C2H6O2 + CHO2 -> C2H5O2 + CH2O2	CCOO,[O]C=O:CCO[O],OC=O	0.00e+00	0.00e+00	1.19e+13	0
R96	C2H5O2 -> C2H4 + HO2	CCO[O]:C=C,O[O]	4.43e+06	1.88e+05	2.11e+07	1
R96*	C2H4 + HO2 -> C2H5O2	C=C,O[O]:CCO[O]	0.00e+00	0.00e+00	2.22e+12	0
R97	C2H5O2 -> C2H4O + HO	CCO[O]:CC=O,[OH]	4.43e+06	1.88e+05	2.11e+07	1
R97*	C2H4O + HO -> C2H5O2	CC=O,[OH]:CCO[O]	0.00e+00	0.00e+00	5.24e+12	0
R98	C5H10O3 + HO -> H2O + O2 + C5H9O	CC[C@H](C=O)COO,[OH]:O,O=O,[CH2][C@H](C=O)CC	9.99e+13	4.23e+12	4.76e+14	1
R98*	H2O + O2 + C5H9O -> C5H10O3 + HO	O,O=O,[CH2][C@H](C=O)CC:CC[C@H](C=O)COO,[OH]	0.00e+00	0.00e+00	3.12e+16	0
R99	C5H9O3 -> C5H9O3	CC[C@H](C=O)CO[O]:CC[C@H]1COO[C@H]1[O]	6.94e+09	1.65e+09	1.84e+10	3
R99*	C5H9O3 -> C5H9O3	CC[C@H]1COO[C@H]1[O]:CC[C@H](C=O)CO[O]	6.86e+10	1.63e+10	1.82e+11	3
R100	C4H9O + O2 -> C4H9O3	CC[C@H](O)[CH2],O=O:CC[C@H](CO[O])O	8.17e+12	2.46e+12	1.93e+13	4
R100*	C4H9O3 -> C4H9O + O2	CC[C@H](CO[O])O:CC[C@H](O)[CH2],O=O	3.65e+10	1.10e+10	8.60e+10	4
R101	C4H9O -> C4H9O	CC[C@H](O)[CH2]:CCC[CH]O	1.95e+10	8.27e+08	9.31e+10	1
R101*	C4H9O -> C4H9O	CCC[CH]O:CC[C@H](O)[CH2]	0.00e+00	0.00e+00	1.96e+11	0
R102	C4H9O3 -> CH2O + C3H6O + HO	CC[C@H](OO)C[O]:C=O,CCC=O,[OH]	2.97e+10	1.26e+09	1.41e+11	1
R102*	CH2O + C3H6O + HO -> C4H9O3	C=O,CCC=O,[OH]:CC[C@H](OO)C[O]	0.00e+00	0.00e+00	8.72e+18	0
R103	C4H9O3 -> C4H9O3	CC[C@H](OO)C[O]:CC[C@H](O[O])CO	5.93e+10	9.01e+09	1.90e+11	2
R103*	C4H9O3 -> C4H9O3	CC[C@H](O[O])CO:CC[C@H](OO)C[O]	2.30e+09	9.75e+07	1.10e+10	1
R104	C5H9O3 -> C5H9O3	CC[C@H](O[O])CC=O:CC[C@H]1C[C@H](OO1)[O]	1.21e+10	5.75e+09	2.19e+10	9
R104*	C5H9O3 -> C5H9O3	CC[C@H]1C[C@H](OO1)[O]:CC[C@H](O[O])CC=O	6.45e+10	2.74e+10	1.26e+11	7
R105	C4H9O3 -> C4H9O + O2	CC[C@H](O[O])CO:CC[CH]CO,O=O	1.04e+11	7.62e+10	1.37e+11	45
R105*	C4H9O + O2 -> C4H9O3	CC[CH]CO,O=O:CC[C@H](O[O])CO	2.86e+12	2.10e+12	3.80e+12	44
R106	C5H9O3 -> C5H9O + O2	CC[C@H]1C[C@H](OO1)[O]:CC[CH]CC=O,O=O	1.84e+10	2.80e+09	5.90e+10	2
R106*	C5H9O + O2 -> C5H9O3	CC[CH]CC=O,O=O:CC[C@H]1C[C@H](OO1)[O]	0.00e+00	0.00e+00	2.53e+11	0
R107	C5H9O -> C5H9O	CC[C@H]1C[C@H]1[O]:CC[CH]CC=O	8.33e+11	4.15e+11	1.47e+12	10
R107*	C5H9O -> C5H9O	CC[CH]CC=O:CC[C@H]1C[C@H]1[O]	5.34e+09	2.27e+09	1.04e+10	7
R108	C3H7 + O2 -> C3H7O2	CC[CH2],O=O:CCCO[O]	1.46e+13	1.23e+13	1.73e+13	131
R108*	C3H7O2 -> C3H7 + O2	CCCO[O]:CC[CH2],O=O	1.21e+10	1.02e+10	1.44e+10	129
R109	C3H7 + HO2 -> C3H7O + HO	CC[CH2],O[O]:CCC[O],[OH]	9.41e+16	3.99e+15	4.48e+17	1
R109*	C3H7O + HO -> C3H7 + HO2	CCC[O],[OH]:CC[CH2],O[O]	0.00e+00	0.00e+00	5.95e+14	0
R110	C3H7 -> C3H7	CC[CH2]:C[CH]C	1.08e+09	4.57e+07	5.14e+09	1
R110*	C3H7 -> C3H7	C[CH]C:CC[CH2]	0.00e+00	0.00e+00	4.23e+09	0
R111	C5H11 + O2 -> C5H11O2	CC[CH]CC,O=O:CCC(O[O])CC	2.56e+12	2.46e+12	2.66e+12	2439
R111*	C5H11O2 -> C5H11 + O2	CCC(O[O])CC:CC[CH]CC,O=O	1.03e+11	9.94e+10	1.08e+11	2436
R112	C5H11 -> C3H6 + C2H5	CC[CH]CC:CC=C,C[CH2]	9.82e+06	4.16e+05	4.68e+07	1
R112*	C3H6 + C2H5 -> C5H11	CC=C,C[CH2]:CC[CH]CC	0.00e+00	0.00e+00	1.52e+12	0
R113	C5H11 -> C4H8 + CH3	CC[CH]CC:CCC=C,[CH3]	7.86e+07	3.56e+07	1.47e+08	8
R113*	C4H8 + CH3 -> C5H11	CCC=C,[CH3]:CC[CH]CC	4.26e+13	6.46e+12	1.36e+14	2
R114	C5H9O + O2 -> C5H9O3	CC[CH]CC=O,O=O:CC[C@H](O[O])CC=O	3.63e+12	2.65e+12	4.83e+12	43
R114*	C5H9O3 -> C5H9O + O2	CC[C@H](O[O])CC=O:CC[CH]CC=O,O=O	5.50e+10	3.98e+10	7.36e+10	41

R115	C5H9O -> C5H9O	CC[CH]CC=O:[CH2][C@H](C=O)CC	1.53e+09	2.31e+08	4.88e+09	2
R115*	C5H9O -> C5H9O	[CH2][C@H](C=O)CC:CC[CH]CC=O	0.00e+00	0.00e+00	2.99e+10	0
R116	C5H11O2 + O2 -> C5H11O4	CC[CH]CCOO,O=O:OOCC[C@H](O[O])CC	2.41e+12	1.39e+12	3.86e+12	15
R116*	C5H11O4 -> C5H11O2 + O2	OOCC[C@H](O[O])CC:CC[CH]CCOO,O=O	1.11e+11	6.23e+10	1.80e+11	14
R117	C5H11O2 + O2 -> C5H11O4	CC[CH]CCOO,O=O:[O]OCC[C@H](OO)CC	1.61e+11	6.82e+09	7.67e+11	1
R117*	C5H11O4 -> C5H11O2 + O2	[O]OCC[C@H](OO)CC:CC[CH]CCOO,O=O	0.00e+00	0.00e+00	2.21e+10	0
R118	C5H11O2 -> CH2O + C4H9O	CC[CH]CCOO:C=O,CC[C@H](O)[CH2]	1.48e+09	6.26e+07	7.04e+09	1
R118*	CH2O + C4H9O -> C5H11O2	C=O,CC[C@H](O)[CH2]:CC[CH]CCOO	0.00e+00	0.00e+00	5.51e+14	0
R119	C5H11O2 -> C5H11O2	CC[CH]CCOO:[O]CC[C@H](CC)O	2.95e+09	4.48e+08	9.46e+09	2
R119*	C5H11O2 -> C5H11O2	[O]CC[C@H](CC)O:CC[CH]CCOO	0.00e+00	0.00e+00	1.09e+11	0
R120	C4H9O + O2 -> C4H9O3	CC[CH]CO,O=O:CC[C@H](OO)C[O]	1.30e+11	1.98e+10	4.17e+11	2
R120*	C4H9O3 -> C4H9O + O2	CC[C@H](OO)C[O]:CC[CH]CO,O=O	0.00e+00	0.00e+00	8.89e+10	0
R121	C4H9O -> C4H9O	CC[CH]CO:OCC(C)[CH2]	5.88e+08	2.49e+07	2.80e+09	1
R121*	C4H9O -> C4H9O	OCC(C)[CH2]:CC[CH]CO	0.00e+00	0.00e+00	3.30e+09	0
R122	C4H8O2 + CH3 -> C5H11O2	CC[CH]CO[O],[CH3]:C[C@H](CC)CO[O]	6.72e+16	2.36e+16	1.46e+17	5
R122*	C5H11O2 -> C4H8O2 + CH3	C[C@H](CC)CO[O]:CC[CH]CO[O],[CH3]	5.01e+06	7.61e+05	1.60e+07	2
R123	C5H11 + O2 -> C5H11O2	CC[C](C)C,O=O:CCC(O[O])(C)C	4.40e+11	3.66e+11	5.23e+11	120
R123*	C5H11O2 -> C5H11 + O2	CCC(O[O])(C)C:CC[C](C)C,O=O	1.77e+11	1.47e+11	2.11e+11	120
R124	C5H11 -> C4H8 + CH3	CC[C](C)C:CC(=C)C,[CH3]	7.03e+07	1.07e+07	2.25e+08	2
R124*	C4H8 + CH3 -> C5H11	CC(=C)C,[CH3]:CC[C](C)C	0.00e+00	0.00e+00	3.15e+14	0
R125	C5H11 -> C5H11	CC[C](C)C:CC(C)([CH2])C	7.03e+07	1.07e+07	2.25e+08	2
R125*	C5H11 -> C5H11	CC(C)([CH2])C:CC[C](C)C	7.35e+09	3.11e+08	3.50e+10	1
R126	C5H11 -> C5H11	CC[C](C)C:CC([CH2])(C)C	3.52e+07	1.49e+06	1.68e+08	1
R126*	C5H11 -> C5H11	CC([CH2])(C)C:CC[C](C)C	0.00e+00	0.00e+00	1.39e+10	0
R127	C2H4O2 + O2 -> CH3O2 + CHO2	COC=O,O=O:CO[O],[O]C=O	6.62e+12	2.80e+11	3.15e+13	1
R127*	CH3O2 + CHO2 -> C2H4O2 + O2	CO[O],[O]C=O:COC=O,O=O	0.00e+00	0.00e+00	2.07e+12	0
R128	C2H4O2 + HO2 -> CH4O2 + CHO2	COC=O,O[O]:COO,[O]C=O	5.74e+14	2.43e+13	2.73e+15	1
R128*	CH4O2 + CHO2 -> C2H4O2 + HO2	COO,[O]C=O:COC=O,O[O]	0.00e+00	0.00e+00	2.00e+12	0
R129	CH4O2 + O2 + C2H3O -> CH3O2 + C2H4O3	COO,O=O,[O]C=C:CO[O],OOCC=O	1.34e+15	5.69e+13	6.40e+15	1
R129*	CH3O2 + C2H4O3 -> CH4O2 + O2 + C2H3O	CO[O],OOCC=O:COO,O=O,[O]C=C	1.87e+13	7.92e+11	8.91e+13	1
R130	CH4O2 + C2H3O -> CH3O2 + C2H4O	COO,[O]C=C:CO[O],OC=C	2.39e+13	3.63e+12	7.66e+13	2
R130*	CH3O2 + C2H4O -> CH4O2 + C2H3O	CO[O],OC=C:COO,[O]C=C	0.00e+00	0.00e+00	4.11e+12	0
R131	C2H5O2 + O2 -> HO2 + CH3 + CHO2	CO[CH]O,O=O:O[O],[CH3],[O]C=O	1.00e+13	4.24e+11	4.77e+13	1
R131*	HO2 + CH3 + CHO2 -> C2H5O2 + O2	O[O],[CH3],[O]C=O:CO[CH]O,O=O	0.00e+00	0.00e+00	2.65e+21	0
R132	C3H5O -> C2H2O + CH3	CO[C]=C:C=C=O,[CH3]	7.81e+10	3.31e+09	3.72e+11	1
R132*	C2H2O + CH3 -> C3H5O	C=C=O,[CH3]:CO[C]=C	0.00e+00	0.00e+00	7.23e+15	0
R133	CH3O2 + CH2O2 -> CH4O2 + CHO2	CO[O],OC=O:COO,[O]C=O	1.26e+13	3.00e+12	3.35e+13	3
R133*	CH4O2 + CHO2 -> CH3O2 + CH2O2	COO,[O]C=O:CO[O],OC=O	0.00e+00	0.00e+00	2.00e+12	0
R134	CH3O2 + H2O2 -> CH4O2 + HO2	CO[O],OO:COO,O[O]	2.97e+13	7.05e+12	7.86e+13	3
R134*	CH4O2 + HO2 -> CH3O2 + H2O2	COO,O[O]:CO[O],OO	8.67e+13	1.32e+13	2.77e+14	2

R135	CH3O2 + C2H4O3 -> CH4O2 + C2H3O3	CO[O],OOCC=O:COO,[O]OCC=O	1.87e+13	7.92e+11	8.91e+13	1
R135*	CH4O2 + C2H3O3 -> CH3O2 + C2H4O3	COO,[O]OCC=O:CO[O],OOCC=O	1.49e+13	6.32e+11	7.11e+13	1
R136	C5H9O3 -> C5H9O + O2	C[C@H](C(=O)C)CO[O]:CC(=O)C([CH2])C,O=O	2.14e+10	1.31e+10	3.26e+10	19
R136*	C5H9O + O2 -> C5H9O3	CC(=O)C([CH2])C,O=O:C[C@H](C(=O)C)CO[O]	1.20e+13	7.23e+12	1.84e+13	18
R137	C5H10O3 + HO -> C5H11O4	C[C@H](C=O)CCOO,[OH]:OOCC[C@H](C([O])O)C	9.41e+16	3.99e+15	4.48e+17	1
R137*	C5H11O4 -> C5H10O3 + HO	OOCC[C@H](C([O])O)C:C[C@H](C=O)CCOO,[OH]	0.00e+00	0.00e+00	9.99e+11	0
R138	C5H11O2 -> C3H6 + C2H5 + O2	C[C@H](CC)CO[O]:CC=C,C[CH2],O=O	7.52e+06	1.79e+06	1.99e+07	3
R138*	C3H6 + C2H5 + O2 -> C5H11O2	CC=C,C[CH2],O=O:C[C@H](CC)CO[O]	0.00e+00	0.00e+00	1.60e+14	0
R139	C5H11O2 -> C5H10 + HO2	C[C@H](CC)CO[O]:CCC(=C)C,O[O]	2.51e+06	1.06e+05	1.19e+07	1
R139*	C5H10 + HO2 -> C5H11O2	CCC(=C)C,O[O]:C[C@H](CC)CO[O]	0.00e+00	0.00e+00	4.95e+12	0
R140	C5H11O2 -> C5H11O2	C[C@H](CC)CO[O]:C[C@H](C[CH2])COO	1.50e+07	5.89e+06	3.07e+07	6
R140*	C5H11O2 -> C5H11O2	C[C@H](C[CH2])COO:C[C@H](CC)CO[O]	0.00e+00	0.00e+00	7.11e+09	0
R141	C5H11O2 -> C5H10O2 + H	C[C@H](CC)CO[O]:C[C@H](C[CH2])CO[O],[H]	2.51e+06	1.06e+05	1.19e+07	1
R141*	C5H10O2 + H -> C5H11O2	C[C@H](C[CH2])CO[O],[H]:C[C@H](CC)CO[O]	0.00e+00	0.00e+00	2.82e+17	0
R142	C5H11O2 -> C5H11O2	C[C@H](CC)CO[O]:C[C@H]([CH]C)COO	7.52e+06	1.79e+06	1.99e+07	3
R142*	C5H11O2 -> C5H11O2	C[C@H]([CH]C)COO:C[C@H](CC)CO[O]	0.00e+00	0.00e+00	2.02e+10	0
R143	C5H11O2 -> C2H5 + C3H6O2	C[C@H](CC)CO[O]:C[CH2],C[CH]CO[O]	2.51e+06	1.06e+05	1.19e+07	1
R143*	C2H5 + C3H6O2 -> C5H11O2	C[CH2],C[CH]CO[O]:C[C@H](CC)CO[O]	0.00e+00	0.00e+00	2.82e+17	0
R144	C5H11O2 -> C4H9 + CH2O2	C[C@H](CC)CO[O]:C[CH]CC,[CH2]O[O]	2.51e+06	1.06e+05	1.19e+07	1
R144*	C4H9 + CH2O2 -> C5H11O2	C[CH]CC,[CH2]O[O]:C[C@H](CC)CO[O]	0.00e+00	0.00e+00	2.82e+17	0
R145	C5H11O2 -> C5H11O2	C[C@H](CC)CO[O]:[CH2][C@H](CC)COO	1.25e+07	4.41e+06	2.72e+07	5
R145*	C5H11O2 -> C5H11O2	[CH2][C@H](CC)COO:C[C@H](CC)CO[O]	0.00e+00	0.00e+00	1.14e+10	0
R146	C4H7O2 -> C2H4O + C2H3O	C[C@H](CC=O)[O]:CC=O,[O]C=C	4.83e+10	2.05e+09	2.30e+11	1
R146*	C2H4O + C2H3O -> C4H7O2	CC=O,[O]C=C:C[C@H](CC=O)[O]	0.00e+00	0.00e+00	1.04e+15	0
R147	C5H11O2 -> C3H7O + C2H4O	C[C@H](C[C@H](O)C)[O]:CC(O)[CH2],CC=O	8.65e+09	2.61e+09	2.04e+10	4
R147*	C3H7O + C2H4O -> C5H11O2	CC(O)[CH2],CC=O:C[C@H](C[C@H](O)C)[O]	5.37e+12	2.27e+11	2.56e+13	1
R148	C5H9O2 -> C4H6O2 + CH3	C[C@H](C[C@H]([O])[CH2])[O]:[CH2][C@H](CC=O)[O],[CH3]	5.00e+12	2.12e+11	2.38e+13	1
R148*	C4H6O2 + CH3 -> C5H9O2	[CH2][C@H](CC=O)[O],[CH3]:C[C@H](C[C@H]([O])[CH2])[O]	0.00e+00	0.00e+00	1.76e+16	0
R149	C5H11O2 + O2 -> C5H11O4	C[C@H](C[CH2])COO,O=O:OOCC[C@H](CO[O])C	2.61e+11	1.10e+10	1.24e+12	1
R149*	C5H11O4 -> C5H11O2 + O2	OOCC[C@H](CO[O])C:C[C@H](C[CH2])COO,O=O	4.01e+08	1.70e+07	1.91e+09	1
R150	C5H11O2 + O2 -> C5H11O4	C[C@H](C[CH2])COO,O=O:[O]OCC[C@H](COO)C	1.51e+13	1.15e+13	1.93e+13	58
R150*	C5H11O4 -> C5H11O2 + O2	[O]OCC[C@H](COO)C:C[C@H](C[CH2])COO,O=O	1.65e+10	1.26e+10	2.12e+10	57
R151	C5H11O2 -> C5H11O2	C[C@H](C[CH2])COO:OCC[C@H](C[O])C	9.49e+09	2.86e+09	2.24e+10	4
R151*	C5H11O2 -> C5H11O2	OCC[C@H](C[O])C:C[C@H](C[CH2])COO	0.00e+00	0.00e+00	2.22e+11	0
R152	C5H11O2 -> C5H11O2	C[C@H](C[CH2])COO:OOCCC([CH2])C	2.37e+09	1.01e+08	1.13e+10	1
R152*	C5H11O2 -> C5H11O2	OOCCC([CH2])C:C[C@H](C[CH2])COO	0.00e+00	0.00e+00	8.54e+09	0
R153	C5H11O2 -> HO2 + C5H10	C[C@H](C[CH2])COO:O[O],[CH2]CC([CH2])C	2.37e+09	1.01e+08	1.13e+10	1
R153*	HO2 + C5H10 -> C5H11O2	O[O],[CH2]CC([CH2])C:C[C@H](C[CH2])COO	0.00e+00	0.00e+00	2.82e+17	0
R154	C5H10O2 + H -> C5H11O2	C[C@H](C[CH2])CO[O],[H]:C[C@H](C[CH2])COO	9.41e+16	3.99e+15	4.48e+17	1
R154*	C5H11O2 -> C5H10O2 + H	C[C@H](C[CH2])COO:C[C@H](C[CH2])CO[O],[H]	0.00e+00	0.00e+00	7.11e+09	0

R155	C5H10O2 + HO2 -> C5H9O2 + H2O2	C[C@H](OO)CC=C,O[O]:C[C@H](O[O])CC=C,OO	1.35e+14	4.07e+13	3.18e+14	4
R155*	C5H9O2 + H2O2 -> C5H10O2 + HO2	C[C@H](O[O])CC=C,OO:C[C@H](OO)CC=C,O[O]	4.51e+13	1.07e+13	1.19e+14	3
R156	C4H8O3 -> C4H7O2 + HO	C[C@H](OO)CC=O:C[C@H](CC=O)[O],[OH]	1.39e+10	5.87e+08	6.60e+10	1
R156*	C4H7O2 + HO -> C4H8O3	C[C@H](CC=O)[O],[OH]:C[C@H](OO)CC=O	0.00e+00	0.00e+00	1.36e+15	0
R157	C5H11O2 + O2 -> HO + C5H10O3	C[C@H](OO)CC[CH2],O=O:[OH],[O]OCCC[C@H]([O])C	7.07e+11	1.07e+11	2.26e+12	2
R157*	HO + C5H10O3 -> C5H11O2 + O2	[OH],[O]OCCC[C@H]([O])C:C[C@H](OO)CC[CH2],O=O	0.00e+00	0.00e+00	9.40e+13	0
R158	C5H11O2 + O2 -> C5H11O4	C[C@H](OO)CC[CH2],O=O:[O]OCCC[C@H](OO)C	1.66e+13	1.23e+13	2.18e+13	47
R158*	C5H11O4 -> C5H11O2 + O2	[O]OCCC[C@H](OO)C:C[C@H](OO)CC[CH2],O=O	1.58e+10	1.14e+10	2.12e+10	41
R159	C5H11O2 -> C5H10O + HO	C[C@H](OO)CC[CH2]:C[C@H]1CCCO1,[OH]	3.26e+09	1.38e+08	1.55e+10	1
R159*	C5H10O + HO -> C5H11O2	C[C@H]1CCCO1,[OH]:C[C@H](OO)CC[CH2]	0.00e+00	0.00e+00	2.09e+13	0
R160	C5H11O2 -> C5H10 + HO2	C[C@H](OO)CC[CH2]:C[CH]CC[CH2],O[O]	3.26e+09	1.38e+08	1.55e+10	1
R160*	C5H10 + HO2 -> C5H11O2	C[CH]CC[CH2],O[O]:C[C@H](OO)CC[CH2]	0.00e+00	0.00e+00	2.82e+17	0
R161	C5H11O2 -> C5H11O2	C[C@H](OO)CC[CH2]:OCCC[C@H]([O])C	9.79e+09	2.32e+09	2.59e+10	3
R161*	C5H11O2 -> C5H11O2	OCCC[C@H]([O])C:C[C@H](OO)CC[CH2]	0.00e+00	0.00e+00	7.97e+09	0
R162	C5H11O2 + O2 -> C5H11O4	C[C@H](OO)C[CH]C,O=O:OO[C@H](C[C@H](O[O])C)C	2.40e+12	1.02e+11	1.14e+13	1
R162*	C5H11O4 -> C5H11O2 + O2	OO[C@H](C[C@H](O[O])C)C:C[C@H](OO)C[CH]C,O=O	4.00e+11	1.69e+10	1.91e+12	1
R163	C5H11O2 -> C3H7O + C2H4O	C[C@H](OO)C[CH]C:CC(O)[CH2],CC=O	2.27e+10	9.63e+08	1.08e+11	1
R163*	C3H7O + C2H4O -> C5H11O2	CC(O)[CH2],CC=O:C[C@H](OO)C[CH]C	0.00e+00	0.00e+00	1.61e+13	0
R164	C5H11O2 -> C5H11O2	C[C@H](OO)C[CH]C:C[C@H](C[C@H](O)C)[O]	6.82e+10	1.62e+10	1.81e+11	3
R164*	C5H11O2 -> C5H11O2	C[C@H](C[C@H](O)C)[O]:C[C@H](OO)C[CH]C	0.00e+00	0.00e+00	6.48e+09	0
R165	C4H9O3 -> C2H5O2 + C2H4O	C[C@H](OO)O[CH]C:CC(O)[O],CC=O	1.11e+12	4.71e+10	5.29e+12	1
R165*	C2H5O2 + C2H4O -> C4H9O3	CC(O)[O],CC=O:C[C@H](OO)O[CH]C	0.00e+00	0.00e+00	4.12e+13	0
R166	C5H9O3 -> C5H9O3	C[C@H](O[O])CC(=O)C:C[C@](O[O])(C[CH]C)[O]	3.08e+09	4.67e+08	9.85e+09	2
R166*	C5H9O3 -> C5H9O3	C[C@](O[O])(C[CH]C)[O]:C[C@H](O[O])CC(=O)C	0.00e+00	0.00e+00	1.50e+13	0
R167	C5H9O2 -> C5H9O2	C[C@H](O[O])CC=C:C[C@H]1OO[C@H](C1)[CH2]	1.60e+09	6.77e+07	7.61e+09	1
R167*	C5H9O2 -> C5H9O2	C[C@H]1OO[C@H](C1)[CH2]:C[C@H](O[O])CC=C	0.00e+00	0.00e+00	7.31e+11	0
R168	C5H9O2 -> C5H9 + O2	C[C@H](O[O])CC=C:C[CH]CC=C,O=O	3.03e+10	1.86e+10	4.62e+10	19
R168*	C5H9 + O2 -> C5H9O2	C[CH]CC=C,O=O:C[C@H](O[O])CC=C	6.48e+12	3.92e+12	9.96e+12	18
R169	C5H11O2 -> C5H10O + HO	C[C@H](O[O])CCC:CCCC(=O)C,[OH]	3.06e+06	1.30e+05	1.46e+07	1
R169*	C5H10O + HO -> C5H11O2	CCCC(=O)C,[OH]:C[C@H](O[O])CCC	0.00e+00	0.00e+00	3.18e+12	0
R170	C5H11O2 -> C5H10 + HO2	C[C@H](O[O])CCC:CCCC=C,O[O]	3.06e+06	1.30e+05	1.46e+07	1
R170*	C5H10 + HO2 -> C5H11O2	CCCC=C,O[O]:C[C@H](O[O])CCC	0.00e+00	0.00e+00	5.32e+11	0
R171	C5H11O2 -> C5H11O2	C[C@H](O[O])CCC:C[C@H](OO)CC[CH2]	3.37e+07	1.74e+07	5.79e+07	11
R171*	C5H11O2 -> C5H11O2	C[C@H](OO)CC[CH2]:C[C@H](O[O])CCC	0.00e+00	0.00e+00	9.77e+09	0
R172	C5H11O2 -> C5H11O2	C[C@H](O[O])CCC:C[C@H](OO)C[CH]C	1.23e+07	3.70e+06	2.89e+07	4
R172*	C5H11O2 -> C5H11O2	C[C@H](OO)C[CH]C:C[C@H](O[O])CCC	0.00e+00	0.00e+00	6.81e+10	0
R173	C5H11O2 -> C5H10O2 + H	C[C@H](O[O])CCC:C[C@H](O[O])[CH]CC,[H]	3.06e+06	1.30e+05	1.46e+07	1
R173*	C5H10O2 + H -> C5H11O2	C[C@H](O[O])[CH]CC,[H]:C[C@H](O[O])CCC	0.00e+00	0.00e+00	2.82e+17	0
R174	C5H11O2 -> C5H11O2	C[C@H](O[O])CCC:[CH2][C@H](OO)CCC	9.19e+06	2.18e+06	2.43e+07	3
R174*	C5H11O2 -> C5H11O2	[CH2][C@H](OO)CCC:C[C@H](O[O])CCC	0.00e+00	0.00e+00	1.86e+11	0

R175	C5H10O2 + H -> C5H11 + O2	C[C@H](O[O])[CH]CC,[H]:C[CH]CCC,O=O	9.41e+16	3.99e+15	4.48e+17	1
R175*	C5H11 + O2 -> C5H10O2 + H	C[CH]CCC,O=O:C[C@H](O[O])[CH]CC,[H]	0.00e+00	0.00e+00	9.60e+08	0
R176	C4H8O2 -> C2H4O + C2H4O	C[C@H]([C@H]([O])C)[O]:CC=O,CC=O	1.00e+13	1.52e+12	3.20e+13	2
R176*	C2H4O + C2H4O -> C4H8O2	CC=O,CC=O:C[C@H]([C@H]([O])C)[O]	0.00e+00	0.00e+00	3.06e+12	0
R177	C5H11O2 + O2 -> C5H11O4	C[C@H]([CH]C)COO,O=O:OOC[C@H]([C@H](O[O])C)C	2.20e+12	5.21e+11	5.82e+12	3
R177*	C5H11O4 -> C5H11O2 + O2	OOC[C@H]([C@H](O[O])C)C:C[C@H]([CH]C)COO,O=O	1.76e+11	4.19e+10	4.68e+11	3
R178	C5H11O2 + O2 -> C5H11O4	C[C@H]([CH]C)COO,O=O:[O]OC[C@H]([C@H](OO)C)C	7.32e+11	3.10e+10	3.49e+12	1
R178*	C5H11O4 -> C5H11O2 + O2	[O]OC[C@H]([C@H](OO)C)C:C[C@H]([CH]C)COO,O=O	0.00e+00	0.00e+00	2.35e+10	0
R179	C5H11O2 -> C5H10 + HO2	C[C@H]([CH]C)COO:C[C@H]1C[C@H]1C,O[O]	6.74e+09	2.85e+08	3.21e+10	1
R179*	C5H10 + HO2 -> C5H11O2	C[C@H]1C[C@H]1C,O[O]:C[C@H]([CH]C)COO	0.00e+00	0.00e+00	9.46e+12	0
R180	C5H11O2 -> C5H11O2	C[C@H]([CH]C)COO:[O]C[C@H]([C@H](O)C)C	2.02e+10	4.80e+09	5.36e+10	3
R180*	C5H11O2 -> C5H11O2	[O]C[C@H]([C@H](O)C)C:C[C@H]([CH]C)COO	0.00e+00	0.00e+00	7.49e+12	0
R181	C5H10O + HO -> C5H11O2	C[C@H]1CCCO1,[OH]:OCCC[C@H]([O])C	6.98e+12	2.96e+11	3.33e+13	1
R181*	C5H11O2 -> C5H10O + HO	OCCC[C@H]([O])C:C[C@H]1CCCO1,[OH]	0.00e+00	0.00e+00	7.97e+09	0
R182	C5H9O3 -> C5H9O3	C[C@H]1COO[C@]1(C)[O]:C[C@H](C(=O)C)CO[O]	4.44e+11	6.74e+10	1.42e+12	2
R182*	C5H9O3 -> C5H9O3	C[C@H](C(=O)C)CO[O]:C[C@H]1COO[C@]1(C)[O]	1.13e+09	4.78e+07	5.37e+09	1
R183	C5H9 + O2 -> C5H9O2	C[C@H]1C[C@H]1[CH2],O=O:C[C@H]1C[C@H]1CO[O]	8.32e+12	3.53e+11	3.97e+13	1
R183*	C5H9O2 -> C5H9 + O2	C[C@H]1C[C@H]1CO[O]:C[C@H]1C[C@H]1[CH2],O=O	2.14e+10	9.07e+08	1.02e+11	1
R184	C5H9O -> C5H9O	C[C@H]1C[C@]1(C)[O]:CC(=O)C([CH2])C	1.52e+11	5.34e+10	3.30e+11	5
R184*	C5H9O -> C5H9O	CC(=O)C([CH2])C:C[C@H]1C[C@]1(C)[O]	1.88e+10	4.48e+09	4.99e+10	3
R185	C5H9O -> C3H6 + C2H3O	C[C@H]1C[C@]1(C)[O]:CC=C,C[C]=O	3.04e+10	1.29e+09	1.45e+11	1
R185*	C3H6 + C2H3O -> C5H9O	CC=C,C[C]=O:C[C@H]1C[C@]1(C)[O]	0.00e+00	0.00e+00	2.56e+16	0
R186	C5H9O2 -> C5H9O2	C[C@H]1C[CH]COO1:C[C@H](O[O])CC=C	2.74e+11	4.16e+10	8.77e+11	2
R186*	C5H9O2 -> C5H9O2	C[C@H](O[O])CC=C:C[C@H]1C[CH]COO1	1.60e+09	6.77e+07	7.61e+09	1
R187	C5H9O2 + O2 -> C5H9O4	C[C@H]1OO[C@H](C1)[CH2],O=O:C[C@H]1C[C@H](OO1)CO[O]	2.73e+13	1.16e+12	1.30e+14	1
R187*	C5H9O4 -> C5H9O2 + O2	C[C@H]1C[C@H](OO1)CO[O]:C[C@H]1OO[C@H](C1)[CH2],O=O	3.29e+10	1.39e+09	1.57e+11	1
R188	C5H9O2 + O2 -> C5H9O4	C[C@H]1OO[C@H](C1)[CH2],O=O:[O]OC[C@H](C[C@H]([O])C)[O]	2.73e+13	1.16e+12	1.30e+14	1
R188*	C5H9O4 -> C5H9O2 + O2	[O]OC[C@H](C[C@H]([O])C)[O]:C[C@H]1OO[C@H](C1)[CH2],O=O	0.00e+00	0.00e+00	1.20e+11	0
R189	C5H9O3 -> C5H9O3	C[C@H]1OO[C@](C1)(C)[O]:C[C@H](O[O])CC(=O)C	1.03e+11	4.03e+10	2.10e+11	6
R189*	C5H9O3 -> C5H9O3	C[C@H](O[O])CC(=O)C:C[C@H]1OO[C@](C1)(C)[O]	7.69e+09	2.70e+09	1.67e+10	5
R190	C5H9O3 -> C5H9O + O2	C[C@H]1OO[C@](C1)(C)[O]:C[CH]CC(=O)C,O=O	1.72e+10	7.27e+08	8.17e+10	1
R190*	C5H9O + O2 -> C5H9O3	C[CH]CC(=O)C,O=O:C[C@H]1OO[C@](C1)(C)[O]	0.00e+00	0.00e+00	5.89e+11	0
R191	C5H9O3 -> C5H9O3	C[C@](O[O])(C[CH]C)[O]:C[C@H]1OO[C@](C1)(C)[O]	1.00e+13	1.52e+12	3.20e+13	2
R191*	C5H9O3 -> C5H9O3	C[C@H]1OO[C@](C1)(C)[O]:C[C@](O[O])(C[CH]C)[O]	0.00e+00	0.00e+00	5.14e+10	0
R192	C2H5 + C3H6O2 -> C5H11 + O2	C[CH2],C[CH]CO[O]:CCC([CH2])C,O=O	9.41e+16	3.99e+15	4.48e+17	1
R192*	C5H11 + O2 -> C2H5 + C3H6O2	CCC([CH2])C,O=O:C[CH2],C[CH]CO[O]	0.00e+00	0.00e+00	8.47e+09	0
R193	C2H5 + O2 + HO -> C2H4O2 + H2O	C[CH2],O=O,[OH]:C[CH]O[O],O	3.20e+14	1.36e+13	9.76e+10	1
R193*	C2H4O2 + H2O -> C2H5 + O2 + HO	C[CH]O[O],O:C[CH2],O=O,[OH]	0.00e+00	0.00e+00	2.85e+15	0
R194	C2H5 + O2 -> C2H5O2	C[CH2],O=O:CCO[O]	5.55e+11	5.24e+11	5.87e+11	1212
R194*	C2H5O2 -> C2H5 + O2	CCO[O]:C[CH2],O=O	5.21e+09	4.92e+09	5.51e+09	1176

R195	C2H5 + O2 -> C2H5O2	C[CH2],O=O:[CH2]COO	4.58e+08	1.94e+07	2.18e+09	1
R195*	C2H5O2 -> C2H5 + O2	[CH2]COO:C[CH2],O=O	0.00e+00	0.00e+00	2.30e+12	0
R196	C3H7 + O2 -> C3H7O2	C[CH]C,O=O:[O]OC(C)C	7.97e+12	6.03e+12	1.03e+13	54
R196*	C3H7O2 -> C3H7 + O2	[O]OC(C)C:C[CH]C,O=O	2.48e+10	1.87e+10	3.21e+10	53
R197	C5H9O + O2 -> C5H9O3	C[CH]CC(=O)C,O=O:C[C@H](O[O])CC(=O)C	5.90e+12	4.03e+12	8.27e+12	30
R197*	C5H9O3 -> C5H9O + O2	C[C@H](O[O])CC(=O)C:C[CH]CC(=O)C,O=O	4.31e+10	2.90e+10	6.11e+10	28
R198	C5H9O -> C5H9O	C[CH]CC(=O)C:C[C@H]1C[C@]1(C)[O]	2.60e+10	1.46e+10	4.23e+10	14
R198*	C5H9O -> C5H9O	C[C@H]1C[C@]1(C)[O]:C[CH]CC(=O)C	3.65e+11	1.94e+11	6.14e+11	12
R199	C4H9 + CH2O2 -> C5H11 + O2	C[CH]CC,[CH2]O[O]:CCC([CH2])C,O=O	9.41e+16	3.99e+15	4.48e+17	1
R199*	C5H11 + O2 -> C4H9 + CH2O2	CCC([CH2])C,O=O:C[CH]CC,[CH2]O[O]	0.00e+00	0.00e+00	8.47e+09	0
R200	C5H9 + O2 -> C5H9O2	C[CH]CC=C,O=O:C[C@H]1C[CH]COO1	3.60e+11	1.52e+10	1.71e+12	1
R200*	C5H9O2 -> C5H9 + O2	C[C@H]1C[CH]COO1:C[CH]CC=C,O=O	0.00e+00	0.00e+00	4.10e+11	0
R201	C5H9 -> C5H9	C[CH]CC=C:C[C@H]1C[C@H]1[CH2]	3.25e+09	1.38e+08	1.55e+10	1
R201*	C5H9 -> C5H9	C[C@H]1C[C@H]1[CH2]:C[CH]CC=C	7.52e+10	3.19e+09	3.58e+11	1
R202	C5H11 + O2 -> C3H6 + C2H5O2	C[CH]CCC,O=O:CC=C,CCO[O]	3.21e+08	1.36e+07	1.53e+09	1
R202*	C3H6 + C2H5O2 -> C5H11 + O2	CC=C,CCO[O]:C[CH]CCC,O=O	0.00e+00	0.00e+00	1.28e+11	0
R203	C5H11 + O2 -> C5H10O + HO	C[CH]CCC,O=O:CCCC(=O)C,[OH]	3.21e+08	1.36e+07	1.53e+09	1
R203*	C5H10O + HO -> C5H11 + O2	CCCC(=O)C,[OH]:C[CH]CCC,O=O	0.00e+00	0.00e+00	3.18e+12	0
R204	C5H11 + O2 -> C5H10 + HO2	C[CH]CCC,O=O:CCCC=C,O[O]	6.41e+08	9.73e+07	2.05e+09	2
R204*	C5H10 + HO2 -> C5H11 + O2	CCCC=C,O[O]:C[CH]CCC,O=O	0.00e+00	0.00e+00	5.32e+11	0
R205	C5H11 + O2 -> C5H11O2	C[CH]CCC,O=O:CCCCCO[O]	3.21e+08	1.36e+07	1.53e+09	1
R205*	C5H11O2 -> C5H11 + O2	CCCCCO[O]:C[CH]CCC,O=O	0.00e+00	0.00e+00	2.95e+07	0
R206	C5H11 + O2 -> C5H11O2	C[CH]CCC,O=O:C[C@H](CC)CO[O]	3.21e+08	1.36e+07	1.53e+09	1
R206*	C5H11O2 -> C5H11 + O2	C[C@H](CC)CO[O]:C[CH]CCC,O=O	2.51e+06	1.06e+05	1.19e+07	1
R207	C5H11 + O2 -> C5H11O2	C[CH]CCC,O=O:C[C@H](OO)CC[CH2]	6.41e+08	9.73e+07	2.05e+09	2
R207*	C5H11O2 -> C5H11 + O2	C[C@H](OO)CC[CH2]:C[CH]CCC,O=O	0.00e+00	0.00e+00	9.77e+09	0
R208	C5H11 + O2 -> C5H11O2	C[CH]CCC,O=O:C[C@H](O[O])CCC	5.77e+12	5.68e+12	5.85e+12	17993
R208*	C5H11O2 -> C5H11 + O2	C[C@H](O[O])CCC:C[CH]CCC,O=O	5.50e+10	5.42e+10	5.58e+10	17959
R209	C5H11 -> C3H6 + C2H5	C[CH]CCC:CC=C,C[CH2]	1.11e+08	7.91e+07	1.51e+08	37
R209*	C3H6 + C2H5 -> C5H11	CC=C,C[CH2]:C[CH]CCC	5.07e+11	2.15e+10	2.42e+12	1
R210	C5H11 -> C5H11	C[CH]CCC:CCC(C)[CH2]	2.20e+08	1.73e+08	2.74e+08	73
R210*	C5H11 -> C5H11	CCC(C)[CH2]:C[CH]CCC	1.00e+09	6.96e+08	1.38e+09	33
R211	C5H11 -> C5H11	C[CH]CCC:CCC([CH2])C	2.50e+08	2.00e+08	3.07e+08	83
R211*	C5H11 -> C5H11	CCC([CH2])C:C[CH]CCC	1.09e+09	7.85e+08	1.45e+09	41
R212	C5H11 -> C5H10 + H	C[CH]CCC:CCC=CC,[H]	6.02e+06	9.13e+05	1.93e+07	2
R212*	C5H10 + H -> C5H11	CCC=CC,[H]:C[CH]CCC	0.00e+00	0.00e+00	1.41e+17	0
R213	C5H11 -> C5H10 + H	C[CH]CCC:CCCC=C,[H]	3.01e+06	1.27e+05	1.43e+07	1
R213*	C5H10 + H -> C5H11	CCCC=C,[H]:C[CH]CCC	0.00e+00	0.00e+00	2.01e+16	0
R214	C5H11 -> C5H11	C[CH]CCC:CCCC[CH2]	5.72e+07	3.51e+07	8.70e+07	19
R214*	C5H11 -> C5H11	CCCC[CH2]:C[CH]CCC	1.06e+08	4.47e+06	5.03e+08	1

R215	C5H11 -> C5H11	C[CH]CCC:CC[CH]CC	4.51e+07	2.59e+07	7.22e+07	15
R215*	C5H11 -> C5H11	CC[CH]CC:C[CH]CCC	7.86e+07	3.56e+07	1.47e+08	8
R216	C5H11O2 + O2 -> C5H10O2 + HO2	C[CH]CCCOO,O=O:OOCCCC=C,O[O]	8.02e+10	3.40e+09	3.82e+11	1
R216*	C5H10O2 + HO2 -> C5H11O2 + O2	OOCCCC=C,O[O]:C[CH]CCCOO,O=O	0.00e+00	0.00e+00	2.63e+13	0
R217	C5H11O2 -> C3H6 + C2H5O2	C[CH]CCCOO:CC=C,[CH2]COO	7.39e+08	3.13e+07	3.52e+09	1
R217*	C3H6 + C2H5O2 -> C5H11O2	CC=C,[CH2]COO:C[CH]CCCOO	0.00e+00	0.00e+00	2.35e+16	0
R218	C5H11O2 -> C5H11O2	C[CH]CCCOO:[O]CCC[C@H](O)C	1.48e+09	2.24e+08	4.73e+09	2
R218*	C5H11O2 -> C5H11O2	[O]CCC[C@H](O)C:C[CH]CCCOO	0.00e+00	0.00e+00	1.52e+10	0
R219	C5H10O -> CH2O + C4H8	C[CH]CCC[O]:C=O,CC1CC1	2.70e+11	1.14e+10	1.29e+12	1
R219*	CH2O + C4H8 -> C5H10O	C=O,CC1CC1:C[CH]CCC[O]	0.00e+00	0.00e+00	1.91e+13	0
R220	C4H9O + O2 -> C4H9O3	C[CH]CCO,O=O:OCC[C@H](O[O])C	6.13e+12	5.24e+12	7.11e+12	165
R220*	C4H9O3 -> C4H9O + O2	OCC[C@H](O[O])C:C[CH]CCO,O=O	6.66e+10	5.69e+10	7.73e+10	163
R221	C4H9O -> C4H9O	C[CH]CCO:OCC(C)[CH2]	3.39e+08	1.43e+07	1.61e+09	1
R221*	C4H9O -> C4H9O	OCC(C)[CH2]:C[CH]CCO	1.10e+09	4.66e+07	5.24e+09	1
R222	C4H9O -> C4H9O	C[CH]CCO:OCC([CH2])C	1.02e+09	2.41e+08	2.69e+09	3
R222*	C4H9O -> C4H9O	OCC([CH2])C:C[CH]CCO	7.27e+08	3.08e+07	3.47e+09	1
R223	C4H9O2 + O2 -> C4H9O4	C[CH]CCOO,O=O:OOCC[C@H](O[O])C	2.32e+13	9.85e+11	1.11e+14	1
R223*	C4H9O4 -> C4H9O2 + O2	OOCC[C@H](O[O])C:C[CH]CCOO,O=O	9.26e+10	3.92e+09	4.41e+11	1
R224	C4H9O2 -> C4H9O2	C[CH]CCOO:[O]CC[C@H](O)C	2.17e+11	9.21e+09	1.04e+12	1
R224*	C4H9O2 -> C4H9O2	[O]CC[C@H](O)C:C[CH]CCOO	0.00e+00	0.00e+00	1.80e+11	0
R225	C5H10 + O2 + HO2 -> C5H11O4	C[CH]CC[CH2],O=O,O[O]:[O]OCCC[C@H](OO)C	1.04e+19	4.41e+17	4.96e+19	1
R225*	C5H11O4 -> C5H10 + O2 + HO2	[O]OCCC[C@H](OO)C:C[CH]CC[CH2],O=O,O[O]	0.00e+00	0.00e+00	1.16e+09	0
R226	C3H7O + O2 -> C3H7O3	C[CH]CO,O=O:OC[C@H](O[O])C	7.00e+12	5.92e+12	8.21e+12	144
R226*	C3H7O3 -> C3H7O + O2	OC[C@H](O[O])C:C[CH]CO,O=O	3.54e+10	2.99e+10	4.15e+10	144
R227	C3H7O + O2 -> C3H7O3	C[CH]CO,O=O:[O]C[C@H](OO)C	1.46e+11	3.47e+10	3.87e+11	3
R227*	C3H7O3 -> C3H7O + O2	[O]C[C@H](OO)C:C[CH]CO,O=O	2.32e+10	3.52e+09	7.43e+10	2
R228	C3H7O2 + O2 -> C3H7O4	C[CH]COO,O=O:OOC[C@H](O[O])C	1.64e+13	2.49e+12	5.26e+13	2
R228*	C3H7O4 -> C3H7O2 + O2	OOC[C@H](O[O])C:C[CH]COO,O=O	5.95e+10	2.52e+09	2.84e+11	1
R229	C3H7O2 -> C3H6 + HO2	C[CH]COO:CC=C,O[O]	1.48e+11	2.25e+10	4.74e+11	2
R229*	C3H6 + HO2 -> C3H7O2	CC=C,O[O]:C[CH]COO	0.00e+00	0.00e+00	3.10e+12	0
R230	C2H5O + O2 -> C2H4O + HO2	C[CH]O,O=O:CC=O,O[O]	2.93e+12	1.24e+11	1.40e+13	1
R230*	C2H4O + HO2 -> C2H5O + O2	CC=O,O[O]:C[CH]O,O=O	0.00e+00	0.00e+00	3.64e+12	0
R231	C2H5O + O2 -> C2H5O3	C[CH]O,O=O:[O]O[C@H](O)C	8.79e+12	2.09e+12	2.33e+13	3
R231*	C2H5O3 -> C2H5O + O2	[O]O[C@H](O)C:C[CH]O,O=O	5.26e+10	7.99e+09	1.68e+11	2
R232	C2H4O2 + O2 -> C2H4O4	C[CH]O[O],O=O:[O]OC(O[O])C	1.09e+13	4.61e+11	5.19e+13	1
R232*	C2H4O4 -> C2H4O2 + O2	[O]OC(O[O])C:C[CH]O[O],O=O	3.33e+12	1.41e+11	1.59e+13	1
R233	C2H4O2 -> C2H4O + O	C[CH]O[O]:CC=O,[O]	1.01e+11	4.28e+09	4.81e+11	1
R233*	C2H4O + O -> C2H4O2	CC=O,[O]:C[CH]O[O]	0.00e+00	0.00e+00	2.49e+15	0
R234	C4H9O + O2 -> C4H9O3	C[CH][C@H](O)C,O=O:[O]O[C@H]([C@H](O)C)C	5.32e+12	4.65e+12	6.05e+12	220
R234*	C4H9O3 -> C4H9O + O2	[O]O[C@H]([C@H](O)C)C:C[CH][C@H](O)C,O=O	6.78e+10	5.92e+10	7.72e+10	217

R235	C5H10O2 + H -> C5H11O2	C[C](CC)CO[O],[H]:C[C@H](CC)CO[O]	9.41e+16	1.43e+16	3.01e+17	2
R235*	C5H11O2 -> C5H10O2 + H	C[C@H](CC)CO[O]:C[C](CC)CO[O],[H]	0.00e+00	0.00e+00	7.51e+06	0
R236	H2O + CHO3 -> HO + HO + CHO2	O,[O]OC=O:[OH],[OH],[O]C=O	2.57e+12	1.09e+11	1.22e+13	1
R236*	HO + HO + CHO2 -> H2O + CHO3	[OH],[OH],[O]C=O:O,[O]OC=O	0.00e+00	0.00e+00	1.19e+14	0
R237	C2H4O + O2 -> C2H4O3	O1CC1,O=O:[O]COC[O]	1.83e+11	7.76e+09	8.73e+11	1
R237*	C2H4O3 -> C2H4O + O2	[O]COC[O]:O1CC1,O=O	0.00e+00	0.00e+00	3.00e+13	0
R238	C5H10O2 + HO2 -> C5H9O2 + H2O2	O=CCC[C@H](O)C,O[O]:O=CCC[C@H]([O])C,OO	1.22e+12	5.18e+10	5.82e+12	1
R238*	C5H9O2 + H2O2 -> C5H10O2 + HO2	O=CCC[C@H]([O])C,OO:O=CCC[C@H](O)C,O[O]	1.99e+14	8.43e+12	9.48e+14	1
R239	C5H10O2 + HO2 -> O2 + C5H11O2	O=CCC[C@H](O)C,O[O]:O=O,O[CH]CC[C@H](O)C	1.22e+12	5.18e+10	5.82e+12	1
R239*	O2 + C5H11O2 -> C5H10O2 + HO2	O=O,O[CH]CC[C@H](O)C:O=CCC[C@H](O)C,O[O]	0.00e+00	0.00e+00	1.58e+13	0
R240	C5H10O2 + HO2 -> C5H11O4	O=CCC[C@H](O)C,O[O]:OO[C@H](CC[C@H](O)C)[O]	2.44e+12	3.71e+11	7.82e+12	2
R240*	C5H11O4 -> C5H10O2 + HO2	OO[C@H](CC[C@H](O)C)[O]:O=CCC[C@H](O)C,O[O]	1.75e+11	7.43e+09	8.36e+11	1
R241	C5H9O2 -> C5H9O2	O=CCC[C@H]([O])C:C[C@H]1CC[C@H](O1)[O]	2.11e+10	8.96e+08	1.01e+11	1
R241*	C5H9O2 -> C5H9O2	C[C@H]1CC[C@H](O1)[O]:O=CCC[C@H]([O])C	3.86e+10	1.64e+09	1.84e+11	1
R242	O2 + O2 + CHO -> CHO5	O=O,O=O,[CH]=O:[O]OC(O[O])[O]	1.31e+14	5.55e+12	6.24e+14	1
R242*	CHO5 -> O2 + O2 + CHO	[O]OC(O[O])[O]:O=O,O=O,[CH]=O	7.14e+11	3.03e+10	3.40e+12	1
R243	O2 + C2H4O + CH3 -> CH4O2 + C2H3O	O=O,OC=C,[CH3]:COO,[O]C=C	2.20e+16	9.32e+14	1.05e+17	1
R243*	CH4O2 + C2H3O -> O2 + C2H4O + CH3	COO,[O]C=C:O=O,OC=C,[CH3]	0.00e+00	0.00e+00	3.59e+13	0
R244	O2 + CH2O2 + CHO2 -> CO2 + H2O2 + CHO2	O=O,OC=O,O[C]=O:O=C=O,OO,[O]C=O	7.02e+16	2.97e+15	3.34e+17	1
R244*	CO2 + H2O2 + CHO2 -> O2 + CH2O2 + CHO2	O=C=O,OO,[O]C=O:O=O,OC=O,O[C]=O	0.00e+00	0.00e+00	9.81e+16	0
R245	O2 + CH2O2 + CH3 -> CH4O2 + CHO2	O=O,OC=O,[CH3]:COO,[O]C=O	1.37e+16	5.80e+14	6.52e+16	1
R245*	CH4O2 + CHO2 -> O2 + CH2O2 + CH3	COO,[O]C=O:O=O,OC=O,[CH3]	0.00e+00	0.00e+00	2.00e+12	0
R246	O2 + C4H9O -> C4H9O3	O=O,OCC(C)[CH2]:OCC[C@H](O[O])C	1.21e+11	5.11e+09	5.75e+11	1
R246*	C4H9O3 -> O2 + C4H9O	OCC[C@H](O[O])C:O=O,OCC(C)[CH2]	0.00e+00	0.00e+00	1.22e+09	0
R247	O2 + C4H9O -> C4H9O3	O=O,OCC(C)[CH2]:[O]OC[C@H](CO)C	1.18e+13	9.64e+12	1.43e+13	98
R247*	C4H9O3 -> O2 + C4H9O	[O]OC[C@H](CO)C:O=O,OCC(C)[CH2]	7.52e+09	6.12e+09	9.13e+09	96
R248	O2 + C4H9O2 -> C4H9O4	O=O,OCC[C](O)C:OCC[C@](O[O])(O)C	4.18e+12	1.77e+11	1.99e+13	1
R248*	C4H9O4 -> O2 + C4H9O2	OCC[C@](O[O])(O)C:O=O,OCC[C](O)C	0.00e+00	0.00e+00	6.36e+10	0
R249	O2 + C3H7O2 -> C3H7O4	O=O,OOC([CH2])C:[O]OC[C@H](OO)C	1.33e+13	2.02e+12	4.27e+13	2
R249*	C3H7O4 -> O2 + C3H7O2	[O]OC[C@H](OO)C:O=O,OOC([CH2])C	1.48e+10	2.25e+09	4.75e+10	2
R250	O2 + C5H11O2 -> C5H11O4	O=O,OOCCC(C)[CH2]:OOCC[C@H](CO[O])C	1.08e+13	4.91e+12	2.03e+13	8
R250*	C5H11O4 -> O2 + C5H11O2	OOCC[C@H](CO[O])C:O=O,OOCCC(C)[CH2]	3.21e+09	1.45e+09	6.01e+09	8
R251	O2 + C5H11O2 -> C5H10O4 + H	O=O,OOCCC([CH2])C:OOCC[C](CO[O])C,[H]	3.16e+11	1.34e+10	1.50e+12	1
R251*	C5H10O4 + H -> O2 + C5H11O2	OOCC[C](CO[O])C,[H]:O=O,OOCCC([CH2])C	0.00e+00	0.00e+00	2.82e+17	0
R252	O2 + C5H11O2 -> C5H11O4	O=O,O[CH]CC[C@H](O)C:OO[C@H](CC[C@H](O)C)[O]	5.27e+12	2.23e+11	2.51e+13	1
R252*	C5H11O4 -> O2 + C5H11O2	OO[C@H](CC[C@H](O)C)[O]:O=O,O[CH]CC[C@H](O)C	0.00e+00	0.00e+00	5.26e+11	0
R253	O2 + C5H11O2 -> C5H11O4	O=O,O[CH]CC[C@H](O)C:[O]O[C@H](CC[C@H](O)C)O	1.58e+13	3.76e+12	4.19e+13	3
R253*	C5H11O4 -> O2 + C5H11O2	[O]O[C@H](CC[C@H](O)C)O:O=O,O[CH]CC[C@H](O)C	6.04e+10	9.17e+09	1.93e+11	2
R254	O2 + C4H9O3 -> C4H9O5	O=O,O[CH]C[C@H](OO)C:OO[C@H](C[C@H](O[O])O)C	1.63e+13	6.89e+11	7.75e+13	1
R254*	C4H9O5 -> O2 + C4H9O3	OO[C@H](C[C@H](O[O])O)C:O=O,O[CH]C[C@H](OO)C	0.00e+00	0.00e+00	1.11e+12	0

R255	O2 + HO2 + C5H10 -> C5H11O4	O=O,O[O],[CH2]CC([CH2])C:[O]OCC[C@H](COO)C	1.04e+19	4.41e+17	4.96e+19	1
R255*	C5H11O4 -> O2 + HO2 + C5H10	[O]OCC[C@H](COO)C:O=O,O[O],[CH2]CC([CH2])C	0.00e+00	0.00e+00	8.69e+08	0
R256	O2 + C2H5O2 -> C2H5O4	O=O,[CH2]C(O)O:OOCC([O])O	4.85e+11	2.05e+10	2.31e+12	1
R256*	C2H5O4 -> O2 + C2H5O2	OOCC([O])O:O=O,[CH2]C(O)O	0.00e+00	0.00e+00	1.22e+11	0
R257	O2 + C4H7O -> C4H7O3	O=O,[CH2]CCC=O:[O][C@H]1CCCOO1	1.77e+11	2.68e+10	5.66e+11	2
R257*	C4H7O3 -> O2 + C4H7O	[O][C@H]1CCCOO1:O=O,[CH2]CCC=O	6.17e+09	2.61e+08	2.94e+10	1
R258	O2 + C3H7O -> C3H6O + HO2	O=O,[CH2]CCO:OCC=C,O[O]	2.13e+11	9.02e+09	1.01e+12	1
R258*	C3H6O + HO2 -> O2 + C3H7O	OCC=C,O[O]:O=O,[CH2]CCO	0.00e+00	0.00e+00	4.63e+13	0
R259	O2 + C3H7O -> C3H7O3	O=O,[CH2]CCO:OCCCO[O]	1.66e+13	1.32e+13	2.06e+13	78
R259*	C3H7O3 -> O2 + C3H7O	OCCCO[O]:O=O,[CH2]CCO	1.15e+10	9.10e+09	1.42e+10	78
R260	O2 + C3H6O2 -> C3H6O4	O=O,[CH2]CCO[O]:[O]OCCCO[O]	7.18e+12	3.04e+11	3.42e+13	1
R260*	C3H6O4 -> O2 + C3H6O2	[O]OCCCO[O]:O=O,[CH2]CCO[O]	1.93e+10	8.16e+08	9.18e+10	1
R261	O2 + C5H10O -> C2H4O + C3H6O2	O=O,[CH2]CC[C@H]([O])C:CC=O,[CH2]CCO[O]	2.63e+13	1.11e+12	1.25e+14	1
R261*	C2H4O + C3H6O2 -> O2 + C5H10O	CC=O,[CH2]CCO[O]:O=O,[CH2]CC[C@H]([O])C	0.00e+00	0.00e+00	1.90e+15	0
R262	O2 + C5H10O -> C5H10O3	O=O,[CH2]CC[C@H]([O])C:[O]OCCC[C@H]([O])C	5.26e+13	7.98e+12	1.68e+14	2
R262*	C5H10O3 -> O2 + C5H10O	[O]OCCC[C@H]([O])C:O=O,[CH2]CC[C@H]([O])C	6.67e+09	1.01e+09	2.13e+10	2
R263	O2 + C2H5O -> C2H5O3	O=O,[CH2]CO:OCCO[O]	1.28e+13	1.01e+13	1.61e+13	70
R263*	C2H5O3 -> O2 + C2H5O	OCCO[O]:O=O,[CH2]CO	1.22e+10	9.52e+09	1.53e+10	69
R264	O2 + C2H5O -> C2H5O3	O=O,[CH2]CO:[O]CCOO	1.83e+11	7.77e+09	8.74e+11	1
R264*	C2H5O3 -> O2 + C2H5O	[O]CCOO:O=O,[CH2]CO	0.00e+00	0.00e+00	4.94e+10	0
R265	O2 + C2H5O2 -> C2H5O4	O=O,[CH2]COO:OOCCO[O]	8.40e+13	3.56e+12	4.00e+14	1
R265*	C2H5O4 -> O2 + C2H5O2	OOCCO[O]:O=O,[CH2]COO	6.92e+09	2.93e+08	3.30e+10	1
R266	O2 + C4H9O -> C4H9O3	O=O,[CH2]C[C@H](O)C:[O]OCC[C@H](O)C	1.63e+13	1.34e+13	1.97e+13	104
R266*	C4H9O3 -> O2 + C4H9O	[O]OCC[C@H](O)C:O=O,[CH2]C[C@H](O)C	1.41e+10	1.15e+10	1.70e+10	102
R267	O2 + C3H6 -> C3H6O2	O=O,[CH2]C[CH2]:[CH2]CCO[O]	1.11e+14	4.69e+12	5.27e+14	1
R267*	C3H6O2 -> O2 + C3H6	[CH2]CCO[O]:O=O,[CH2]C[CH2]	0.00e+00	0.00e+00	2.02e+11	0
R268	O2 + C5H9O -> C5H9O3	O=O,[CH2][C@H](C=O)CC:CC[C@H](C=O)CO[O]	1.11e+13	5.51e+12	1.95e+13	10
R268*	C5H9O3 -> O2 + C5H9O	CC[C@H](C=O)CO[O]:O=O,[CH2][C@H](C=O)CC	2.31e+10	1.15e+10	4.08e+10	10
R269	O2 + C5H11O2 -> C5H11O4	O=O,[CH2][C@H](CC)COO:OOCC(CO[O])CC	1.22e+13	8.26e+12	1.72e+13	29
R269*	C5H11O4 -> O2 + C5H11O2	OOCC(CO[O])CC:O=O,[CH2][C@H](CC)COO	1.03e+10	6.87e+09	1.47e+10	27
R270	O2 + C4H6O2 -> C4H6O4	O=O,[CH2][C@H](CC=O)[O]:[O]OC[C@H](CC=O)[O]	7.00e+13	2.97e+12	3.34e+14	1
R270*	C4H6O4 -> O2 + C4H6O2	[O]OC[C@H](CC=O)[O]:O=O,[CH2][C@H](CC=O)[O]	0.00e+00	0.00e+00	3.62e+10	0
R271	O2 + CH3 -> CH3O2	O=O,[CH3]:CO[O]	1.50e+13	1.24e+13	1.80e+13	112
R271*	CH3O2 -> O2 + CH3	CO[O]:O=O,[CH3]	1.10e+09	8.94e+08	1.34e+09	94
R272	O2 + CHO -> CHO3	O=O,[CH]=O:[O]OC=O	3.71e+13	2.55e+13	5.18e+13	31
R272*	CHO3 -> O2 + CHO	[O]OC=O:O=O,[CH]=O	7.91e+09	5.36e+09	1.12e+10	29
R273	O2 + H -> HO2	O=O,[H]:O[O]	1.29e+14	3.89e+13	3.04e+14	4
R273*	HO2 -> O2 + H	O[O]:O=O,[H]	2.24e+07	3.39e+06	7.16e+07	2
R274	O2 + C2H3O -> C2H3O3	O=O,[O]C=C:[O]OCC=O	4.04e+12	2.77e+12	5.63e+12	31
R274*	C2H3O3 -> O2 + C2H3O	[O]OCC=O:O=O,[O]C=C	4.69e+10	3.22e+10	6.54e+10	31

R275	O2 + CH2O2 -> HO2 + CHO2	O=O,[O]C[O]:O[O],[O]C=O	2.70e+14	1.15e+13	1.29e+15	1
R275*	HO2 + CHO2 -> O2 + CH2O2	O[O],[O]C=O:O=O,[O]C[O]	0.00e+00	0.00e+00	1.33e+14	0
R276	CH2O2 + HO2 -> H2O2 + CHO2	OC=O,O[O]:OO,[O]C=O	5.99e+13	2.54e+12	2.85e+14	1
R276*	H2O2 + CHO2 -> CH2O2 + HO2	OO,[O]C=O:OC=O,O[O]	0.00e+00	0.00e+00	5.04e+12	0
R277	CH2O2 + HO -> H2O + CHO2	OC=O,[OH]:O,[O]C=O	1.34e+14	5.68e+12	6.39e+14	1
R277*	H2O + CHO2 -> CH2O2 + HO	O,[O]C=O:OC=O,[OH]	0.00e+00	0.00e+00	2.77e+12	0
R278	C4H9O -> C4H8O + H	OCC([CH2])C:CC(=C)CO,[H]	1.45e+09	2.21e+08	4.66e+09	2
R278*	C4H8O + H -> C4H9O	CC(=C)CO,[H]:OCC([CH2])C	0.00e+00	0.00e+00	1.41e+17	0
R279	C4H9O -> C3H6 + CH3O	OCC([CH2])C:CC=C,[CH2]O	2.18e+09	5.18e+08	5.78e+09	3
R279*	C3H6 + CH3O -> C4H9O	CC=C,[CH2]O:OCC([CH2])C	0.00e+00	0.00e+00	1.52e+15	0
R280	C3H6O + HO2 -> C3H7O3	OCC=C,O[O]:OC[CH]COO	1.55e+13	6.55e+11	7.37e+13	1
R280*	C3H7O3 -> C3H6O + HO2	OC[CH]COO:OCC=C,O[O]	7.69e+11	3.26e+10	3.67e+12	1
R281	C3H6O + CH3 -> C4H9O	OCC=C,[CH3]:CC[CH]CO	2.35e+16	9.97e+14	1.12e+17	1
R281*	C4H9O -> C3H6O + CH3	CC[CH]CO:OCC=C,[CH3]	0.00e+00	0.00e+00	1.76e+09	0
R282	C3H6O + CH3 -> C4H9O	OCC=C,[CH3]:OCC([CH2])C	4.70e+16	7.14e+15	1.51e+17	2
R282*	C4H9O -> C3H6O + CH3	OCC([CH2])C:OCC=C,[CH3]	7.27e+08	3.08e+07	3.47e+09	1
R283	C5H11O2 -> C2H4O + C3H7O	OCCCO[CH]C:CC=O,[CH2]CCO	2.50e+12	1.06e+11	1.19e+13	1
R283*	C2H4O + C3H7O -> C5H11O2	CC=O,[CH2]CCO:OCCCO[CH]C	0.00e+00	0.00e+00	5.42e+13	0
R284	C5H11O2 -> C4H8O2 + CH3	OCCC[C@H]([O])C:OCCCC=O,[CH3]	5.32e+09	8.08e+08	1.70e+10	2
R284*	C4H8O2 + CH3 -> C5H11O2	OCCCC=O,[CH3]:OCCC[C@H]([O])C	0.00e+00	0.00e+00	3.31e+14	0
R285	C5H11O2 -> C5H11O2	OCCC[C@H]([O])C:OCCCO[CH]C	2.66e+09	1.13e+08	1.27e+10	1
R285*	C5H11O2 -> C5H11O2	OCCCO[CH]C:OCCC[C@H]([O])C	0.00e+00	0.00e+00	7.49e+12	0
R286	C5H11O2 -> C5H11O2	OCCC[C@H]([O])C:O[CH]CC[C@H](O)C	2.66e+09	1.13e+08	1.27e+10	1
R286*	C5H11O2 -> C5H11O2	O[CH]CC[C@H](O)C:OCCC[C@H]([O])C	0.00e+00	0.00e+00	1.43e+11	0
R287	C5H11O2 -> C3H6O + C2H5O	OCC[C@H](CC)[O]:CCC=O,[CH2]CO	1.49e+10	2.26e+09	4.77e+10	2
R287*	C3H6O + C2H5O -> C5H11O2	CCC=O,[CH2]CO:OCC[C@H](CC)[O]	0.00e+00	0.00e+00	4.77e+13	0
R288	C5H11O2 -> C2H5 + C3H6O2	OCC[C@H](CC)[O]:C[CH2],OCCC=O	7.45e+09	3.16e+08	3.55e+10	1
R288*	C2H5 + C3H6O2 -> C5H11O2	C[CH2],OCCC=O:OCC[C@H](CC)[O]	0.00e+00	0.00e+00	2.11e+14	0
R289	C5H11O2 -> CH2O + C4H9O	OCC[C@H](C[O])C:C=O,C[CH]CCO	2.22e+11	5.28e+10	5.89e+11	3
R289*	CH2O + C4H9O -> C5H11O2	C=O,C[CH]CCO:OCC[C@H](C[O])C	0.00e+00	0.00e+00	9.55e+12	0
R290	C5H11O2 -> C5H11O2	OCC[C@H](C[O])C:[O]CC[C@H](CO)C	7.41e+10	3.14e+09	3.53e+11	1
R290*	C5H11O2 -> C5H11O2	[O]CC[C@H](CO)C:OCC[C@H](C[O])C	0.00e+00	0.00e+00	4.94e+10	0
R291	C4H9O3 -> C4H9O3	OCC[C@H](O[O])C:O[CH]C[C@H](OO)C	4.08e+08	1.73e+07	1.95e+09	1
R291*	C4H9O3 -> C4H9O3	O[CH]C[C@H](OO)C:OCC[C@H](O[O])C	0.00e+00	0.00e+00	4.61e+11	0
R292	C4H9O3 -> C4H9O3	OCC[C@H](O[O])C:[O]CC[C@H](OO)C	4.08e+08	1.73e+07	1.95e+09	1
R292*	C4H9O3 -> C4H9O3	[O]CC[C@H](OO)C:OCC[C@H](O[O])C	0.00e+00	0.00e+00	2.36e+11	0
R293	C4H9O2 -> C4H9O2	OCC[C@H]([O])C:OCC[C](O)C	1.49e+11	6.32e+09	7.11e+11	1
R293*	C4H9O2 -> C4H9O2	OCC[C](O)C:OCC[C@H]([O])C	0.00e+00	0.00e+00	1.17e+11	0
R294	C4H9O4 -> C4H8O2 + HO2	OCC[C@](OO)([O])C:CC(=O)CCO,O[O]	1.62e+10	6.85e+08	7.71e+10	1
R294*	C4H8O2 + HO2 -> C4H9O4	CC(=O)CCO,O[O]:OCC[C@](OO)([O])C	0.00e+00	0.00e+00	2.75e+14	0

R295	C4H9O4 -> C4H9O4	OCC[C@](O[O])(O)C:OCC[C@](OO)([O])C	8.49e+10	2.56e+10	2.00e+11	4
R295*	C4H9O4 -> C4H9O4	OCC[C@](OO)([O])C:OCC[C@](O[O])(O)C	4.85e+10	1.15e+10	1.29e+11	3
R296	C3H7O3 -> C3H7O3	OC[C@H](O[O])C:[O]C[C@H](OO)C	7.38e+08	1.75e+08	1.95e+09	3
R296*	C3H7O3 -> C3H7O3	[O]C[C@H](OO)C:OC[C@H](O[O])C	3.48e+10	8.27e+09	9.22e+10	3
R297	CH3O2 -> H2O + CHO	OC[O]:O,[CH]=O	1.03e+08	4.36e+06	4.91e+08	1
R297*	H2O + CHO -> CH3O2	O,[CH]=O:OC[O]	0.00e+00	0.00e+00	3.12e+14	0
R298	CH3O2 -> CH2O2 + H	OC[O]:OC=O,[H]	1.03e+08	4.36e+06	4.91e+08	1
R298*	CH2O2 + H -> CH3O2	OC=O,[H]:OC[O]	0.00e+00	0.00e+00	2.82e+17	0
R299	C5H11O4 -> C5H11O4	OOCC(CO[O])CC:OO[CH]C(COO)CC	3.81e+08	1.61e+07	1.81e+09	1
R299*	C5H11O4 -> C5H11O4	OO[CH]C(COO)CC:OOCC(CO[O])CC	0.00e+00	0.00e+00	3.00e+13	0
R300	C5H11O4 -> HO + C5H10O3	OOCC(CO[O])CC:[OH],[O]OC[C@H](C[O])CC	3.81e+08	1.61e+07	1.81e+09	1
R300*	HO + C5H10O3 -> C5H11O4	[OH],[O]OC[C@H](C[O])CC:OOCC(CO[O])CC	0.00e+00	0.00e+00	2.82e+16	0
R301	C2H5O4 -> C2H5O4	OOCC(O)[O]:[O]OCC(O)O	6.91e+10	2.94e+10	1.35e+11	7
R301*	C2H5O4 -> C2H5O4	[O]OCC(O)O:OOCC(O)[O]	2.18e+09	8.56e+08	4.46e+09	6
R302	C2H5O4 -> C2H5O4	OOCC([O])O:[O]OCC(O)O	8.16e+10	1.24e+10	2.61e+11	2
R302*	C2H5O4 -> C2H5O4	[O]OCC(O)O:OOCC([O])O	3.64e+08	1.54e+07	1.73e+09	1
R303	C2H4O3 + C2H3O -> C2H4O + C2H3O3	OOCC=O,[O]C=C:OC=C,[O]OCC=O	1.94e+14	8.24e+12	9.26e+14	1
R303*	C2H4O + C2H3O3 -> C2H4O3 + C2H3O	OC=C,[O]OCC=O:OOCC=O,[O]C=C	0.00e+00	0.00e+00	6.09e+13	0
R304	C5H11O2 -> C5H11O2	OOCCC(C)[CH2]:[O]CC[C@H](CO)C	1.28e+10	5.43e+08	6.11e+10	1
R304*	C5H11O2 -> C5H11O2	[O]CC[C@H](CO)C:OOCCC(C)[CH2]	0.00e+00	0.00e+00	4.94e+10	0
R305	C5H11O2 -> C5H11O2	OOCCC([CH2])C:[O]CC[C@H](CO)C	2.85e+09	1.21e+08	1.36e+10	1
R305*	C5H11O2 -> C5H11O2	[O]CC[C@H](CO)C:OOCCC([CH2])C	0.00e+00	0.00e+00	4.94e+10	0
R306	C5H10O2 + HO2 -> O2 + H2O2 + C5H9	OOCCCC=C,O[O]:O=O,OO,[CH2]CCC=C	1.75e+13	2.66e+12	5.62e+13	2
R306*	O2 + H2O2 + C5H9 -> C5H10O2 + HO2	O=O,OO,[CH2]CCC=C:OOCCCC=C,O[O]	4.56e+15	1.93e+14	2.17e+16	1
R307	C5H10O2 + HO2 -> H2O2 + C5H9O2	OOCCCC=C,O[O]:OO,[O]OCCCC=C	1.75e+13	2.66e+12	5.62e+13	2
R307*	H2O2 + C5H9O2 -> C5H10O2 + HO2	OO,[O]OCCCC=C:OOCCCC=C,O[O]	1.23e+13	1.87e+12	3.94e+13	2
R308	C5H11O4 -> C5H11O2 + O2	OOCCC[C@H](O[O])C:C[C@H](OO)CC[CH2],O=O	1.02e+09	4.31e+07	4.85e+09	1
R308*	C5H11O2 + O2 -> C5H11O4	C[C@H](OO)CC[CH2],O=O:OOCCC[C@H](O[O])C	0.00e+00	0.00e+00	1.06e+12	0
R309	C5H11O4 -> C5H11O2 + O2	OOCCC[C@H](O[O])C:C[CH]CCCOO,O=O	5.80e+10	4.42e+10	7.45e+10	57
R309*	C5H11O2 + O2 -> C5H11O4	C[CH]CCCOO,O=O:OOCCC[C@H](O[O])C	4.57e+12	3.48e+12	5.86e+12	57
R310	C5H11O4 -> C5H11O4	OOCCC[C@H](O[O])C:OOC[CH]C[C@H](OO)C	1.02e+09	4.31e+07	4.85e+09	1
R310*	C5H11O4 -> C5H11O4	OOC[CH]C[C@H](OO)C:OOCCC[C@H](O[O])C	0.00e+00	0.00e+00	5.99e+12	0
R311	C5H11O4 -> HO + C5H10O3	OOCCC[C@H](O[O])C:[OH],[O]CCC[C@H](O[O])C	1.02e+09	4.31e+07	4.85e+09	1
R311*	HO + C5H10O3 -> C5H11O4	[OH],[O]CCC[C@H](O[O])C:OOCCC[C@H](O[O])C	0.00e+00	0.00e+00	9.40e+15	0
R312	C5H11O4 -> C4H9O2 + CH2O2	OOCC[C@H](C([O])O)C:C[CH]CCOO,OC=O	3.33e+11	1.41e+10	1.59e+12	1
R312*	C4H9O2 + CH2O2 -> C5H11O4	C[CH]CCOO,OC=O:OOCC[C@H](C([O])O)C	0.00e+00	0.00e+00	6.13e+15	0
R313	C5H11O4 -> O2 + C5H11O2	OOCC[C@H](CO[O])C:O=O,OOCCC([CH2])C	1.65e+10	1.19e+10	2.20e+10	41
R313*	O2 + C5H11O2 -> C5H11O4	O=O,OOCCC([CH2])C:OOCC[C@H](CO[O])C	1.26e+13	9.10e+12	1.70e+13	40
R314	C5H11O4 -> C5H11O4	OOCC[C@H](O[O])CC:[O]OCC[C@H](OO)CC	7.92e+09	3.36e+08	3.78e+10	1
R314*	C5H11O4 -> C5H11O4	[O]OCC[C@H](OO)CC:OOCC[C@H](O[O])CC	0.00e+00	0.00e+00	2.21e+10	0

R315	C4H9O3 -> C2H4O + C2H5O2	OOCC[C@H]([O])C:CC=O,[CH2]COO	1.24e+10	5.27e+08	5.93e+10	1
R315*	C2H4O + C2H5O2 -> C4H9O3	CC=O,[CH2]COO:OOCC[C@H]([O])C	0.00e+00	0.00e+00	2.82e+17	0
R316	C4H9O3 -> O2 + C4H9O	OOCC[C@H]([O])C:O=O,[CH2]C[C@H](O)C	1.24e+10	5.27e+08	5.93e+10	1
R316*	O2 + C4H9O -> C4H9O3	O=O,[CH2]C[C@H](O)C:OOCC[C@H]([O])C	1.57e+11	6.66e+09	7.49e+11	1
R317	C5H10O4 + H -> C5H11O4	OOCC[C](CO[O])C,[H]:OOCC[C@H](CO[O])C	9.41e+16	3.99e+15	4.48e+17	1
R317*	C5H11O4 -> C5H10O4 + H	OOCC[C@H](CO[O])C:OOCC[C](CO[O])C,[H]	0.00e+00	0.00e+00	1.20e+09	0
R318	C4H9O3 -> CH2O + C3H7O2	OOC[C@H](C[O])C:C=O,C[CH]COO	4.66e+10	7.07e+09	1.49e+11	2
R318*	CH2O + C3H7O2 -> C4H9O3	C=O,C[CH]COO:OOC[C@H](C[O])C	0.00e+00	0.00e+00	1.04e+15	0
R319	C3H7O4 -> C3H7O4	OOC[C@H](O[O])C:[O]OC[C@H](OO)C	5.95e+10	2.52e+09	2.84e+11	1
R319*	C3H7O4 -> C3H7O4	[O]OC[C@H](OO)C:OOC[C@H](O[O])C	0.00e+00	0.00e+00	2.22e+10	0
R320	C3H7O3 -> C3H7O + O2	OOC[C@H]([O])C:CC(O)[CH2],O=O	3.63e+10	5.51e+09	1.16e+11	2
R320*	C3H7O + O2 -> C3H7O3	CC(O)[CH2],O=O:OOC[C@H]([O])C	0.00e+00	0.00e+00	1.80e+11	0
R321	C5H11O4 -> C5H10O2 + HO2	OOC[CH]C[C@H](OO)C:C[C@H](OO)CC=C,O[O]	2.00e+12	8.47e+10	9.53e+12	1
R321*	C5H10O2 + HO2 -> C5H11O4	C[C@H](OO)CC=C,O[O]:OOC[CH]C[C@H](OO)C	0.00e+00	0.00e+00	1.01e+14	0
R322	CH3O3 -> CH3O3	OOC[O]:[O]OCO	2.63e+11	3.99e+10	8.42e+11	2
R322*	CH3O3 -> CH3O3	[O]OCO:OOC[O]	9.46e+09	4.01e+08	4.51e+10	1
R323	C5H11O2 -> C5H10 + HO2	OO[C@H](C([CH2])C)C:C[C@H]1C[C@H]1C,O[O]	1.11e+11	4.71e+09	5.29e+11	1
R323*	C5H10 + HO2 -> C5H11O2	C[C@H]1C[C@H]1C,O[O]:OO[C@H](C([CH2])C)C	0.00e+00	0.00e+00	9.46e+12	0
R324	C5H11O4 -> C5H11O4	OO[C@H](CC[C@H](O)C)[O]:OO[C@H](CC[C@H]([O])C)O	1.75e+11	7.43e+09	8.36e+11	1
R324*	C5H11O4 -> C5H11O4	OO[C@H](CC[C@H]([O])C)O:OO[C@H](CC[C@H](O)C)[O]	0.00e+00	0.00e+00	3.26e+11	0
R325	C5H11O4 -> C5H11O4	OO[C@H](CC[C@H](O)C)[O]:[O]O[C@H](CC[C@H](O)C)O	3.51e+11	5.32e+10	1.12e+12	2
R325*	C5H11O4 -> C5H11O4	[O]O[C@H](CC[C@H](O)C)O:OO[C@H](CC[C@H](O)C)[O]	3.02e+10	1.28e+09	1.44e+11	1
R326	C5H11O4 -> C5H10O2 + HO2	OO[C@H](CC[C@H]([O])C)O:O=CCC[C@H](O)C,O[O]	1.09e+11	4.60e+09	5.18e+11	1
R326*	C5H10O2 + HO2 -> C5H11O4	O=CCC[C@H](O)C,O[O]:OO[C@H](CC[C@H]([O])C)O	0.00e+00	0.00e+00	3.66e+12	0
R327	C4H9O5 -> C4H8O3 + HO2	OO[C@H](C[C@H](O[O])O)C:C[C@H](OO)CC=O,O[O]	3.70e+11	1.57e+10	1.76e+12	1
R327*	C4H8O3 + HO2 -> C4H9O5	C[C@H](OO)CC=O,O[O]:OO[C@H](C[C@H](O[O])O)C	0.00e+00	0.00e+00	3.90e+14	0
R328	C4H9O3 -> C4H8O2 + HO	OO[C@H]([C@H]([O])C)C:C[C@H]([C@H]([O])C)[O],[OH]	2.38e+10	3.61e+09	7.61e+10	2
R328*	C4H8O2 + HO -> C4H9O3	C[C@H]([C@H]([O])C)[O],[OH]:OO[C@H]([C@H]([O])C)C	0.00e+00	0.00e+00	1.41e+17	0
R329	C4H9O3 -> C4H9O + O2	OO[C@H]([C@H]([O])C)C:C[CH][C@H](O)C,O=O	2.38e+10	3.61e+09	7.61e+10	2
R329*	C4H9O + O2 -> C4H9O3	C[CH][C@H](O)C,O=O:OO[C@H]([C@H]([O])C)C	2.42e+10	1.02e+09	1.15e+11	1
R330	C2H5O3 -> C2H4O + HO2	OO[C@H]([O])C:CC=O,O[O]	9.43e+10	1.43e+10	3.02e+11	2
R330*	C2H4O + HO2 -> C2H5O3	CC=O,O[O]:OO[C@H]([O])C	2.43e+12	3.69e+11	7.77e+12	2
R331	C2H5O3 -> C2H5O + O2	OO[C@H]([O])C:C[CH]O,O=O	1.42e+11	3.36e+10	3.75e+11	3
R331*	C2H5O + O2 -> C2H5O3	C[CH]O,O=O:OO[C@H]([O])C	2.93e+12	1.24e+11	1.40e+13	1
R332	C5H11O4 -> C5H10O3 + HO	OO[CH]C(COO)CC:CC[C@H](C=O)COO,[OH]	1.00e+13	4.24e+11	4.77e+13	1
R332*	C5H10O3 + HO -> C5H11O4	CC[C@H](C=O)COO,[OH]:OO[CH]C(COO)CC	0.00e+00	0.00e+00	2.99e+14	0
R333	CHO2 -> CO + HO	O[C]=O:[C]=O,[OH]	6.29e+10	2.66e+09	3.00e+11	1
R333*	CO + HO -> CHO2	[C]=O,[OH]:O[C]=O	0.00e+00	0.00e+00	2.82e+17	0
R334	HO2 + CHO2 -> O2 + CH2O2	O[O],[O]C=O:O=O,OC=O	4.42e+13	1.87e+12	2.11e+14	1
R334*	O2 + CH2O2 -> HO2 + CHO2	O=O,OC=O:O[O],[O]C=O	0.00e+00	0.00e+00	7.71e+10	0

R335	HO2 + C2H4O4 -> O2 + C2H5O4	O[O],[O]OCC(O)[O]:O=O,OOCC(O)[O]	8.96e+14	3.80e+13	4.27e+15	1
R335*	O2 + C2H5O4 -> HO2 + C2H4O4	O=O,OOCC(O)[O]:O[O],[O]OCC(O)[O]	0.00e+00	0.00e+00	3.16e+12	0
R336	HO2 + C2H3O3 -> O2 + C2H4O3	O[O],[O]OCC=O:O=O,OOCC=O	2.09e+14	8.86e+12	9.96e+14	1
R336*	O2 + C2H4O3 -> HO2 + C2H3O3	O=O,OOCC=O:O[O],[O]OCC=O	0.00e+00	0.00e+00	5.02e+11	0
R337	C4H7O -> C4H7O	[CH2]CCC=O:C1CC[CH]O1	3.38e+09	1.02e+09	7.98e+09	4
R337*	C4H7O -> C4H7O	C1CC[CH]O1:[CH2]CCC=O	2.03e+11	6.13e+10	4.79e+11	4
R338	C3H6O2 -> C3H6 + O2	[CH2]CCO[O]:C1CC1,O=O	6.76e+10	2.86e+09	3.22e+11	1
R338*	C3H6 + O2 -> C3H6O2	C1CC1,O=O:[CH2]CCO[O]	0.00e+00	0.00e+00	6.59e+10	0
R339	C3H6O2 -> C3H6O2	[CH2]CCO[O]:C1COOC1	6.76e+10	2.86e+09	3.22e+11	1
R339*	C3H6O2 -> C3H6O2	C1COOC1:[CH2]CCO[O]	0.00e+00	0.00e+00	8.34e+08	0
R340	C2H5O2 -> C2H4 + HO2	[CH2]COO:C=C,O[O]	3.08e+12	9.28e+11	7.25e+12	4
R340*	C2H4 + HO2 -> C2H5O2	C=C,O[O]:[CH2]COO	7.40e+11	3.13e+10	3.52e+12	1
R341	C3H6O2 -> CH2O + C2H4O	[CH2]OCC[O]:C=O,O1CC1	1.00e+13	4.24e+11	4.77e+13	1
R341*	CH2O + C2H4O -> C3H6O2	C=O,O1CC1:[CH2]OCC[O]	0.00e+00	0.00e+00	4.83e+13	0
R342	C5H9O -> C5H9O	[CH2][C@H](C=O)CC:CC[C@H]1C[C@H]1[O]	6.99e+10	2.97e+10	1.36e+11	7
R342*	C5H9O -> C5H9O	CC[C@H]1C[C@H]1[O]:[CH2][C@H](C=O)CC	3.33e+11	1.01e+11	7.86e+11	4
R343	C5H11O2 -> C5H11O2	[CH2][C@H](CC)COO:CCC(C[O])CO	3.80e+09	1.61e+08	1.81e+10	1
R343*	C5H11O2 -> C5H11O2	CCC(C[O])CO:[CH2][C@H](CC)COO	0.00e+00	0.00e+00	3.00e+13	0
R344	C5H11O2 -> C5H10 + HO2	[CH2][C@H](CC)COO:CCC1CC1,O[O]	3.80e+09	1.61e+08	1.81e+10	1
R344*	C5H10 + HO2 -> C5H11O2	CCC1CC1,O[O]:[CH2][C@H](CC)COO	0.00e+00	0.00e+00	2.82e+16	0
R345	C5H11O2 -> C5H11O2	[CH2][C@H](CC)COO:CC[CH]CCOO	3.80e+09	1.61e+08	1.81e+10	1
R345*	C5H11O2 -> C5H11O2	CC[CH]CCOO:[CH2][C@H](CC)COO	0.00e+00	0.00e+00	4.43e+09	0
R346	C5H11O2 -> C5H10 + HO2	[CH2][C@H](OO)CCC:CCCC=C,O[O]	1.86e+11	4.43e+10	4.94e+11	3
R346*	C5H10 + HO2 -> C5H11O2	CCCC=C,O[O]:[CH2][C@H](OO)CCC	0.00e+00	0.00e+00	5.32e+11	0
R347	CH3 + CHO2 -> C2H4O2	[CH3],[O]C=O:COC=O	1.06e+14	4.47e+12	5.03e+14	1
R347*	C2H4O2 -> CH3 + CHO2	COC=O:[CH3],[O]C=O	0.00e+00	0.00e+00	1.75e+11	0
R348	CO + HO -> CHO2	[C]=O,[OH]:[O]C=O	9.41e+16	3.99e+15	4.48e+17	1
R348*	CHO2 -> CO + HO	[O]C=O:[C]=O,[OH]	0.00e+00	0.00e+00	1.06e+08	0
R349	HO + HO -> H2O + O	[OH],[OH]:O,[O]	6.59e+13	2.79e+12	3.14e+14	1
R349*	H2O + O -> HO + HO	O,[O]:[OH],[OH]	1.65e+15	6.99e+13	7.87e+15	1
R350	HO + C2H3O3 -> C2H4O4	[OH],[O]OCC=O:[O]OCC(O)[O]	9.41e+16	3.99e+15	4.48e+17	1
R350*	C2H4O4 -> HO + C2H3O3	[O]OCC(O)[O]:[OH],[O]OCC=O	0.00e+00	0.00e+00	2.85e+11	0
R351	HO + C5H10O3 -> CH4O + C4H7O3	[OH],[O]OCCC[C@H]([O])C:CO,[O]OCCCC=O	6.28e+13	9.52e+12	2.01e+14	2
R351*	CH4O + C4H7O3 -> HO + C5H10O3	CO,[O]OCCCC=O:[OH],[O]OCCC[C@H]([O])C	0.00e+00	0.00e+00	2.37e+12	0
R352	CHO2 -> CHO2	[O]C=O:O[C]=O	7.06e+07	1.07e+07	2.26e+08	2
R352*	CHO2 -> CHO2	O[C]=O:[O]C=O	0.00e+00	0.00e+00	1.88e+11	0
R353	C5H11O2 -> CH2O + C4H9O	[O]CCC[C@H](O)C:C=O,[CH2]C[C@H](O)C	1.02e+10	1.54e+09	3.25e+10	2
R353*	CH2O + C4H9O -> C5H11O2	C=O,[CH2]C[C@H](O)C:[O]CCC[C@H](O)C	0.00e+00	0.00e+00	4.00e+13	0
R354	C5H11O2 -> C5H11O2	[O]CCC[C@H](O)C:OCCC[C@H]([O])C	2.54e+10	8.94e+09	5.52e+10	5
R354*	C5H11O2 -> C5H11O2	OCCC[C@H]([O])C:[O]CCC[C@H](O)C	1.33e+10	4.68e+09	2.89e+10	5

R355	C5H10O3 -> C5H10O + O2	[O]CCC[C@H](O[O])C:C[CH]CCC[O],O=O	3.33e+11	1.41e+10	1.59e+12	1
R355*	C5H10O + O2 -> C5H10O3	C[CH]CCC[O],O=O:[O]CCC[C@H](O[O])C	0.00e+00	0.00e+00	8.96e+13	0
R356	C3H6O2 -> C3H6O2	[O]CCC[O]:[CH2]OCC[O]	7.29e+09	3.09e+08	3.47e+10	1
R356*	C3H6O2 -> C3H6O2	[CH2]OCC[O]:[O]CCC[O]	0.00e+00	0.00e+00	3.00e+13	0
R357	C2H5O3 -> C2H5O3	[O]CCOO:OCCO[O]	4.94e+10	1.17e+10	1.31e+11	3
R357*	C2H5O3 -> C2H5O3	OCCO[O]:[O]CCOO	3.53e+08	5.36e+07	1.13e+09	2
R358	C5H11O2 -> CH2O + C4H9O	[O]CC[C@H](CC)O:C=O,CC[C@H](O)[CH2]	3.62e+10	1.53e+09	1.73e+11	1
R358*	CH2O + C4H9O -> C5H11O2	C=O,CC[C@H](O)[CH2]:[O]CC[C@H](CC)O	0.00e+00	0.00e+00	5.51e+14	0
R359	C5H11O2 -> C5H11O2	[O]CC[C@H](CC)O:OCC[C@H](CC)[O]	2.17e+11	8.52e+10	4.44e+11	6
R359*	C5H11O2 -> C5H11O2	OCC[C@H](CC)[O]:[O]CC[C@H](CC)O	3.73e+10	1.31e+10	8.10e+10	5
R360	C5H11O2 -> CH2O + C4H9O	[O]CC[C@H](CO)C:C=O,OCC(C)[CH2]	3.30e+10	5.01e+09	1.06e+11	2
R360*	CH2O + C4H9O -> C5H11O2	C=O,OCC(C)[CH2]:[O]CC[C@H](CO)C	0.00e+00	0.00e+00	3.10e+13	0
R361	C5H11O2 -> CH2O + C4H9O	[O]CC[C@H](CO)C:C=O,OCC([CH2])C	1.65e+10	6.99e+08	7.86e+10	1
R361*	CH2O + C4H9O -> C5H11O2	C=O,OCC([CH2])C:[O]CC[C@H](CO)C	0.00e+00	0.00e+00	2.05e+13	0
R362	C4H9O2 -> C4H9O2	[O]CC[C@H](O)C:OCC[C@H]([O])C	1.20e+11	1.83e+10	3.86e+11	2
R362*	C4H9O2 -> C4H9O2	OCC[C@H]([O])C,[O]CC[C@H](O)C	1.49e+11	6.32e+09	7.11e+11	1
R363	C4H9O3 -> C4H9O + O2	[O]C:[C@H](OO)C,C[CH]CCO,O=O	7.87e+10	3.34e+09	3.75e+11	1
R363*	C4H9O + O2 -> C4H9O3	C[CH]CCO,O=O:[O]CC[C@H](OO)C	0.00e+00	0.00e+00	1.11e+11	0
R364	C2H4O3 -> CH2O + CH2O + O	[O]COC[O]:C=O,C=O,[O]	1.00e+13	4.24e+11	4.77e+13	1
R364*	CH2O + CH2O + O -> C2H4O3	C=O,C=O,[O]:[O]COC[O]	0.00e+00	0.00e+00	5.53e+19	0
R365	C5H11O4 -> CH2O + C4H9O3	[O]COO[C@H]([C@H](O)C)C:C=O,[O]O[C@H]([C@H](O)C)C	1.67e+12	7.06e+10	7.94e+12	1
R365*	CH2O + C4H9O3 -> C5H11O4	C=O,[O]O[C@H]([C@H](O)C)C:[O]COO[C@H]([C@H](O)C)C	0.00e+00	0.00e+00	8.81e+12	0
R366	C3H7O3 -> CH2O + C2H4O + HO	[O]C[C@H](OO)C:C=O,CC=O,[OH]	1.16e+10	4.91e+08	5.53e+10	1
R366*	CH2O + C2H4O + HO -> C3H7O3	C=O,CC=O,[OH]:[O]C[C@H](OO)C	0.00e+00	0.00e+00	7.70e+16	0
R367	C5H11O2 -> CH2O + C4H9O	[O]C[C@H]([C@H](O)C)C:C=O,C[CH][C@H](O)C	7.50e+12	1.78e+12	1.99e+13	3
R367*	CH2O + C4H9O -> C5H11O2	C=O,C[CH][C@H](O)C:[O]C[C@H]([C@H](O)C)C	0.00e+00	0.00e+00	6.20e+12	0
R368	CHO5 -> O2 + CHO3	[O]OC(O[O])[O]:O=O,[O]OC=O	2.86e+12	8.62e+11	6.74e+12	4
R368*	O2 + CHO3 -> CHO5	O=O,[O]OC=O:[O]OC(O[O])[O]	1.20e+11	3.63e+10	2.84e+11	4
R369	C2H5O4 -> O2 + C2H5O2	[O]OCC(O)O:O=O,[CH2]C(O)O	9.46e+09	6.27e+09	1.36e+10	26
R369*	O2 + C2H5O2 -> C2H5O4	O=O,[CH2]C(O)O:[O]OCC(O)O	1.21e+13	7.96e+12	1.75e+13	25
R370	C5H9O2 -> C5H9O2	[O]OCCCC=C:C1[CH]COOCC1	6.54e+08	2.77e+07	3.11e+09	1
R370*	C5H9O2 -> C5H9O2	C1[CH]COOCC1:[O]OCCCC=C	4.55e+11	1.93e+10	2.17e+12	1
R371	C5H9O2 -> O2 + C5H9	[O]OCCCC=C:O=O,[CH2]CCC=C	1.70e+10	1.13e+10	2.44e+10	26
R371*	O2 + C5H9 -> C5H9O2	O=O,[CH2]CCC=C:[O]OCCCC=C	1.26e+13	8.35e+12	1.81e+13	26
R372	C4H7O3 -> O2 + C4H7O	[O]OCCCC=O:O=O,[CH2]CCC=O	1.22e+10	1.03e+10	1.43e+10	145
R372*	O2 + C4H7O -> C4H7O3	O=O,[CH2]CCC=O:[O]OCCCC=O	1.27e+13	1.08e+13	1.49e+13	144
R373	C5H11O4 -> C5H11O2 + O2	[O]OCCC[C@H](OO)C:C[CH]CCCOO,O=O	7.71e+08	1.17e+08	2.47e+09	2
R373*	C5H11O2 + O2 -> C5H11O4	C[CH]CCCOO,O=O:[O]OCCC[C@H](OO)C	0.00e+00	0.00e+00	2.40e+11	0
R374	C5H11O4 -> O2 + C5H10O + HO	[O]OCCC[C@H](OO)C:O=O,[CH2]CC[C@H]([O])C.[OH]	3.86e+08	1.63e+07	1.84e+09	1
R374*	O2 + C5H10O + HO -> C5H11O4	O=O,[CH2]CC[C@H]([O])C,[OH]:[O]OCCC[C@H](OO)C	0.00e+00	0.00e+00	7.41e+17	0

R375	C5H11O4 -> C5H11O4	[O]OCCC[C@H](OO)C:OOCCC[C@H](O[O])C	3.08e+09	1.40e+09	5.77e+09	8
R375*	C5H11O4 -> C5H11O4	OOCCC[C@H](O[O])C:[O]OCCC[C@H](OO)C	5.09e+09	1.79e+09	1.11e+10	5
R376	C5H11O4 -> HO + C5H10O3	[O]OCCC[C@H](OO)C:[OH],[O]OCCC[C@H]([O])C	3.86e+08	1.63e+07	1.84e+09	1
R376*	HO + C5H10O3 -> C5H11O4	[OH],[O]OCCC[C@H]([O])C:[O]OCCC[C@H](OO)C	0.00e+00	0.00e+00	9.40e+13	0
R377	C5H10O3 -> C2H4O + O2 + C3H6	[O]OCCC[C@H]([O])C:CC=O,O=O,[CH2]C[CH2]	3.33e+09	1.41e+08	1.59e+10	1
R377*	C2H4O + O2 + C3H6 -> C5H10O3	CC=O,O=O,[CH2]C[CH2]:[O]OCCC[C@H]([O])C	0.00e+00	0.00e+00	3.12e+18	0
R378	C5H10O3 -> C5H10O3	[O]OCCC[C@H]([O])C:C[C@H]1CCCOOO1	3.33e+09	1.41e+08	1.59e+10	1
R378*	C5H10O3 -> C5H10O3	C[C@H]1CCCOOO1:[O]OCCC[C@H]([O])C	1.14e+11	4.81e+09	5.41e+11	1
R379	C5H11O4 -> C5H10O3 + HO	[O]OCC[C@H](COO)C:C[C@H](C=O)CCOO,[OH]	2.90e+08	1.23e+07	1.38e+09	1
R379*	C5H10O3 + HO -> C5H11O4	C[C@H](C=O)CCOO,[OH]:[O]OCC[C@H](COO)C	0.00e+00	0.00e+00	2.82e+17	0
R380	C5H11O4 -> O2 + C5H11O2	[O]OCC[C@H](COO)C:O=O,OOCCC(C)[CH2]	2.90e+08	1.23e+07	1.38e+09	1
R380*	O2 + C5H11O2 -> C5H11O4	O=O,OOCCC(C)[CH2]:[O]OCC[C@H](COO)C	0.00e+00	0.00e+00	4.06e+12	0
R381	C5H11O4 -> C5H11O4	[O]OCC[C@H](COO)C:OOCC[C@H](CO[O])C	2.90e+09	1.44e+09	5.11e+09	10
R381*	C5H11O4 -> C5H11O4	OOCC[C@H](CO[O])C:[O]OCC[C@H](COO)C	4.01e+09	2.00e+09	7.07e+09	10
R382	C4H9O3 -> C4H9O3	[O]OCC[C@H](O)C:OOCC[C@H]([O])C	6.92e+08	2.43e+08	1.50e+09	5
R382*	C4H9O3 -> C4H9O3	OOCC[C@H]([O])C:[O]OCC[C@H](O)C	4.98e+10	1.50e+10	1.17e+11	4
R383	C5H11O4 -> C5H11O2 + O2	[O]OCC[C@H](OO)CC:CCC(OO)C[CH2],O=O	4.42e+10	1.73e+10	9.04e+10	6
R383*	C5H11O2 + O2 -> C5H11O4	CCC(OO)C[CH2],O=O:[O]OCC[C@H](OO)CC	1.14e+13	3.45e+12	2.69e+13	4
R384	CH3O3 -> CH2O + HO2	[O]OCO:C=O,O[O]	9.46e+09	4.01e+08	4.51e+10	1
R384*	CH2O + HO2 -> CH3O3	C=O,O[O]:[O]OCO	0.00e+00	0.00e+00	1.78e+12	0
R385	C4H6O4 -> C2H3O + C2H3O3	[O]OC[C@H](CC=O)[O]:[O]C=C,[O]OCC=O	1.21e+10	5.12e+08	5.76e+10	1
R385*	C2H3O + C2H3O3 -> C4H6O4	[O]C=C,[O]OCC=O:[O]OC[C@H](CC=O)[O]	0.00e+00	0.00e+00	4.71e+14	0
R386	C4H9O3 -> O2 + C4H9O	[O]OC[C@H](CO)C:O=O,OCC([CH2])C	1.15e+10	9.75e+09	1.35e+10	147
R386*	O2 + C4H9O -> C4H9O3	O=O,OCC([CH2])C:[O]OC[C@H](CO)C	1.15e+13	9.77e+12	1.35e+13	145
R387	C4H9O3 -> O2 + C3H6O + CH3	[O]OC[C@H](CO)C:O=O,OCC=C.[CH3]	7.84e+07	3.32e+06	3.73e+08	1
R387*	O2 + C3H6O + CH3 -> C4H9O3	O=O,OCC=C,[CH3]:[O]OC[C@H](CO)C	2.57e+18	1.09e+17	1.23e+19	1
R388	C4H9O3 -> C4H9O3	[O]OC[C@H](CO)C:OOC[C@H](C[O])C	1.57e+08	2.38e+07	5.02e+08	2
R388*	C4H9O3 -> C4H9O3	OOC[C@H](C[O])C:[O]OC[C@H](CO)C	0.00e+00	0.00e+00	6.98e+10	0
R389	C5H9O4 -> C5H9O2 + O2	[O]OC[C@H](C[C@H]([O])C)[O]:C[C@H](C[C@H]([O])[CH2])[O],O=O	4.02e+10	1.70e+09	1.91e+11	1
R389*	C5H9O2 + O2 -> C5H9O4	C[C@H](C[C@H]([O])[CH2])[O],O=O:[O]OC[C@H](C[C@H]([O])C)[O]	0.00e+00	0.00e+00	1.68e+15	0
R390	C5H10O3 -> CH2O + C4H8 + O2	[O]OC[C@H](C[O])CC:C=O,CCC=C,O=O	1.00e+12	4.24e+10	4.77e+12	1
R390*	CH2O + C4H8 + O2 -> C5H10O3	C=O,CCC=C,O=O:[O]OC[C@H](C[O])CC	0.00e+00	0.00e+00	8.12e+15	0
R391	C3H7O3 -> C3H7O3	[O]OC[C@H](O)C:OOC[C@H]([O])C	4.98e+08	1.95e+08	1.02e+09	6
R391*	C3H7O3 -> C3H7O3	OOC[C@H]([O])C:[O]OC[C@H](O)C	7.26e+10	2.19e+10	1.71e+11	4
R392	C3H7O4 -> C3H7O2 + O2	[O]OC[C@H](OO)C:C[CH]COO,O=O	7.42e+09	3.14e+08	3.53e+10	1
R392*	C3H7O2 + O2 -> C3H7O4	C[CH]COO,O=O:[O]OC[C@H](OO)C	0.00e+00	0.00e+00	2.46e+13	0
R393	C5H11O4 -> O2 + C5H11O2	[O]OC[C@H]([C@H](OO)C)C:O=O,OO[C@H](C([CH2])C)C	1.57e+10	2.38e+09	5.02e+10	2
R393*	O2 + C5H11O2 -> C5H11O4	O=O,OO[C@H](C([CH2])C)C:[O]OC[C@H]([C@H](OO)C)C	1.23e+13	5.21e+11	5.86e+13	1
R394	C5H11O4 -> C5H10O2 + HO2	[O]O[C@H](CC[C@H](O)C)O:O=CCC[C@H](O)C,O[O]	6.04e+10	9.17e+09	1.93e+11	2
R394*	C5H10O2 + HO2 -> C5H11O4	O=CCC[C@H](O)C,O[O]:[O]O[C@H](CC[C@H](O)C)O	0.00e+00	0.00e+00	3.66e+12	0

R395	C2H5O3 -> C2H5O3	[O]O[C@H](O)C:OO[C@H]([O])C	1.05e+11	3.18e+10	2.48e+11	4
R395*	C2H5O3 -> C2H5O3	OO[C@H]([O])C:[O]O[C@H](O)C	9.43e+10	1.43e+10	3.02e+11	2
R396	C4H9O3 -> C4H9O3	[O]O[C@H]([C@H](O)C)C:C[C@H](OO)O[CH]C	3.12e+08	1.32e+07	1.49e+09	1
R396*	C4H9O3 -> C4H9O3	C[C@H](OO)O[CH]C:[O]O[C@H]([C@H](O)C)C	0.00e+00	0.00e+00	3.33e+12	0
R397	C4H9O3 -> C4H9O3	[O]O[C@H]([C@H](O)C)C:OO[C@H]([C@H]([O])C)C	2.19e+09	9.29e+08	4.26e+09	7
R397*	C4H9O3 -> C4H9O3	OO[C@H]([C@H]([O])C)C:[O]O[C@H]([C@H](O)C)C	4.76e+10	1.44e+10	1.12e+11	4
R398	C4H7O3 -> C4H7O3	[O][C@H]1CCCOO1:[O]OCCCC=O	7.40e+10	3.95e+10	1.25e+11	12
R398*	C4H7O3 -> C4H7O3	[O]OCCCC=O:[O][C@H]1CCCOO1	9.24e+08	4.77e+08	1.59e+09	11

A.1.2. Geometries

The geometries of the most stable conformers of the 2-pentyl radical and the 2-hydroperoxide-pent-4-yl radical, of the transition states of the 2-pentyl → iso-pentyl isomerization, and of the OH-migration of the 2-hydroperoxide-pent-4-yl are given below in Å. The geometries have been optimized with the CBS-QB3 method implemented in Gaussian09.e01.

16			
2-pentyl radical			
0	2		
C	-2.61476800	-0.22335300	-0.07678300
C	-1.29363100	0.45425800	0.05388600
H	-2.91474200	-0.73100700	0.85612700
H	-3.41301800	0.48122400	-0.32395700
H	-2.59514800	-1.00000100	-0.85130500
H	-1.26998200	1.50147600	0.34168600
C	-0.02036900	-0.32243000	0.10792900
C	1.24404800	0.52649800	-0.08743200
H	-0.04549900	-1.12354500	-0.64587200
H	0.05913900	-0.85284900	1.07586100
C	2.53733000	-0.28877200	-0.01276600
H	1.26410400	1.31558600	0.67352600
H	1.18379000	1.03681800	-1.05512200
H	3.41705600	0.34520000	-0.15268200
H	2.63694600	-0.78571300	0.95744500
H	2.56169800	-1.06440100	-0.78471500

16

TS pentyl radical isomerization

0	2		
C	-0.66404900	-0.04568300	-0.47297800
C	-1.87461200	-0.70461300	0.11539500
H	-0.36498700	-0.37971700	-1.46211200
H	-1.89755100	-1.77866200	-0.08952600
H	-1.90594800	-0.57217000	1.20500400
H	-2.80558300	-0.26657600	-0.27717500
C	-0.35348300	1.35625000	-0.13963800
H	-0.91208800	1.85613000	0.64315700
H	0.16563200	1.98978400	-0.84878800
C	0.82791800	0.09651200	0.61763700
C	1.94319300	-0.54998400	-0.20633900
H	0.42980100	-0.56814300	1.39256800
H	1.30617000	0.91541900	1.19564100
H	2.84597100	-0.65745800	0.40497200
H	1.65828400	-1.53249000	-0.58662900
H	2.20649400	0.07898900	-1.06157900

18			
2-hydroperoxide-pent-4-yl			
0	2		
C	1.57009500	1.56662100	-0.03715200
C	0.63455800	0.38858600	-0.28360700
H	2.58671200	1.30884300	-0.33869100
H	1.25681500	2.43733600	-0.61964500
H	1.57423900	1.84055100	1.02204000
C	-0.79394500	0.62915700	0.21652900
H	0.61105900	0.14011600	-1.35077300
H	-0.74692700	0.75925800	1.31305000

H	-1.13232900	1.59820300	-0.17432800
C	-1.76304400	-0.44429400	-0.15213400
C	-3.23224300	-0.22428800	-0.04158100
H	-1.38122600	-1.45278900	-0.26086300
H	-3.79930200	-1.00588600	-0.55332100
H	-3.57299500	-0.21914100	1.00809400
H	-3.53152000	0.74377100	-0.46262000
O	1.06612800	-0.78355200	0.42717900
O	2.27649900	-1.27244800	-0.21618000
H	2.90193600	-1.19695600	0.51672800

18			
TS OH-migration			
0	2		
C	-1.46230900	-0.28853700	0.56878900
O	-0.55269300	1.60803700	-0.09782000
O	1.09227300	1.18937700	-0.01784900
C	0.94297800	-0.19419300	-0.28427000
H	-1.98345300	-0.00676200	1.47793800
H	-0.55005000	2.13351100	0.71642500
H	0.55441600	-0.32797900	-1.30371100
C	-0.05824000	-0.81586200	0.71543500
H	0.31579500	-0.62322500	1.72567200
H	-0.06008300	-1.90869200	0.57618200
C	2.34764000	-0.79417700	-0.20894900
H	2.33581100	-1.84436400	-0.51199900
H	3.01849900	-0.25045700	-0.87702600
H	2.74426900	-0.72588500	0.80729100
C	-2.29414500	-0.65947400	-0.60964000

H	-3.20862900	-0.06548400	-0.66279100
H	-1.74649400	-0.52653900	-1.54748800
H	-2.59226100	-1.71997900	-0.56334100

A.1.3. Carbonyl-Peroxide Formation

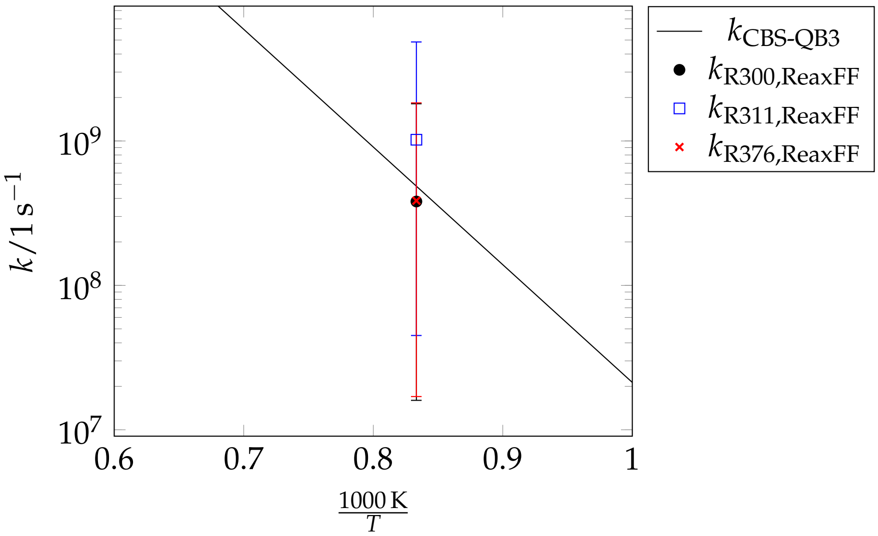


Figure A.1.: The CBS-QB3 reaction rate constant corresponds to the carbonyl-peroxide + OH formation of the 2-hydroperoxide-4-peroxide-pentane, taken from Asatryan and Bozzelli 2010[65]. With the same isomer, the reaction did not occur in the simulations. Therefore, reactions of $\dot{O}OQOOH$ isomers are shown. R300, R311, R376 are the corresponding reaction IDs that can be found in the reaction list in section A.1.1. The statistical errors of the rMD reaction rate constants comprise approx. one order of magnitude.

A.1.4. Error Propagation

For a property x, the error in it's parameters p_i is propagated via

$$\Delta x = \sqrt{\sum_i \left| \frac{\delta x}{\delta p_i} \Delta p_i \right|^2} \tag{A.1}$$

$$k_i = A \exp\left(-\frac{\Delta E}{RT}\right) \tag{A.2}$$

Applying eq A.1 to eq. A.2, yields

$$\left|\frac{\Delta k_i}{k_i}\right| = \left|-\frac{\Delta\Delta E}{RT}\right| \tag{A.3}$$

with respect to errors in the electronic energy barrier heights.

The branching ratio BR_i is defined as the ratio of the reaction rate of reaction i devided by the sum over all $N^{\text{par. reactions}}$ parallel reaction rates. In the simple cases of unimolecular, or pseudo-unimolecular reactions, such as O_2-addition reactions with excess oxygen, BR_i can be written as

$$BR_i = \frac{k_i}{k_j + \sum_{k \neq j}^{N^{\text{par. reactions}}} (k_k)} \tag{A.4}$$

In order to apply eq A.1, 2 cases have to be distinguished. First, the error in k_i, second the error in any other parallel reaction k_j. The following two reactions are the corresponding replacements for the partial derivatives $\frac{\delta x}{\delta p_i}$ in eq A.1.

$$\frac{\delta BR_i}{\delta k_i} = \frac{k_j + \sum_{k \neq j}^{N^{\text{par. reactions}}} (k_k) - k_i}{\left(k_j + \sum_{k \neq j}^{N^{\text{par. reactions}}} (k_k)\right)^2} \tag{A.5}$$

$$\frac{\delta BR_i}{\delta k_j} = -\frac{k_i}{\left(k_j + \sum_{k \neq j}^{N^{\text{par. reactions}}} (k_k)\right)^2} \tag{A.6}$$

The two eqs A.5, A.6 can be further simplified. The error in BR_i reads then:

$$\Delta BR_i = \sqrt{\left| \frac{\sum_{k}^{N^{\text{par. reactions}}} (k_k) - k_i}{\left(\sum_{k}^{N^{\text{par. reactions}}} (k_k)\right)^2} \Delta k_i \right|^2 + \sum_{j \neq i}^{N^{\text{par. reactions}}} \left| -\frac{k_i}{\left(\sum_{k}^{N^{\text{par. reactions}}} (k_k)\right)^2} \Delta k_j \right|^2} \tag{A.7}$$

A.1.5. Ignition Delay Times

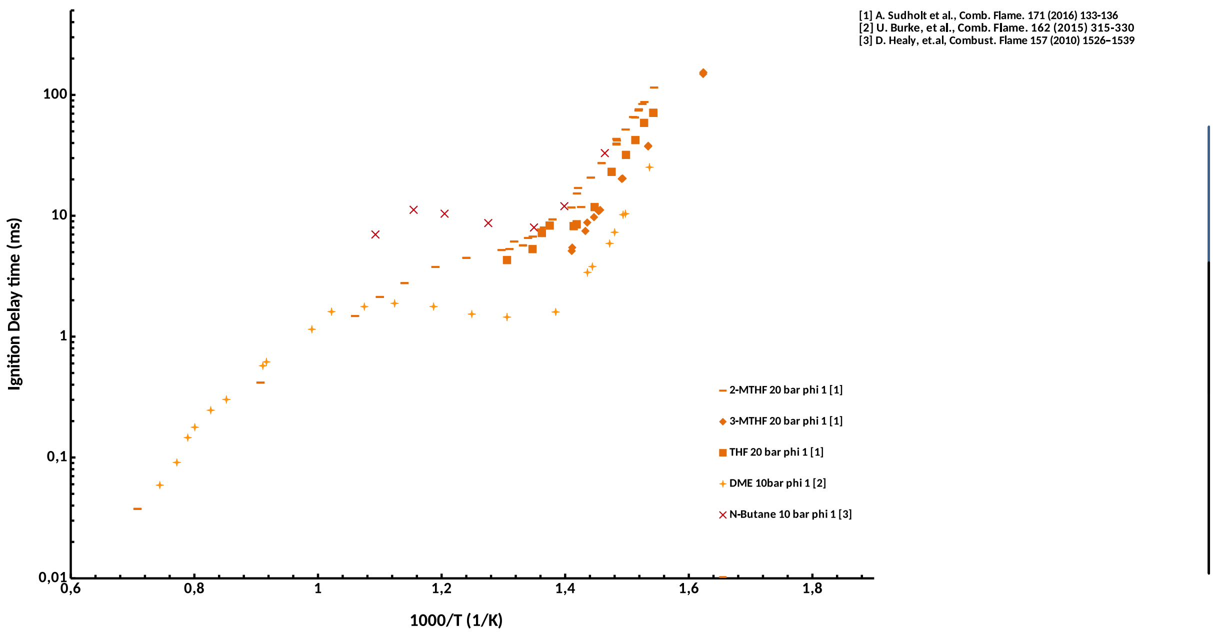
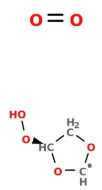
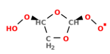
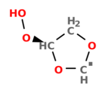
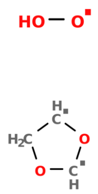
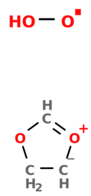
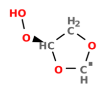
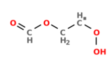
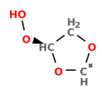
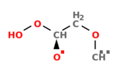


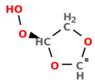
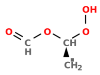
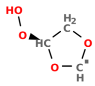
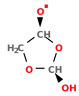
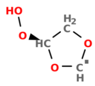
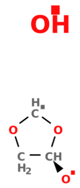
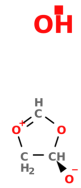
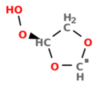
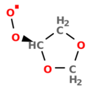
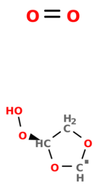
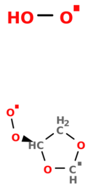
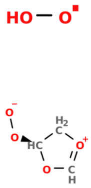
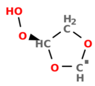
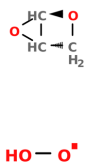
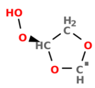
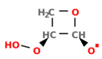
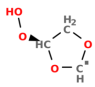
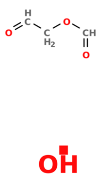
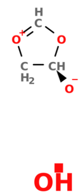
Figure A.2.: Comparison of different fuels shows that the change from the low-temperature to the intermediate regime occurs at similar temperatures for similar pressures.

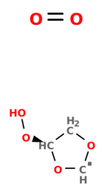
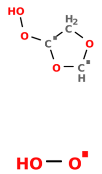
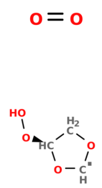
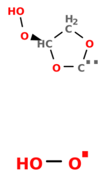
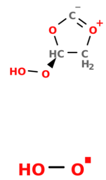
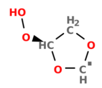
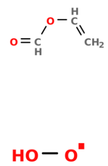
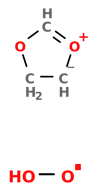
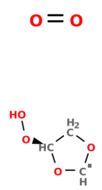
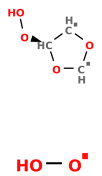
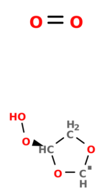
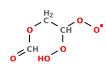
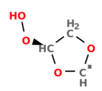
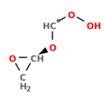
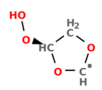
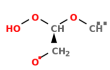
A.2. Supplementary Information to Chapter 3

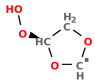
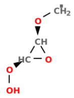
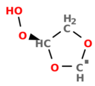
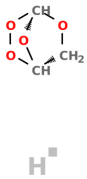
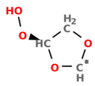
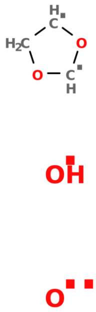
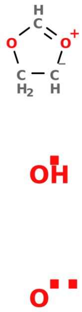
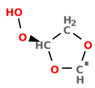
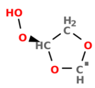
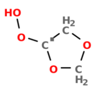
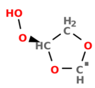
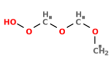
A.2.1. Reaction List

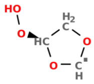
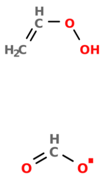
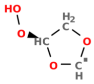
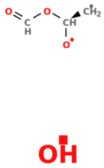
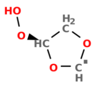
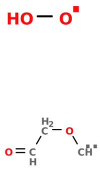
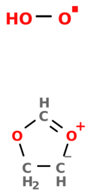
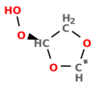
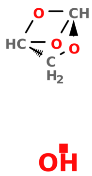
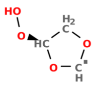
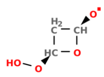
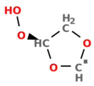
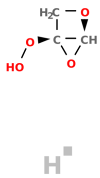
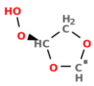
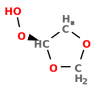
Table A.3.: List of all observed types of reactions. $V_{\mathrm{DS}}^{\mathrm{min}}$ is the minimum potential energy difference between the reactive snapshot and the optimized reactants from all reaction events of the same reaction type. Opt. InChi test and Opt. SMILES test mark the reaction identification test result. Checkmarks mark truncated InChi codes, or SMILES codes, respectively, of the products that are retrieved after the optimization with ReaxFF. Reactions marked with crosses are not correctly identified by the ChemTraYzer DS. In the last column, you find the optimized products of some of the reaction events if the new SMILES representations of the optimized products from all reaction events with a different SMILES code than the original are unambiguous.

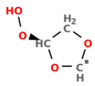
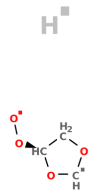
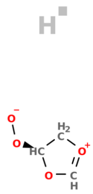
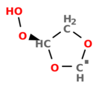
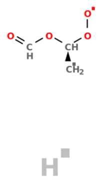
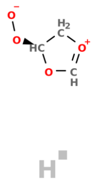
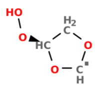
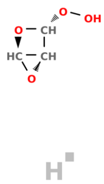
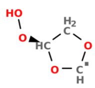
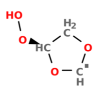
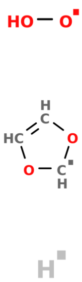
$\frac{V_{\mathrm{DS}}^{\mathrm{min}}}{\mathrm{kcal\,mol^{-1}}}$	reactant(s)	product(s)	Opt. InChi test	Opt. SMILES test	(some) products changed to
-12			✓	✓	ambiguous
-4			✓	×	
-4			×	×	reactant
-4			×	×	reactant

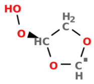
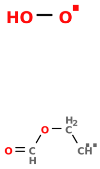
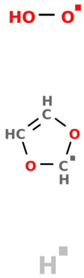
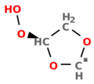
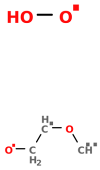
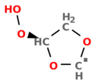
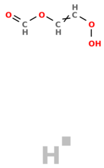
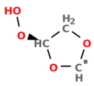
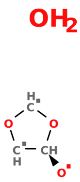
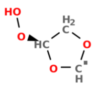
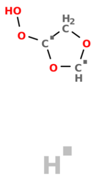
-3			×	×	reactant
8			✓	✓	reactant
17			✓	×	
19			✓	✓	reactant
20			✓	×	
22			✓	✓	ambiguous
22			×	×	ambiguous
24			✓	✓	

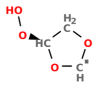
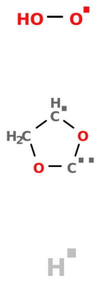
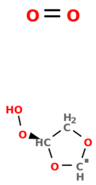
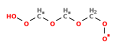
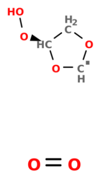
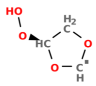
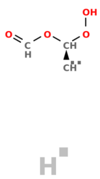
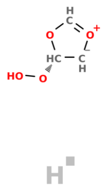
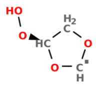
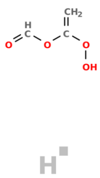
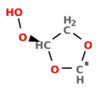
25			✓	×	ambiguous
31			✓	×	
33			×	×	
34			✓	×	ambiguous
35			✓	✓	ambiguous
35			×	×	reactant
36			×	×	reactant

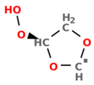
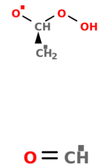
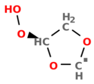
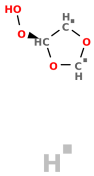
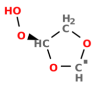
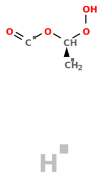
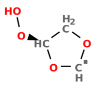
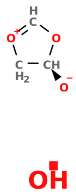
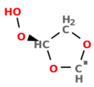
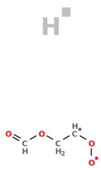
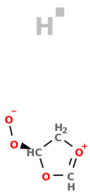
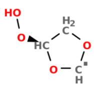
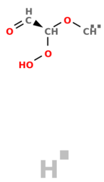
36			×	×	reactant
36			✓	✓	
38			✓	×	
39			✓	✓	
39			✓	✓	reactant
39			×	×	reactant

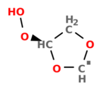
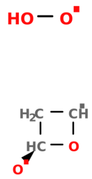
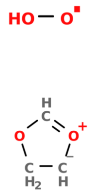
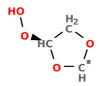
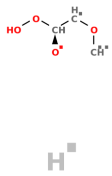
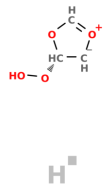
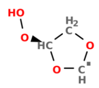
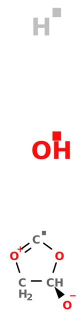
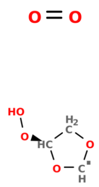
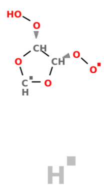
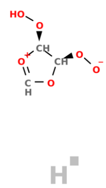
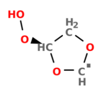
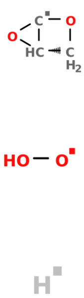
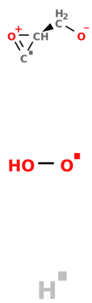
40			✓	✓	
40			✓	×	ambiguous
42			×	×	
42			✓	✓	
44			×	×	reactant
44			✓	✓	ambiguous
45			✓	✓	reactant

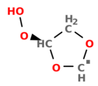
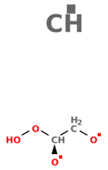
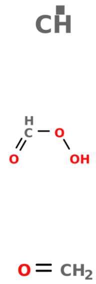
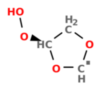
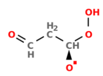
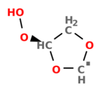
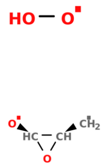
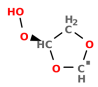
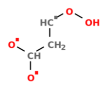
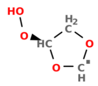
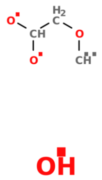
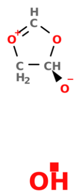
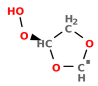
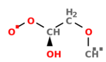
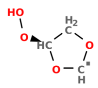
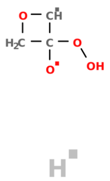
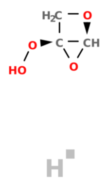
45			✓	×	
48			×	×	
48			✓	✓	ambiguous
50			✓	×	ambiguous
51			✓	✓	

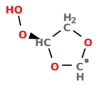
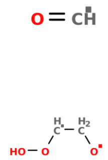
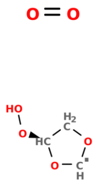
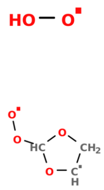
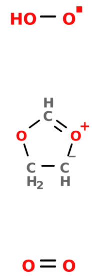
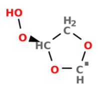
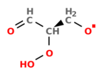
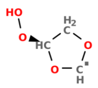
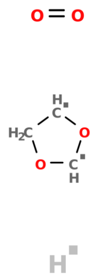
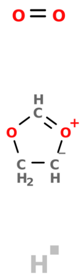
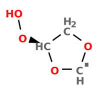
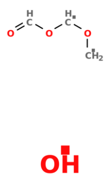
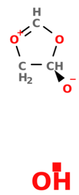
51			×	×	
53			×	×	ambiguous
55			×	×	ambiguous
57			✓	×	ambiguous
57			✓	×	ambiguous

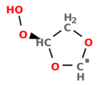
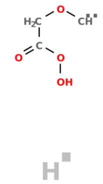
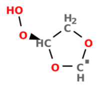
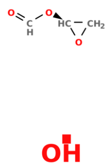
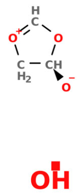
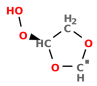
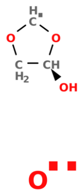
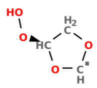
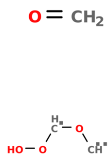
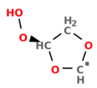
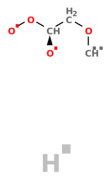
58			✓	×	ambiguous
61			×	×	
62			×	×	
62			×	×	ambiguous
62			×	×	

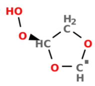
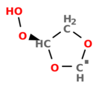
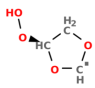
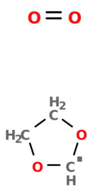
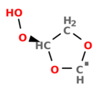
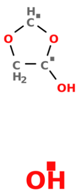
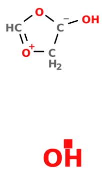
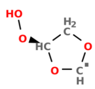
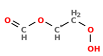
62			✓	×	ambiguous
63			✓	×	ambiguous
63			×	×	ambiguous
66			×	×	
66			✓	×	
67			✓	×	ambiguous

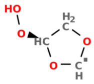
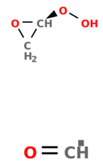
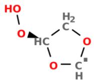
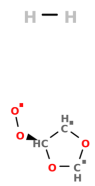
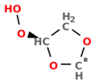
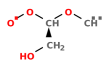
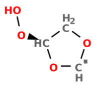
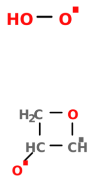
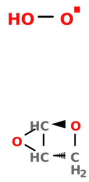
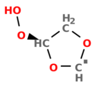
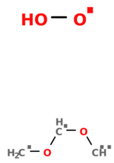
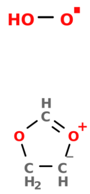
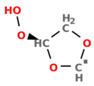
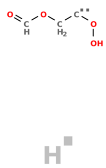
70			×	×	
70			×	×	
72			✓	×	
72			✓	×	
73			✓	✓	

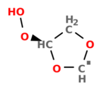
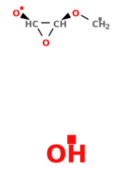
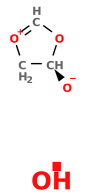
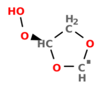
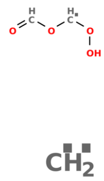
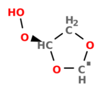
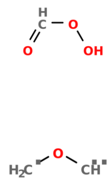
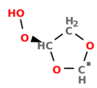
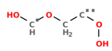
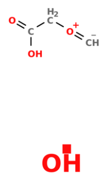
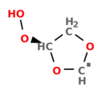
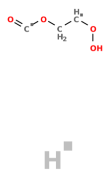
74			×	×	
75			×	×	reactant
75			×	×	ambiguous
76			×	×	ambiguous
77			×	×	
77			×	×	reactant
77			×	×	

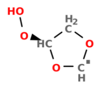
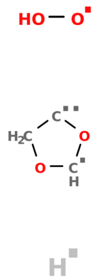
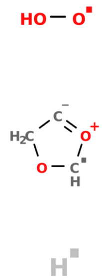
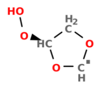
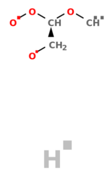
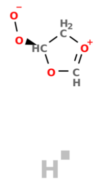
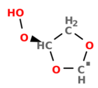
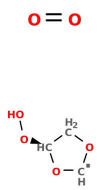
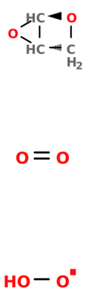
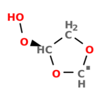
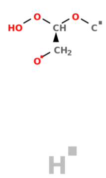
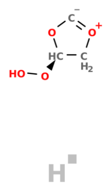
79			✓	×	ambiguous
79			×	×	
79			×	×	reactant
79			✓	×	
79			×	×	

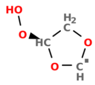
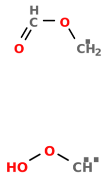
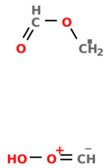
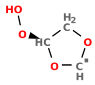
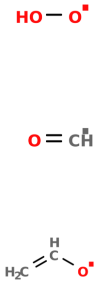
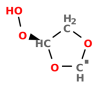
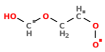
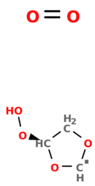
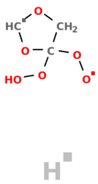
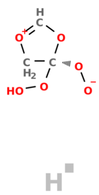
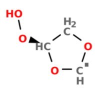
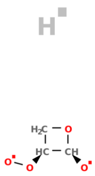
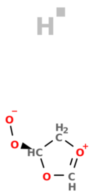
80			×	×	ambiguous
81			✓	✓	
82			✓	✓	
84			✓	×	ambiguous
86			×	×	ambiguous

86			×	×	
87			✓	✓	
87			✓	✓	
91			✓	×	
92			×	×	reactant

93			✓	✓	
93			✓	×	ambiguous
93			×	×	reactant
93			×	×	
94			×	×	
94			✓	×	ambiguous

94			×	×	
96			✓	✓	
96			✓	×	ambiguous
96			✓	×	
97			×	×	ambiguous

97			✓	×	
97			×	×	
98			✓	×	
98			✓	✓	ambiguous
99			×	×	

99			✓	×	
100			✓	✓	
100			×	×	reactant
100			✓	×	
101			×	×	
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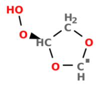
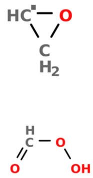
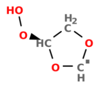
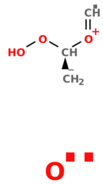
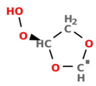
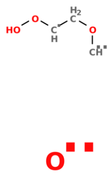
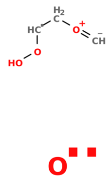
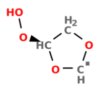
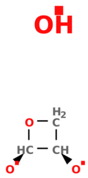
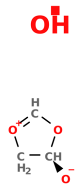
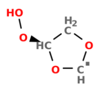
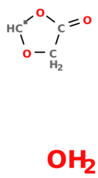
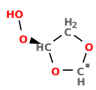
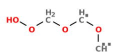
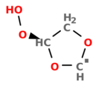
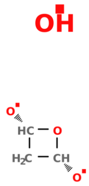
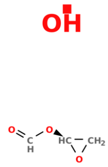
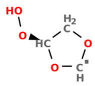
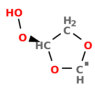
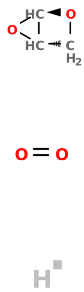
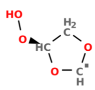
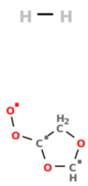
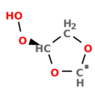
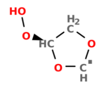
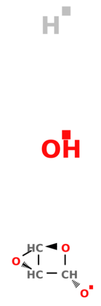
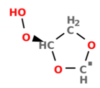
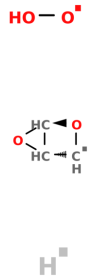
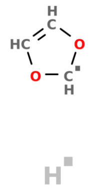
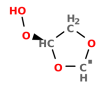
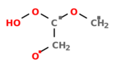
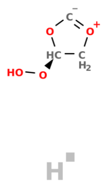
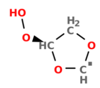
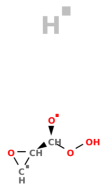
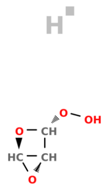
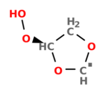
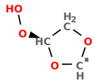
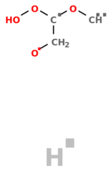
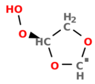
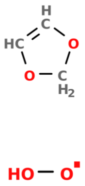
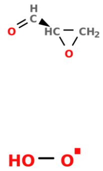
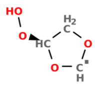
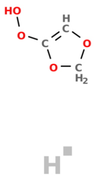
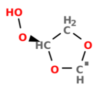
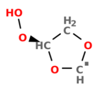
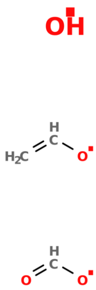
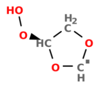
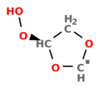
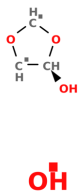
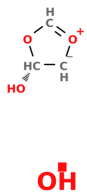
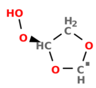
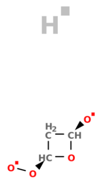
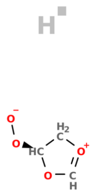
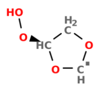
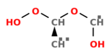
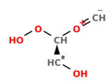
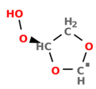
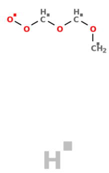
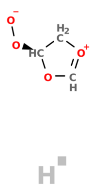
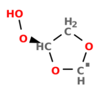
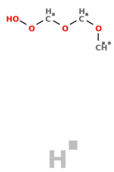
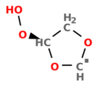
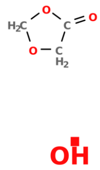
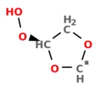
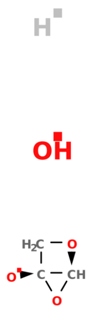
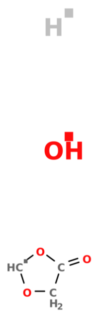
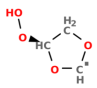
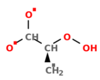
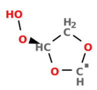
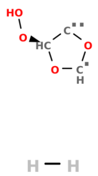
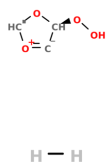
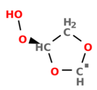
102			✓	✓	
102			✓	×	
102			✓	×	
103			✓	×	
103			✓	✓	
103			×	×	reactant

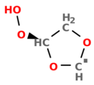
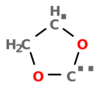
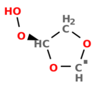
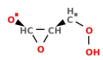
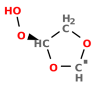
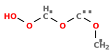
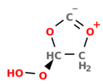
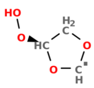
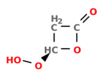
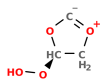
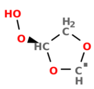
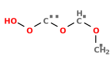
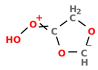
105			×	×	
106			✓	×	ambiguous
107			✓	✓	ambiguous
108			✓	×	ambiguous
111			✓	×	

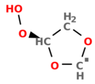
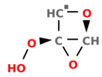
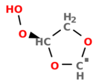
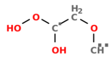
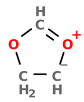
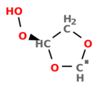
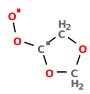
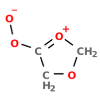
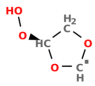
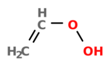
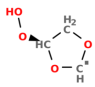
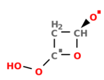
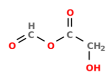
112			×	×	
113			×	×	
113			×	×	
113			×	×	
116			✓	✓	

117			×	×	
118			×	×	
119			✓	✓	
119			✓	×	ambiguous
119			✓	✓	

119			✓	×	
120			✓	×	
123			×	×	
124			×	×	
125			×	×	
125			×	×	ambiguous

127			✓	✓	
127			×	×	
129			×	×	reactant
135			✓	×	
136			✓	×	ambiguous

137		   	✓	×	ambiguous
137		 	×	×	ambiguous
137		 	×	×	 
139		 	×	×	 
146		 	×	×	 

149		 $H^\bullet$ $H^\bullet$	✓	✓	
149			×	×	 $HO-O^\bullet$
157		$H^\bullet$ 	✓	×	$H^\bullet$ 
160		 $O=CH^\bullet$ $O^{\bullet\bullet}$	✓	✓	
167		 $H^\bullet$	×	×	 $H^\bullet$

A.2.2. R9 TS Geometry

Table A.4.: *XYZ* coordinates in Å of the R9 (concerted β-scission plus O_2-addition) TS at the B3PLYPD3/6-311+g(d,p) level-of-theory in the douplet state.

C	-0.15757200	-0.64175100	-0.67378400
C	0.01932700	0.21333100	0.56574500
O	0.20394200	1.53883000	0.47912900
O	1.10682200	1.79972000	-0.63444200
O	1.38183400	-0.46438700	1.15438300
C	1.89952500	-1.07024400	0.16081200
H	2.95432400	-1.31887600	0.11375400
O	1.16220500	-1.27878400	-0.88701900
H	-0.37769300	-0.07536600	-1.56925400
H	-0.88854600	-1.43438700	-0.49905800
H	-0.71430400	0.04017500	1.37752000
H	0.62083600	2.51413400	-1.07358100
O	-2.46696500	0.35977700	-0.16504200
O	-2.87913600	-0.79686800	0.21974000

A.3. Supplementary Information to Chapter 4

A.3.1. Preliminary Force Field Parameters

The preliminary version differs from the published parameters [150] only by the parameters for Cl−Cl interactions. The preliminary parameters are contributed by Leonid Komissarov.

40-2 > ’C.C.O.Cl’, ’C.O.Cl’

41 ! Number of general parameters

50.0000

9.5469

26.5405

0.6863

2.7295

70.0000

1.0588

4.1262

12.1176

13.3056

-68.9784

0.0000

10.0000

0.0000

33.8667

6.0891

1.0563

2.0384

6.1431

6.9290

0.3989

3.9954

0.0000

5.7796

10.0000

1.9487

0.0000

2.1645

1.5591

0.1000

2.1365

0.6991

50.0000

1.8512

0.0000

0.0000

0.0000

1.0000

2.6962

0.0000

0.0000

4 ! Number of atoms

C 1.3674 4.0000 12.0000 2.0453 0.1444 0.9500 1.1706 4.0000

9.0000 1.5000 4.0000 27.5134 79.5548 5.0191 7.0500 0.0000

1.1168 0.0000 181.0000 14.2732 24.4406 6.7313 0.8563 0.0000

-4.1021 5.0000 1.0564 4.0000 2.9663 0.0000 0.0000 0.0000

H 0.9479 1.0000 1.0080 1.1364 0.0232 0.9900 -0.1000 1.0000

9.0643 4.7746 1.0000 0.0000 121.1250 4.7757 9.7732 1.0000

-0.1000 0.0000 62.4879 2.5194 2.3785 0.2223 1.0698 0.0000

-15.7683 2.1488 1.0338 1.0000 2.8793 0.0000 0.0000 0.0000

O 1.1939 2.0000 15.9990 1.9289 0.1201 0.9900 1.0981 6.0000

10.4842 8.2916 4.0000 28.8967 116.0768 7.9703 7.0500 2.0000

1.0479 20.0000 60.8726 10.0338 2.2024 0.9942 0.9745 0.0000

-3.6141 2.7025 1.0493 4.0000 2.9225 0.0000 0.0000 0.0000

Cl 1.4622 1.0000 35.4500 2.1699 0.1143 0.5854 -1.0000 7.0000

9.4263 6.2402 1.0000 2.3692 0.0000 8.7026 18.3815 2.0000

-1.0000 5.0303 28.0000 10.5240 4.0008 0.2045 0.8656 0.0000

-11.9991 2.6929 1.0338 4.0000 2.0389 0.0000 0.0000 0.0000

10 ! Number of bonds

1 1 80.8865 107.9944 52.0636 0.5218 -0.3636 1.0000 34.9876 0.7769

6.1244 -0.1693 8.0804 1.0000 -0.0586 8.1850 1.0000 0.0000

1 2 179.5195 0.0000 0.0000 -0.5242 0.0000 1.0000 6.0000 0.7187

5.4740 1.0000 0.0000 1.0000 -0.1144 6.7029 0.0000 0.0000

2 2 113.9232 0.0000 0.0000 -0.5971 0.0000 1.0000 6.0000 0.9093

1.7152 1.0000 0.0000 1.0000 -0.0450 6.0710 0.0000 0.0000

1 3 136.4945 164.1201 5.5000 -0.9159 -0.1075 1.0000 10.6519 0.8644

0.6858 -0.4602 9.5754 1.0000 -0.1745 4.5987 0.0000 0.0000

3 3 148.0798 155.2406 20.1160 -1.0000 -0.1254 1.0000 33.0027 0.7790

0.7673 -0.1697 7.0028 1.0000 -0.1300 5.1959 1.0000 0.0000

2 3 169.1351 0.0000 0.0000 -0.8810 0.0000 1.0000 6.0000 0.5757

1.5482 1.0000 0.0000 1.0000 -0.1788 4.6622 0.0000 0.0000

1 4 110.2423 0.0000 0.0000 0.2779 0.0000 1.0535 6.0000 0.1394

18.0983 -0.2500 0.0000 1.0000 -0.4777 3.2360 0.0000 0.0000

2 4 148.8040 0.0000 0.0000 -0.2898 0.0000 0.8642 6.0000 1.7483

2.6898 -0.2000 0.0000 1.0000 -0.1352 3.6144 0.0000 0.0000

3 4 68.6034 0.0000 0.0000 -0.9994 0.0000 0.0051 6.0000 0.8167

2.7959 -0.2000 0.0000 1.0000 -0.1428 8.9863 0.0000 0.0000

4 4 91.7101 0.0000 0.0000 0.0713 0.0000 0.4976 6.0000 4.9999

-0.0523 -0.2500 0.0000 1.0000 -0.0100 7.2315 0.0000 0.0000

6 ! Number of off-diagonal terms

1 2 0.1253 1.5717 9.9736 1.2057 -1.0000 -1.0000

2 3 0.1125 1.6311 8.7528 1.0929 -1.0000 -1.0000

1 3 0.0953 1.7397 8.8986 1.4256 1.1067 1.1265

1 4 0.2182 1.4549 12.1783 1.8100 -1.0000 -1.0000

2 4 0.2059 0.9935 12.8651 1.2900 -1.0000 -1.0000

3 4 0.2175 1.1010 15.9842 1.7300 -1.0000 -1.0000

29 ! Number of angles

1 1 1 76.1370 34.6920 1.1328 0.0000 0.0050 0.3556 1.8065

1 1 2 68.0572 9.9461 4.7000 0.0000 0.4566 0.0000 1.8532

2 1 2 65.6815 35.0000 1.8622 0.0000 0.0490 0.0000 1.0937

1 2 2 0.0000 4.0000 7.2043 0.0000 0.0000 0.0000 1.0728

1 2 1 0.0000 3.4110 7.7350 0.0000 0.0000 0.0000 1.0400

2 2 2 0.0000 30.0000 5.6235 0.0000 0.0000 0.0000 1.0400

1 1 3 78.3624 13.0773 9.0480 0.0000 0.1270 52.1129 2.3964

3 1 3 76.7101 24.3833 5.8613 -21.8559 2.6395 -32.6534 3.6179

2 1 3 79.1288 30.0000 1.4632 0.0000 0.2065 0.0000 2.0000

1 3 1 80.7352 16.4130 4.9987 0.0000 0.0843 0.0000 1.0137

1 3 3 85.4436 14.4937 3.9928 0.0000 1.4350 44.5320 1.1348

3 3 3 89.9282 32.1199 2.7181 0.0000 0.3323 57.6122 1.0000

1 3 2 82.9640 32.4874 0.8777 0.0000 0.9627 0.0000 1.0010

2 3 3 85.7838 17.3139 1.9157 0.0000 3.6306 0.0000 2.1858

2 3 2 84.2527 33.1226 0.6730 0.0000 0.7238 0.0000 2.4348

1 2 3 0.0000 14.4588 3.1507 0.0000 3.4571 0.0000 1.0149

3 2 3 0.0000 0.9696 3.6303 0.0000 0.0000 0.0000 1.6987

2 2 3 0.0000 0.5797 1.9739 0.0000 0.0000 0.0000 2.4494

1 4 4 0.0000 97.9606 8.0433 0.0000 0.0000 0.0000 3.8708

1 1 4 55.8900 20.7732 1.7644 0.0000 4.8904 0.0000 1.6417

1 3 4 2.6434 -9.1685 9.0474 0.0000 1.5171 0.0000 6.1281

2 1 4 75.5068 -0.5924 9.9246 0.0000 4.7767 0.0000 5.8248

2 3 4 126.8583 41.5246 0.8000 0.0000 0.5282 0.0000 1.2442

3 1 4 61.8784 72.7241 0.3156 0.0000 4.9506 0.0000 3.4766

3 2 4 0.0000 91.4780 4.2256 0.0000 0.0000 0.0000 1.5737

3 3 4 73.5892 20.0466 1.1863 0.0000 4.9940 0.0000 4.8561

4 1 4 51.0157 39.4737 0.8047 0.0000 2.1395 0.0000 1.1603

4 3 4 10.9340 -29.7636 1.4274 0.0000 1.2533 0.0000 7.9693

4 4 4 0.0000 42.8758 4.2241 0.0000 0.0000 0.0000 1.6690

32 ! Number of torsions

1 1 1 1 2.0474 32.6719 0.5282 -9.0000 -2.6449 0.0000 0.0000

1 1 1 2 1.6328 78.4995 -0.1514 -6.9161 -2.9986 0.0000 0.0000

2 1 1 2 2.4142 78.7025 0.3506 -8.8640 -6.9283 0.0000 0.0000

1 1 1 3 -0.7104 22.6038 0.5309 -2.0000 -0.6614 0.0000 0.0000

2 1 1 3 1.9323 52.9368 0.6554 -8.8118 -3.9854 0.0000 0.0000

3 1 1 3 -1.2500 1.1248 -0.1230 -9.9453 -3.9000 0.0000 0.0000

1 1 3 1 -0.6848 56.7751 -1.2733 -2.2937 -4.0000 0.0000 0.0000

1 1 3 2 -1.4557 78.6279 0.9945 -3.2742 -2.4240 0.0000 0.0000

2 1 3 1 0.6928 78.1546 0.5608 -3.1713 -3.7301 0.0000 0.0000

2 1 3 2 -1.4343 77.0699 0.9875 -3.4139 -1.4053 0.0000 0.0000

1 1 3 3 0.5153 2.1584 0.2000 -6.5859 -3.0000 0.0000 0.0000

2 1 3 3 0.2018 80.0000 0.3778 -2.5000 -2.8750 0.0000 0.0000

3 1 3 1 -1.9875 79.2591 1.0000 -2.4206 -3.9342 0.0000 0.0000

3 1 3 2 -1.1000 78.8002 -1.0000 -2.6282 -4.0000 0.0000 0.0000

3 1 3 3 -1.0000 83.5323 4.3660 -2.6805 -1.2938 0.0000 0.0000

1 3 3 1 3.4682 0.0781 0.9887 -6.1195 -0.5004 0.0000 0.0000

1 3 3 2 1.0000 16.5478 -1.0313 -2.0000 -2.6888 0.0000 0.0000

2 3 3 2 4.0818 -3.2744 -0.9664 -7.1634 -3.0000 0.0000 0.0000

1 3 3 3 4.2014 -10.0642 1.8690 -2.4805 -2.5000 0.0000 0.0000

2 3 3 3 1.0000 -10.0500 -1.0000 -2.1946 -0.5300 0.0000 0.0000

3 3 3 3 1.0000 1.6871 3.0000 -6.2660 -0.5500 0.0000 0.0000

1 1 1 4 4.1485 38.1222 2.2912 -10.8271 19.5577 0.0000 0.0000

1 3 1 4 13.5963 143.0497 -0.6551 -10.3770 50.3494 0.0000 0.0000

1 1 3 4 4.6153 -41.2146 2.6095 -2.3656 -59.9900 0.0000 0.0000

1 3 3 4 24.0662 -13.6312 -9.8252 1.2433 -59.9340 0.0000 0.0000

2 1 1 4 -2.5298 120.3131 0.8292 -5.3840 19.6104 0.0000 0.0000

2 1 3 4 3.1162 64.2411 1.3957 -4.6578 -59.4257 0.0000 0.0000

2 3 1 4 0.8541 -25.0788 -0.8283 -3.7322 36.5262 0.0000 0.0000

2 3 3 4 6.0246 29.4544 -0.4845 -6.3106 47.2012 0.0000 0.0000

3 1 1 4 -1.8658 158.9111 1.1132 -7.2577 -0.4250 0.0000 0.0000

3 1 3 4 2.7912 54.0974 -2.1091 -2.9448 -58.0874 0.0000 0.0000

4 1 1 4 -13.2905 144.9389 3.9884 -4.2634 -30.4780 0.0000 0.0000

2 ! Number of hydrogen bonds

3 2 3 1.8130 -3.5409 2.3815 21.9463

4 2 3 1.8319 -7.9316 3.2138 3.2323

A.3.2. Illustration of the Undertermined Rotational Alignment Problem

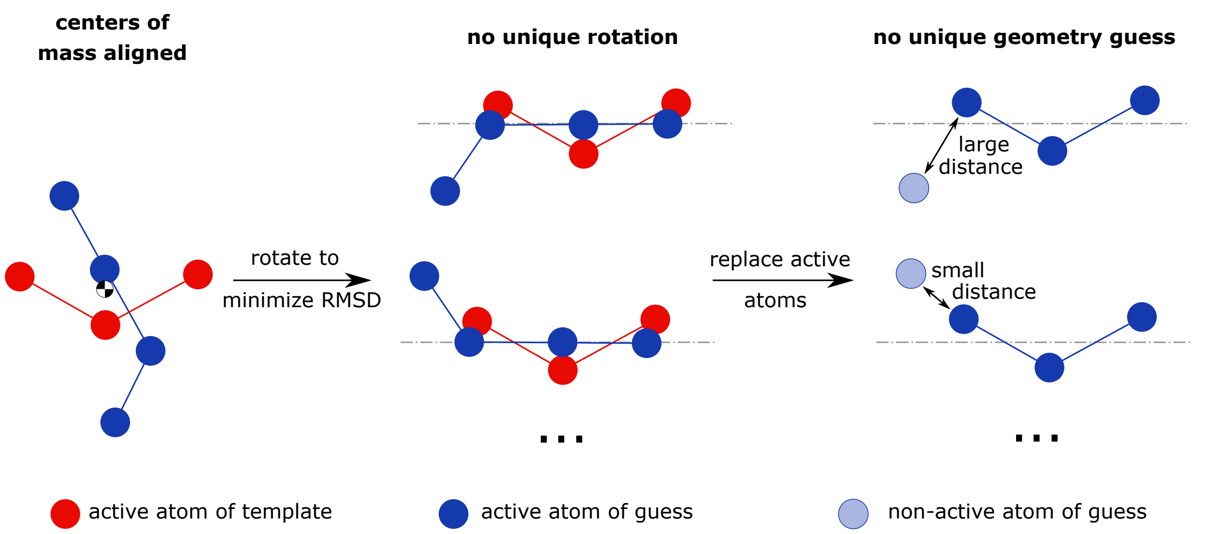
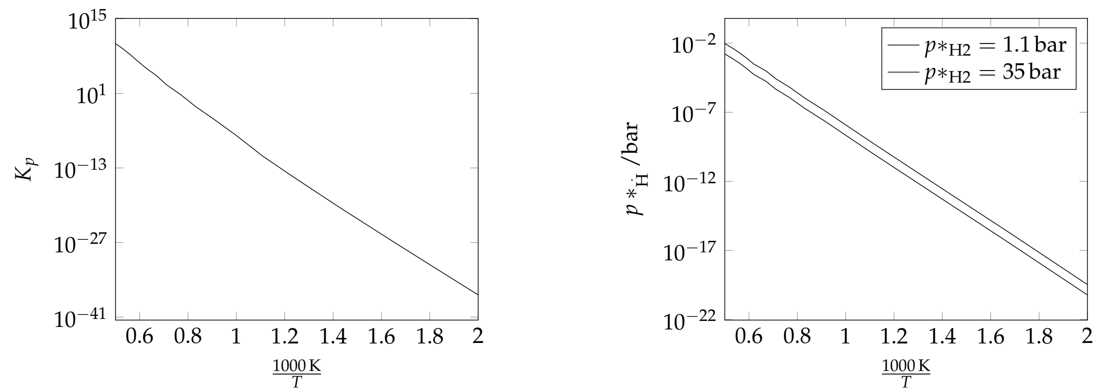


Figure A.3.: The rotation of the guess (blue) around the dash-dotted line is ambiguous. This leads to different TS geometry guesses, of which only one is chosen arbitrarily by the current implementation.

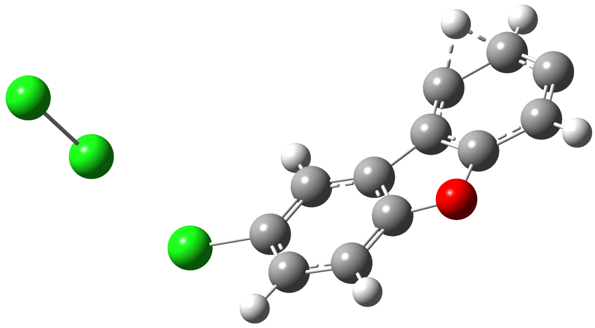
A.3.3. $H_2 \rightleftharpoons 2\,H$ Reaction Equilibrium



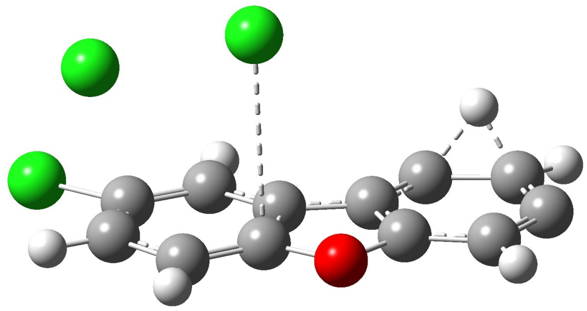
(a) Reaction equilibrium constant obtained at the B3LYP level of theory. Comparison of the B3LYP reaction Gibbs free energy with the value reported in the Chemkin Thermodynamic Data Base [169] at 298 K shows a deviation of less than 1 kcal mol^{-1}.

(b) $\dot{\mathrm{H}}$ partial pressure in reaction equilibrium. The partial pressures of H_2 are chosen in accordance with the experimental conditions of the thermal hydrogenolysis study by Cieplik *et al.* [163].

A.3.4. TS Snapshot of Reactions Involving Chlorine



(a) Doublet TS of the intramolecular H-shift reaction R8 with an inert Cl_2. The closest distance between the Cl_2 and the dibenzofuranyl radical is 3.2 Å. The reaction barrier is $78\,\mathrm{kcal\,mol^{-1}}$. The reactants are in singlet (Cl_2) and doublet states (dibenzofuranyl radical).



(b) Quadruplet TS of the chlorination reaction discussed in chapter 4. The reactants and products are shown in chapter 4, Fig. 4.8. The distance between the two Cl radicals is 2.58 Å. The dashed active C – Cl bond has a length of 3.29 Å. The reaction barrier is submerged [170] and has a height of $-11\,\mathrm{kcal\,mol^{-1}}$. The reactants are in doublet ($\dot{\mathrm{Cl}}$ radicals) and triplet states (dibenzofuranyl radical).

Figure A.5.: The two TS of reactions involving chlorine discussed in chapter 4. They both originate from the same reactive rMD trajectory snapshot.

A.3.5. Reaction List

Table A.5.: The complete list of reactions from the dibenzofurane studies in Chapter 4. Molecular properties, including optimized geometries with B3LYP, of reactants, products and TS can be obtained from the sqlite database files in the electronic Supporting Information of [122] after publication. The reactions can be found in the database files by searching for the reaction SMILES given below in the reaction column. The reaction SMILES are colon-seperated lists of the reactant and product SMILES. If multiple reactants or products are present, they are seperated by comma. Energies are ZPE-corrected electronic energies on the B3LYP level of theory using the TZVP basis set.

reaction	multiplicities			V_{barrier}	V_{reaction}
	reactants	TS	products	kcal mol^{-1}	kcal mol^{-1}
[C]#CC=c1c(=C)oc2c1cccc2:C#CC=C(c1ccccc1[O])C#C	2	2	2	57.85	-1.38
[C]#CC=c1oc2c(c1=C)cc(cc2)Cl:Clc1ccc2c(c1)C1=CC=C=C[C]1O2	2	2	2	28.22	-68.56
[C-]#CC=CC1=C(Cl)[O+]=C2C1=CC=[C]C2:[C]#CC=Cc1c(Cl)oc2c1cccc2	2	2	2	5.09	-85.80
[C-]#CC=CC1=C(Cl)[O+]=C2C1=CC=[C]C2:[C]#CC=Cc1c(Cl)oc2c1cccc2	4	4	4	60.14	-30.37
[C-]#CC=CC1=C=[O+]C2=CC=C[C@H]([C]12)Cl:ClC#CC=CC1=[C]Oc2c1cccc2	2	2	2	12.65	-72.90
[C-]#CC=Cc1c(Cl)oc(=CC=[C+])c1=C:Cl[C@@]12C#CC=CC2=C2C(=CC=[C]C2)O1	2	2	2	3.28	-63.16
[C]#CC=Cc1c(Cl)oc2c1cccc2:Cl[C@@]12C#CC=CC2=C2C(=CC=[C]C2)O1	2	2	2	89.82	48.37
[C],[Cl]#CC=CC(=[C]c1cccc[c+]1)[O-]:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	3.2		2	-196.09	-196.09
[C][C@H]1C=c2c(=C1)c1c(o2)cccc1:[C]#CC=c1c(=C)oc2c1cccc2	2	2	2	23.74	-30.65
[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl:ClC#CC=CC1=[C]Oc2c1cccc2	2	2	2	5.26	-47.41
[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl:ClC#CC=CC1=[C]Oc2c1cccc2	4	4	4	10.97	-23.58
[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl:ClC#CC=CC1=[C]Oc2c1cccc2	2	2	2	3.28	-47.64
[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl:ClC1=[C-][C@@H]2C=CC=C3[C]2C(=C=[O+]3)C=C1	4	4	4	16.27	-19.94
[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl:ClC1=C[C@H]([C-]1)C1=C=[O+]c2c1cccc2	4	4	4	19.44	0.14
[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl:ClC1=C2[C@H]1[C@H]2C1=[C]Oc2c1cccc2	2	2	2	4.90	-0.31
[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl:O=C=C(C1=C=C[CH-]C=C1)C=CC(=[C+])Cl	4	4	4	24.28	1.85
[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl:O=C=C(C1=C=C[CH-]C=C1)C=CC(=[C+])Cl	4	4	4	24.28	1.85
[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl:O=C=C(C1=C=C[CH-]C=C1)C=CC(=[C+])Cl	2	2	2	26.15	2.02
[C-]=C[CH+][C@]12Oc3c(C2=C1)c(Cl)ccc3:[cH-]1ccc(c(c1=O)C1=CC1=CC=[C+])Cl	2	2	2	5.29	-28.20
[C-]=C[CH+]C1=C=COc2c1cccc2:[cH-]1cccc(c1=O)C(=CC=[C+])C#C	4	4	4	27.04	7.53
[C-]=CC=C(OC1=C=CC(=C[CH+]1)Cl)C#C:[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl	4	4	4	-0.07	-41.14
[C-]=CC=C(OC1=C=CC(=C[CH+]1)Cl)C#C:[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl	2	2	2	1.19	-39.37
[C-]=CC=C=C1C=CC=CC1=[O+]C#C:[C-]=CC=c1c(=[CH+])oc2c1cccc2	4	4	4	0.59	-47.92
[C-]=CC=C1[O+]=C2C(=C=CC=C2)C1=CCl:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	4	4	4	41.32	-1.60
[C-]=CC=C1[O+]=C2C(=C=CC=C2)C1=CCl:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2	2	2	40.39	-0.49
[C-]=CC=C1[O+]=C2C(=C=CC=C2)C1=CCl:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2	2	2	40.39	-1.12
[C-]=CC=C1[O+]=C2C(=C=CC=C2)C1=CCl:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	4	4	4	41.32	-1.60
[C-]=CC=c1c(=[CH+])oc2c1cccc2:[C-]=C[C+]=c1c(=C)oc2c1cccc2	4	4	4	23.53	-9.04
[C-]=CC=c1c(=[CH+])oc2c1cccc2:[C+]=CC=C(C1=C=C[CH-]C=C1)C(=[CH+])[O-].[H+]	2	2	2	74.22	72.63

[C-]=CC=c1c(=[CH+])oc2c1cccc2:[CH]=C1Oc2c([C@@]31[C@@H]1C3=C1)cccc2	2	2	2	13.11	10.29
[C-]=CC=c1c(=[CH+])oc2c1cccc2:[CH]=c1oc2c3c1cccc3ccc2	2	2	2	12.34	-94.00
[C-]=CC=c1c(=[CH+])oc2c1cccc2:[CH-]=CC=c1c(=[C+])oc2c1cccc2	4	4	4	6.87	-8.78
[C-]=CC=c1c(=[CH+])oc2c1cccc2:[CH-]=CC=C1C(=[CH])[O+]=C2C1=C=CC=C2	4	4	4	15.85	3.82
[C-]=CC=c1c(=[CH+])oc2c1cccc2:[cH-]1cccc(c1=O)C(=CC=[C+])C#C	4	4	4	19.67	8.27
[C-]=CC=c1c(=[CH+])oc2c1cccc2:[CH+]=c1oc2=C[CH-]C=Cc2c1[C@@H]1[C]=C1	2	2	2	4.83	4.80
[C-]=CC=c1c(=[CH+])oc2c1cccc2:C=C1Oc2c([C]1C1=C=C1)cccc2	2	2	2	25.08	-49.06
[C-]=CC=c1c(=[CH+])oc2c1cccc2:c1ccc2c(c1)[C]1C(=CC3=C[C@@H]13)O2	2	2	2	4.99	-49.10
[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2:[C-]=CC=c1oc([c-]c1=[CH+])C=CC=C=[Cl+]	2	2	2	144.56	142.45
[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2:C#CC=c1oc2c(c1=[CH])c(Cl)ccc2	2	2	2	-0.63	-45.23
[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl:[cH-]1cc(Cl)cc(c1=O)C1=CC1=CC=[C+]	2	2	2	58.22	7.42
[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl:C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	4	4	4	0.69	-44.53
[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl:C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	2	2	2	-0.91	-46.74
[C-]=CC=c1oc2c(c1=[CH+])ccc(c2)Cl:C#CC=c1oc2c(c1=[CH])ccc(c2)Cl	2	2	2	-0.63	-45.86
[C-]=CC=c1oc2c(c1=[CH+])cccc2Cl:[C]#CC=c1oc2c(c1=C)cccc2Cl	4	4	4	2.79	-23.03
[C-]=CC=c1oc2c(c1=[CH+])cccc2Cl:[CH-]=C1[C]=CC=C=[O+]c2c1cccc2Cl	4	4	4	30.01	0.24
[C-]=CC=c1oc2c(c1=[CH+])cccc2Cl:[CH-]=C1[C]=CC=C=[O+]c2c1cccc2Cl	2	2	2	35.20	34.01
[C-]=CC=c1oc2c(c1=[CH+])cccc2Cl:C#CC=c1oc2c(c1=[CH])cccc2Cl	2	2	2	0.44	-47.62
[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2:[C-]=CC1=[O+]C2=[C+]CC=CC2=C1C(=[CH-])Cl	4	4	4	95.35	49.48
[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2:[C-]=CC1=[O+]C2=[C+]CC=CC2=C1C(=[CH-])Cl	2	2	2	82.56	76.55
[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2:[C+]=CC(=C(C1=C=C[CH-]C=C1)C(=[CH+])Cl)[O-].[H+]	4	4	4	50.18	35.29
[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2:[H],[C-]#Cc1oc2c(c1C(=[CH+])Cl)cccc2	4	4	2,3	42.21	36.44
[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2:ClC(=[CH])C(=C1C=CC=CC1=O)C1=C=C1	2	2	2	25.43	25.20
[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2:ClC(=[CH+])C1=C2C=C[CH-]C=C2O[C@]21[C]=C2	4	4	4	12.68	0.34
[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2:ClC=C=C1C(=[O+]c2c1cccc2)C=[C-]	4	4	4	5.96	-9.22
[C-]1[C@@H]2[C@]31C(=[O+]c1c3cccc1Cl)C=C2:Clc1cccc2c1oc1=CC=C[C]=c21	2	2	2	0.96	-89.16
[C-]1=CC2=c3cc(Cl)ccc3=[O+][C@]32[C@@H]1[CH]3:ClC1=CC2=C3C=C=CC=C3O[C]2C=C1	4	4	4	1.73	-47.15
[C+]#CC=CC(=C(Cl)[O-])C1=CC=CC=[C]1:[C]#CC=Cc1c(Cl)oc2c1cccc2	2	2	2	11.45	-37.90
[C+]#CC=CC(=C(Cl)[O-])C1=CC=CC=[C]1:[C]#CC=Cc1c(Cl)oc2c1cccc2	4	4	4	15.27	-10.09
[C+]=c1oc2=C[CH-]C=Cc2c1C1C=C1:[CH-]=CC=c1c(=[C+])oc2c1cccc2	4	4	4	10.38	-25.01
[C+]=CC(=C(C1=C=C[CH-]C=C1)C(=[CH+])Cl)[O-].[H+]:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	4	4	4	11.70	-35.29
[CH-](C(=O)C(=[C+])C1=C=C(C=C[CH-]1)Cl)C=[CH+]:[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl	4	4	4	4.87	-38.13
[CH],[C-]=CC=C1[C]=c2c(=[O+]1)c(Cl)ccc2:[C-]=CC=c1oc2c(c1=[CH+])cccc2Cl	2,3		2	-140.00	-140.00
[CH],[C-]=CC=C1[C]=c2c(=[O+]1)c(Cl)ccc2:[C-]=CC=c1oc2c(c1=[CH+])cccc2Cl	2,3		2	-140.00	-140.00
[CH],[C+]=CC=C1O[C-]2C(=[C]1)C(=CC=C2)Cl:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2,3		2	-141.81	-141.81
[CH],[C+]=CC=C1O[C-]2C(=[C]1)C(=CC=C2)Cl:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2,3		2	-141.81	-141.81
[CH-]=C[C@@H]1[C]=C(C1=O)C1=C=CC(=C[CH+]1)Cl:[CH-]=C[CH-]C(=O)C(=[C+])C1=C=CC(=[Cl+])C=C1	2	2	2	26.15	23.55
[CH-]=C[C@H]1[C]=C(C1=O)C1=C=CC(=C[CH+]1)Cl:[CH-]=C[CH-]C(=O)C(=[C+])C1=C=CC(=[Cl+])C=C1	2	2	2	26.15	23.55
[CH-]=C[C]=c1oc2c(c1=[CH+])c(Cl)ccc2:[C-]=C[C+]=c1oc2c(c1=C)c(Cl)ccc2	4	4	4	21.16	7.63
[CH-]=C[C]=c1oc2c(c1=[CH+])ccc(c2)Cl:Clc1ccc2=C3C(=[C]C=[C-]C3)[O+]=c2c1	4	4	4	20.75	-45.12

[CH-]=C[C]=c1oc2c(c1=[CH+])cccc2Cl:[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl	4	4	4	0.00	-29.46
[CH-]=C[CH-]C(=O)C(=[C+])C1=C=CC(=[Cl+])C=C1:[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl	4	4	4	3.58	-38.73
[CH-]=C[CH-]C(=O)C(=[C+])C1=C=CC=CC1=[Cl+]:[CH+]=C[C@@H]1[C]=C(C1=O)C1=C=CC=C[C-]1Cl	2	2	2	4.14	-21.02
[CH-]=c1c(=[C+])oc2c1cccc2,C#C:[CH-]=CC=c1c(=[C+])oc2c1cccc2	2,3	2	2	-88.21	-131.20
[CH-]=c1c(=[C+])oc2c1cccc2,C#C:[CH-]=CC=c1c(=[C+])oc2c1cccc2	4,3	2	2	-98.87	-141.85
[CH-]=c1c(=[C+])oc2c1cccc2,C#C:[CH-]=CC=c1c(=[C+])oc2c1cccc2	4,1	4	4	4.21	-45.48
[CH]=C1C=C(c2c1c1ccccc1o2)Cl:C#Cc1c(oc2c1cccc2)C(=[CH])Cl	2	2	2	26.74	18.44
[CH]=C1C=c2c(=C1)c1c(o2)cccc1:c1ccc2c(c1)[C]1C(=CC3=C[C@@H]13)O2	2	2	2	26.22	22.06
[CH]=c1oc2c(c1=C1C=C1)cccc2:[O-]C(=[CH+])C(=C1C=C1)C1=CC=CC=[C]1	2	2	2	66.31	65.31
[CH]=c1oc2c3c1cccc3ccc2:[C-]=CC=c1c(=[CH+])oc2c1cccc2	2	2	2	107.54	94.00
[CH]=c1oc2c3c1cccc3ccc2:[C-]=CC=c1c(=[CH+])oc2c1cccc2	2	2	2	106.66	94.54
[CH-]=CC(=[C+])Cl,c1ccc2c(c1)C#CO2:[C-]=C(C(=CC1=C=[O+]c2c1cccc2)Cl	2,3		2	-122.50	-122.50
[CH-]=CC(=[C+])Cl,c1ccc2c(c1)C#CO2:[C-]=C(C(=CC1=C=[O+]c2c1cccc2)Cl	2,3		2	-122.50	-122.50
[CH-]=CC(=[C+])Cl,c1ccc2c(c1)C#CO2:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	2,3		2	-127.09	-127.09
[CH-]=CC=C(C1=C=[O+]1)C1=CC=CC=[C]1:[CH-]=CC=c1c(=[C+])oc2c1cccc2	4	4	4	14.47	-45.77
[CH]=CC=C1C#Cc2c(O1)c(Cl)ccc2:[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl	2	2	2	27.67	10.38
[CH]=CC=C1C#Cc2c(O1)ccc(c2)Cl:[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl	2	2	2	26.49	11.86
[CH]=CC=C1C#Cc2c(O1)cccc2Cl:[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2	2	2	2	26.41	8.84
[CH]=CC=C1C#COc2c1cccc2:[CH-]=CC=c1c(=[C+])oc2c1cccc2	2	2	2	28.62	-1.22
[CH]=CC=C1C#COc2c1cccc2:[CH-]=CC=c1c(=[C+])oc2c1cccc2	4	4	4	19.26	0.03
[CH]=CC=C1C#COc2c1cccc2:C#CC=c1c(=[CH])oc2c1cccc2	2	2	2	30.70	-54.40
[CH-]=CC=c1c(=[C+])oc2c1cccc2:[C]#CC(=CC=[CH+])c1ccccc1[O-]	4	4	4	25.69	19.58
[CH-]=CC=c1c(=[C+])oc2c1cccc2:[CH-][C@H]1C=c2c(=[C+]1)oc1c2cccc1	4	4	4	19.96	-3.35
[CH-]=CC=c1c(=[C+])oc2c1cccc2:[CH-]=CC=C(C1=C=[O+]1)C1=CC=CC=[C]1	4	4	4	53.33	45.77
[CH-]=CC=c1c(=[C+])oc2c1cccc2:[CH-]=CC=C(C1=C=[O+]1)C1=CC=CC=[C]1	4	4	4	53.33	45.77
[CH-]=CC=c1c(=[C+])oc2c1cccc2:[CH-]=CC=C=C1C=CC=CC1=[O+]C#[C]	4	4	4	57.07	56.97
[CH-]=CC=c1c(=[C+])oc2c1cccc2:C#CC=c1c(=[CH])oc2c1cccc2	4	4	4	10.79	-35.52
[CH-]=CC=c1c(=[C+])oc2c1cccc2:C#CC=c1c(=[CH])oc2c1cccc2	4	4	4	10.79	-35.71
[CH-]=CC=C1C(=[CH])[O+]=C2C1=C=CC=C2:[C-]=CC=c1c(=[CH+])oc2c1cccc2	4	4	4	11.95	-2.60
[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2:[CH-]=C[CH-]C(=O)C(=[C+])C1=C=CC=CC1=[Cl+]	4	4	4	40.43	34.73
[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2:C#C,Clc1cccc2c1c(=[C-])c(=[CH+])o2	2	2	3,4	44.50	158.15
[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2:C#C,Clc1cccc2c1c(=[C-])c(=[CH+])o2	2	2	3,2	44.50	135.48
[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2:C#C.Clc1cccc2c1c(=[C-])c(=[CH+])o2	2	2	1,2	44.50	41.75
[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2:Cl[C]=C1C(=CC=[CH+])O[C-]2C1=C=CC=C2	4	4	4	40.24	3.62
[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl:[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl	4	4	4	15.56	10.37
[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl:[CH]=CC=C1C#Cc2c(O1)ccc(c2)Cl	4	4	4	29.61	-4.88
[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl:C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	4	4	4	11.45	-34.06
[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl:Clc1cc[c+]c(c1)C(=[C-])c1ccco1	4	4	4	46.54	2.46
[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl:[CH-]=C[CH-]C(=O)C(=[C+])C1=C=CC(=[Cl+])C=C1	4	4	4	42.31	38.73
[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl:[CH-]=CC=c1oc2=CC#CC=c2c1=[C+],Cl	4	4	4,1	97.78	77.76

[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl:C#C,Clc1ccc2c(c1)oc(=[CH+])c2=[C-]	2	2	3,2	45.21	136.28
[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl:C#C,Clc1ccc2c(c1)oc(=[CH+])c2=[C-]	2	2	3,4	45.21	156.01
[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl:C#C,Clc1ccc2c(c1)oc(=[CH+])c2=[C-]	2	2	1,2	45.21	42.56
[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl:C=C=CC1=[O+]c2c(C1=[C-])ccc(c2)Cl	4	4	4	46.98	-1.83
[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl:Clc1cc2OC(=CC=[CH])C#Cc2cc1	2	2	2	14.75	-11.77
[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl:Clc1cc2OC(=CC=[CH])C#Cc2cc1	2	2	2	16.13	-11.76
[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl:Clc1ccc2c(c1)OC=CC=[C+]C2=[C-]	4	4	4	49.20	40.61
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:[C-]=CC=c1oc2c(c1=[CH+])cccc2Cl	4	4	4	14.10	8.87
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:[CH-](C(=O)C(=[C+])C1=C=C(C=C[CH-]1)Cl)C=[CH+]	2	2	2	57.39	55.74
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:[CH-](C(=O)C(=[C+])C1=C=C(C=C[CH-]1)Cl)C=[CH+]	4	4	4	44.03	39.17
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:[CH]=CC=C1C#Cc2c(O1)c(Cl)ccc2	2	2	2	16.38	-10.38
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:C#C,ClC1=CC=CC2=C3[C@]4(O[C]12)C=C34	4	4	3,2	46.89	96.95
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:C#C,ClC1=CC=CC2=C3[C@]4(O[C]12)C=C34	4	4	3,2	47.93	101.14
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:C#C,Clc1cccc2c1oc(=[CH+])c2=[C-]	4	4	3,4	46.89	140.76
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:C#C,Clc1cccc2c1oc(=[CH+])c2=[C-]	4	4	1,4	46.89	47.03
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:C#C,Clc1cccc2c1oc(=[CH+])c2=[C-]	4	4	1,4	47.93	48.07
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:C#C,Clc1cccc2c1oc(=[CH+])c2=[C-]	4	4	3,4	47.93	144.95
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:C#CC=c1oc2c(c1=[CH])cccc2Cl	4	4	4	10.01	-35.41
[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl:Clc1cccc2c1oc1=C[C@H]([C+]=c21)[CH-]	4	4	4	18.81	-5.67
[CH-]=CC1=[O+]c2c([C@]31[C]=C3Cl)cccc2:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	4	4	4	4.18	-15.21
[CH-]=CC1=C=[O+]c2c1cccc2,ClC#[C]:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	3,2		2	-86.93	-86.93
[CH-]=CC1=C=[O+]c2c1cccc2,ClC#[C]:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	3,2		2	-86.70	-86.70
[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2:C#Cc1oc2c(c1C(=[CH])Cl)cccc2	4	4	4	10.72	-25.00
[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2:C#Cc1oc2c(c1C(=[CH])Cl)cccc2	2	2	2	3.21	-49.28
[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2:ClC#Cc1c(C=[CH])oc2c1cccc2	4	4	4	5.67	-26.12
[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2:ClC#Cc1c(C=[CH])oc2c1cccc2	2	2	2	1.38	-47.57
[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2:ClC=Cc1oc2c(c1C#[C])cccc2	4	4	4	5.80	-8.79
[CH-]1C(=C=CC=C1Cl)OC(=CC=[C+])C#C:[C-]=CC=c1oc2c(c1=[CH+])ccc(c2)Cl	2	2	2	1.02	-38.77
[CH-]1C(=C=CC=C1Cl)OC(=CC=[C+])C#C:[C-]=CC=c1oc2c(c1=[CH+])ccc(c2)Cl	4	4	4	0.01	-46.92
[CH-]1C(=C=CC=C1Cl)OC(=CC=[C+])C#C:[C-]=CC=c1oc2c(c1=[CH+])ccc(c2)Cl	2	2	2	1.20	-39.27
[CH-]1C(=C=CC=C1Cl)OC(=CC=[C+])C#C:[C-]=CC=c1oc2c(c1=[CH+])ccc(c2)Cl	4	4	4	-0.24	-43.35
[cH-]1c(Cl)ccc(c1=O)C1=CC1=CC=[C+]:[C-]=C[CH+][C@]12Oc3c(C2=C1)ccc(c3)Cl	2	2	2	33.97	28.80
[cH-]1c(Cl)ccc(c1=O)C1=CC1=CC=[C+]:[C-]=C[CH+][C@]12Oc3c(C2=C1)ccc(c3)Cl	2	2	2	33.97	28.80
[CH-]1C=CC(=C=C1Cl)OC(=CC=[C+])C#C:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2	2	2	0.49	-39.16
[CH-]1C=CC(=C=C1Cl)OC(=CC=[C+])C#C:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2	2	2	0.49	-39.80
[cH-]1ccc(c(c1=O)C1=CC1=CC=[C+])Cl:[C-]=C[CH+][C@@]12Oc3c(C2=C1)c(Cl)ccc3	2	2	2	33.49	28.20
[cH-]1cccc(c1=O)C(=CC=[C+])C#C:[C-]=CC=c1c(=[CH+])oc2c1cccc2	4	4	4	9.17	-13.18
[CH+]=c1oc2=C[CH-]C=Cc2c1[C@H]1[C]=C1:[C-]=CC=c1c(=[CH+])oc2c1cccc2	4	4	4	8.43	1.12
[CH+]=c1oc2=C[CH-]C=Cc2c1[C@H]1[C]=C1:C#CC=c1c(=[CH])oc2c1cccc2	2	2	2	0.34	-52.98
[CH+]=Cc1oc2c3c1C(=[C][C@H]3[CH-]C=C2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	2	2	2	-0.04	-32.23

[CH+]=Cc1oc2c3c1C(=[C][C@H]3[CH-]C=C2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	4	4	4	17.26	-10.15
[Cl],[C-]=CC=[C]C(=[CH+])C1=C=CC=CC1=O:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2,3		2	-145.75	-145.75
[Cl],[C-]=CC=C=[O+]C1=CC=C=C[C]1C#C:[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl	2,3		2	-121.23	-121.23
[Cl],[C-]=CC=C=[O+]C1=CC=C=C[C]1C#C:[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl	2,3		2	-121.23	-121.23
[Cl],[C-]=CC=C1[O+]=C2C(=C=CC=C2)C1=[CH]:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2,3		2	-88.81	-88.81
[Cl],[C-]=CC=C1[O+]=C2C(=CC=C=C2)C1=[CH]:[C-]=CC=c1oc2c(c1=[CH+])ccc(c2)Cl	2,3		2	-89.77	-89.77
[Cl],[C-]=CC=C1OC2=CC=C=C[C]2C1=[CH+]:[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl	2,3		2	-87.89	-87.89
[Cl],[C]=CC=c1oc2c(c1=CCl)cccc2Cl:[Cl],C#CC=c1oc2c(c1=CCl)cccc2Cl	2,1	2	2,3	-7.63	-16.35
[Cl],[C]=CC=c1oc2c(c1=CCl)cccc2Cl:[Cl],C#CC=c1oc2c(c1=CCl)cccc2Cl	2,3	2	2,3	-37.54	-46.26
[Cl],[C]=CC=c1oc2c(c1=CCl)cccc2Cl:[Cl],C#CC=c1oc2c(c1=CCl)cccc2Cl	2,3	2	2,1	-37.54	-80.45
[Cl],[C]=CC=c1oc2c(c1=CCl)cccc2Cl:[Cl],C#CC=c1oc2c(c1=CCl)cccc2Cl	2,1	2	2,1	-7.63	-50.54
[Cl],[C+]#CC=CC#Cc1ccccc1[O-]:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	2,3		2	-40.51	-40.51
[Cl],[C+]#CC=CC(=[C])OC1=C=C[CH-]C=C1:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	2,3		2	-144.64	-144.64
[Cl],[C+]=CC=C1OC2=C=CC=C[C-]2C1=[CH]:[C][C@@H]1C=c2c(=C1)c1c(o2)c(Cl)ccc1	2,3		2	-100.37	-100.37
[Cl],[CH-]=C(c1c[c+]c[cH-]c1=O)[C]=CC=[C+]:[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl	2,3		2	-151.40	-151.40
[Cl],[CH-]=C1C=c2c(=C1)c1c(o2)cc[c+]c1:Clc1ccc2c(c1)[C]1C(=C[C@@H]3C1=C3)O2	2,3		2	-65.31	-65.31
[Cl],[CH-]=CC=C1[O+]=C2C(=C=CC=C2)C1=[C]:[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2	2,3		2	-88.36	-88.36
[Cl],[CH-]=CC=C1[O+]=C2C(=C=CC=C2)C1=[C]:[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2	2,3		2	-88.36	-88.36
[Cl],[CH-]=CC=C1[O+]=C2C(=CC=C=C2)C1=[C]:[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl	2,3		2	-85.78	-85.78
[Cl],[CH-]=CC=C1[O+]=C2C(=CC=C=C2)C1=[C]:Clc1ccc2c(c1)[O+]=C1C2=[C][C@H]2[C@@H]1[CH-]2	2,3		2	-102.15	-102.15
[Cl],[CH-]=CC=C1OC2=CC=C=C[C]2C1=[C+]:[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl	2,3		2	-85.82	-85.82
[Cl],[CH-]1C=CC2=[C-][C@@]3(OC2=C1)C=[C+]C=[C+]3:C#CC=C=C1[C]=C2[C@@]1(OCl)C=CC=C2	2,3		2	-26.34	-26.34
[Cl],[CH-]1C=Cc2c(=C1)oc1=[C+]C=C=Cc21:C#CC=[C+][C@@]1(Cl)OC2=C[CH-]C=CC2=[C]1	2,3		2	34.35	34.35
[Cl],[CH-]1C=Cc2c(=C1)oc1=[C+]C=C=Cc21:ClC(=[CH+])C=C=C1OC2=C[CH-]C=CC2=[C]1	2,3		2	33.20	33.20
[Cl],[CH-]1C=Cc2c(=C1)oc1=[C+]C=C=Cc21:ClC=[C]C=C=C=C=C1C=CC=CC1=O	2,3		2	-16.35	-16.35
[Cl],[CH+]=c1cc2-c(c1)[c-]1c(=C=CC=C1)o2:[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl	2,3		2	-13.27	-13.27
[Cl],[CH+]=CC=C1OC2=C=CC=C[C-]2C1=[C]:[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl	2,3		2	-86.76	-86.76
[Cl],[Cl],Clc1cc2-c(c1)c2:Clc1cc(Cl)cc(c1)Cl	2,2,3		1	-231.06	-231.06
[Cl],[Cl],Clc1cc2-c(c1)c2:Clc1cc(Cl)cc(c1)Cl	2,2,3		1	-227.98	-227.98
[Cl],[Cl],Clc1ccc2c(c1)C1=CC=C=C[C]1O2:[Cl],[Cl],C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	2,2,4	4	2,2,4	-24.43	29.97
[Cl],[Cl],Clc1ccc2c(c1)C1=CC=C=C[C]1O2:[Cl],[Cl],C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	2,2,2	4	2,2,2	42,67	63,49
[Cl],[Cl],Clc1ccc2c(c1)C1=CC=C=C[C]1O2:[Cl],[Cl],C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	2,2,2	4	2,2,4	42.67	97.08
[Cl],[Cl],Clc1ccc2c(c1)C1=CC=C=C[C]1O2:[Cl],[Cl],C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	2,2,4	4	2,2,2	-24.43	-3.62
[Cl],[Cl],Clc1ccc2c(c1)C1=CC=C=C[C]1O2:[Cl],Clc1ccc2c(c1)C1=[C+]C[C-]=C[C@@]1(O2)Cl	2,2,4	4	2,3	-10.57	-23.27
[Cl],[Cl],Clc1ccc2c(c1)C1=CC=C=C[C]1O2:[Cl],Clc1ccc2c(c1)C1=[C+]C[C-]=C[C@@]1(O2)Cl	2,2,2	4	2,3	56.54	43.84
[Cl],[Cl]:ClCl	2,2		1	-48.13	-48.13
[Cl],[Cl]:ClCl	2,2		1	-48.13	-48,13
[Cl],[H],ClC1=CC(=C=C=C1)Cl:Clc1cc(Cl)cc(c1)Cl	2,2,3		1	-208.71	-208.71
[Cl],[O-]C1=CC=C[C@H]([CH+]1)c1cc(Cl)cc(c1)Cl:Cl[C@H]1C(=CC=C[C@H]1c1cc(Cl)cc(c1)Cl)[O]	2,3	4	4	27.96	14.31
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	2,2,2	4	4,1	9.65	-9.15

[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	2,2,4	4	4,3	-68.04	-11.51
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	2,2,4	4	4,1	-68.04	-86.84
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	2,2,2	4	4,3	9.65	66.18
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,4	4	3,2	-68.04	-91.30
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,2	4	3,2	9.65	-13.61
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,4	4	3,2	-64.51	-88.82
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,4,4	4	3,2	-144.89	-169.19
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,4	4	1,4	-64.51	-83.85
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,4	4	3,4	-64.51	-8.44
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,4,4	4	3,4	-144.89	-88.82
[Cl],[O]c1ccccc1,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,4,4	4	1,4	-144.89	-164.23
[Cl],C#CC=C=C1[C-]=[O+]c2c1cccc2:[CH-]=C(C=C=C1[C]=[O+]c2c1cccc2)Cl	2,3	4	4	-6.53	-14.83
[Cl],C#CC=C=c1oc2=C[CH+]C=Cc2[c-]1:ClC(=[CH+])C=C=C1OC2=C[CH-]C=CC2=[C]1	2,3		2	-13.92	-13.92
[Cl],C#CC=c1oc2c(c1=CCl)cccc2Cl:[Cl],[C]=CC=c1oc2c(c1=CCl)cccc2Cl	2,3	4	2,3	42.19	46.26
[Cl],C#Cc1c(C#C)oc2c1cccc2:C#C[C@]1(Cl)c2ccccc2O[C@@]21[C]=C2	2,1	2	2	38.23	33.51
[Cl],C#Cc1c(C#C)oc2c1cccc2:C#C[C@]1(Cl)c2ccccc2O[C@@]21[C]=C2	2,3	2	2	-11.08	-15.80
[Cl],C1#CC=c2c(=C1)oc1c2cccc1:ClC1=C=C[C]2C(=C1)c1ccccc1O2	2,3	2	2	-43.37	-93.44
[Cl],C1#CC=c2c(=C1)oc1c2cccc1:ClC1=C=C[C]2C(=C1)c1ccccc1O2	2,1	2	2	-13.28	-63.35
[Cl],C1=C[C-]=[O+]C#CC=C=CC#CC=C1:ClC1=[C][O+]=C=CC=CC=C=C=C[C-]=C1	2,3		2	-26.63	-26.63
[Cl],C1=C=C=C=C[C@H]2[C-]=C2[O+]=C=CC=C1:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	2,3		2	-158.45	-158.45
[Cl],C1=CC=c2[c-](C=1)[o+]c1c2=CC=C=C1:Clc1ccc2c(c1)O[C]1C2=CC=C=C1	2,3		2	10.90	10.90
[Cl],c1ccc2c(c1)oc1=C=CC=C=c21:C#Cc1c(oc2c1cccc2)C(=[CH])Cl	2,3		2	-31.58	-31.58
[Cl],c1ccc2c(c1)oc1=C=CC=C=c21:C#Cc1c(oc2c1cccc2)C(=[CH])Cl	2,3		2	-31.58	-31.58
[Cl],Cl[C]=CC(=CC#C)Cl:ClC=C1C=C(C=C1Cl)Cl	2,2	1	1	-20.43	-113.08
[Cl],Cl[C]=CC(=CC#C)Cl:Clc1cc(Cl)cc(c1)Cl	2,2	1	1	-20.43	-145.80
[Cl],Cl[C]1C=C=CC(=C1)Cl,[O]c1ccccc1:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,2	4	1,4	13.10	-6.24
[Cl],Cl[C]1C=C=CC(=C1)Cl,[O]c1ccccc1:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,2	4	3,2	13.10	-11.21
[Cl],Cl[C]1C=C=CC(=C1)Cl,[O]c1ccccc1:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,4	4	3,4	-67.28	-11.21
[Cl],Cl[C]1C=C=CC(=C1)Cl,[O]c1ccccc1:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,4	4	3,2	-67.28	-91.58
[Cl],Cl[C]1C=C=CC(=C1)Cl,[O]c1ccccc1:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,4	4	1,4	-67.28	-86.62
[Cl],Cl[C]1C=C=CC(=C1)Cl,[O]c1ccccc1:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2,2,2	4	3,4	13.10	69.17
[Cl],Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl	2,2		1	-86.62	-86.62
[Cl],O=C1C=CC=C[C@H]1c1cc(Cl)cc(c1)Cl:[O]C1=CC=C[C@H]([C@H]1c1cc(Cl)cc(c1)Cl)Cl	2,3	4	4	7.06	6.85
[H],[C-]#C[C+]=Cc1c(Cl)oc2c1cccc2:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	2,3		2	-38.50	-38.50
[H],[Cl+]=C1C=C[C@]23[C@@]41[C@H]1[C-]5[C@H]1[C@]1([O+]=C24)[C@]35[CH-]1:Cl[C]1C=CC#CC[C-]=CC#C[O+]=C=C1	2,3		2	-85.61	-85.61
[H],[O-]C1=C2C=C=C[C@]31C1=CC=C[C@@H]2[C+]31:[C-]1=CC#CC=CC=C[C]=[O+]C#CC1	2,3		2	-43.86	-43.86
[H],[O-]c1ccccc1C1=[C+]C(=C=C=C1)Cl:ClC1=[C][O+]=C=CC=CC=C=C=C[C-]=C1	2,3		2	40.63	40.63
[H],C#Cc1oc2c(c1C(=[C])Cl)cccc2:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	2,3	4	4	5.04	-41.02
[H],Cl[C@]12[C@@H]3C1=C3c1c2c2ccccc2o1:ClC=C=C1Oc2c([C]1C#C)cccc2	2,3		2	-152.78	-152.78
[H],ClC1=[C]C(=CC(=C1)Cl)Cl:Clc1cc(Cl)cc(c1)Cl	2,4	3	3	-71.30	-114.24

[H],ClC1=[C]C(=CC(=C1)Cl)Cl:Clc1cc(Cl)cc(c1)Cl	2,2	3	3	4.43	-38.51
[H],ClC1=[C]C(=CC(=C1)Cl)Cl:Clc1cc(Cl)cc(c1)Cl	2,2	3	3	5.14	-37.60
[H],ClC1=[C]C(=CC(=C1)Cl)Cl:Clc1cc(Cl)cc(c1)Cl	2,4	3	3	-89.94	-132.68
[H],ClC1=C=C([C+]2C1=Cc1c2cccc1)[O-]:C#C[C]1Oc2c(C1=C=CCl)cccc2	2,3		2	-62.71	-62.71
[H],ClC1=C=C2C(=c3cc[cH-]c3=CO2)C1=[CH+]:ClC=C=C1Oc2c([C]1C#C)cccc2	2,3		2	-131.57	-131.57
[H],ClC1=CC=C=c2c1c1C=C=CC=c1o2:ClC=C=C1Oc2c([C]1C#C)cccc2	2,3		2	-72.41	-72.41
[H],Clc1ccc(c(c1)c1[c-]c[c+]c[c+]1)[O-]:Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2	2,3	4	4	4.09	-74.76
[H],Clc1ccc2c(c1)C1=[C-]C=C=CC1=[O+]2:Clc1ccc2c(c1)C1=CC=C=C[C]1O2	2,3	4	4	5.43	-45.76
[H],Clc1ccc2c(c1)C1=[C-]C=C=CC1=[O+]2:Clc1ccc2c(c1)C1=CC=C=C[C]1O2	2,3	4	4	5.43	-45.76
[H],Clc1cccc2c1[O+]=C1C2=[C-]C=C=C1:Clc1cccc2c1O[C]1C2=CC=C=C1	2,3	4	4	5.42	-45.53
[H],Clc1cccc2c1[O+]=C1C2=[C-]C=C=C1:Clc1cccc2c1O[C]1C2=CC=C=C1	2,3	4	4	5.42	-45.53
[H],O=C1[C@H]2[CH-]C=CC1=C1C(=[C+]2)C=C=C1:[C-]1=CC#CC=CC=C[C]=[O+]C#CC1	2,3		2	10.81	10.81
[H],O=C1[C@H]2[CH-]C=CC1=C1C(=[C+]2)C=C=C1:[C-]1=CC#CC=CC=C[C]=[O+]C#CC1	2,3		2	10.81	10.81
[H],O=C1[C@H]2C=C[C-]=C1C(=C1C=C1)C(=[C+]2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	2,3		2	-126.45	-126.45
[O],[C-]=C([CH+]C=C1[C@]23[C@@]1([C]3)C=CC=C2)Cl:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	3,2		2	-189.50	-189.50
[O],[O-]C1=C=C=CC=CC=C2[C@@H](C(=C1)Cl)[C]=[O+]2:[O-]OC1=C=C=CC=CC=C=[O+][C]=CC(=C1)Cl	3,2		2	9.61	9.61
[O],[O]c1cccc2c1c1ccccc1o2:[O]Oc1cccc2c1c1ccccc1o2	3,4	2	2	-78.14	-106.32
[O],[O]c1cccc2c1c1ccccc1o2:[O]Oc1cccc2c1c1ccccc1o2	3,2	2	2	-2.26	-30.45
[O],[O]Oc1cccc2=c3c(=c12)cc(Cl)cc3:O=[O+]c1cccc2c1[C]1C=C(Cl)C=C[C@]21[O-]	3,4	4	4	-45.33	-69.22
[O],[O]Oc1cccc2=c3c(=c12)cc(Cl)cc3:O=[O+]c1cccc2c1[C]1C=C(Cl)C=C[C@]21[O-]	3,2	4	4	-1.69	-25.58
[O],C1=[C]C=CC=C1:[O]c1ccccc1	3,2		2	-123.09	-123.09
[O],C1=[C]C=CC=C1:[O]c1ccccc1	3,2	4	4	-0.59	-42.71
[O],C1=[C]C=CC=C1:[O]c1ccccc1	3,4	4	4	-103.46	-145.59
[O],ClC(=[C])C=CC(=[C+])C1=C=C[CH-]C=C1:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	3,2		2	-170.83	-170.83
[O],ClC(=[C])C=CC(=[C+])C1=C=C[CH-]C=C1:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	3,2		2	-170.83	-170.83
[O],ClC1=Cc2c3C1=[C]C=Cc(c3)cc2:ClC1=C2c3ccccc3O[C@]32[C]2[C@@H]1[C@@H]32	3,2		2	-133.56	-133.56
[O],Clc1cc([O])c2c(c1)oc1c2cccc1:ClC1=CC(=O)c2c(=C1)oc1=CC=C[C@@H](c21)[O]	3,2	4	4	1.77	-13.98
[O],Clc1cc([O])c2c(c1)oc1c2cccc1:ClC1=CC(=O)c2c(=C1)oc1=CC=C[C@@H](c21)[O]	3,4	4	4	-68.27	-84.02
[O],Clc1ccc2c(c1[O])c1ccccc1o2:ClC1=CC=c2c(C1=O)c1[C@@H]([O])C=CC=c1o2	3,4	4	4	-71.41	-87.37
[O],Clc1ccc2c(c1[O])c1ccccc1o2:ClC1=CC=c2c(C1=O)c1[C@@H]([O])C=CC=c1o2	3,2	4	4	1.25	-14.71
[O-]C(=[CH+])C(=C1C=C1)C1=CC=CC=[C]1:[CH]=c1oc2c(c1=C1C=C1)cccc2	2	2	2	1.00	-65.31
[O-]C(=CC=[CH+])C(=[C-])c1cc(Cl)cc[c+]1:[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl	4	4	4	5.17	-40.63
[O-]C1=CC=[C+]C[CH]1:[H],[O-]C1=CC=C=C[CH+]1	4	4	2,3	29.27	26.52
[O]c1ccccc1,[Cl],Cl[C]1C=C=CC(=C1)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	4,2,4	4	4,3	-147.83	-91.30
[O]c1ccccc1,[Cl],Cl[C]1C=C=CC(=C1)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	4,2,2	4	4,3	-70.14	-13.61
[O]c1ccccc1,[Cl],Cl[C]1C=C=CC(=C1)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	4,2,4	4	4,1	-147.83	-166.63
[O]c1ccccc1,[Cl],Cl[C]1C=C=CC(=C1)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	4,2,2	4	4,1	-70.14	-88.94
[O]c1ccccc1,[Cl],Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	4,2,4	4	3,2	-147.83	-171.09
[O]c1ccccc1,[Cl],Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	4,2,2	4	3,2	-70.14	-93.41
[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl:Clc1cc(Cl)cc(c1)Cl,[CH]=CC=CC=C=O	4,1	4	3,4	27.90	103.67

[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl:Clc1cc(Cl)cc(c1)Cl,[CH]=CC=CC=C=O	4,3	4	1,4	-47.43	-46.99
[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl:Clc1cc(Cl)cc(c1)Cl,[CH]=CC=CC=C=O	4,3	4	3,4	-47.43	28.34
[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl:Clc1cc(Cl)cc(c1)Cl,[CH]=CC=CC=C=O	4,1	4	1,4	27.90	28.34
[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	4,1	4	3,2	27.90	-4.46
[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	4,3	4	3,2	-47.43	-79.79
[O]c1ccccc1:[O-]C1=CC=[C+]C[CH]1	2	2	2	90.49	78.98
[O]c1ccccc1C1=C=CC=C=C1:C#CC=C(c1ccccc1[O])C#C	2	2	2	21.81	-7.75
[O]c1ccccc1c1ccc2-c1c2Cl:ClC1=[C]C=Cc2c1oc1c2cccc1	2	2	2	4.47	-56.38
[O-]OC(=c1c(=[CH])oc2c1cccc2)C(=[CH+])Cl:[O]Oc1c(Cl)ccc2c1c1ccccc1o2	4	4	4	2.16	-67.96
[O-]OC1=CC=CC(=[C]1)Oc1[c+]ccc(c1)Cl:[O]Oc1cccc2c1c1ccc(cc1o2)Cl	4	4	4	7.82	-49.22
[O-]OC1=CC=CC(=[C]1)Oc1cccc[c+]1:[O]Oc1cccc2c1c1ccccc1o2	4	4	4	8.66	-46.58
[O]Oc1c(Cl)ccc2c1c1ccccc1o2:Clc1ccc2c3c1OO[C@@H]1c3c(o2)C=C[CH]1	2	2	2	37.05	30.97
[O]Oc1cc(Cl)cc2c1c1ccccc1o2:[O],Clc1cc([O])c2c(c1)oc1c2cccc1	2	2	3,2	28.37	30.16
[O]Oc1cc(Cl)cc2c1c1ccccc1o2:[O],Clc1cc([O])c2c(c1)oc1c2cccc1	2	2	3,4	28.37	100.20
[O]Oc1cc(Cl)cc2c1c1ccccc1o2:[O-]OC1=C=C(Cl)C=C2C1=c1ccccc1=[O+]2,[H]	4	4	3,2	48.50	41.89
[O]Oc1cc(Cl)cc2c1c1ccccc1o2:Clc1cc2OO[C@@H]3c4c2c(c1)oc4C=C[CH]3	2	2	2	38.67	32.89
[O]Oc1cccc2c1c1c(Cl)cccc1o2:O=[O+]C1=CC=C[C]=C1c1c([O-])cccc1Cl	2	2	2	78.52	77.11
[O]Oc1cccc2c1c1c(Cl)cccc1o2:O=[O+]c1cccc(c1C1=[C]C=CC=C1Cl)[O-]	2	2	2	78.23	78.75
[O]Oc1cccc2c1c1ccc(cc1o2)Cl:Clc1ccc2c(c1)oc1=CC=C[C]=c21,O=O	4	4	4,1	8.28	76.90
[O]Oc1cccc2c1c1ccc(cc1o2)Cl:O=O,Clc1ccc2c(c1)oc1=CC=C[C]=c21	4	4	3,2	8.28	-28.11
[O]Oc1cccc2c1c1ccc(cc1o2)Cl:O=O,Clc1ccc2c(c1)oc1=CC=C[C]=c21	4	4	3,4	8.28	38.15
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	3,2	2	2	-94.88	-143.09
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	3,4	2	2	-117.22	-165.42
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	3,2		2	-143.09	-143.09
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	1,2	2	2	-1.16	-49.36
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	3,4	4	4	-97.36	-145.55
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	3,4	2	2	-115.88	-167.25
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	1,4	4	4	-0.48	-48.67
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	1,2	2	2	0.95	-50.43
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	3,2	4	4	-77.42	-125.60
C#C,[C-]=C(C1=C=[O+]c2c1cccc2)Cl:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	3,2	2	2	-95.93	-147.31
C#C,[C-]=c1c(=[CH+])oc2c1cccc2:[CH]=c1oc2c(c1=C1C=C1)cccc2	3,2		2	-261.88	-261.88
C#C,[C-]=c1c(=[CH+])oc2c1cccc2:[CH]=c1oc2c(c1=C1C=C1)cccc2	3,2		2	-265.03	-265.03
C#C,[C-]=c1c(=[CH+])oc2c1cccc2:[CH]=CC1=Cc2c1c1ccccc1o2	3,2		2	-243.54	-243.54
C#C,[C-]=c1c(=[CH+])oc2c1cccc2:[CH]=CC1=Cc2c1c1ccccc1o2	3,2		2	-246.69	-246.69
C#C,[CH-]=c1c(=[C+])oc2c1cccc2:[CH-]=CC=c1c(=[C+])oc2c1cccc2	1,2	2	2	5.52	-37.47
C#C,[CH-]=c1c(=[C+])oc2c1cccc2:[CH-]=CC=c1c(=[C+])oc2c1cccc2	3,2	4	4	-81.01	-130.69
C#C,[CH-]=c1c(=[C+])oc2c1cccc2:[CH-]=CC=c1c(=[C+])oc2c1cccc2	3,4	4	4	-92.67	-142.36
C#C,[CH-]=c1c(=[C+])oc2c1cccc2:[CH+]=C[C@H]1[C]=c2c1c1C=C[CH-]C=c1o2	3,2		2	-111.63	-111.63
C#C,[CH-]1C=CC2=C3[C@@]4(OC2=C1)[C@@H]3[C+]4:[CH-]=CC=c1c(=[C+])oc2c1cccc2	3,2		2	-216.09	-216.09

C#C,Cl[C]=C1C(=[CH+])O[C-]2C1=C=CC=C2:[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2	3,2		2	-159.36	-159.36
C#C,Clc1ccc2c(c1)[C]1C3=C[C@@]13O2:[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl	3,2		2	-115.82	-115.82
C#C,Clc1ccc2c(c1)c(=[C-])c(=[CH+])o2:[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl	1,4	4	4	-0.15	-48.48
C#C,Clc1ccc2c(c1)c(=[C-])c(=[CH+])o2:[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl	3,2	4	4	-76.17	-124.50
C#C,Clc1ccc2c(c1)c(=[C-])c(=[CH+])o2:[CH-]=CC=c1oc2c(c1=[C+])cc(cc2)Cl	3,4	4	4	-97.03	-145.36
C#C,Clc1ccc2c(c1)oc(=[CH+])c2=[C-]:[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl	3,2		2	-136.28	-136.28
C#C,Clc1ccc2c(c1)oc(=[CH+])c2=[C-]:[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl	3,2		2	-233.96	-233.96
C#C,Clc1ccc2c(c1)oc(=[CH+])c2=[C-]:[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl	3,2		2	-233.96	-233.96
C#C,Clc1cccc2c1c(=[C-])c(=[CH+])o2:[CH]=C[C@@H]1C2=C3[C@]12Oc1c3c(Cl)ccc1	3,2		2	-30.59	-30.59
C#C,Clc1cccc2c1c(=[C-])c(=[CH+])o2:[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2	3,2		2	-135.48	-135.48
C#C,Clc1cccc2c1c(=[C-])c(=[CH+])o2:[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2	1,2	2	2	2.75	-43.24
C#C,Clc1cccc2c1c(=[C-])c(=[CH+])o2:[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2	3,2	2	2	-94.12	-140.12
C#C,Clc1cccc2c1c(=[C-])c(=[CH+])o2:[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2	3,4	2	2	-116.79	-162.79
C#C,Clc1cccc2c1oc(=[C+])c2=[CH-]:[CH-]=C[C]=c1oc2c(c1=[CH+])cccc2Cl	3,2		2	-130.85	-130.85
C#C,Clc1cccc2c1oc(=[C+])c2=[CH-]:Clc1cccc2c1oc(=C1C=C1)c2=[CH]	3,2		2	-169.11	-169.11
C#C,Clc1cccc2c1oc(=[CH+])c2=[C-]:[CH]=C1C=c2c(=C1)c1c(o2)c(Cl)ccc1	3,2		2	-310.24	-310.24
C#C[C@H]([C@]1(Cl)C=C1Cl)Cl:C#C[C@H](C(=CC#[Cl])Cl)Cl	1	1	1	35.42	26.56
C#C[C@H]([C@]1(Cl)C=C1Cl)Cl:C#C[C@H](C(=CC#[Cl])Cl)Cl	1	1	1	43.42	34.19
C#C[C]=C(c1ccccc1[O-])C(=[CH+])Cl:C#Cc1oc2c(c1C(=[CH])Cl)cccc2	4	4	4	13.39	-14.71
C#C[CH][C@@]12Oc3c(C2=C1)c(Cl)ccc3:C#CC=C1C=C1c1c([O])cccc1Cl	2	2	2	6.50	-22.23
C#C[CH][C@]12Oc3c(C2=C1)ccc(c3)Cl:C#CC=C1C=C1c1ccc(cc1[O])Cl	2	2	2	6.35	-22.74
C#CC(=C(C(=[CH+])Cl)[O-])C1=CC=CC=[C]1:C#Cc1c(oc2c1cccc2)C(=[CH])Cl	4	4	4	14.27	-22.89
C#CC(=C1C=C1)OC1=C=C[C](C=C1)Cl:Clc1ccc2c(c1)c(=[CH])c(=C1C=C1)o2	2	2	2	3.12	-35.07
C#CC(=COc1ccccc1C#[C])Cl:ClC1=C=CC#Cc2c(O[CH]1)cccc2	2	2	2	7.94	-43.70
C#CC=C(C(=[CH])C1=C=CC(=[Cl+])C=C1)[O-]:C#CC=c1oc2c(c1=[CH])ccc(c2)Cl	2	2	2	0.42	-63.60
C#CC=C(C(=[CH])C1=C=CC=CC1=[Cl+])[O-]:ClC1=CC=C[C]=C1C(=[CH-])C1=CC=C=[O+]1	4	4	4	45.98	12.25
C#CC=C(C(=[CH+])C1=[C]C(=CC=C1)Cl)[O-]:C#CC=c1oc2c(c1=[CH])cccc2Cl	2	2	2	2.34	-64.02
C#CC=C(C(=[CH+])C1=[C]C(=CC=C1)Cl)[O-]:ClC1=CC=CC(=[C]1)C(=[CH-])C1=CC=C=[O+]1	4	4	4	46.23	11.96
C#CC=C(OC1=CC=CC(=[C]1)Cl)C#C:C#CC=c1oc2c(c1=[CH])c(Cl)ccc2	2	2	2	0.33	-39.54
C#CC=C=C(C1=CC=CC=[C]1)C(=O)Cl:C#CC=C=C1[C](Cl)Oc2c1cccc2	4	4	4	20.92	-3.79
C#CC=C=C1[C](Cl)Oc2c1cccc2:C#CC=C=C(C1=CC=CC=[C]1)C(=O)Cl	4	4	4	24.72	3.79
C#CC=c1c(=[CH])oc2c1cccc2:[C-]=CC=c1c(=[CH+])oc2c1cccc2	4	4	4	47.07	46.17
C#CC=c1c(=[CH])oc2c1cccc2:C1=CC=C2[C](C=1)Oc1c2cccc1	2	2	2	45.57	-64.42
C#CC=c1c(=[CH])oc2c1cccc2:C1=CC=c2c(=[C]1)oc1c2cccc1	2	2	2	63.76	-62.44
C#CC=C1c2c1c1c(o2)cccc1:[CH-]=C=CC1=C=C1Oc1cccc[c+]1	3	3	3	35.61	27.76
C#CC=c1oc2=CC=C[C]=c2c1=CCl:C#CC=c1oc2c(c1=[CH])c(Cl)ccc2	2	2	2	40.43	-0.53
C#CC=c1oc2=CC=C[C]=c2c1=CCl:C#CC=c1oc2c(c1=[CH])c(Cl)ccc2	4	4	4	42.21	-0.25
C#CC=c1oc2c(c1=[CH])c(Cl)ccc2:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	4	4	4	46.36	42.77
C#CC=c1oc2c(c1=[CH])c(Cl)ccc2:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2	2	2	54.29	45.87
C#CC=c1oc2c(c1=[CH])c(Cl)ccc2:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2	2	2	54.94	45.23

C#CC=c1oc2c(c1=[CH])c(Cl)ccc2:Clc1cccc2c1C(=[CH-])C(=[O+]2)[C@@H]1[C]=C1	4	4	4	49.82	45.38
C#CC=c1oc2c(c1=[CH])cc(cc2)Cl:[C]#CC=c1oc2c(c1=C)cc(cc2)Cl	4	4	4	38.66	21.30
C#CC=c1oc2c(c1=[CH])ccc(c2)Cl:[C-]=CC=c1oc2c(c1=[CH+])ccc(c2)Cl	2	2	2	54.61	47.24
C#CC=c1oc2c(c1=[CH])ccc(c2)Cl:[C-]=CC=c1oc2c(c1=[CH+])ccc(c2)Cl	2	2	2	46.47	45.86
C#CC=c1oc2c(c1=[CH])ccc(c2)Cl:[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl	4	4	4	46.12	35.06
C#CC=c1oc2c(c1=[CH])ccc(c2)Cl:C#CC=C(OC1=[C]C=CC(=C1)Cl)C#C	2	2	2	40.67	39.69
C#CC=c1oc2c(c1=[CH])cccc2Cl:[C-]=CC=c1oc2c(c1=[CH+])cccc2Cl	2	2	2	45.86	46.66
C#CC=c1oc2c(c1=[CH])cccc2Cl:[C-]=CC=c1oc2c(c1=[CH+])cccc2Cl	2	2	2	47.10	47.62
C#CC=c1oc2c(c1=[CH])cccc2Cl:[CH]=C1C=c2c(=C1)c1c(o2)c(Cl)ccc1	2	2	2	7.35	-22.16
C#CC=c1oc2c(c1=[CH])cccc2Cl:[CH]=C1C=c2c(=C1)c1c(o2)c(Cl)ccc1	4	4	4	7.98	-27.84
C#CC=c1oc2c(c1=[CH])cccc2Cl:[CH-]=CC=c1oc2c(c1=[C+])cccc2Cl	4	4	4	45.44	34.39
C#CC=c1oc2c(c1=[CH])cccc2Cl:[O]c1c(cccc1Cl)C1=CC1=CC#C	2	2	2	57.54	6.24
C#CC=c1oc2c(c1=[CH])cccc2Cl:Clc1cccc2c1O[C]1C2=CC=C=C1	2	2	2	4.44	-63.43
C#CC=c1oc2c(c1=[CH])cccc2Cl:Clc1cccc2c1O[C]1C2=CC=C=C1	4	4	4	12.06	-29.85
C#Cc1c(=[C+])oc2=C[CH-]C=Cc12,C#[Cl]:ClC=C=C1Oc2c([C]1C#C)cccc2	2,3		2	-242.47	-242.47
C#Cc1c(=[CH+])oc2=CC=C[C@H](c12)[CH-]:C#Cc1c(=[CH])oc2c1c(=C)ccc2	4	4	4	27.49	-75.59
C#Cc1oc2c(c1C(=[CH])Cl)cccc2:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	2	2	2	46.32	45.33
C#Cc1oc2c(c1C(=[CH])Cl)cccc2:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	2	2	2	47.04	45.81
C#Cc1oc2c(c1C(=[CH])Cl)cccc2:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	2	2	2	51.77	48.56
C#Cc1oc2c(c1C(=[CH])Cl)cccc2:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	4	4	4	35.71	25.00
C1=C2C1=C2,[O-]C#Cc1cccc[c+]1:[CH-]1C=Cc2c(=C1)oc1=[C+]C=C=Cc21	3,3		3	-156.36	-156.36
C1=C2C1=C2,ClC1=[C]c2c(O1)cccc2:Cl[C]1Oc2c3C1=C1C=C1Cc3ccc2	3,2		2	-123.36	-123.36
C1=CC=C2[C](C=1)Oc1c2cccc1:[CH-]=C1Oc2c([C@]31[C]=C[CH+]3)cccc2	4	4	4	83.82	73.65
C1=CC=C2[C](C=1)Oc1c2cccc1:C#CC=c1c(=[CH])oc2c1cccc2	2	2	2	109.99	64.43
C1=CC=C2[C](C=1)Oc1c2cccc1:C1=CC=c2c(=[C]1)oc1c2cccc1	2	2	2	109.75	1.98
C1=Cc2c(=C[CH-]1)oc1=C[C@H]3[C@@H](c21)[C+]3:[CH+]=c1oc2=C[CH-]C=Cc2c1[C@H]1[C]=C1	4	4	4	38.85	36.64
c1ccc2c(c1)C#CO2,[C-]=CC(=[CH+])Cl:C#CC=C=C1[C](Cl)Oc2c1cccc2	3,2		2	-184.71	-184.71
c1ccc2c(c1)C#CO2,[CH-]=CC(=[C+])Cl:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	3,2		2	-122.73	-122.73
c1ccc2c(c1)C#CO2,[CH-]=CC(=[C+])Cl:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	3,2		2	-127.09	-127.09
c1ccc2c(c1)C#CO2,C1=C2C1=C2:[CH-]1C=Cc2c(=C1)oc1=[C+]C=C=Cc21	3,3		3	-188.49	-188.49
c1ccc2c(c1)C#CO2,C1=C2C1=C2:[CH-]1C=Cc2c(=C1)oc1=[C+]C=C=Cc21	3,3		3	-188.49	-188.49
c1ccc2c(c1)C#CO2,C1=C2C1=C2:[CH-]1C=Cc2c(=C1)oc1=[C+]C=C=Cc21	3,3		3	-188.49	-188.49
c1ccc2c(c1)C#CO2,ClC(=[CH])C#C:OC1=C2C(=[C]C=C=C=C2C=CC=C1)Cl	3,2		2	-84.14	-84.14
c1ccc2c(c1)c1=C=CC=c1o2,[CH]:[CH]=CC1=Cc2c1c1ccccc1o2	3,2		2	-81.22	-81.22
c1ccc2c(c1)c1=C=CC=c1o2,[CH]:C1=C[C]=c2c(=C1)oc1c2cccc1	3,2		2	-186.95	-186.95
c1ccc2c(c1)OC1=CC=C=C[C]21:[C]#CC=c1c(=C)oc2c1cccc2	2	2	2	95.84	66.18
c1ccc2c(c1)OC1=CC=C=C[C]21:[CH-]1C=Cc2c(=C1)oc1=[C]C[C+]=Cc21	4	4	4	64.01	16.88
Cl[C@@H]1C#C[C]2C(=C1)Oc1c2cccc1:ClC=Cc1oc2c(c1C#[C])cccc2	2	2	2	36.45	15.38
Cl[C@@H]1C=CC=CC1=O,Cl[C]1C=C=CC(=C1)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	1,2	2	4,3	1.00	105.58
Cl[C@@H]1C=CC=CC1=O,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	1,2	2	1,2	1.00	-49.55

Cl[C@@H]1C=CC=CC1=O,Cl[C]1C=C=CC(=C1)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	1,2	2	3,2	1.00	25.79
Cl[C@]12[CH]C1=Cc1c2oc2c1cccc2:Cl[C]1C=C=Cc2c1oc1c2cccc1	2	2	2	1.29	-60.99
Cl[C@H]1[C]=C=Cc2c1oc1c2cccc1:[H],ClC1=C=C=Cc2c1oc1c2cccc1	2	2	2,1	39.01	32.23
Cl[C@H]1[C]=C=Cc2c1oc1c2cccc1:[H],ClC1=C=C=Cc2c1oc1c2cccc1	2	2	2,3	39.01	64.37
Cl[C]=C1C(=CC=[CH+])O[C-]2C1=C=CC=C2:[CH-]=CC=c1oc2c(c1=[C+])c(Cl)ccc2	4	4	4	36.62	-3.62
Cl[C]=C1C=Cc2c1c1ccccc1o2:ClC#CC=CC1=[C]c2c(O1)cccc2	2	2	2	40.47	34.38
Cl[C]=C1C=Cc2c1c1ccccc1o2:ClC#CC=CC1=[C]c2c(O1)cccc2	2	2	2	40.47	33.42
Cl[C]=C1C=Cc2c1oc1c2cccc1:ClC#CC=CC1=[C]Oc2c1cccc2	2	2	2	39.84	34.77
Cl[C]=Cc1oc2c(c1C#C)cccc2:[C-]=CC1=C(C(=C)Cl)C2=C=CC=CC2=[O+]1	2	2	2	68.48	50.48
Cl[C]=Cc1oc2c(c1C#C)cccc2:[Cl],C#Cc1c(C=[C])oc2c1cccc2	4	4	2,3	48.62	54.48
Cl[C]1C=C=CC(=C1)Cl,Cl[C@H]1C=CC=CC1=O:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	4,1	4	3,4	1.19	28.74
Cl[C]1C=C=CC(=C1)Cl,Cl[C@H]1C=CC=CC1=O:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	4,1	4	1,4	1.19	-46.67
Cl[C]1C=C=CC(=C1)Cl,Cl[C@H]1C=CC=CC1=O:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	4,1	4	3,2	1.19	-51.64
Cl[C]1C=C=Cc2c1oc1c2cccc1:C#CC=c1c(=[C]Cl)oc2c1cccc2	4	4	4	33.60	23.89
Cl[C]1C=C=Cc2c1oc1c2cccc1:C#CC=c1c(=[C]Cl)oc2c1cccc2	2	2	2	64.31	60.75
Cl[C]1C=C=Cc2c1oc1c2cccc1:C#Cc1c(oc2c1cccc2)C(=[CH])Cl	2	2	2	103.48	56.46
Cl[C]1C=C=Cc2c1oc1c2cccc1:Cl[C@]12[CH]C1=Cc1c2oc2c1cccc2	4	4	4	45.19	45.39
Cl[C]1C=C=Cc2c1oc1c2cccc1:ClC1=[C]C=Cc2c1oc1c2cccc1	4	4	4	51.86	0.24
Cl[C]1C=C=Cc2c1oc1c2cccc1:ClC1=C=C(C=C=C1)c1ccccc1[O]	4	4	4	27.15	6.00
Cl[C]1C=CC(=C=C1)C(=[CH-])C1=CC=C=[O+]1:[C-]=CC=c1oc2c(c1=[CH+])ccc(c2)Cl	2	2	2	15.93	-31.14
Cl[C+]1C=[C-][C@@H](C(=C1)Cl)Cl:Clc1cc(Cl)cc(c1)Cl	1	1	1	7.01	-87.96
ClC#[C],[CH+]=CC1=C=C2C(=C[CH-]C=C2)O1:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	2,3		2	-93.87	-93.87
ClC#[C],[CH+]=CC1=C=C2C(=C[CH-]C=C2)O1:[CH-]=Cc1oc2c(c1C(=[C+])Cl)cccc2	2,3		2	-93.87	-93.87
ClC#[C],[CH+]=CC1=C=C2C(=C[CH-]C=C2)O1:Cl[C@]12[CH]C1=Cc1c2c2ccccc2o1	2,3		2	-143.01	-143.01
ClC#[C],[CH+]=CC1=C=C2C(=C[CH-]C=C2)O1:Cl[C@]12[CH]C1=Cc1c2c2ccccc2o1	2,3		2	-143.01	-143.01
ClC#[C],[CH+]=CC1=C=C2C(=C[CH-]C=C2)O1:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	2,3		2	-88.11	-88.11
ClC#[C],[CH+]=CC1=C=C2C(=C[CH-]C=C2)O1:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	2,3		2	-88.11	-88.11
ClC#[C],[CH+]=CC1=C=C2C(=C[CH-]C=C2)O1:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	2,3		2	-88.11	-88.11
ClC#[C],[CH+]=CC1=C=C2C(=C[CH-]C=C2)O1:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	2,3		2	-88.11	-88.11
ClC#C,[C+]=CC1=C=C2C(=C[CH-]C=C2)O1:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	3,2	4	4	-39.96	-93.29
ClC#C,[C+]=CC1=C=C2C(=C[CH-]C=C2)O1:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	3,4	4	4	-72.46	-125.79
ClC#C,[C+]=CC1=C=C2C(=C[CH-]C=C2)O1:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	1,4	4	4	3.47	-49.86
ClC#C,[C+]=CC1=C=C2C(=C[CH-]C=C2)O1:Cl[C@@]12[C]3[C@@H]1[C@H]3c1c2c2ccccc2o1	3,2		2	-154.21	-154.21
ClC#C,C#CC1=[C]c2c(O1)cccc2:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	3,2		2	-80.94	-80.94
ClC#C,C#CC1=[C]Oc2c1cccc2:[CH-]=C1[C@H](Cl)[C]=C2C1=[O+]c1c2cccc1	3,2		2	-86.42	-86.42
ClC#C,C#CC1=[C]Oc2c1cccc2:C#Cc1c(oc2c1cccc2)C(=[CH])Cl	3,2		2	-128.93	-128.93
ClC#CC=CC1=[C]Oc2c1cccc2:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	2	2	2	49.63	47.41
ClC#CC=CC1=[C]Oc2c1cccc2:Cl[C]=C1C=Cc2c1oc1c2cccc1	2	2	2	5.07	-34.77
ClC#Cc1c(C=[CH])oc2c1cccc2:ClC1=C=CC=C2[C]1c1ccccc1O2	2	2	2	49.68	-60.33
ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1:[C-]=C(C=CC1=C=C=CC=CC=C=[O+]1)Cl	4	4	4	70.51	68.71

ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1:ClC#CC=CC1=[C]c2c(O1)cccc2	2	2	2	2.68	-46.84
ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1:ClC(=[C+])C=CC(=O)[C+]=C1[C-]=C[CH-]C=C1	2	2	2	64.49	59.67
ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1:ClC1=C=[O+]C(=[C]c2[c-]cccc2)C=C1	4	4	4	46.82	-3.14
ClC(=[CH])C(=C1C=CC=CC1=O)C1=C=C1:C#C[C]=C(c1ccccc1[O-])C(=[CH+])Cl	4	4	4	25.82	-28.13
ClC(=[CH+])[C](C12OC2=C1)C1=C=C[CH-]C=C1:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	2	2	2	5.95	-76.19
ClC(=[CH+])C1=C2C=C[CH-]C=C2O[C@]21[C]=C2:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	4	4	4	11.92	-0.90
ClC=C=C1C(=[O+]c2c1cccc2)C=[C-]:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	4	4	4	13.48	7.52
ClC=C=C1C(=[O+]c2c1cccc2)C=[C-]:[C-]=Cc1oc2c(c1C(=[CH+])Cl)cccc2	4	4	4	15.18	9.22
ClC=C1C=C(C=C1Cl)Cl:Clc1cc(Cl)cc(c1)Cl	1	1	1	94.70	-32.47
ClC=Cc1oc2c(c1C#[C])cccc2:Cl[C@H]1C=c2c(C1=[C+])c1c(=C[CH-]C=C1)o2	4	4	4	7.73	-9.58
ClC1=[C][C@@]2(C=C1)[C-]=C1C(=C[CH+]C=C1)O2:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	4	4	4	29.18	9.74
ClC1=[C][C@@]2(C=C1)[C-]=C1C(=C[CH+]C=C1)O2:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	4	4	4	29.18	9.74
ClC1=[C-][C@@H]2C=CC=C3[C]2C(=C=[O+]3)C=C1:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	2	2	2	24.53	16.27
ClC1=[C-][C@H]2C=CC=C3[C]2C(=C=[O+]3)C=C1:[C-]=C(C=CC1=C=[O+]c2c1cccc2)Cl	2	2	2	24.53	16.04
ClC1=[C]C=Cc2c1oc1c2cccc1:[CH-]=Cc1c2ccccc2oc1C(=[C+])Cl	4	4	4	67.94	62.78
ClC1=[C]C=Cc2c1oc1c2cccc1:Cl[C]1C=C=Cc2c1oc1c2cccc1	2	2	2	103.86	-0.93
ClC1=[C+]C=C=C([CH]1)Oc1[c-]cccc1:ClC1=C=C[C]2C(=C1)Oc1c2cccc1	4	4	4	9.08	-52.92
ClC1=C[C@@H]([O])c2c(=C1)oc1=CC=CC(=O)c21:[O],Clc1ccc2c(c1)oc1c2c([O])ccc1	4	4	3,4	15.78	87.66
ClC1=C[C@@H]([O])c2c(=C1)oc1=CC=CC(=O)c21:[O],Clc1ccc2c(c1)oc1c2c([O])ccc1	4	4	3,2	15.78	13.39
ClC1=C[C@]2([C@@]([CH-]1)([CH+]2)Cl)Cl:Clc1cc(Cl)cc(c1)Cl	3	3	3	1.50	-30.01
ClC1=C[C@]2([C@@]([CH-]1)([CH+]2)Cl)Cl:Clc1cc(Cl)cc(c1)Cl	3	3	3	1.20	-31.16
ClC1=C[CH+]C(=C=C1)C(=[C-])c1ccco1:[CH-]=CC=c1oc2c(c1=[C+])ccc(c2)Cl	4	4	4	44.61	-0.70
ClC1=C=C(C=C[CH-]1)Oc1cccc[c+]1:Clc1cccc2c1c1ccccc1o2	3	3	3	8.68	-48.86
ClC1=C=C(C=C1)C12OC32C1=CC=C[CH]3:ClC1=[C]C(=C=C2C=CC=CC2=O)C=C1	4	4	4	26.25	2.08
ClC1=C=C[C]2C(=C1)c1ccccc1O2:ClC1=C=C[C]2C(=C1)Oc1c2cccc1	4	4	4	5.49	-9.97
ClC1=C=C[C]2C(=C1)Oc1c2cccc1:[H],ClC1=C=C=Cc2c1oc1c2cccc1	4	4	2,3	59.43	52.70
ClC1=C=C[C]2C(=C1)Oc1c2cccc1:ClC1=C=C[C]2C(=C1)c1ccccc1O2	4	4	4	15.46	9.97
ClC1=C=C2[C](C=C1)c1c(O2)cccc1:[C]#CC=Cc1c(Cl)oc2c1cccc2	2	2	2	127.67	72.77
ClC1=C=C2[C](C=C1)c1c(O2)cccc1:Cl[C]=C1C=Cc2c1oc1c2cccc1	4	4	4	61.62	-3.07
ClC1=C=C2[C](C=C1)c1c(O2)cccc1:ClC1=[C]C=Cc2c1oc1c2cccc1	2	2	2	48.68	-0.46
ClC1=C=CC=C2[C]1c1ccccc1O2:ClC(=[C+])C1=C2C=C[CH-]C=C2OC21C=C2	2	2	2	128.22	114.38
ClC1=C=CC=C2[C]1c1ccccc1O2:ClC1=C=CC=C=C1c1ccccc1[O]	4	4	4	30.62	9.92
ClC1=C=CC=C2[C]1c1ccccc1O2:ClC1=C2[C@]31[C](C=C2)Oc1c3cccc1	4	4	4	51.17	51.21
ClC1=C2[C](C=C=C1)Oc1c2cccc1:C#Cc1oc2c(c1C(=[CH])Cl)cccc2	2	2	2	113.66	60.76
ClC1=C2[C](C=C=C1)Oc1c2cccc1:ClC=C=C1C(=[O+]c2c1cccc2)C=[C-]	4	4	4	77.38	63.84
ClC1=c2oc3=CC=C[C@H](c3c2C(=O)C=C1)[O]:[O],[O]c1ccc(c2c1c1ccccc1o2)Cl	4	4	3,4	15.87	90.21
ClC1=c2oc3=CC=C[C@H](c3c2C(=O)C=C1)[O]:[O],[O]c1ccc(c2c1c1ccccc1o2)Cl	4	4	3,2	15.87	14.31
ClC1=c2oc3=CC=CC(=O)c3c2[C@H](C=C1)[O]:[O],ClC1=CC=CC2=C3C(=CC=CC3=O)O[C]12	4	4	3,4	16.00	87.12
ClC1=c2oc3=CC=CC(=O)c3c2[C@H](C=C1)[O]:[O],ClC1=CC=CC2=C3C(=CC=CC3=O)O[C]12	4	4	3,2	16.00	14.88
ClC1=CC(=C=C=C1)Cl,[Cl],[H]:Clc1cc(Cl)cc(c1)Cl	3,2,2		1	-206.02	-206,02

ClC1=CC(=C=C=C1)Cl,Cl:Clc1cc(Cl)cc(c1)Cl	1,1	1	1	14.94	-79.42
ClC1=CC(=C=C=C1)Cl,Cl:Clc1cc(Cl)cc(c1)Cl	1,1	1	1	15.94	-75.39
ClC1=CC(=C=C=C1)Cl,Cl:Clc1cc(Cl)cc(c1)Cl	1,1	1	1	17.47	-75.39
ClC1=CC(=CC([CH]1)(Cl)[C@@H]1C=CC=CC1=O)Cl:[O]c1ccccc1,Clc1cc(Cl)cc(c1)Cl	2	2	4,3	6.84	123.99
ClC1=CC(=CC([CH]1)(Cl)[C@@H]1C=CC=CC1=O)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2	2	3,2	6.84	44.20
ClC1=CC(=CC([CH]1)(Cl)[C@@H]1C=CC=CC1=O)Cl:Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1	2	2	1,2	6.84	-31.13
ClC1=CC=C(C#C1)OC1=CC=CC=[C]1:ClC1=C=C2[C](C=C1)Oc1c2cccc1	4	4	4	4.44	-56.73
ClC1=CC=C[C]=C1C(=[CH-])C1=CC=C=[O+]1:[C-]=CC=c1oc2c(c1=[CH+])c(Cl)ccc2	2	2	2	14.90	-32.98
ClC1=CC=C[C-]2C1=C1C[C+]=C[C]=C1O2:Clc1cccc2c1c1=CC=C[C]=c1o2	4	4	4	48.86	-12.06
ClC1=CC=C[C-]2C1=C1C[C+]=C[C]=C1O2:Clc1cccc2c1c1=CC=C[C]=c1o2	2	2	2	10.12	-71.09
ClC1=CC=C=C([CH]1)C(=[CH+])C(=C1C=C1)[O-]:Clc1ccc2c(c1)c(=[CH])c(=C1C=C1)o2	2	2	2	0.49	-60.12
ClC1=CC=C=C([CH]1)C1=CC2(C1=O)C=C2:ClC1=CC=C=C([CH]1)C(=[CH+])C(=C1C=C1)[O-]	4	4	4	17.04	-5.54
ClC1=CC=C=C2[C]1Oc1c2cccc1:C#Cc1c(oc2c1cccc2)C(=[CH])Cl	4	4	4	41.69	30.27
ClC1=CC=C=C2[C]1Oc1c2cccc1:C#Cc1c(oc2c1cccc2)C(=[CH])Cl	2	2	2	62.18	55.85
ClC1=CC=CC(=[C]1)C1=CC(=[CH+])C=C1[O-]:[CH]=C1C=c2c(=C1)c1c(o2)c(Cl)ccc1	2	2	2	4.94	-55.47
ClC1=CC=CC2=C=C3C(=CC=C3)O[C]12:Clc1cccc2c1oc1=CC=C[C]=c21	2	2	2	104.74	-28.73
ClC1=CC=CC2=C3C(=CC=C=C3)O[C]12:[C-]1=CC2=c3cccc(c3=[O+][C@]32[C@@H]1[CH]3)Cl	4	4	4	48.74	47.12
ClC1=CC=CC2=C3C(=CC=C=C3)O[C]12:Clc1cccc2c1oc1=CC=C[C]=c21	2	2	2	106.05	-0.01
ClC1=CC2=C3C=C=CC=C3O[C]2C=C1:C#CC=c1c(=[CH])oc2c1cc(Cl)cc2	2	2	2	68.33	64.36
ClC1=CC2=C3C=C=CC=C3O[C]2C=C1:C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	2	2	2	109.07	63.11
ClC1=Cc2c3[C@@H]([CH]1)OOc1c3c(o2)ccc1:[O]Oc1cccc2c1c1ccc(cc1o2)Cl	2	2	2	6.10	-34.00
Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1:ClC1=C[C@]2([C@H](C(=C1)Cl)OC1=CC=C[CH][C@H]21)Cl	3,2		2	-31.12	-31.12
Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1:Clc1cc(Cl)cc(c1)Cl,[CH]=CC=CC=C=O	3,2	4	1,4	32.36	32.80
Clc1cc(Cl)cc(c1)Cl,[O]c1ccccc1:Clc1cc(Cl)cc(c1)Cl,[CH]=CC=CC=C=O	3,2	4	3,4	32.36	108.13
Clc1cc(Cl)cc(c1)Cl:[H],ClC1=[C]C(=CC(=C1)Cl)Cl	3	3	2,4	42.94	114.24
Clc1cc(Cl)cc(c1)Cl:[H],ClC1=[C]C(=CC(=C1)Cl)Cl	3	3	2,2	42.94	38.51
Clc1cc(Cl)cc(c1)Cl:[H],ClC1=[C]C(=CC(=C1)Cl)Cl	3	3	2,2	42.74	37.60
Clc1cc(Cl)cc(c1)Cl:[H],ClC1=[C]C(=CC(=C1)Cl)Cl	3	3	2,4	42.74	132.68
Clc1cc(Cl)cc(c1)Cl:Cl[C+]=CC(=CC(=[CH-])Cl)Cl	3	3	3	62.28	61.67
Clc1cc(Cl)cc(c1)Cl:ClC=C1C=C(C(=C1Cl)Cl	1	1	1	127.17	32.47
Clc1cc(Cl)cc(c1)Cl:ClC=C1C=C(C(=C1Cl)Cl	1	1	1	125.37	32.72
Clc1cc(Cl)cc(c1)Cl:ClC1=C[C@]2([C@@]([CH-]1)([CH+]2)Cl)Cl	3	3	3	32.36	31.16
Clc1cc(Cl)cc(c1)Cl:ClC1=C[C@]2([C@@H]1C(=C2)Cl)Cl	1	1	1	191.74	76.96
Clc1cc(Cl)cc(c1)Cl:ClC1=CC(=C=C=C1)Cl,Cl	1	1	1,1	94.35	79.42
Clc1cc(Cl)cc(c1)Cl:ClC1=CC(=C=C=C1)Cl,Cl	1	1	1,1	92.86	75.39
Clc1cc2C3=[C]CC4=C[C@@]34Oc2cc1:Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2	2	2	2	3.01	-37.40
Clc1ccc(c(c1)c1ccc2-c1c2)[O]:Clc1ccc2c(c1)C1=CC=C=C[C]1O2	2	2	2	4.32	-59.89
Clc1ccc[c-]c1C1=C[C@]21C=C[C]=[O+]2:[C+]=C[C@H]1C=C(C1=O)C1=C=CC=C[C-]1Cl	2	2	2	29.90	2.52
Clc1ccc2c(c1)C(=[CH-])C(=[O+]2)[C@@H]1[C]=C1:[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl	4	4	4	7.97	-0.29
Clc1ccc2c(c1)C(=[CH-])C(=[O+]2)[C@H]1[C]=C1:[C-]=CC=c1oc2c(c1=[CH+])cc(cc2)Cl	4	4	4	7.97	-0.39

Clc1ccc2c(c1)c(=[CH])c(=C1C=C1)o2:C#Cc1cc(Cl)ccc1O[C]=C1C=C1	2	2	2	37.63	36.13
Clc1ccc2c(c1)c(=[CH])c(=C1C=C1)o2:ClC1=CC(=C2C=C2[C]2C=C2)C(=O)C=C1	2	2	2	50.75	5.85
Clc1ccc2c(c1)c(=[CH])c(=C1C=C1)o2:ClC1=CC=C=C([CH]1)C(=[CH+])C(=C1C=C1)[O-]	2	2	2	65.10	59.69
Clc1ccc2c(c1)c(=[CH])c(=C1C=C1)o2:Clc1ccc2c(c1)C1=C[C@@H]3C(=C3)[C]1O2	2	2	2	17.74	-23.48
Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2:[C-](=C1[C]=[O+]c2c1cc(Cl)cc2)CC#C	4	4	4	49.50	43.21
Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2:[C-](=C1[C]=[O+]c2c1cc(Cl)cc2)CC#C	2	2	2	49.15	40.89
Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2:[C-]1=C(c2cc(Cl)c[cH-]c2=O)[C+]C[C+]=C1	4	4	4	51.40	50.62
Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2:[C-]1=CC(=O)C(=[C]C1)c1cc(Cl)cc[c+]1	2	2	2	37.94	29.37
Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2:Clc1ccc2c(c1)c1=[C]C=CC=c1o2	2	2	2	12.13	-69.43
Clc1ccc2c(c1)c1=[C]C=CC=c1o2:[H],Clc1ccc2c(c1)C1=[C-]C=C=CC1=[O+]2	4	4	2,3	50.77	45.07
Clc1ccc2c(c1)c1=[C]C=CC=c1o2:ClC1=CC2=C3C=C=CC=C3O[C]2C=C1	2	2	2	106.15	0.05
Clc1ccc2c(c1)C1=CC=C=C[C]1O2,ClCl:Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2,ClCl	4,3	2	4,3	-24.18	21.16
Clc1ccc2c(c1)C1=CC=C=C[C]1O2,ClCl:ClCl,Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2	4,3	2	1,2	-24.18	-32.57
Clc1ccc2c(c1)C1=CC=C=C[C]1O2,ClCl:ClCl,Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2	4,3	2	3,2	-24.18	2.65
Clc1ccc2c(c1)C1=CC=C=C[C]1O2:[C]#CC=c1oc2c(c1=C)cc(cc2)Cl	2	2	2	96.78	68.56
Clc1ccc2c(c1)C1=CC=C=C[C]1O2:C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	2	2	2	67.88	63.49
Clc1ccc2c(c1)C1=CC=C=C[C]1O2:C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	4	4	4	41.96	29.97
Clc1ccc2c(c1)C1=CC=C=C[C]1O2:C#CC=c1oc2c(c1=[CH])cc(cc2)Cl	4	4	4	41.96	29.97
Clc1ccc2c(c1)C1=CC=C=C[C]1O2:Clc1c[cH-]c(=O)c(c1)C1=CC=C=C[C+]1	4	4	4	29.76	8.37
Clc1ccc2c(c1)C1=CC=C=C[C]1O2:Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2	2	2	2	79.05	69.76
Clc1ccc2c(c1)O[CH]C=CC1=C=C21:Clc1ccc2c(c1)OC=CC=[C+]C2=[C-]	2	2	2	52.30	48.96
Clc1ccc2c(c1)oc1=[C]C=CC=c21:Clc1ccc(c(c1)[O])C1=CC=CC#C1	2	2	2	48.66	46.52
Clc1ccc2c(c1)oc1=[C]C=CC=c21:Clc1ccc2=C3C(=[C]C=[C-]C3)[O+]=c2c1	4	4	4	62.84	13.69
Clc1ccc2c(c1)oc1=[C]C=CC=c21:Clc1ccc2=C3C(=[C]C=[C-]C3)[O+]=c2c1	2	2	2	81.64	71.54
Clc1ccc2c3c1oc1c3[C@H](OO2)[CH]C=C1:[O]Oc1ccc(c2c1c1ccccc1o2)Cl	2	2	2	6.18	-32.77
Clc1cccc2c1[O+]=C1C2=[C][C@H]2[C@@H]1[CH-]2:Clc1cccc2c1[O+]=C(C2=[C-])C1C=C1	2	2	2	22.31	17.14
Clc1cccc2c1[O+]=C1C2=[C]C[C-]=C1:Clc1cccc2c1O[C]1C2=CC=C=C1	2	2	2	9.13	-69.98
Clc1cccc2c1O[C]1C2=CC=C=C1:C#CC=c1oc2c(c1=[CH])cccc2Cl	2	2	2	67.87	63.43
Clc1cccc2c1oc(=[CH])c2=C1C=C1:ClC1=CC=CC2=C3C(=C[C@@H]4C3=C4)O[C]12	2	2	2	16.61	-23.33
Clc1cccc2c1oc(=C1C=C1)c2=[CH]:ClC1=CC=CC(=[C]1)C(=[CH+])C(=C1C=C1)[O-]	2	2	2	59.25	59.06
ClCl,Clc1ccc2c(c1)C1=CC=C=C[C]1O2:Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2,ClCl	3,2	2	4,3	42.93	88.27
ClCl,Clc1ccc2c(c1)C1=CC=C=C[C]1O2:Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2,ClCl	1,2	2	4,3	78.14	123.48
ClCl,Clc1ccc2c(c1)C1=CC=C=C[C]1O2:ClCl,Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2	3,2	2	1,2	42.93	34.54
ClCl,Clc1ccc2c(c1)C1=CC=C=C[C]1O2:ClCl,Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2	1,2	2	1,2	78.14	69.76
ClCl,Clc1ccc2c(c1)C1=CC=C=C[C]1O2:ClCl,Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2	3,2	2	3,2	42.93	69.76
ClCl,Clc1ccc2c(c1)C1=CC=C=C[C]1O2:ClCl,Clc1ccc2c(c1)C1=[C]C[C-]=CC1=[O+]2	1,2	2	3,2	78.14	104.97
O=c1[cH-]cccc1C#CC=CC(=[C+])Cl:ClC(=[C+])C=CC1=C=C2C(=C[CH-]C=C2)O1	4	4	4	27.34	0.78
O=C1C(=CC=CC1=C1C=C1[C]1C=C1)Cl:Clc1cccc2c1oc(=C1C=C1)c2=[CH]	2	2	2	45.68	-4.51
O=C1C(=CC=CC1=C1C=C1[C]1C=C1)Cl:Clc1cccc2c1oc(=C1C=C1)c2=[CH]	2	2	2	45.68	-4.51
O=O,ClC1=[C][O+]=[C-]C=C[C@@H]2C(=C2)C#CC=C1:[O-]OC1=C=C=CC=CC=C=[O+][C]=C(C=C1)Cl	3,2		2	-44.83	-44.83

O=O,ClC1=C=C=C=C[CH-][C@@H]2C(=C2)[O+]=[C]C=C1:[O-]OC1=C=C=CC=CC=C=[O+][C]=CC=C1Cl	3,2		2	-31.87	-31.87
O=O,ClC1=C=C=C=C[CH-][C@@H]2C(=C2)[O+]=[C]C=C1:[O-]OC1=C=C=CC=CC=C=[O+][C]=CC=C1Cl	3,2		2	-31.87	-31.87
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:[O-]OC1=C=C=CC=CC=C=[O+][C]=CC=C1Cl	3,2		2	175.85	175.85
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:Cl[C@]12C=C[C]=[O+][C-]=CC=CC=C=C=C2OO1	3,2		2	170.64	170.64
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:O=O,[O]c1ccccc1c1c2-c1c(cc2)Cl	1,4	4	3,2	-41.74	-47.34
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:O=O,[O]c1ccccc1c1c2-c1c(cc2)Cl	3,4	4	1,4	-2.99	87.03
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:O=O,[O]c1ccccc1c1c2-c1c(cc2)Cl	3,2	4	3,4	64.71	115.98
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:O=O,[O]c1ccccc1c1c2-c1c(cc2)Cl	1,4	4	1,4	-41.74	48.28
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:O=O,[O]c1ccccc1c1c2-c1c(cc2)Cl	3,2	4	1,4	64.71	154.73
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:O=O,[O]c1ccccc1c1c2-c1c(cc2)Cl	1,4	4	3,4	-41.74	9.54
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:O=O,[O]c1ccccc1c1c2-c1c(cc2)Cl	3,4	4	3,4	-2.99	48.28
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:O=O,[O]c1ccccc1c1c2-c1c(cc2)Cl	3,2	4	3,2	64.71	59.10
O=O,ClC1=C=C2[C](C=C1)Oc1c2cccc1:O=O,[O]c1ccccc1c1c2-c1c(cc2)Cl	3,4	4	3,2	-2.99	-8.60
O=O,ClC1=C=C2C(=C[CH+]1)C1=[C-]C[C@@H]3[C@H](O2)[C]13:[O-]OC1=C=C=CC=C(Cl)C=[C][O+]=C=CC=C1	3,2		2	-2.12	-2.12
O=O,ClC1=C=C2C(=C[CH+]1)C1=[C-]C[C@@H]3[C@H](O2)[C]13:ClC1=CC=C=C=C2OO[C@@H]2C=C[C-]=[O+][C]=C1	3,2		2	-6.99	-6.99

A.4. Discussion of the Consequences of Hypersensitive Reaction Detection for CTY-TAD and CTY TS Searching and Possible Solutions

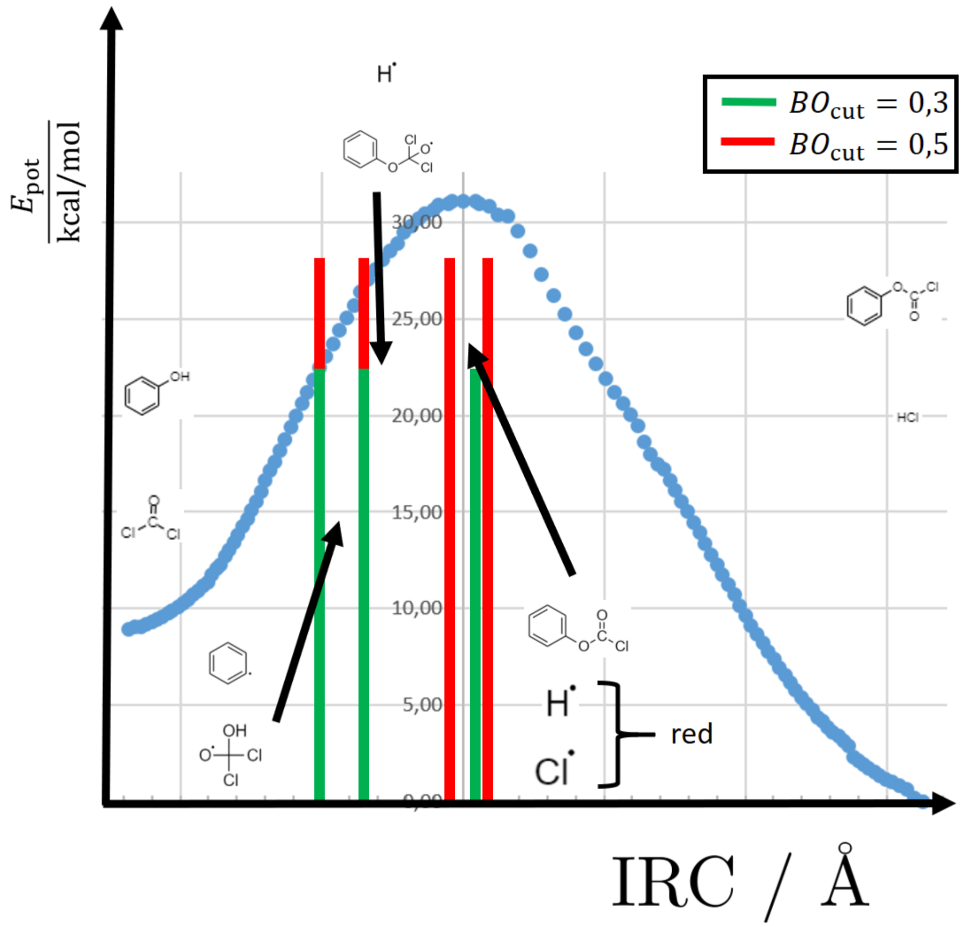


Figure A.6.: IRC scan of the first phenol phosgenation reaction step using ReaxFF. The reactants phenol and phosgene are shown on the very left of the IRC scan and the products phenylchloroformate and HCl on the very right. The two species at the top show intermediates that the reaction detection schemes find at positions on the IRC between the first two vertical bars. The vertical bars show the positions on the IRC, at which a reaction event was detected using the CTY bond-order processing. The green and the red vertical bars differ by the bond-order cut-off BO_{cut}. The BO_{cut} of 0.5 produces one additional intermediate reaction step for the formation of HCl.

In Fig. A.6, the IRC scan of the first phenol phosgenation reaction step is shown. The IRC scan was produced with the ReaxFF implementation in the Amsterdam Modeling Suite (AMS) and the C/H/O/Cl force field parametrization given in the appendix of chapter 4, section A.3.1. The vertical bars were obtained using the CTY bond-order processing for reaction detection with different bond-order cut-offs BO_{cut}. Bond-order cutoffs are used in the CTY bond-order processing to distinguish bonded from non-bonded interactions. Before the CTY bond-order processing could be employed, a one-timestep *NVE* simulation with the ReaxFF implementation in LAMMPS [71, 72] and zeroed velocities was run to obtain the bond orders corresponding to the atomic *XYZ* coordinates contained in each IRC frame. Traversing the IRC scan from left to right, different SMILES are obtained every time a vertical bar is crossed. One example of intermediate products after crossing the first bars on the left is given in the figure. The reaction modeler, on the other hand, expects only one concerted reaction event in the vicinity of the TS (*i. e.* the maximum of the IRC scan), in which all four bond changes in the phosgenation reaction happen at the same time. In the phosgenation reaction, the O−H bond of the phenolic hydroxy group and a C−Cl of the phosgene molecule break. An H−Cl bond and an C−O bond are formed. As the changes in bond lengths and therefore also of the bond orders are a continuous process, the reaction detection scheme detects between three and four ”sub reactions” instead of the one expected concerted reaction. Even though the choice of the bond-order cut-off has an impact on the position and the number of the vertical bars in the figure, the choice does not affect the core issue, *i. e.* the fact that the reaction is split into multiple sub reactions.

Consequences for Reported Reactions Using CTY-TAD, Placement of Virtual Walls in CTY-TAD, and TS Searching Using The CTY rMD-QM Coupling Scheme

1. ”Incomplete” reactions may be reported by CTY-TAD, *i. e.* the sub reactions.

2. As the reaction detection scheme may report a wrong set of bond changes, other unknown reactions may be blocked by the virtual walls in CTY-TAD if they involve the reported set of bond changes. The figure illustrates the issue. We can imagine that the first reaction event, which produces the intermediates shown in the figure, is not only a sub reaction of the phosgenation reaction but may also be an independent reaction of its own. Let’s imagine that this reaction that produces the depicted intermediates has already been detected in the past by

CTY-TAD. As the trajectory progresses towards the phosgenation reaction, CTY-TAD detects the already known reaction as a sub reaction and builds a virtual wall.

3. I summarize the CTY TS searching process again briefly before I discuss the consequences for the TS searches. First, the geometry from the trajectory at the time of the detected reaction event is cut out and serves as a TS guess. In a second step, the geometry is relaxed with a frozen active core to reduce the number of imaginary frequencies, thereby producing a better TS guess. The active core is the geometry of the active atoms, *i. e.* the atoms involved in the detected bond changes. In the third step, the relaxed geometry is used as TS guess for the eigenvector-following algorithm that searches the TS.
   If, for example, only a subset of the active atoms is detected, the relaxation in step two may break the geometry of the active core. In that case, chances are that either the eigenvector-following algorithm in step three will not be able to converge a TS at all or converge a TS that belongs to a different reaction.

Solutions

- Solution for Consequence 1: The reaction identification test described in chapter three is a simple post-processing step that sorts out incomplete reactions from the reported reaction list.

- Solutions for Consequence 2: The reduction of the trajectory sampling frequency of the bond-order processing is not a viable solution. Such a solution requires a new user parameter, *i. e.* the sampling frequency, for which I believe it is difficult to find a generally suitable value. There is a risk that, dependent on the sampling frequency, but also dependent on the simulation temperature, multiple elementary reaction steps are being detected as one concerted reaction. The exit channels from the to-be-confined reactant configuration space then become less well defined, and ”holes” in the virtual walls may appear. A simple compensation strategy has already been proposed in chapter three: Parallel simulations that do not share information about the observed reactions can be run as a compensation. The solution relies on the fact that the waiting times of reaction events are stochastic variables [54]. That means, the reactions may be observed in different chronological order in a set of independent parallel simulations. A reaction that is unintendedly blocked in one simulation can then be observed in a different simulation.

- I am not aware of similarly simple compensation strategies for Consequence 3 as for the first two consequences. Therefore, the reaction detection scheme should be improved to reduce the negative impact of reaction detection errors on the TS searching success rate. The alternative reaction detection scheme described by Hutchings *et al.* [154] may be an approach that improves the consistency of the detected reactions with transitions between PES basins. Hutchings *et al.* suggest to analyze first derivatives of bond orders [154] instead of comparing connectivities to user-defined thresholds as applied in CTY. They demonstrate that their approach accurately locates REs in rMD trajectories. However, their reaction detection scheme still needs to be demonstrated for trajectories containing more than one molecule [154], which is essential for practical applications.

  An alternative ansatz for improvement is the integration of information about the local curvature of the PES at the time of a RE into the reaction detection procedure by analyzing frequencies. Reactions with a reaction barrier could only then be be returned by the method if not only the connectivities have changed but also if only one dominant imaginary frequency is present. That way, the returned geometry from the trajectory snapshot would make for an excellent TS search start geometry for a TS search on the force field level of theory. The molecular identifiers of the reactants and products could then be derived directly from the optimized geometries that are connected by an IRC scan. This way, the detected reactions would be inherently consistent with transitions between PES basins.

I have shown that the hypersensitivity of the reaction detection scheme towards bond changes has a negative impact on the quality of the reported results, the functionality of the basin confinement method in CTY-TAD, but also for the CTY TS searching success rate. I have pointed out approaches to improve the reaction detection methodology. Most of the discussed problems that arise as consequences of the reaction detection hypersensitivity can already be compensated with simple procedures that have been described and used in chapter three.

A.5. Source Code Availability

The CTY-TAD and CTY codes described in chapters three and four are available at sourceforge.net/projects/chemtrayzer.

A.6. Journal Publications

L. Krep, W. A. Kopp, L. C. Kröger, M. Döntgen, K. Leonhard, *Exploring the Chemistry of Low-Temperature Ignition by Pressure-Accelerated Dynamics*, ChemSys- temsChem 2(4):e1900043, 2020, doi:10.1002/syst.201900043

F. Solbach, A. Bernardi, S. Bansal, M. S. Budamagunta, L. Krep, K. Leonhard, J. C. Voss, K. S. Lam, and R. Faller, *Determining structure and action mechanism of LBF14 by molecular simulation*, J. Biomol. Struct. Dyn. 0(0):1–12, 2021, doi:10.1080/07391102.2021.1967783

L. Krep, I. S. Roy, W. Kopp, F. Schmalz, C. Huang, K. Leonhard, *Efficient Reaction Space Exploration with ChemTraYzer-TAD*, JCIM 62(4):890–902, 2022, doi: 10.1021/acs.jcim.1c01197

T. Nevolianis, N. Wolter, L. Kaven, L. Krep, C. Huang, A. Mhamdi, A. Mitsos, A. Pich, K. Leonhard, *Kinetic Modeling of a Poly(N-vinylcaprolactam-co-glycidyl methacrylate) Microgel Synthesis – A Hybrid In-Silico and Experimental Approach*, Ind. Eng. Chem. Res., 2023, 10.1021/acs.iecr.2c03291

L. Komissarov, L. Krep, F. Schmalz, W. A. Kopp, K. Leonhard, T. Verstraelen, *A Reactive Molecular Dynamics Study of Chlorinated Organic Compounds. Part I: Force Field Development*, ChemPhysChem, 2022, doi:10.1002/cphc.202200786

L. Krep, F. Schmalz, F. Solbach, L. Komissarov, T. Nevolianis, W. A. Kopp, T. Verstraelen, K. Leonhard, *A Reactive Molecular Dynamics Study of Chlorinated Organic Compounds. Part II: A ChemTraYzer Study of Chlorinated Dibenzofuran Formation and Decomposition Processes*, ChemPhysChem, 2022, doi:10.1002/cphc.202200783

Bibliography

[1] J. Bugler, K. P. Somers, E. J. Silke, and H. J. Curran, *Revisiting the Kinetics and Thermodynamics of the Low-Temperature Oxidation Pathways of Alkanes: A Case Study of the Three Pentane Isomers*, J. Phys. Chem. A 119(28):7510–7527, 2015, doi:10.1021/acs.jpca.5b00837

[2] J. Bugler, B. Marks, O. Mathieu, R. Archuleta, A. Camou, C. Grégoire, K. A. Heufer, E. L. Petersen, and H. J. Curran, *An ignition delay time and chemical kinetic modeling study of the pentane isomers*, Combust. Flame 163:138–156, 2016, doi:10.1016/j.combustflame.2015.09.014

[3] J. Bugler, A. Rodriguez, O. Herbinet, F. Battin-Leclerc, C. Togbé, G. Dayma, P. Dagaut, and H. J. Curran, *An experimental and modelling study of n-pentane oxidation in two jet-stirred reactors: The importance of pressure-dependent kinetics and new reaction pathways*, Proc. Combust. Inst. 36(1):441–448, 2017, doi:10.1016/j.proci.2016.05.048

[4] E. Ranzi, A. A. Frassoldati, R. Grana, A. Cuoci, T. Faravelli, A. P. Kelley, and C. K. Law, *Hierarchical and Comparative Kinetic Modeling of Laminar Flame Speeds of Hydrocarbon and Oxygenated Fuels*, Prog. Energy Combust. Sci. 38(4):468–501, 2012, doi:10.1016/j.pecs.2012.03.004

[5] M. L. Coote, E. I. Izgorodina, G. E. Cavigliasso, M. Roth, M. Busch, and C. Barner-Kowollik, *Addition-Fragmentation Kinetics of Fluorodithioformates (F-RAFT) in Styrene, Vinyl Acetate, and Ethylene Polymerization: An Ab Initio Investigation*, Macromolecules 39(13):4585–4591, 2006, doi:10.1021/ma060470z

[6] F. Jung, F. A. Janssen, A. Caspari, H. Spütz, L. Kröger, K. Leonhard, A. Mhamdi, and A. Mitsos, *Dynamic Optimization of a Fed-Batch Microgel Synthesis **We thank the Deutsche Forschungsgemeinschaft (DFG) in the Collaborative Research Center SFB 985 Functional Microgels and Microgel Systems within Project B4 for financial support. The authors gratefully acknowledge the financial support of the Kopernikus project SynErgie by the Federal Ministry of Education and Research (BMBF)*, IFAC-PapersOnLine 52(1):394–399, 2019, doi:10.1016/j.ifacol.2019.06.094

[7] S. Vranckx, K. A. Heufer, C. Lee, H. Olivier, L. Schill, W. A. Kopp, K. Leonhard, C. A. Taatjes, and R. X. Fernandes, *Role of peroxy chemistry in the high-pressure ignition of n-butanol – Experiments and detailed kinetic modelling*, Combust. Flame 158:1444–1455, 2011, doi:10.1016/j.combustflame.2010.12.028

[8] J. Zádor, C. A. Taatjes, and R. X. Fernandes, *Kinetics of Elementary Reactions in Low-Temperature Autoignition Chemistry*, Prog. Energy Combust. Sci. 37(4):371–421, 2011, doi:10.1016/j.pecs.2010.06.006

[9] C. Waldron, A. Pankajakshan, M. Quaglio, E. Cao, F. Galvanin, and A. Gavriilidis, *Closed-Loop Model-Based Design of Experiments for Kinetic Model Discrimination and Parameter Estimation: Benzoic Acid Esterification on a Heterogeneous Catalyst*, Ind. Eng. Chem. Res. 58(49):22165–22177, 2019, doi:10.1021/acs.iecr.9b04089

[10] S. Maeda and Y. Harabuchi, *Exploring paths of chemical transformations in molecular and periodic systems: An approach utilizing force*, WIREs Comput. Mol. Sci. 11(6):e1538, 2021, doi:10.1002/wcms.1538

[11] P. L. Bhoorasingh, B. L. Slakman, F. Seyedzadeh Khanshan, J. Y. Cain, and R. H. West, *Automated Transition State Theory Calculations for High-Throughput Kinetics*, J. Phys. Chem. A 121(37):6896–6904, 2017, doi:10.1021/acs.jpca.7b07361

[12] G. N. Simm and M. Reiher, *Context-Driven Exploration of Complex Chemical Reaction Networks*, JCTC 13(12):6108–6119, 2017, doi:10.1021/acs.jctc.7b00945

[13] V. Warth, F. Battin-Leclerc, R. Fournet, P. Glaude, G. Côme, and G. Scacchi, *Computer Based Generation of Reaction Mechanisms for Gas-Phase Oxidation*, Comput. Chem. 24(5):541–560, 2000, doi:10.1016/S0097-8485(99)00092-3

[14] M. Bergeler, G. N. Simm, J. Proppe, and M. Reiher, *Heuristics-Guided Exploration of Reaction Mechanisms*, JCTC 11(12):5712–5722, 2015, doi:10.1021/acs.jctc.5b00866

[15] D. Rappoport, C. J. Galvin, D. Y. Zubarev, and A. Aspuru-Guzik, *Complex Chemical Reaction Networks from Heuristics-Aided Quantum Chemistry*, JCTC 10(3):897–907, 2014, doi:10.1021/ct401004r

[16] D. Rappoport and A. Aspuru-Guzik, *Predicting Feasible Organic Reaction Pathways Using Heuristically Aided Quantum Chemistry*, JCTC 15(7):4099–4112, 2019, doi:10.1021/acs.jctc.9b00126

[17] R. Van de Vijver, K. M. Van Geem, G. B. Marin, and J. Zàdor, *Decomposition and Isomerization of 1-Pentanol Radicals and the Pyrolysis of 1-Pentanol*, Combust. Flame 196:500–514, 2018, doi:10.1016/j.combustflame.2018.05.011

[18] R. Van de Vijver and J. Zàdor, *KinBot: Automated Stationary Point Search on Potential Energy Surfaces*, Comput. Phys. Commun. 248:106947, 2020, doi:10.1016/j.cpc.2019.106947

[19] S. Rangarajan, A. Bhan, and P. Daoutidis, *Rule-Based Generation of Thermochemical Routes to Biomass Conversion*, Ind. Eng. Chem. Res. 49(21):10459–10470, 2010, doi: 10.1021/ie100546t

[20] C. W. Gao, J. W. Allen, W. H. Green, and R. H. West, *Reaction Mechanism Generator: Automatic Construction of Chemical Kinetic Mechanisms*, Comput. Phys. Commun. 203:212–225, 2016, doi:10.1016/j.cpc.2016.02.013

[21] Y. Kim, J. W. Kim, Z. Kim, and W. Y. Kim, *Efficient Prediction of Reaction Paths through Molecular Graph and Reaction Network Analysis*, Chem. Sci. 9:825–835, 2018, doi:10.1039/C7SC03628K

[22] P. M. Zimmerman, *Automated Discovery of Chemically Reasonable Elementary Reaction Steps*, J. Comput. Chem. 34(16):1385–1392, 2013, doi:10.1002/jcc.23271

[23] A. L. Dewyer and P. M. Zimmerman, *Finding reaction mechanisms, intuitive or otherwise*, Org. Biomol. Chem. 15:501–504, 2017, doi:10.1039/C6OB02183B

[24] Y. V. Suleimanov and W. H. Green, *Automated Discovery of Elementary Chemical Reaction Steps Using Freezing String and Berny Optimization Methods*, JCTC 11(9):4248–4259, 2015, doi:10.1021/acs.jctc.5b00407

[25] S. Habershon, *Sampling Reactive Pathways with Random Walks in Chemical Space: Applications to Molecular Dissociation and Catalysis*, J. Chem. Phys. 143(9):094106, 2015, doi:10.1063/1.4929992

[26] S. Habershon, *Automated Prediction of Catalytic Mechanism and Rate Law Using Graph-Based Reaction Path Sampling*, JCTC 12(4):1786–1798, 2016, doi:10.1021/acs.jctc.6b00005

[27] C. Robertson and S. Habershon, *Fast screening of homogeneous catalysis mechanisms using graph-driven searches and approximate quantum chemistry*, Catal. Sci. Technol. 9:6357–6369, 2019, doi:10.1039/C9CY01997A

[28] L.-P. Wang, A. Titov, R. McGibbon, F. Liu, V. S. Pande, and T. J. Martínez, *Discovering Chemistry with an Ab Initio Nanoreactor*, Nat. Chem. 6:1044–1048, 2014, doi:doi:10.1038/nchem.2099

[29] J. Ford, S. Seritan, X. Zhu, M. N. Sakano, M. M. Islam, A. Strachan, and T. J. Martínez, *Nitromethane Decomposition via Automated Reaction Discovery and an Ab Initio Corrected Kinetic Model*, JPCA 125(7):1447–1460, 2021, doi:10.1021/acs.jpca.0c09168

[30] L.-P. Wang, R. T. McGibbon, V. S. Pande, and T. J. Martinez, *Automated Discovery and Refinement of Reactive Molecular Dynamics Pathways*, JCTC 12(2):638–649, 2016, doi:10.1021/acs.jctc.5b00830

[31] E. Martìnez-Núñez, *An Automated Method to Find Transition States Using Chemical Dynamics Simulations*, J. Comput. Chem. 36(4):222–234, 2015, doi:10.1002/jcc.23790

[32] E. Martìnez-Núñez, *An Automated Transition State Search Using Classical Trajectories Initialized at Multiple Minima*, Phys. Chem. Chem. Phys. 17:14912–14921, 2015, doi:10.1039/C5CP02175H

[33] A. Rodrìguez, R. Rodrìguez-Fernàndez, S. A. Vázquez, G. L. Barnes, J. J. P. Stewart, and E. Martínez-Núñez, *tsscds2018: A Code for Automated Discovery of Chemical Reaction Mechanisms and Solving the Kinetics*, J. Comput. Chem. 39(23):1922–1930, 2018, doi:10.1002/jcc.25370

[34] M. Döntgen, M.-D. Przybylski-Freund, L. C. Kröger, W. A. Kopp, A. E. Ismail, and K. Leonhard, *Automated Discovery of Reaction Pathways, Rate Constants and Transition States Using Reactive Molecular Dynamics Simulations*, J. Chem. Theory Comput. 11:2517–2524, 2015, doi:10.1021/acs.jctc.5b00201

[35] M. Döntgen, F. Schmalz, W. A. Kopp, L. C. Kröger, and K. Leonhard, *Automated Chemical Kinetic Modeling via Hybrid Reactive Molecular Dynamics and Quantum Chemistry Simulations*, J. Chem. Inf. Model. 58(7):1343–1355, 2018, doi:10.1021/acs.jcim.8b00078

[36] C. A. Grambow, A. Jamal, Y.-P. Li, W. H. Green, J. Zàdor, and Y. V. Suleimanov, *Unimolecular Reaction Pathways of a γ-Ketohydroperoxide from Combined Application of Automated Reaction Discovery Methods*, JACS 140(3):1035–1048, 2018, doi:10.1021/jacs.7b11009

[37] A. L. Dewyer, A. J. Argüelles, and P. M. Zimmerman, *Methods for Exploring Reaction Space in Molecular Systems*, WIREs Comput. Mol. Sci. 8(2):e1354, 2018, doi:10.1002/wcms.1354

[38] A. L. Dewyer, A. J. Argüelles, and P. M. Zimmerman, *Methods for exploring reaction space in molecular systems*, WIREs Comput Mol Sci 8(2):e1354, 2018, doi:10.1002/wcms.1354

[39] G. N. Simm, A. C. Vaucher, and M. Reiher, *Exploration of Reaction Pathways and Chemical Transformation Networks*, J. Phys. Chem. A 123(2):385–399, 2019, doi:10.1021/acs.jpca.8b10007

[40] J. P. Unsleber and M. Reiher, *The Exploration of Chemical Reaction Networks*, Annu. Rev. Phys. Chem. 71(1):121–142, 2020, doi:10.1146/annurev-physchem-071119-040123

[41] J. A. Miller, R. Sivaramakrishnan, Y. Tao, C. F. Goldsmith, M. P. Burke, A. W. Jasper, N. Hansen, N. J. Labbe, P. Glarborg, and J. Zádor, *Combustion Chemistry in the Twenty-First Century: Developing Theory-Informed Chemical Kinetics Models*, Prog. Energy. Combust. Sci. 83:100886, 2021, doi:10.1016/j.pecs.2020.100886

[42] S. A. Grimmel and M. Reiher, *The electrostatic potential as a descriptor for the protonation propensity in automated exploration of reaction mechanisms*, Faraday Discuss. 220:443–463, 2019, doi:10.1039/C9FD00061E

[43] S. A. Grimmel and M. Reiher, *On the Predictive Power of Chemical Concepts*, CHIMIA 75(4):311–318, 2021, doi:10.2533/chimia.2021.311

[44] K. M. Bal and E. C. Neyts, *Merging Metadynamics into Hyperdynamics: Accelerated Molecular Simulations Reaching Time Scales from Microseconds to Seconds*, J. Chem. Theory Comput. 11(10):4545–4554, 2015, doi:10.1021/acs.jctc.5b00597

[45] A. Laio and F. L. Gervasio, *Metadynamics: A Method to Simulate Rare Events and Reconstruct the Free Energy in Biophysics, Chemistry and Material science*, Rep. Prog. Phys. 71:126601, 2008, doi:10.1088/0034-4885/71/12/126601

[46] A. F. Voter, *A Method for Accelerating the Molecular Dynamics Simulation of Infrequent Events*, J. Chem. Phys. 106(11):4665–4677, 1997, doi:10.1063/1.473503

[47] K. M. Bal and E. C. Neyts, *Direct Observation of Realistic-Temperature Fuel Combustion Mechanisms in Atomistic Simulations*, Chem. Sci. 7:5280–5286, 2016, doi:10.1039/C6SC00498A

[48] A. Barducci, M. Bonomi, and M. Parrinello, *Metadynamics*, WIREs Comput Mol Sci 1(5):826–843, 2011, doi:10.1002/wcms.31

[49] S. Grimme, *Exploration of Chemical Compound, Conformer, and Reaction Space with Meta-Dynamics Simulations Based on Tight-Binding Quantum Chemical Calculations*, JCTC 15(5):2847–2862, 2019, doi:10.1021/acs.jctc.9b00143

[50] E. Trizio and M. Parrinello, *From Enhanced Sampling to Reaction Profiles*, J. Phys. Chem. Lett. 12(35):8621–8626, 2021, doi:10.1021/acs.jpclett.1c02317

[51] G. Piccini, D. Mendels, and M. Parrinello, *Metadynamics with Discriminants: A Tool for Understanding Chemistry*, JCTC 14(10):5040–5044, 2018, doi:10.1021/acs.jctc.8b00634

[52] M. Yang, L. Bonati, D. Polino, and M. Parrinello, *Using metadynamics to build neural network potentials for reactive events: the case of urea decomposition in water*, Catal , 2021, doi:10.1016/j.cattod.2021.03.018

[53] D. Mandelli, B. Hirshberg, and M. Parrinello, *Metadynamics of Paths*, Phys. Rev. Lett. 125:026001, 2020, doi:10.1103/PhysRevLett.125.026001

[54] L. C. Kröger, W. A. Kopp, M. Döntgen, and K. Leonhard, *Assessing Statistical Uncertainties of Rare Events in Reactive Molecular Dynamics Simulations*, JCTC 13(9):3955–3960, 2017, doi:10.1021/acs.jctc.7b00524

[55] G. Schmitz, z. Yönder, B. Schnieder, R. Schmid, and C. Hättig, *An automatized workflow from molecular dynamic simulation to quantum chemical methods to identify elementary reactions and compute reaction constants*, J. Comput. Chem. 42(32):2264–2282, 2021, doi:10.1002/jcc.26757

[56] M. Altarawneh, B. Z. Dlugogorski, E. M. Kennedy, and J. C. Mackie, *Mechanisms for formation, chlorination, dechlorination and destruction of polychlorinated dibenzo-p-dioxins and dibenzofurans (PCDD/Fs)*, Prog. Energy Combust. Sci. 35(3):245–274, 2009, doi:10.1016/j.pecs.2008.12.001

[57] L. Krep, W. A. Kopp, L. C. Kröger, M. Döntgen, and K. Leonhard, *Exploring the Chemistry of Low-Temperature Ignition by Pressure-Accelerated Dynamics*, ChemSystemsChem 2(4):e1900043, 2020, doi:10.1002/syst.201900043

[58] A. C. T. van Duin, S. Dasgupta, F. Lorant, and W. A. Goddard, *ReaxFF: A Reactive Force Field for Hydrocarbons*, J. Phys. Chem. A 105(41):9396–9409, 2001, doi:10.1021/jp004368u

[59] K. Chenoweth, A. C. T. van Duin, and W. A. Goddard, *ReaxFF Reactive Force Field for Molecular Dynamics Simulations of Hydrocarbon Oxidation*, J. Phys. Chem. A 112(5):1040–1053, 2008, doi:10.1021/jp709896w

[60] S. Kylasa, H. Aktulga, and A. Grama, *PuReMD-GPU: A reactive molecular dynamics simulation package for GPUs*, J. Comput. Phys. 272:343–359, 2014, doi: https://doi.org/10.1016/j.jcp.2014.04.035

[61] H. Curran, P. Gaffuri, W. Pitz, and C. Westbrook, *A Comprehensive Modeling Study of n-Heptane Oxidation*, Combust. Flame 114(1):149–177, 1998, doi:10.1016/S0010-2180(97)00282-4

[62] V. P. Zhukov, *Kinetic model of alkane oxidation at high pressure from methane to n-heptane*, Combust. Theor. Model 13(3):427–442, 2009, doi:10.1080/13647830902767302

[63] D. G. Goodwin, H. K. Moffat, and R. L. Speth, *"Cantera: An Object-oriented Software Toolkit for Chemical Kinetics, Thermodynamics, and Transport Processes"*, http://www.cantera.org, 2017, doi:10.5281/zenodo.170284, version 2.3.0

[64] C. Ashraf and A. C. van Duin, *Extension of the ReaxFF Combustion Force Field toward Syngas Combustion and Initial Oxidation Kinetics*, J. Phys. Chem. A 121(5):1051–1068, 2017, doi:10.1021/acs.jpca.6b12429

[65] R. Asatryan and J. W. Bozzelli, *Chain Branching and Termination in the Low-Temperature Combustion of n-Alkanes: 2-Pentyl Radical + O2, Isomerization and Association of the Second O2*, J. Phys. Chem. A 114:7693–7708, 2010, doi:10.1021/jp101159h

[66] A. Comandini, I. A. Awan, and J. A. Manion, *Thermal decomposition of 1-pentyl radicals at high pressures and temperatures*, Chem. Phys. Lett. 552:20–26, 2012, doi:10.1016/j.cplett.2012.09.039

[67] A. Sudholt, C. Lee, J. Klankermayer, R. X. Fernandes, and H. Pitsch, *Ignition characteristics of saturated and unsaturated furans*, Combust. Flame 171:133 – 136, 2016, doi:10.1016/j.combustflame.2016.05.016

[68] U. Burke, K. P. Somers, P. O'Toole, C. M. Zinner, N. Marquet, G. Bourque, E. L. Petersen, W. K. Metcalfe, Z. Serinyel, and H. J. Curran, *An ignition delay and kinetic modeling study of methane, dimethyl ether, and their mixtures at high pressures*, Combust. Flame 162(2):315 – 330, 2015, doi:10.1016/j.combustflame.2014.08.014

[69] D. Healy, N. Donato, C. Aul, E. Petersen, C. Zinner, G. Bourque, and H. Curran, *n-Butane: Ignition delay measurements at high pressure and detailed chemical kinetic simulations*, Combust. Flame 157(8):1526 – 1539, 2010, doi:10.1016/j.combustflame.2010.01.016

[70] L. Martínez, R. Andrade, E. G. Birgin, and J. M. Martínez, *Packmol: A Package for Building Initial Configurations for Molecular Dynamics Simulations*, J. Comput. Chem. 30:2157–2164, 2009, doi:10.1002/jcc.21224

[71] H. Aktulga, J. Fogarty, S. Pandit, and A. Grama, *Parallel Reactive Molecular Dynamics: Numerical Methods and Algorithmic Techniques*, Parallel Comput. 38(4):245–259, 2012, doi:10.1016/j.parco.2011.08.005

[72] S. Plimpton, *Fast Parallel Algorithms for Short-Range Molecular Dynamics*, J. Comput. Phys. 117(1):1–19, 1995, doi:10.1006/jcph.1995.1039

[73] J. Ding, L. Zhang, Y. Zhang, and K.-L. Han, *A Reactive Molecular Dynamics Study of n-Heptane Pyrolysis at High Temperature*, J. Phys. Chem. A 117(16):3266–3278, 2013, doi:10.1021/jp311498u

[74] F. Castro-Marcano and A. C. van Duin, *Comparison of thermal and catalytic cracking of 1-heptene from ReaxFF reactive molecular dynamics simulations*, Combust. Flame 160(4):766–775, 2013, doi:10.1016/j.combustflame.2012.12.007

[75] A. Bharti and T. Banerjee, *Reactive force field simulation studies on the combustion behavior of n-octanol*, Fuel Process Technol. 152:132–139, 2016, doi:10.1016/j.fuproc.2016.06.027

[76] X.-M. Cheng, Q.-D. Wang, J.-Q. Li, J.-B. Wang, and X.-Y. Li, *ReaxFF Molecular Dynamics Simulations of Oxidation of Toluene at High Temperatures*, J. Phys. Chem. A 116(40):9811–9818, 2012, doi:10.1021/jp304040q

[77] F. Castro-Marcano, A. M. Kamat, M. F. Russo, A. C. van Duin, and J. P. Mathews, *Combustion of an Illinois No. 6 coal char simulated using an atomistic char representation and the ReaxFF reactive force field*, Combust. Flame 159(3):1272–1285, 2012, doi:10.1016/j.combustflame.2011.10.022

[78] M. Zheng, X. Li, J. Liu, Z. Wang, X. Gong, L. Guo, and W. Song, *Pyrolysis of Liulin Coal Simulated by GPU-Based ReaxFF MD with Cheminformatics Analysis*, Energ. Fuel 28(1):522–534, 2014, doi:10.1021/ef402140n

[79] T. Zhang, X. Li, X. Qiao, M. Zheng, L. Guo, W. Song, and W. Lin, *Initial Mechanisms for an Overall Behavior of Lignin Pyrolysis through Large-Scale ReaxFF Molecular Dynamics Simulations*, Energy Fuels 30(4):3140–3150, 2016, doi:10.1021/acs.energyfuels.6b00247

[80] S. G. Srinivasan and A. C. T. van Duin, *Molecular-Dynamics-Based Study of the Collisions of Hyperthermal Atomic Oxygen with Graphene Using the ReaxFF Reactive Force Field*, J. Phys. Chem. A 115(46):13269–13280, 2011, doi:10.1021/jp207179x

[81] P. Ganesh, P. R. C. Kent, and V. Mochalin, *Formation, characterization, and dynamics of onion-like carbon structures for electrical energy storage from nanodiamonds using reactive force fields*, J. Appl. Phys. 110(7):073506, 2011, doi:10.1063/1.3641984

[82] L. Goerigk, A. Hansen, C. Bauer, S. Ehrlich, A. Najibi, and S. Grimme, *A Look at the Density Functional Theory Zoo with the Advanced GMTKN55 Database for General Main Group Thermochemistry, Kinetics and Noncovalent Interactions*, Phys. Chem. Chem. Phys. 19:32184–32215, 2017, doi:10.1039/C7CP04913G

[83] A. Ghysels, T. Verstraelen, K. Hemelsoet, M. Waroquier, and V. Van Speybroeck, *TAMkin: A Versatile Package for Vibrational Analysis and Chemical Kinetics*, J. Chem. Inf. Model 50(9):1736–1750, 2010, doi:10.1021/ci100099g

[84] X. Li and M. J. Frisch, *Energy-Represented Direct Inversion in the Iterative Subspace within a Hybrid Geometry Optimization Method*, J Chem. Theory. Comput. 2(3):835–839, 2006, doi:10.1021/ct050275a

[85] J. A. Montgomery, M. J. Frisch, J. W. Ochterski, and G. A. Petersson, *A complete basis set model chemistry. VI. Use of density functional geometries and frequencies*, J. Chem. Phys. 110(6):2822–2827, 1999, doi:10.1063/1.477924

[86] J. A. Montgomery, M. J. Frisch, J. W. Ochterski, and G. A. Petersson, *A complete basis set model chemistry. VII. Use of the minimum population localization method*, J. Chem. Phys. 112(15):6532–6542, 2000, doi:10.1063/1.481224

[87] J. Zheng, Y. Zhao, and D. G. Truhlar, *The DBH24/08 Database and Its Use to Assess Electronic Structure Model Chemistries for Chemical Reaction Barrier Heights*, J. Chem. Theory Comput. 5(4):808–821, 2009, doi:10.1021/ct800568m

[88] W. A. Kopp, R. T. Langer, M. Döntgen, and K. Leonhard, *Hydrogen Abstraction from n-Butyl Formate by H• and HO2•*, J. Phys. Chem. A 117:6757–6770, 2013, doi:10.1021/jp4063675

[89] J. Pieper, C. Hemken, R. Büttgen, I. Graf, N. Hansen, K. A. Heufer, and K. Kohse-Höinghaus, *A high-temperature study of 2-pentanone oxidation: experiment and kinetic modeling*, P. Combust. Inst. 37(2):1683 – 1690, 2019, doi: 10.1016/j.proci.2018.05.039

[90] C. F. Goldsmith, W. H. Green, and S. J. Klippenstein, *Role of O2 + QOOH in Low-Temperature Ignition of Propane. 1. Temperature and Pressure Dependent Rate Coefficients*, J. Phys. Chem. A 116(13):3325–3346, 2012, doi:10.1021/jp210722w

[91] H. Jin, J. Pieper, C. Hemken, E. Bräuer, L. Ruwe, and K. Kohse-Höinghaus, *Chemical interaction of dual-fuel mixtures in low-temperature oxidation, comparing n-pentane/dimethyl ether and n-pentane/ethanol*, Combust. Flame 193:36–53, 2018, doi:10.1016/j.combustflame.2018.03.003

[92] W. H. Green, C. D. Wijaya, P. E. Yelvington, and R. Sumathi, *Predicting Chemical Kinetics with Computational Chemistry: Is QOOH->HOQO Important in Fuel Ignition?*, Mol. Phys. 102(4):371–380, 2004, doi:10.1080/00268970410001675617

[93] S. M. Villano, L. K. Huynh, H.-H. Carstensen, and A. M. Dean, *High-Pressure Rate Rules for Alkyl + O2 Reactions. 2. The Isomerization, Cyclic Ether Formation, and β-Scission Reactions of Hydroperoxy Alkyl Radicals*, J. Phys. Chem. A 116(21):5068–5089, 2012, doi:10.1021/jp3023887

[94] L. Krep, I. S. Roy, W. Kopp, F. Schmalz, C. Huang, and K. Leonhard, *Efficient Reaction Space Exploration with ChemTraYzer-TAD*, JCIM 62(4):890–902, 2022, doi:10.1021/acs.jcim.1c01197

[95] M. R. Sørensen and A. F. Voter, *Temperature-Accelerated Dynamics for Simulation of Infrequent Events*, J. Chem. Phys. 112:9599–9606, 2000, doi:10.1063/1.481576

[96] S. Chekmarev and S. Krivov, *Confinement of the Molecular Dynamics Trajectory to a Specified Catchment Area on the Potential Surface: Application to Clusters*, Eur. Phys. J. D 9:201–204, 1999, doi:10.1007/s100530050426

[97] S. Krivov and S. Chekmarev, *Self-Diffusion in Clusters*, Eur. Phys. J. D 9:205–209, 1999, doi:10.1007/s100530050427

[98] S. Chekmarev and S. Krivov, *Confinement of the Molecular Dynamics Trajectory to a Specified Catchment Area on the Potential Surface*, Chem. Phys. Lett. 287(5):719–724, 1998, doi:10.1016/S0009-2614(98)00209-7

[99] P. Pracht and S. Grimme, *Calculation of Absolute Molecular Entropies and Heat Capacities Made Simple*, Chem. Sci. 12:6551–6568, 2021, doi:10.1039/D1SC00621E

[100] E. Martìnez-Núñez and D. V. Shalashilin, *Acceleration of Classical Mechanics by Phase Space Constraints*, J. Chem. Theory Comput. 2(4):912–919, 2006, doi:10.1021/ct060042z

[101] D. R. Glowacki, E. Paci, and D. V. Shalashilin, *Boxed Molecular Dynamics: A Simple and General Technique for Accelerating Rare Event Kinetics and Mapping Free Energy in Large Molecular Systems*, J. Phys. Chem. B 113(52):16603–16611, 2009, doi:10.1021/jp9074898

[102] D. R. Glowacki, E. Paci, and D. V. Shalashilin, *Boxed Molecular Dynamics: Decorrelation Time Scales and the Kinetic Master Equation*, JCTC 7(5):1244–1252, 2011, doi:10.1021/ct200011e

[103] J. Booth, S. Vazquez, E. Martinez-Nunez, A. Marks, J. Rodgers, D. R. Glowacki, and D. V. Shalashilin, *Recent Applications of Boxed Molecular Dynamics: A Simple Multiscale Technique for Atomistic Simulations*, Phil. Trans. R. Soc. A 372(2021):20130384, 2014, doi:10.1098/rsta.2013.0384

[104] M. O'Connor, E. Paci, S. McIntosh-Smith, and D. R. Glowacki, *Adaptive Free Energy Sampling in Multidimensional Collective Variable Space Using Boxed Molecular Dynamics*, Faraday Discuss. 195:395–419, 2016, doi:10.1039/C6FD00138F

[105] J. Kästner, *Umbrella Sampling*, WIREs Comput. Mol. Sci. 1(6):932–942, 2011, doi:10.1002/wcms.66

[106] R. J. Shannon, S. Amabilino, M. O'Connor, D. V. Shalishilin, and D. R. Glowacki, *Adaptively Accelerating Reactive Molecular Dynamics Using Boxed Molecular Dynamics in Energy Space*, JCTC 14(9):4541–4552, 2018, doi:10.1021/acs.jctc.8b00515

[107] R. A. Jara-Toro, G. A. Pino, D. R. Glowacki, R. J. Shannon, and E. Martínez-Núñez, *Enhancing Automated Reaction Discovery with Boxed Molecular Dynamics in Energy Space*, ChemSystemsChem 2(1):e1900024, 2020, doi:10.1002/syst.201900024

[108] A. Wildenberg, Y. Fenard, M. Carbonnier, A. Kéromnès, B. Lefort, Z. Serinyel, G. Dayma, L. Le Moyne, P. Dagaut, and K. A. Heufer, *An Experimental and Kinetic Modeling Study on the Oxidation of 1,3-Dioxolane*, Proc. Combust. Inst. 38(1):543–553, 2021, doi:10.1016/j.proci.2020.06.362

[109] A. Wildenberg and K. A. Heufer, Personal Communication

[110] J. Song, V. Zello, A. Boehman, and F. Waller, *Comparison of the Impact of Intake Oxygen Enrichment and Fuel Oxygenation on Diesel Combustion and Emissions*, Energy Fuels 18(5):1282–1290, 2004, doi:10.1021/ef034103p

[111] N. M. O'Boyle, M. Banck, C. A. James, C. Morley, T. Vandermeersch, and G. R. Hutchison, *Open Babel: An Open Chemical Toolbox*, J. Cheminf. 3(1):33, 2011, doi:10.1186/1758-2946-3-33

[112] S. F. Chekmarev, *Taboo Search by Successive Confinement: Surveying a Potential Energy Surface*, Phys. Rev. E 64:036703, 2001, doi:10.1103/PhysRevE.64.036703

[113] J. M. Simmie, G. Black, H. J. Curran, and J. P. Hinde, *Enthalpies of Formation and Bond Dissociation Energies of Lower Alkyl Hydroperoxides and Related Hydroperoxy and Alkoxy Radicals*, J. Phys. Chem. A 112(22):5010–5016, 2008, doi:10.1021/jp711360z

[114] Y. Liu, X. Wei, W. Sun, and L. Zhao, *ReaxFF MD Investigation of the High-Temperature Combustion of Six Octane Isomers*, Energy Fuels 35(20):16778–16790, 2021, doi:10.1021/acs.energyfuels.1c02462

[115] X. Li, M. Zheng, C. Ren, and L. Guo, *ReaxFF Molecular Dynamics Simulations of Thermal Reactivity of Various Fuels in Pyrolysis and Combustion*, Energy Fuels 35(15):11707–11739, 2021, doi:10.1021/acs.energyfuels.1c01266

[116] L. Wang, W. Sun, and Q. Yang, *Exploration of the Influences of the PODE3 Additive on the Initial Pyrolysis of Diesel by ReaxFF Molecular Dynamics Simulations*, Energy Fuels 35(12):9825–9835, 2021, doi:10.1021/acs.energyfuels.1c00074

[117] D. Furman and D. J. Wales, *Transforming the Accuracy and Numerical Stability of ReaxFF Reactive Force Fields*, J. Phys. Chem. Lett. 10(22):7215–7223, 2019, doi: 10.1021/acs.jpclett.9b02810

[118] S. Grimme, J. Antony, S. Ehrlich, and H. Krieg, *A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu*, Chem. Phys. 132(15):154104, 2010, doi:10.1063/1.3382344

[119] M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, G. A. Petersson, H. Nakatsuji, X. Li, M. Caricato, A. V. Marenich, J. Bloino, B. G. Janesko, R. Gomperts, B. Mennucci, H. P. Hratchian, J. V. Ortiz, A. F. Izmaylov, *et al.*, *Gaussian~16 Revision C.01*, 2016, gaussian Inc. Wallingford CT

[120] W. Quapp, *How Does a Reaction Path Branching Take Place? A Classification of Bifurcation Events*, J. Mol. Struct. 695-696:95–101, 2004, doi:10.1016/j.molstruc.2003.10.034

[121] H. J. Curran, P. Gaffuri, P. W. J., and C. K. Westbrook, *A Comprehensive Modeling Study of n-Heptane Oxidation*, Combust. Flame 114:149–177, 1998, doi:10.1016/s0010-2180(97)00282-4

[122] L. Krep, F. Schmalz, F. Solbach, L. Komissarov, T. Nevolianis, W. A. Kopp, T. Verstraelen, and K. Leonhard, *A Reactive Molecular Dynamics Study of Chlorinated Organic Compounds. Part II: A ChemTraYzer Study of Chlorinated Dibenzofuran Formation and Decomposition Processes*, ChemPhysChem doi:10.1002/cphc.202200783

[123] J.-C. Lizardo-Huerta, B. Sirjean, L. Verdier, R. Fournet, and P.-A. Glaude, *Thermal Decomposition of Phosgene and Diphosgene*, J. Phys. Chem. A 122(1):249–257, 2018, doi:10.1021/acs.jpca.7b09554

[124] H. Babad and A. G. Zeiler, *Chemistry of phosgene*, Chem. Rev. 73(1):75–91, 1973, doi:10.1021/cr60281a005

[125] D. Noort, A. G. Hulst, A. Fidder, R. A. van Gurp, L. P. A. de Jong, and H. P. Benschop, *In Vitro Adduct Formation of Phosgene with Albumin and Hemoglobin in Human Blood*, Chem. Res. Toxicol. 13(8):719–726, 2000, doi:10.1021/tx000022z

[126] V. B. Arce, C. O. Della Védova, A. J. Downs, S. Parsons, and R. M. Romano, *Trichloromethyl Chloroformate ("Diphosgene"), ClC(O)OCCl3: Structure and Conformational Properties in the Gaseous and Condensed Phases*, J. Org. Chem. 71(9):3423–3428, 2006, doi:10.1021/jo052260a

[127] P. S. Kulkarni, J. G. Crespo, and C. A. Afonso, *Dioxins sources and current remediation technologies — A review*, Environ. Int. 34(1):139–153, 2008, doi: 10.1016/j.envint.2007.07.009

[128] D. Consonni, A. C. Pesatori, C. Zocchetti, R. Sindaco, L. C. D'Oro, M. Rubagotti, and P. A. Bertazzi, *Mortality in a Population Exposed to Dioxin after the Seveso, Italy, Accident in 1976: 25 Years of Follow-Up*, Am. J. Epidemiol. 167(7):847–858, 2008, doi:10.1093/aje/kwm371

[129] R. Madaj, E. Sobiecka, and H. Kalinowska, *Lindane, kepone and pentachlorobenzene: chloropesticides banned by Stockholm convention*, IJEST 15(2):471–480, 2018, doi:10.1007/s13762-017-1417-9

[130] I. Amato, *The crusade against chlorine*, Science 261(5118):152–154, 1993, doi:10.1126/science.8327884

[131] M. Altarawneh, B. Z. Dlugogorski, E. M. Kennedy, and J. C. Mackie, *Quantum Chemical Study of Low Temperature Oxidation Mechanism of Dibenzofuran*, J. Phys. Chem. A 110(50):13560–13567, 2006, doi:10.1021/jp065135r

[132] M. Altarawneh, B. Z. Dlugogorski, E. M. Kennedy, and J. C. Mackie, *Theoretical Study of Reaction Pathways of Dibenzofuran and Dibenzo-p-Dioxin under Reducing Conditions*, J. Phys. Chem. A 111(30):7133–7140, 2007, doi:10.1021/jp071166q

[133] M. Altarawneh, E. M. Kennedy, B. Z. Dlugogorski, and J. C. Mackie, *Computational Study of the Oxidation and Decomposition of Dibenzofuran under Atmospheric Conditions*, J. Phys. Chem. A 112(30):6960–6967, 2008, doi:10.1021/jp800093j

[134] M. Altarawneh, M. W. Radny, P. V. Smith, J. C. Mackie, E. M. Kennedy, B. Z. Dlugogorski, A. Soon, and C. Stampfl, *Adsorption of 2-chlorophenol on Cu2O(111)–CuCUS: A first-principles density functional study*, Appl. Surf. Sci. 256(15):4764–4770, 2010, doi:10.1016/j.apsusc.2010.01.101

[135] M. Altarawneh and B. Z. Dlugogorski, *Formation of dibenzofuran, dibenzo-p-dioxin and their hydroxylated derivatives from catechol*, Phys. Chem. Chem. Phys. 17:1822–1830, 2015, doi:10.1039/C4CP04168B

[136] N. Bai, W. Wang, Y. Zhao, W. Feng, and P. Li, *Theoretical Insights into the Reaction Mechanism between 2,3,7,8-Tetrachlorodibenzofuran and Hydrogen Peroxide: A DFT Study*, ACS Omega 4(1):358–367, 2019, doi:10.1021/acsomega.8b00724

[137] K. Zhang, S. Sun, and H. Zhang, *Mechanism and kinetic study on the ring-opening degradation of 2,3,7,8-tetrachlorinated dibenzofuran initiated by OH radicals in waste incineration*, RSC Adv. 5:81153–81161, 2015, doi:10.1039/C5RA13804C

[138] W. Wang, Y. Wang, W. Feng, W. Wang, and P. Li, *Theoretical Investigations on the Reactivity of Hydrogen Peroxide toward 2,3,7,8-Tetrachlorodibenzo-p-dioxin*, Molecules 23(11), 2018, doi:10.3390/molecules23112826

[139] Z. Wen, Z. Wang, J. Xu, Y. Liu, and K. Cen, *Quantum chemistry study on the destruction mechanism of 2,3,7,8-TCDD by OH and O3 radicals*, Chemosphere 92(3):293–298, 2013, doi:10.1016/j.chemosphere.2013.03.027

[140] Z. Wen, P. Kang, Z. Wang, J. Xu, and K. Cen, *Mechanism of Polychlorinated Dibenzo-p-dioxins Degradation by Ozone*, Asian J. Chem. 27(3), 2015, doi: 10.14233/ajchem.2015.18266

[141] Z. Wen, S. Li, H. Li, and Y. Li, *A theoretical study on the mechanism and kinetics of toxic PCDD destruction by OH radicals*, IJEP 65(4):337–349, 2019, doi:10.1504/IJEP.2019.103749

[142] N. Zeinali, I. Oluwoye, M. Altarawneh, and B. Z. Dlugogorski, *Destruction of dioxin and furan pollutants via electrophilic attack of singlet oxygen*, Ecotoxicol. Environ. Saf. 184:109605, 2019, doi:10.1016/j.ecoenv.2019.109605

[143] Y. Fernández Pulido, E. Suárez, R. López, and M. I. Menéndez, *The role of CuCl on the mechanism of dibenzo-p-dioxin formation from poly-chlorophenol precursors: A computational study*, Chemosphere 145:77–82, 2016, doi:10.1016/j.chemosphere.2015.11.042

[144] X. Sun, C. Zhang, Y. Zhao, J. Bai, Q. Zhang, and W. Wang, *Atmospheric Chemical Reactions of 2,3,7,8-Tetrachlorinated Dibenzofuran Initiated by an OH Radical: Mechanism and Kinetics Study*, Environ. Sci. Technol. 46(15):8148–8155, 2012, doi: 10.1021/es301413v

[145] C. J. E. Bajamundi, P. Vainikka, M. Hedman, and J. Konttinen, *Towards controlling PCDD/F production in a multi-fuel fired BFB boiler using two sulfur addition strategies. Part II: Thermodynamic analysis*, Fuel 134:688–697, 2014, doi: https://doi.org/10.1016/j.fuel.2014.05.033

[146] O. V. Dorofeeva, V. S. Iorish, and N. F. Moiseeva, *Thermodynamic Properties of Dibenzo-p-dioxin, Dibenzofuran, and Their Polychlorinated Derivatives in the Gaseous and Condensed Phases. 1. Thermodynamic Properties of Gaseous Compounds*, J. Chem. Eng. Data 44(3):516–523, 1999, doi:10.1021/je980238r

[147] D. Thompson and B. C. R. Ewan, *A Group Additivity Algorithm for Polychlorinated Dibenzofurans Derived from Selected DFT Analyses*, J. Phys. Chem. A 111(23):5043–5047, 2007, doi:10.1021/jp0709499

[148] M. Takaoka, A. Shiono, K. Nishimura, T. Yamamoto, T. Uruga, N. Takeda, T. Tanaka, K. Oshita, T. Matsumoto, and H. Harada, *Dynamic Change of Copper in Fly Ash during de Novo Synthesis of Dioxins*, Environ. Sci. Technol. 39(15):5878–5884, 2005, doi:10.1021/es048019f

[149] P.-M. Marquaire, R. Bounaceur, and F. Baronnet, *Detailed mechanism of dibenzofuran oxidation*, Organohalogen Compounds 63:377–380, 2003

[150] L. Komissarov, L. Krep, F. Schmalz, W. Kopp, K. Leonhard, and T. Verstraelen, *A Reactive Molecular Dynamics Study of Chlorinated Organic Compounds. Part I: Force Field Development*, ChemPhysChem doi:10.1002/cphc.202200786

[151] M. Altarawneh, B. Dlugogorski, E. M. Kennedy, and J. C. Mackie, *MECHANISMS FOR PCDF AND PCB FORMATION FROM FIRES: PATHWAYS FROM OXIDATION OF CHLOROBENZENES*, AOFST, 2007

[152] R. J. Kee, F. M. Rupley, and J. A. Miller, *Chemkin-II: A Fortran chemical kinetics package for the analysis of gas-phase chemical kinetics*, 1989, doi:10.2172/5681118

[153] Rüger, Franchini, Trnka, Yakovlev, van Lenthe, Philipsen, van Vuren, B. Klumpers, and T. Soini, *Amsterdam Modeling Suite*, URL https://scm.com

[154] M. Hutchings, J. Liu, Y. Qiu, C. Song, and L.-P. Wang, *Bond-Order Time Series Analysis for Detecting Reaction Events in Ab Initio Molecular Dynamics Simulations*, JCTC 16(3):1606–1617, 2020, doi:10.1021/acs.jctc.9b01039

[155] T. R. Cundari, *Computational Organometallic Chemistry -*, CRC Press, Boca Raton, Fla, 2001, ISBN 978-0-824-70478-0

[156] G. Henkelman, B. P. Uberuaga, and H. Jónsson, *A climbing image nudged elastic band method for finding saddle points and minimum energy paths*, J. Chem. Phys. 113(22):9901–9904, 2000, doi:10.1063/1.1329672

[157] P. M. Zimmerman, *Navigating Molecular Space for Reaction Mechanisms: An Efficient, Automated Procedure*, Mol. Simul. 41(1-3):43–54, 2015, doi:10.1080/08927022.2014.894999

[158] L. Pattanaik, J. B. Ingraham, C. A. Grambow, and W. H. Green, *Generating transition states of isomerization reactions with deep learning*, PCCP 22(41):23618–23626, 2020, doi:10.1039/d0cp04670a

[159] W. Kabsch, *A Solution for the Best Rotation to Relate Two Sets of Vectors*, Acta Crystallogr., Sect. A: Found. Crystallogr. 32(5):922–923, 1976, doi:10.1107/S0567739476001873

[160] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, *et al.*, *SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python*, Nat. Methods 17(3):261–272, 2020, doi:10.1038/s41592-019-0686-2

[161] M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, G. A. Petersson, H. Nakatsuji, X. Li, M. Caricato, A. Marenich, J. Bloino, B. G. Janesko, R. Gomperts, B. Mennucci, H. P. Hratchian, J. V. Ortiz, A. F. Izmaylov, *et al.*, *Gaussian 09, Revision D.01*, 2013

[162] M. Saeys, M.-F. c. Reyniers, G. B. Marin, V. Van Speybroeck, and M. Waroquier, *Ab Initio Calculations for Hydrocarbons: Enthalpy of Formation, Transition State Geometry, and Activation Energy for Radical Reactions*, J. Phys. Chem. A 107(43):9147–9159, 2003, doi:10.1021/jp021706d

[163] M. K. Cieplik, O. J. Epema, and R. Louw, *Thermal Hydrogenolysis of Dibenzo-p-dioxin and Dibenzofuran*, Eur. J. Org. Chem. 2002(16):2792–2799, 2002, doi:10.1002/1099-0690(200208)2002:16<2792::AID-EJOC2792>3.0.CO;2-H

[164] B. Yang, L. Gagliardi, and D. G. Truhlar, *Transition states of spin-forbidden reactions*, Phys. Chem. Chem. Phys. 20:4129–4136, 2018, doi:10.1039/C7CP07227A

[165] A. Behn, P. M. Zimmerman, A. T. Bell, and M. Head-Gordon, *Efficient exploration of reaction paths via a freezing string method*, J. Chem. Phys. 135(22):224108, 2011, doi:10.1063/1.3664901

[166] S. Maeda, T. Taketsugu, and K. Morokuma, *Exploring transition state structures for intramolecular pathways by the artificial force induced reaction method*, J. Comput. Chem. 35(2):166–173, 2014, doi:10.1002/jcc.23481

[167] S. Maeda, Y. Harabuchi, M. Takagi, T. Taketsugu, and K. Morokuma, *Artificial Force Induced Reaction (AFIR) Method for Exploring QuantumChemical Potential Energy Surfaces*, Chem. Rec. 16(5):2232–2248, 2016, doi:10.1002/tcr.201600043

[168] Z. G. Nicolaou and A. E. Motter, *Missing links as a source of seemingly variable constants in complex reaction networks*, Phys. Rev. Research 2:043135, 2020, doi:10.1103/PhysRevResearch.2.043135

[169] R. J. Kee, F. M. Rupley, and J. A. Miller, *The Chemkin Thermodynamic Data Base* , 1990, doi:10.2172/7073290

[170] N. M. Donahue, *Reaction Barriers: Origin and Evolution*, Chem. Rev. 103(12):4593–4604, 2003, doi:10.1021/cr020650g